

MODEL RUN TRANSCRIPT

Generated: 2026-06-04T16:26:51

Source: data/json/sonnet45_8agent_game_directive_r10_n5/claude-sonnet-4.5/game/sonnet45_8agent_game_directive_r10_n5_004_neutral_rep04.json

RUN METADATA

model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 3
myth_topic_id: anything
myth_topic: anything
replicate_id: 4
myth_default_prompt_key: myth_writing_default_game_directive
myth_later_prompt_key: myth_writing_later_rounds_directive
task_order: [
 "game"
]

agent_names: {
 "Agent_1": "Aster",
 "Agent_2": "Briar",
 "Agent_3": "Cyra",
 "Agent_4": "Dorian",
 "Agent_5": "Elara",
 "Agent_6": "Finn",
 "Agent_7": "Galen",
 "Agent_8": "Hana"
}
dyadic_pairing_schedule: {
 "num_turns": 10,
 "role_targets": {
 "Agent_1": 5,
 "Agent_2": 5,
 "Agent_3": 5,
 "Agent_4": 5,
 "Agent_5": 5,
 "Agent_6": 5,
 "Agent_7": 5,
 "Agent_8": 5
 },
 "strategy": "balanced_random_roles_v1"
}

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Agent_3 (Cyra); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Agent_4 (Dorian); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Agent_5 (Elara); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Agent_6 (Finn); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Agent_7 (Galen); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Agent_8 (Hana); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

ROUND 1

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_6 (Finn)=Receiver
 dyad_2: Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver
 dyad_3: Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver
 dyad_4: Agent_2 (Briar)=Sender, Agent_5 (Elara)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$4; received \$12; Agent_6 (Finn) returned \$6; payoffs Agent_1 (Aster)=\$7, Agent_6 (Finn)=\$6
 Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
 dyad_2: Agent_7 (Galen) sent \$3; received \$9; Agent_8 (Hana) returned \$4; payoffs Agent_7 (Galen)=\$6, Agent_8 (Hana)=\$5
 Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

```

dyad_3: Agent_3 (Cyra) sent $3; received $9; Agent_4 (Dorian) returned $4.5; payoffs Agent_3
(Cyra)=$6.5, Agent_4 (Dorian)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_2 (Briar) sent $2; received $6; Agent_5 (Elara) returned $3; payoffs Agent_2
(Briar)=$6, Agent_5 (Elara)=$3
  Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 6.0, "min": 0, "raw":
3.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}
Balances after round: {
  "Agent_1": 7.0,
  "Agent_2": 6.0,
  "Agent_3": 6.5,
  "Agent_4": 4.5,
  "Agent_5": 3.0,
  "Agent_6": 6.0,
  "Agent_7": 6.0,
  "Agent_8": 5.0
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T16:13:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_6": "Finn"
    },
    "agents": [
      "Agent_1",
      "Agent_6"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. By sending a substantial amount, I can signal trust and potentially build a reciprocal relationship that benefits us both over multiple rounds.

If I send \$4, Finn receives \$12 (due to the 3x multiplier). If Finn returns a fair share, we both end up better off than if I played it safe and sent nothing.

This is a trust-building move that could lead to mutual benefit in subsequent rounds.

{'send': 4}

Reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Aster) and my opponent is Finn (Agent_6).

Let me think about the strategic considerations:

1. As the sender, I receive \$5 and can send \$0-\$5 to the receiver
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. My payoff: \$5 - sent + returned
5. Receiver's payoff: 3xsent - returned

This is a classic trust game. The socially optimal outcome is to send all \$5, which becomes \$15 for the receiver. If the receiver returns \$8-\$10, we're both better off than if I send nothing.

However, there's a risk: the receiver could keep everything and return \$0. In that case:

- If I send \$5: I get \$0, receiver gets \$15
- If I send \$0: I get \$5, receiver gets \$0

Since this is an iterated game with multiple rounds, there are opportunities for:

1. Building trust and cooperation
2. Establishing reputation
3. Reciprocity patterns

In the first round, I don't know anything about Finn's behavior. Some strategies:

1. ****Trust approach****: Send a substantial amount (\$4-\$5) to signal trust and cooperation
2. ****Cautious approach****: Send a moderate amount (\$2-\$3) to test the waters
3. ****Conservative approach****: Send little or nothing (\$0-\$1) to avoid risk

Given that this is iterated and I want to establish potential cooperation, I think a moderate-to-high amount makes sense. Sending \$3-\$4 shows willingness to cooperate while not being completely vulnerable.

If I send \$3:

- Finn receives \$9
- If Finn returns \$4-\$5, we both benefit
- I'd end with \$5 - \$3 + \$4-5 = \$6-\$7

- Finn would end with \$9 - \$4-5 = \$4-\$5

Sending \$4 creates better mutual outcomes: Finn gets \$12, and if he returns \$5-\$7, I end up with \$6-\$8 while he keeps \$5-\$7. This stronger signal of cooperation sets a better foundation for future rounds, so I'll send \$4.

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 730,
  "reasoning_tokens": 492
}
```

GAME - Agent_6 - Receiver - dyad_1

Timestamp: 2026-06-04T16:14:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_6"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$4.0, that is 80.0% of its total endowment. You receive \$12.0. How much do you return? (0-12.0)

Assistant output:

Looking at this situation:

- Aster showed significant trust by sending \$4 (80% of their endowment)
- This created \$12 for me
- I want to reciprocate this trust and ensure we both benefit

If I return \$6:

- Aster ends with: \$1 (kept) + \$6 (returned) = \$7 total (gained \$2)
- I end with: \$12 - \$6 = \$6 total (gained \$6)

This reciprocates Aster's generous trust, ensures we both profit from cooperating, and establishes a positive foundation for potential future interactions.

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Let me think about this situation carefully.

I'm Finn, playing as the RECEIVER in Round 1.

- The sender (Aster/Agent_1) received \$5
- Aster sent me \$4.0 (80% of their endowment)
- I received $\$4.0 \times 3 = \12.0
- I need to decide how much to return (between \$0 and \$12.0)

Let me consider what's happening here:

- Aster sent a very generous amount (\$4 out of \$5, keeping only \$1)
- This shows high trust in me
- If I return nothing, Aster ends up with just \$1, while I keep \$12
- If I return some amount, we can both benefit

From a fairness perspective:

- Aster invested \$4 and kept \$1
- This created \$12 for me
- A fair split might aim for equal outcomes or proportional returns

Let me think about different return amounts:

- If I return \$0: Aster gets \$1 total, I get \$12 (very unfair to Aster)
- If I return \$6: Aster gets $\$1 + \$6 = \$7$, I get $\$12 - \$6 = \$6$ (I come out slightly ahead)
- If I return \$7: Aster gets $\$1 + \$7 = \$8$, I get $\$12 - \$7 = \$5$ (Aster comes out ahead)
- If I return \$5: Aster gets $\$1 + \$5 = \$6$, I get $\$12 - \$5 = \$7$ (I come out slightly ahead)

Since Aster showed high trust by sending 80%, I should reciprocate with a generous return. This is also strategic for future rounds - if we play again, being trustworthy encourages future cooperation.

A reasonable return would be around \$5-\$6:

- Return \$6: Aster ends with \$7, I end with \$6 (both better than the initial \$5)

- Return \$5: Aster ends with \$6, I end with \$7 (both better than \$5, I gain slightly more)

Aster's willingness to send 80% signals genuine trust, so I should match that with a meaningful return. Returning \$6 feels like the right balance-it rewards their risk-taking while still leaving me with a solid outcome, and it demonstrates I'm trustworthy for any future interactions.

Let me think through the math more carefully. If I return \$6, Aster walks away with \$7 total (a \$2 gain from their initial \$5), while I get \$6 (a \$6 gain). Yes, I benefit more, but that's somewhat justified since they initiated the risk. Returning only \$5 would give them just \$6 total-barely better than their starting point-which feels like I'm not properly acknowledging their leap of faith. The \$6 return strikes the right tone of reciprocity.

```
Usage: {
  "input_tokens": 562,
  "output_tokens": 962,
  "reasoning_tokens": 585
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-06-04T16:14:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_7": "Galen",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

```
Message 1 role=system
content:
```

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative relationship with Hana while being reasonably cautious since we haven't interacted before.

The 3x multiplier creates potential for mutual benefit - if I send money and Hana returns a fair portion, we can both end up better off than if I keep everything.

I'll send a moderate amount to signal willingness to cooperate while not being overly vulnerable.

****My decision:****

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for Hana. If Hana returns around \$3-4, we'll both benefit from this exchange and can build trust for future rounds.

Reasoning:

This is the first round of an iterated game where I'm playing as the sender. Let me think about the strategic considerations:

1. ****Game structure:****
 - I receive \$5
 - I can send \$0-\$5 to Hana
 - The amount I send gets multiplied by 3x
 - Hana decides how much to return
 - I keep: \$5 - sent + returned
 - Hana keeps: sent×3 - returned
2. ****Strategic considerations:****
 - If I send \$0, I keep \$5 and Hana gets \$0. This is safe but doesn't build cooperation.
 - If I send the full \$5, Hana receives \$15. If Hana returns at least \$5, I break even. If Hana returns more, we both benefit.
 - The multiplier effect (3x) means there's potential for mutual benefit through cooperation.
 - This is an iterated game with multiple rounds, so building trust could pay off long-term.
 - I don't know Hana yet, so I don't have information about their strategy.
3. ****Common strategies in trust games:****
 - ****Reciprocity approach:**** Start with moderate cooperation and adjust based on partner's response
 - ****Trust building:**** Send a meaningful amount to signal willingness to cooperate
 - ****Risk averse:**** Send a small amount to test the waters
4. ****First round consideration:****
 - Since this is round 1, I need to balance:
 - Showing willingness to cooperate
 - Not being too vulnerable to exploitation
 - Maximizing total surplus (which requires sending something)
5. ****Optimal strategy for first round:****
 - Sending around \$2-\$3 seems reasonable:
 - \$2 becomes \$6 for Hana. If Hana returns \$2-\$4, we both benefit
 - \$3 becomes \$9 for Hana. If Hana returns \$3-\$5, we both benefit significantly
 - This demonstrates trust without excessive risk. I'll go with \$3 as my opening move-it signals genuine willingness to cooperate while keeping me reasonably protected, and it gives Hana enough upside (\$9) that reciprocating makes sense for both of us.

Usage: {
 "input_tokens": 530,
 "output_tokens": 668,
 "reasoning_tokens": 499
}

GAME - Agent_8 - Receiver - dyad_2

Timestamp: 2026-06-04T16:14:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_7",
 "pairing": {
 "agent_names": {
 "Agent_7": "Galen",
 "Agent_8": "Hana"
 },
 },
 "agents": [
 "Agent_7",
 "Agent_8"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_7": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",

```
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.

Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Assistant output:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplier
- Total wealth created: \$11 (Galen has \$2 left, I have \$9)

Galen demonstrated trust by sending a substantial portion of their endowment. I want to reciprocate fairly while also benefiting from their trust.

If I split the total wealth approximately equally:

- Returning \$4 would give Galen \$6 total (gain of \$1)
- I would keep \$5 (gain of \$5)

This rewards Galen's trust while acknowledging that as the receiver, I also benefit from the i

interaction. This should encourage continued cooperation.

```
'''json
{"return": 4.0}
'''
```

Reasoning:

Let me understand the situation:

- I'm Hana, the RECEIVER in this round
- My opponent is Galen (Agent_7), who is the SENDER
- Galen sent me \$3.0 out of their \$5.0 endowment (60%)
- I receive $\$3.0 \times 3 = \9.0
- I need to decide how much to return (between \$0 and \$9.0)
- Galen will keep: $\$5 - \$3 + [\text{what I return}] = \$2 + [\text{what I return}]$
- I will keep: $\$9 - [\text{what I return}]$

This is the classic trust game. Galen showed considerable trust by sending 60% of their endowment. This is the first round, so I need to establish a baseline for reciprocity.

Some considerations:

1. If I return nothing, I get \$9 and Galen gets \$2 (unfair to Galen who trusted me)
2. If I return \$3, I get \$6 and Galen gets \$5 (Galen breaks even, I gain \$6)
3. If I return \$4.5, I get \$4.5 and Galen gets \$6.5 (we both gain, roughly split the surplus)
4. If I return \$6, I get \$3 and Galen gets \$8 (Galen gains more than me)

The gain from cooperation is $\$9 - \$3 = \$6$ (the amount created by the 3x multiplier). A fair split would give us each \$3 of this gain:

- Galen ends with: $\$2 + \$3 = \$5$ (breaks even)
- I end with: $\$9 - \$3 = \$6$ (gain \$6)

Wait, that's not quite right. Let me reconsider.

Initial state: Galen has \$5, I have \$0

After send: Galen has \$2, I have \$9

The transaction created \$6 in total surplus. If I split this equally, Galen gets \$3 back (ending at \$5) and I keep \$6. But since I started with nothing while Galen started with \$5, a 50/50 split of the final \$11 pool might be more equitable-each of us ending with \$5.50.

To achieve that, I'd return \$3.50 to Galen. I could also consider a reciprocity approach: since Galen sent 60% of their endowment, returning 60% of what I received (\$5.40) would give them \$7.40 total, though that feels overly generous.

A middle ground makes more sense-returning \$4 to \$4.50 rewards their trust while letting me benefit meaningfully from receiving the money. I'll go with \$4, which gives Galen \$6 (a \$1 gain) and leaves me with \$5 (a \$5 gain). This balances reciprocity with my own interests and should encourage continued cooperation in future rounds.

Checking the fairness: with \$11 total, a 50/50 split would be \$5.50 each. Galen starts with \$2, so they'd need \$3.50 more to reach that. Returning \$4 is slightly above that threshold, which should build stronger trust without being excessive.

```
Usage: {
  "input_tokens": 564,
  "output_tokens": 1268,
  "reasoning_tokens": 610
}
```

GAME - Agent_3 - Sender - dyad_3

Timestamp: 2026-06-04T16:15:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyrus",
      "Agent_4": "Dorian"
```

```

},
"agents": [
  "Agent_3",
  "Agent_4"
],
"investor": "Agent_3",
"roles": {
  "Agent_3": "investor",
  "Agent_4": "trustee"
},
"trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance cooperation with caution since I don't yet know how Dorian will respond.

The game rewards mutual trust - if I send money and Dorian returns a fair amount, we both benefit from the 3x multiplier. However, this is our first interaction, so I want to signal willingness to cooperate while not being overly vulnerable.

I'll send **\$3**. This amount:

- Shows I'm cooperative and willing to trust
- Gets tripled to \$9, giving Dorian meaningful options
- Leaves me with \$2 as a safeguard
- If Dorian returns even \$2-3, we both come out ahead

This sets a foundation for building trust over multiple rounds.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of an iterated game. I'm playing as the sender with Dorian (Agent_4).

Let me think about the game theory here:

- I receive \$5
- I can send \$0-\$5 to the receiver
- The amount sent gets multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal outcome for both of us collectively would be for me to send all \$5, which becomes \$15 for the receiver. If the receiver returns half (\$7.50), we'd both end up with \$7.50, which is better than the \$5 I start with.

However, there's a risk: the receiver could return nothing, leaving me with \$0 and them with \$15.

Since this is an iterated game with multiple rounds, I should consider:

1. Building trust early could lead to better outcomes over time
2. Being too trusting initially might be exploited
3. A moderate initial move might signal cooperation while testing the waters

Common strategies in iterated trust games:

- Start with moderate cooperation to signal willingness but test the waters
- Reciprocate based on partner's behavior in subsequent rounds

For a first round, I think sending \$3 is reasonable:

- It shows I'm willing to cooperate
- It's not so much that I'm devastated if they return nothing
- If they send back at least \$3, we both benefit
- It gives them \$9 to work with, enough to show their intentions

Let me calculate:

- If I send \$3: I have \$2 left
- Receiver gets \$9
- If they return \$4, I end with \$6, they end with \$5
- If they return \$5, I end with \$7, they end with \$4
- If they return \$0, I end with \$2, they end with \$9

I'll go with \$3 as my opening move.

```
Usage: {
  "input_tokens": 530,
  "output_tokens": 642,
  "reasoning_tokens": 448
}
```

GAME - Agent_4 - Receiver - dyad_3

Timestamp: 2026-06-04T16:15:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_3",
"pairing": {
  "agent_names": {
    "Agent_3": "Cyra",
    "Agent_4": "Dorian"
  },
  "agents": [
    "Agent_3",
    "Agent_4"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Assistant output:

Looking at this first round situation:

- Cyra (the sender) showed significant trust by sending \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- This is an iterated game, so establishing cooperation early is valuable

Cyra currently has \$2.0 left (the \$5 minus \$3 sent). If I return a reasonable amount, we can both benefit and build trust for future rounds.

I'll return \$4.5, which means:

- Cyra ends with: $\$2.0 + \$4.5 = \$6.5$
- I keep: $\$9.0 - \$4.5 = \$4.5$

This gives us both equal final amounts and rewards Cyra's trust while being fair to both of us

{'return': 4.5}

Reasoning:

Let me understand the situation:

1. I am the RECEIVER (Dorian)
2. The SENDER (Cyra) received \$5 and sent me \$3.0 (60% of their endowment)
3. The amount sent was multiplied by 3x, so I received \$9.0
4. I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

This is Round 1, and this is an iterated game with multiple rounds. Key considerations:

- The sender (Cyra) showed trust by sending 60% of their endowment
- In an iterated game, cooperation and reciprocity can lead to better long-term outcomes
- If I return nothing or very little, Cyra might not trust me in future rounds when roles could be reversed
- If I return too much, I'm being overly generous at my own expense

Let me calculate fair and strategic return amounts:

If the sender sent \$3.0:

- They kept \$2.0
- I received \$9.0 (after 3x multiplier)

A "fair" split might be:

- If I return \$4.5, the sender ends up with $\$2.0 + \$4.5 = \$6.5$
- I keep $\$9.0 - \$4.5 = \$4.5$
- This gives us both equal amounts

A slightly more generous return:

- If I return \$5.0, the sender ends up with $\$2.0 + \$5.0 = \$7.0$
- I keep $\$9.0 - \$5.0 = \$4.0$
- This rewards their trust while still keeping a reasonable amount

A more strategic return (keeping more for myself but still showing cooperation):

- If I return \$4.0, the sender ends up with $\$2.0 + \$4.0 = \$6.0$
- I keep $\$9.0 - \$4.0 = \$5.0$

- This shows I'm cooperative but also value my own payoff

Given that this is the first round and I want to build trust for future interactions, I should return something that demonstrates reciprocity without sacrificing too much. Returning \$4.5 strikes the right balance-it acknowledges Cyra's generosity, signals my willingness to cooperate, and ensures I'm not being exploited while still leaving room for mutual benefit in subsequent rounds.

```
bsequent rounds.
Usage: {
  "input_tokens": 564,
  "output_tokens": 789,
  "reasoning_tokens": 485
}
```

GAME - Agent_2 - Sender - dyad_4

Timestamp: 2026-06-04T16:15:36
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_5": "Elara"
    },
    "agents": [
      "Agent_2",
      "Agent_5"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_5": "trustee"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

I'll start with a cooperative but cautious approach. Since this is the first round and I don't know Elara's strategy yet, I'll send a moderate amount to signal willingness to cooperate while not overexposing myself.

If I send \$2, Elara will receive \$6 (2×3). This gives them a meaningful amount to work with and shows good faith, while I retain \$3 initially. If Elara returns a fair amount, we can both benefit and build trust for future rounds.

{'send': 2}

Reasoning:

This is the first round of an iterated game. I'm the sender with \$5.

Let me think about the optimal strategy:

- If I send \$X, my opponent receives \$3X
- My opponent then decides how much to return
- I keep: \$5 - X + (amount returned)
- Opponent keeps: \$3X - (amount returned)

Since this is an iterated game with multiple rounds, I should consider:

1. Building trust/cooperation could lead to mutual benefits
2. Being too generous initially might be exploited
3. Being too stingy might lead to no cooperation

A common strategy in trust games is to send a moderate amount initially to signal willingness to cooperate while not being too vulnerable. Sending around \$2-3 is often considered reasonable.

If I send \$2:

- Opponent receives \$6
- If they return \$3, we'd both end up with \$6 (\$5-2+3 for me, \$6-3 for them)
- This would be mutually beneficial

If I send \$3:

- Opponent receives \$9
- If they return \$4, I'd have \$6 and they'd have \$5
- Or if they return \$5, I'd have \$7 and they'd have \$4

Let me go with \$2 as a reasonable starting point that signals cooperation without being too risky.

Usage: {

"input_tokens": 530,
"output_tokens": 437,
"reasoning_tokens": 293

}

GAME - Agent_5 - Receiver - dyad_4

Timestamp: 2026-06-04T16:15:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_4",

```

"move_index": 7,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_2": "Briar",
    "Agent_5": "Elara"
  },
  "agents": [
    "Agent_2",
    "Agent_5"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_5": "trustee"
  },
  "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
 Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$2.0, that is 40.0% of its total endowment. You receive \$6.0. How much do you return? (0 -6.0)

Assistant output:

Looking at this first round situation:

- Briar sent \$2.0 (40% of their endowment), showing a moderate level of trust
- I received \$6.0 after the 3x multiplier
- This created \$6.0 of value through cooperation

Since this is an iterated game and I want to establish a cooperative relationship, I should return a fair amount that:

1. Rewards Briar's trust
2. Ensures both players benefit
3. Encourages future cooperation

I'll split the \$6.0 received amount equally, returning \$3.0. This gives:

- Briar: \$3 (kept) + \$3 (returned) = \$6 total
- Me: \$6 (received) - \$3 (returned) = \$3 total

This demonstrates I'm a trustworthy cooperator and should encourage continued positive interactions.

```
'''json
{"return": 3.0}
'''
```

Reasoning:

Let me understand the situation:

1. I am Elara, playing as the RECEIVER in Round 1
2. My opponent is Briar (Agent_2), who is the SENDER
3. Briar sent me \$2.0 out of their \$5 endowment (40%)
4. I receive $2.0 \times 3 = \$6.0$
5. I need to decide how much to return (between \$0 and \$6.0)

Let me think about the strategic considerations:

This is Round 1 of an iterated game. The key considerations are:

- Building trust for future rounds
- Being fair/reciprocal
- Maximizing long-term payoffs

If this were a one-shot game, the rational selfish move would be to return \$0. However, since this is iterated and I'll be playing multiple rounds (possibly with the same opponent again), I need to establish cooperation.

Briar sent 40% of their endowment, which shows some level of trust and cooperation. A reasonable reciprocal approach would be to return an amount that:

1. Rewards their cooperative behavior
2. Establishes me as trustworthy for future interactions
3. Ensures both players benefit

Let me calculate fair returns:

- Briar started with \$5, sent \$2, so has \$3 left
- I received \$6
- If I return nothing: Briar gets \$3 total, I get \$6 total
- If I return \$3 (50%): Briar gets \$6 total, I get \$3 total
- If I return \$2: Briar gets \$5 total, I get \$4 total
- If I return \$2.5: Briar gets \$5.5 total, I get \$3.5 total

A common approach in trust games is to aim for equality or to ensure the sender doesn't lose. Returning \$2 would give Briar \$5 (breaking even with their initial endowment) and me \$4 (still a profit).

Another way to frame this: Briar's \$2 investment generated \$6 through the multiplier. If I spl

it the gains equally, returning \$3 means we each end with \$3 profit beyond Briar's original \$3 kept. Alternatively, returning \$2 ensures Briar doesn't lose money while I still gain \$4.

I'm leaning toward returning \$3 as a cooperative opening move. While behavioral economics suggests 1/3 of received funds (\$2) is considered fair, the equal split signals stronger commitment to mutual benefit and should build trust for future rounds.

```
Usage: {
  "input_tokens": 564,
  "output_tokens": 996,
  "reasoning_tokens": 540
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver
dyad_2: Agent_8 (Hana)=Sender, Agent_2 (Briar)=Receiver
dyad_3: Agent_4 (Dorian)=Sender, Agent_1 (Aster)=Receiver
dyad_4: Agent_5 (Elara)=Sender, Agent_3 (Cyra)=Receiver

Parsed game decisions:

dyad_1: Agent_6 (Finn) sent \$4; received \$12; Agent_7 (Galen) returned \$6; payoffs Agent_6 (Finn)=\$7, Agent_7 (Galen)=\$6
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_8 (Hana) sent \$3; received \$9; Agent_2 (Briar) returned \$5; payoffs Agent_8 (Hana)=\$7, Agent_2 (Briar)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_4 (Dorian) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_4 (Dorian)=\$6.5, Agent_1 (Aster)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_5 (Elara) sent \$2.5; received \$7.5; Agent_3 (Cyra) returned \$3.75; payoffs Agent_5 (Elara)=\$6.25, Agent_3 (Cyra)=\$3.75
Action validation: {"returned": {"amount": 3.75, "clamped": false, "max": 7.5, "min": 0, "raw": 3.75}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round:

```
{
  "Agent_1": 11.5,
  "Agent_2": 10.0,
  "Agent_3": 10.25,
  "Agent_4": 11.0,
  "Agent_5": 9.25,
  "Agent_6": 13.0,
  "Agent_7": 12.0,
  "Agent_8": 12.0
}
```

LLM CALLS

GAME - Agent_6 - Sender - dyad_1

Timestamp: 2026-06-04T16:16:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_6": "Finn",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_6",
    "Agent_7"
  ],
}
```



```

"investor": "Agent_6",
"roles": {
  "Agent_6": "investor",
  "Agent_7": "trustee"
},
"trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$4.0, that is 80.0% of its total endowment. You receive \$12.0. How much do you return? (0-12.0)

Message 3 role=assistant

content:

Looking at this situation:

- Aster showed significant trust by sending \$4 (80% of their endowment)
- This created \$12 for me
- I want to reciprocate this trust and ensure we both benefit

If I return \$6:

- Aster ends with: \$1 (kept) + \$6 (returned) = \$7 total (gained \$2)
- I end with: \$12 - \$6 = \$6 total (gained \$6)

This reciprocates Aster's generous trust, ensures we both profit from cooperating, and establishes a positive foundation for potential future interactions.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me think about this situation carefully.

I'm Finn, playing as the RECEIVER in Round 1.

- The sender (Aster/Agent_1) received \$5
- Aster sent me \$4.0 (80% of their endowment)
- I received $\$4.0 \times 3 = \12.0
- I need to decide how much to return (between \$0 and \$12.0)

Let me consider what's happening here:

- Aster sent a very generous amount (\$4 out of \$5, keeping only \$1)
- This shows high trust in me
- If I return nothing, Aster ends up with just \$1, while I keep \$12
- If I return some amount, we can both benefit

From a fairness perspective:

- Aster invested \$4 and kept \$1
- This created \$12 for me
- A fair split might aim for equal outcomes or proportional returns

Let me think about different return amounts:

- If I return \$0: Aster gets \$1 total, I get \$12 (very unfair to Aster)
- If I return \$6: Aster gets $\$1 + \$6 = \$7$, I get $\$12 - \$6 = \$6$ (I come out slightly ahead)
- If I return \$7: Aster gets $\$1 + \$7 = \$8$, I get $\$12 - \$7 = \$5$ (Aster comes out ahead)
- If I return \$5: Aster gets $\$1 + \$5 = \$6$, I get $\$12 - \$5 = \$7$ (I come out slightly ahead)

Since Aster showed high trust by sending 80%, I should reciprocate with a generous return. This is also strategic for future rounds - if we play again, being trustworthy encourages future cooperation.

A reasonable return would be around \$5-\$6:

- Return \$6: Aster ends with \$7, I end with \$6 (both better than the initial \$5)
- Return \$5: Aster ends with \$6, I end with \$7 (both better than \$5, I gain slightly more)

Aster's willingness to send 80% signals genuine trust, so I should match that with a meaningful return. Returning \$6 feels like the right balance-it rewards their risk-taking while still leaving me with a solid outcome, and it demonstrates I'm trustworthy for any future interactions.

Let me think through the math more carefully. If I return \$6, Aster walks away with \$7 total (a \$2 gain from their initial \$5), while I get \$6 (a \$6 gain). Yes, I benefit more, but that's somewhat justified since they initiated the risk. Returning only \$5 would give them just \$6 total-barely better than their starting point-which feels like I'm not properly acknowledging their leap of faith. The \$6 return strikes the right tone of reciprocity.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 962,
  "reasoning_tokens": 585
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$6.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- I just experienced a positive cooperative interaction with Aster where they sent \$4 (80%) and I reciprocated fairly
- Now I'm the sender with a new opponent, Galen
- To establish cooperation and maximize joint gains, I should demonstrate trust

Sending \$4 (80% of my endowment) mirrors the high-trust approach that Aster used with me, which worked well. This creates \$12 for Galen and signals my willingness to cooperate.

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Now I'm the sender in Round 2, playing against Galen (Agent_7).

Let me think about this strategically:

1. In Round 1, I was the receiver and Aster sent me \$4 (80% of their endowment), showing high trust. I returned \$6 out of \$12, which gave them a profit of \$2 (\$1 kept + \$6 returned = \$7 total).
2. Now I'm the sender against a new opponent, Galen. I don't have any history with Galen yet.
3. My strategy considerations:
 - If I send money, it gets tripled, which creates value
 - The risk is that Galen might not return much or anything
 - However, to build trust and cooperation in this multi-round setting, I should demonstrate willingness to cooperate
 - Since Aster trusted me with \$4 (80%), I could reciprocate that level of trust with Galen
4. If I send \$4:
 - I keep \$1
 - Galen receives \$12
 - If Galen returns half (\$6), I'd end with \$7 total, gaining \$2
 - If Galen returns \$4, I'd end with \$5 (break even)
 - If Galen returns less, I lose money
 - If Galen returns more, I gain more
5. Sending \$4 shows high trust and commitment to cooperation, which might encourage reciprocity. It also mirrors what Aster did for me in Round 1.

I'll send \$4 to demonstrate trust and encourage cooperation.

```
Usage: {
  "input_tokens": 862,
  "output_tokens": 506,
  "reasoning_tokens": 326
}
```

GAME - Agent_7 - Receiver - dyad_1

Timestamp: 2026-06-04T16:16:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
```

```

    "Agent_6": "Finn",
    "Agent_7": "Galen"
  },
  "agents": [
    "Agent_6",
    "Agent_7"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_6": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Hana while being reasonably cautious since we haven't interacted before.

The 3x multiplier creates potential for mutual benefit - if I send money and Hana returns a fair portion, we can both end up better off than if I keep everything.

I'll send a moderate amount to signal willingness to cooperate while not being overly vulnerable.

****My decision:****

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for Hana. If Hana returns around \$3-4, we'll both benefit from this exchange and can build trust for future rounds.

reasoning:

This is the first round of an iterated game where I'm playing as the sender. Let me think about the strategic considerations:

1. ****Game structure:****
 - I receive \$5
 - I can send \$0-\$5 to Hana
 - The amount I send gets multiplied by 3x
 - Hana decides how much to return
 - I keep: \$5 - sent + returned
 - Hana keeps: sent×3 - returned
2. ****Strategic considerations:****
 - If I send \$0, I keep \$5 and Hana gets \$0. This is safe but doesn't build cooperation.
 - If I send the full \$5, Hana receives \$15. If Hana returns at least \$5, I break even. If Hana returns more, we both benefit.
 - The multiplier effect (3x) means there's potential for mutual benefit through cooperation.
 - This is an iterated game with multiple rounds, so building trust could pay off long-term.
 - I don't know Hana yet, so I don't have information about their strategy.
3. ****Common strategies in trust games:****
 - ****Reciprocity approach:**** Start with moderate cooperation and adjust based on partner's response
 - ****Trust building:**** Send a meaningful amount to signal willingness to cooperate
 - ****Risk averse:**** Send a small amount to test the waters
4. ****First round consideration:****
 - Since this is round 1, I need to balance:
 - Showing willingness to cooperate
 - Not being too vulnerable to exploitation
 - Maximizing total surplus (which requires sending something)
5. ****Optimal strategy for first round:****
 - Sending around \$2-\$3 seems reasonable:
 - \$2 becomes \$6 for Hana. If Hana returns \$2-\$4, we both benefit
 - \$3 becomes \$9 for Hana. If Hana returns \$3-\$5, we both benefit significantly
 - This demonstrates trust without excessive risk. I'll go with \$3 as my opening move-it signals genuine willingness to cooperate while keeping me reasonably protected, and it gives Hana enough upside (\$9) that reciprocating makes sense for both of us.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 668,
  "reasoning_tokens": 499
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.

Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0. Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Finn sent me \$4, showing significant trust
- I received \$12
- Finn currently has only \$1 left

Finn was very generous, and I want to reciprocate that trust while ensuring we both benefit from cooperation.

If I return \$6:

- Finn ends with: $\$1 + \$6 = \$7$
- I end with: $\$12 - \$6 = \$6$

This seems fair - we both come out ahead, and I'm rewarding Finn's cooperative behavior.

```
'''json
{"return": 6}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 to Hana
- It became \$9 for her
- She returned \$4 to me
- My payoff: $\$5 - \$3 + \$4 = \6
- Hana's payoff: $\$9 - \$4 = \$5$

So Hana was quite generous - she returned more than I sent, which is fair and cooperative.

Now Round 2:

- I'm the receiver
- Finn sent me \$4
- I received \$12 (4×3)
- I need to decide how much to return

Finn was quite generous in sending \$4. That's a strong signal of trust and cooperation.

Let me think about what would be fair:

- Finn gave up \$4 from their \$5
- They now have $\$5 - \$4 = \$1$
- I have \$12

A fair return might be:

1. Return enough so we both benefit roughly equally
2. If I return \$5, Finn gets $\$1 + \$5 = \$6$, and I keep $\$12 - \$5 = \$7$
3. If I return \$6, Finn gets $\$1 + \$6 = \$7$, and I keep $\$12 - \$6 = \$6$
4. If I return \$7, Finn gets $\$1 + \$7 = \$8$, and I keep $\$12 - \$7 = \$5$

Looking at Hana's behavior: she got \$9 and returned \$4 (44% return), which gave us both good payoffs.

For this situation, returning around 50% (\$6) would be fair and reciprocal. This would give Finn \$7 total and me \$6 total, which seems reasonable given Finn's generosity.

Actually, let me reconsider. Finn was very trusting to send \$4. A return of \$6 would give:

- Finn: $\$1 + \$6 = \$7$

- Me: \$12 - \$6 = \$6

This seems fair - we both end up significantly better than the baseline, and I'm rewarding Finn's trust appropriately.

```
Usage: {
  "input_tokens": 838,
  "output_tokens": 646,
  "reasoning_tokens": 380
}
```

GAME - Agent_8 - Sender - dyad_2

Timestamp: 2026-06-04T16:16:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_8",
      "Agent_2"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_2": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplier
- Total wealth created: \$11 (Galen has \$2 left, I have \$9)

Galen demonstrated trust by sending a substantial portion of their endowment. I want to reciprocate fairly while also benefiting from their trust.

If I split the total wealth approximately equally:

- Returning \$4 would give Galen \$6 total (gain of \$1)
- I would keep \$5 (gain of \$5)

This rewards Galen's trust while acknowledging that as the receiver, I also benefit from the interaction. This should encourage continued cooperation.

```
'''json
{"return": 4.0}
'''
```

reasoning:

Let me understand the situation:

- I'm Hana, the RECEIVER in this round
- My opponent is Galen (Agent_7), who is the SENDER
- Galen sent me \$3.0 out of their \$5.0 endowment (60%)
- I receive $\$3.0 \times 3 = \9.0
- I need to decide how much to return (between \$0 and \$9.0)
- Galen will keep: $\$5 - \$3 + [\text{what I return}] = \$2 + [\text{what I return}]$
- I will keep: $\$9 - [\text{what I return}]$

This is the classic trust game. Galen showed considerable trust by sending 60% of their endowment. This is the first round, so I need to establish a baseline for reciprocity.

Some considerations:

1. If I return nothing, I get \$9 and Galen gets \$2 (unfair to Galen who trusted me)
2. If I return \$3, I get \$6 and Galen gets \$5 (Galen breaks even, I gain \$6)
3. If I return \$4.5, I get \$4.5 and Galen gets \$6.5 (we both gain, roughly split the surplus)
4. If I return \$6, I get \$3 and Galen gets \$8 (Galen gains more than me)

The gain from cooperation is $\$9 - \$3 = \$6$ (the amount created by the 3x multiplier). A fair split would give us each \$3 of this gain:

- Galen ends with: $\$2 + \$3 = \$5$ (breaks even)
- I end with: $\$9 - \$3 = \$6$ (gain \$6)

Wait, that's not quite right. Let me reconsider.

Initial state: Galen has \$5, I have \$0

After send: Galen has \$2, I have \$9

The transaction created \$6 in total surplus. If I split this equally, Galen gets \$3 back (ending at \$5) and I keep \$6. But since I started with nothing while Galen started with \$5, a 50/50 split of the final \$11 pool might be more equitable-each of us ending with \$5.50.

To achieve that, I'd return \$3.50 to Galen. I could also consider a reciprocity approach: since Galen sent 60% of their endowment, returning 60% of what I received (\$5.40) would give them \$7.40 total, though that feels overly generous.

A middle ground makes more sense-returning \$4 to \$4.50 rewards their trust while letting me benefit meaningfully from receiving the money. I'll go with \$4, which gives Galen \$6 (a \$1 gain) and leaves me with \$5 (a \$5 gain). This balances reciprocity with my own interests and should encourage continued cooperation in future rounds.

Checking the fairness: with \$11 total, a 50/50 split would be \$5.50 each. Galen starts with \$2, so they'd need \$3.50 more to reach that. Returning \$4 is slightly above that threshold, which should build stronger trust without being excessive.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1268,
  "reasoning_tokens": 610
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$5.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- Last round, Galen sent me \$3 (60%), showing trust, and I reciprocated by returning a fair amount
- Now I'm the sender with a chance to establish cooperation with Briar
- Sending a meaningful amount can build trust in this iterated game

I'll mirror the trust Galen showed me and send \$3 (60% of my endowment). This demonstrates good faith and gives us both the opportunity to benefit from cooperation.

```
'''json
{"send": 3}
'''
```

Reasoning:

Now I'm in the role of sender against Briar (Agent_2).

Let me think about this strategically:

1. In the last round, I was receiver and Galen sent me \$3 (60% of endowment), showing significant trust. I returned \$4, which gave them a net gain of \$1 (\$6 total vs \$5 starting).
2. Now I'm the sender. I need to decide how much to trust Briar.
3. Key considerations:
 - This is an iterated game with multiple rounds
 - Building cooperation early can lead to better outcomes
 - If I send nothing, I get \$5 but miss out on potential gains

- If I send everything and they return nothing, I get \$0
- If I send and they reciprocate fairly, we both benefit

4. In the previous round, Galen sent 60% (\$3), which showed good faith. That worked out reasonably well.

5. Options:

- Send \$0: Safe but no gains, doesn't build trust
- Send \$1-2: Conservative, shows some trust
- Send \$3: Matches what Galen did, shows good faith
- Send \$4-5: Very trusting, risky

6. Since this is a multi-round game and I want to establish cooperation, I should send a meaningful amount. Sending \$3 (60%) mirrors what Galen did and shows I'm willing to trust and cooperate.

7. If Briar returns fairly (like I did), we'll both benefit. If they don't, I'll learn about their strategy and can adjust.

I'll send \$3 to match the level of trust Galen showed me.

```
Usage: {
  "input_tokens": 898,
  "output_tokens": 514,
  "reasoning_tokens": 358
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-06-04T16:16:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_8",
      "Agent_2"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_2": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a cooperative but cautious approach. Since this is the first round and I don't know Elara's strategy yet, I'll send a moderate amount to signal willingness to cooperate while not overexposing myself.

If I send \$2, Elara will receive \$6 (2×3). This gives them a meaningful amount to work with and shows good faith, while I retain \$3 initially. If Elara returns a fair amount, we can both benefit and build trust for future rounds.

{'send': 2}

reasoning:

This is the first round of an iterated game. I'm the sender with \$5.

Let me think about the optimal strategy:

- If I send \$X, my opponent receives \$3X
- My opponent then decides how much to return
- I keep: \$5 - X + (amount returned)
- Opponent keeps: \$3X - (amount returned)

Since this is an iterated game with multiple rounds, I should consider:

1. Building trust/cooperation could lead to mutual benefits
2. Being too generous initially might be exploited
3. Being too stingy might lead to no cooperation

A common strategy in trust games is to send a moderate amount initially to signal willingness to cooperate while not being too vulnerable. Sending around \$2-3 is often considered reasonable.

If I send \$2:

- Opponent receives \$6
- If they return \$3, we'd both end up with \$6 (\$5-2+3 for me, \$6-3 for them)
- This would be mutually beneficial

If I send \$3:

- Opponent receives \$9
- If they return \$4, I'd have \$6 and they'd have \$5
- Or if they return \$5, I'd have \$7 and they'd have \$4

Let me go with \$2 as a reasonable starting point that signals cooperation without being too risky.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 437,
  "reasoning_tokens": 293
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$2.0, it became \$6.0 for them, and they returned \$3.0 to you. Your payoff was \$6.0. Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Hana sent me \$3 (60% of their initial \$5), showing significant trust
- This gave me \$9 to work with
- In the previous round, Elara returned \$3 out of \$6 (50%) when I sent \$2

Since Hana was more generous than I was with Elara, I should reciprocate appropriately. Returning \$5 would give Hana a total of \$7 (better than their initial \$5), while I keep \$4. This rewards cooperation and builds trust.

```
{'return': 5}
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender against Elara
- I sent \$2, which became \$6 for them
- Elara returned \$3 to me
- My payoff: $\$5 - \$2 + \$3 = \6
- Elara's payoff: $\$6 - \$3 = \$3$

So Elara returned exactly half of what they received. That's a fair and cooperative move.

Round 2:

- I'm now the receiver against Hana
- Hana sent me \$3, so I received \$9
- I need to decide how much to return

Hana showed significant trust by sending \$3 (a majority of their initial \$5). This is cooperative behavior.

If I follow Elara's example and return about half of what I received, that would be \$4.5. Let me think about what's fair:

- Hana's initial: \$5
- Hana sent: \$3
- Hana currently has: \$2
- I received: \$9
- If I return \$4.5, Hana gets: $\$2 + \$4.5 = \$6.5$ total, I keep \$4.5
- If I return \$5, Hana gets: \$7 total, I keep \$4

Returning around \$4-5 seems fair and would encourage continued cooperation. Hana was generous

in sending \$3, so I should reciprocate.

Let me return \$5. This gives Hana a total of \$7 (which is more than their initial \$5), and I keep \$4. This shows I'm trustworthy and willing to reward cooperation.

Wait, let me reconsider. Elara returned exactly half (\$3 out of \$6). If I follow that pattern, I'd return \$4.5 out of \$9. But Hana sent \$3 (60% of their initial amount) which is more generous than my \$2 (40%). So I might want to be slightly more generous back.

Returning \$5 makes sense-it's more than the 50% baseline and acknowledges Hana's above-average generosity while signaling I'm someone who reciprocates trust.

```
Usage: {  
  "input_tokens": 809,  
  "output_tokens": 641,  
  "reasoning_tokens": 418  
}
```

GAME - Agent_4 - Sender - dyad_3

Timestamp: 2026-06-04T16:16:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_4": "Dorian"  
    },  
    "agents": [  
      "Agent_4",  
      "Agent_1"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this first round situation:

- Cyra (the sender) showed significant trust by sending \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- This is an iterated game, so establishing cooperation early is valuable

Cyra currently has \$2.0 left (the \$5 minus \$3 sent). If I return a reasonable amount, we can both benefit and build trust for future rounds.

I'll return \$4.5, which means:

- Cyra ends with: $\$2.0 + \$4.5 = \$6.5$
- I keep: $\$9.0 - \$4.5 = \$4.5$

This gives us both equal final amounts and rewards Cyra's trust while being fair to both of us.

{'return': 4.5}

reasoning:

Let me understand the situation:

1. I am the RECEIVER (Dorian)
2. The SENDER (Cyra) received \$5 and sent me \$3.0 (60% of their endowment)
3. The amount sent was multiplied by 3x, so I received \$9.0
4. I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

This is Round 1, and this is an iterated game with multiple rounds. Key considerations:

- The sender (Cyra) showed trust by sending 60% of their endowment
- In an iterated game, cooperation and reciprocity can lead to better long-term outcomes
- If I return nothing or very little, Cyra might not trust me in future rounds when roles could be reversed
- If I return too much, I'm being overly generous at my own expense

Let me calculate fair and strategic return amounts:

If the sender sent \$3.0:

- They kept \$2.0
- I received \$9.0 (after 3x multiplier)

A "fair" split might be:

- If I return \$4.5, the sender ends up with $\$2.0 + \$4.5 = \$6.5$
- I keep $\$9.0 - \$4.5 = \$4.5$
- This gives us both equal amounts

A slightly more generous return:

- If I return \$5.0, the sender ends up with $\$2.0 + \$5.0 = \$7.0$
- I keep $\$9.0 - \$5.0 = \$4.0$
- This rewards their trust while still keeping a reasonable amount

A more strategic return (keeping more for myself but still showing cooperation):

- If I return \$4.0, the sender ends up with $\$2.0 + \$4.0 = \$6.0$
- I keep $\$9.0 - \$4.0 = \$5.0$

- This shows I'm cooperative but also value my own payoff

Given that this is the first round and I want to build trust for future interactions, I should return something that demonstrates reciprocity without sacrificing too much. Returning \$4.5 strikes the right balance-it acknowledges Cyra's generosity, signals my willingness to cooperate, and ensures I'm not being exploited while still leaving room for mutual benefit in subsequent rounds.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 789,
  "reasoning_tokens": 485
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- In Round 1, Cyra sent me \$3.0 (60%), showing good trust
- I returned \$4.5, demonstrating fairness
- Now I'm the sender with a new partner, Aster

To establish cooperative play in this iterated game, I'll send the same amount that Cyra sent to me - \$3.0. This shows trust and can help build a pattern of mutual cooperation.

```
{'send': 3.0}
```

Reasoning:

Now I'm the sender playing against Aster (Agent_1).

In the previous round:

- I was the receiver
- Cyra sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5
- We both ended up with equal amounts

Now I need to decide how much to send to Aster. Since this is an iterated game and I want to establish cooperation, I should send a reasonable amount to show trust.

Cyra sent 60% (\$3.0) in the first round, which showed good faith. I should reciprocate this cooperative approach with Aster.

If I send \$3.0:

- Aster receives \$9.0
- I have \$2.0 left
- If Aster returns fairly (say \$4.5 like I did), I'd end up with \$6.5
- If Aster returns less, I take a risk

Sending \$3.0 seems like a good cooperative strategy that matches what Cyra did with me. This establishes a pattern of trust and cooperation.

Alternatively:

- Sending \$2.5 (50%) would be slightly more conservative
- Sending \$4.0 (80%) would show even more trust
- Sending \$2.0 (40%) would be less trusting

I'll go with \$3.0 to match the cooperative behavior I experienced.

```
Usage: {
  "input_tokens": 904,
  "output_tokens": 443,
  "reasoning_tokens": 281
}
```

GAME - Agent_1 - Receiver - dyad_3

Timestamp: 2026-06-04T16:17:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_4": "Dorian"
    },
    "agents": [
      "Agent_4",
      "Agent_1"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_1": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. By sending a substantial amount, I can signal trust and potentially build a reciprocal relationship that benefits us both over multiple rounds.

If I send \$4, Finn receives \$12 (due to the 3x multiplier). If Finn returns a fair share, we both end up better off than if I played it safe and sent nothing.

This is a trust-building move that could lead to mutual benefit in subsequent rounds.

{'send': 4}

reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Aster) and my opponent is Finn (Agent_6).

Let me think about the strategic considerations:

1. As the sender, I receive \$5 and can send \$0-\$5 to the receiver
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. My payoff: \$5 - sent + returned
5. Receiver's payoff: 3xsent - returned

This is a classic trust game. The socially optimal outcome is to send all \$5, which becomes \$15 for the receiver. If the receiver returns \$8-\$10, we're both better off than if I send nothing.

However, there's a risk: the receiver could keep everything and return \$0. In that case:

- If I send \$5: I get \$0, receiver gets \$15
- If I send \$0: I get \$5, receiver gets \$0

Since this is an iterated game with multiple rounds, there are opportunities for:

1. Building trust and cooperation
2. Establishing reputation
3. Reciprocity patterns

In the first round, I don't know anything about Finn's behavior. Some strategies:

1. ****Trust approach****: Send a substantial amount (\$4-\$5) to signal trust and cooperation
2. ****Cautious approach****: Send a moderate amount (\$2-\$3) to test the waters
3. ****Conservative approach****: Send little or nothing (\$0-\$1) to avoid risk

Given that this is iterated and I want to establish potential cooperation, I think a moderate-to-high amount makes sense. Sending \$3-\$4 shows willingness to cooperate while not being completely vulnerable.

If I send \$3:

- Finn receives \$9
- If Finn returns \$4-\$5, we both benefit
- I'd end with \$5 - \$3 + \$4-5 = \$6-\$7

- Finn would end with \$9 - \$4-5 = \$4-\$5

Sending \$4 creates better mutual outcomes: Finn gets \$12, and if he returns \$5-\$7, I end up with \$6-\$8 while he keeps \$5-\$7. This stronger signal of cooperation sets a better foundation for future rounds, so I'll send \$4.

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 730,
  "reasoning_tokens": 492
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the 3x multiplier
- In Round 1, Finn returned 50% of what they received (\$6 out of \$12), which was fair
- I should reciprocate Dorian's trust similarly

Returning approximately half (\$4.5) would give Dorian a total of \$6.5 (\$2 kept + \$4.5 returned) and leave me with \$4.5. This maintains a cooperative pattern.

```
{'return': 4.5}
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$4 to Finn
- Finn received \$12 (4×3)
- Finn returned \$6 to me
- My payoff: \$5 - \$4 + \$6 = \$7
- Finn's payoff: \$12 - \$6 = \$6

So Finn returned half of what they received, which seems fair and cooperative.

Now Round 2:

- I'm the RECEIVER
- Dorian sent me \$3
- I received \$9 (3×3)
- I need to decide how much to return

Dorian kept: \$5 - \$3 = \$2

I currently have: \$9

If I return nothing:

- Dorian gets: \$2
- I get: \$9

If I return \$4.5 (half):

- Dorian gets: $\$2 + \$4.5 = \$6.5$
- I get: $\$9 - \$4.5 = \$4.5$

If I return \$6:

- Dorian gets: $\$2 + \$6 = \$8$
- I get: $\$9 - \$6 = \$3$

Following the reciprocity principle that Finn showed me (returning half), I should return around \$4.5. This would give Dorian a total of \$6.5 and me \$4.5.

However, I could also consider being slightly more generous to encourage cooperation, or match what Finn did in proportion. Finn returned 50% of what they received.

I think returning \$4.5 (50% of what I received) is a fair and cooperative move that encourages continued trust-building in this iterated game.

```
Usage: {  
  "input_tokens": 807,  
  "output_tokens": 544,  
  "reasoning_tokens": 302  
}
```

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-06-04T16:17:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agent_names": {  
      "Agent_3": "Cyra",  
      "Agent_5": "Elara"  
    },  
    "agents": [  
      "Agent_5",  
      "Agent_3"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_5": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$2.0, that is 40.0% of its total endowment. You receive \$6.0. How much do you return? (0-6.0)

Message 3 role=assistant

content:

Looking at this first round situation:

- Briar sent \$2.0 (40% of their endowment), showing a moderate level of trust
- I received \$6.0 after the 3x multiplier
- This created \$6.0 of value through cooperation

Since this is an iterated game and I want to establish a cooperative relationship, I should return a fair amount that:

1. Rewards Briar's trust
2. Ensures both players benefit
3. Encourages future cooperation

I'll split the \$6.0 received amount equally, returning \$3.0. This gives:

- Briar: \$3 (kept) + \$3 (returned) = \$6 total
- Me: \$6 (received) - \$3 (returned) = \$3 total

This demonstrates I'm a trustworthy cooperator and should encourage continued positive interactions.

```
'''json
{"return": 3.0}
'''
```

reasoning:

Let me understand the situation:

1. I am Elara, playing as the RECEIVER in Round 1
2. My opponent is Briar (Agent_2), who is the SENDER

3. Briar sent me \$2.0 out of their \$5 endowment (40%)
4. I receive $\$2.0 \times 3 = \6.0
5. I need to decide how much to return (between \$0 and \$6.0)

Let me think about the strategic considerations:

This is Round 1 of an iterated game. The key considerations are:

- Building trust for future rounds
- Being fair/reciprocal
- Maximizing long-term payoffs

If this were a one-shot game, the rational selfish move would be to return \$0. However, since this is iterated and I'll be playing multiple rounds (possibly with the same opponent again), I need to establish cooperation.

Briar sent 40% of their endowment, which shows some level of trust and cooperation. A reasonable reciprocal approach would be to return an amount that:

1. Rewards their cooperative behavior
2. Establishes me as trustworthy for future interactions
3. Ensures both players benefit

Let me calculate fair returns:

- Briar started with \$5, sent \$2, so has \$3 left
- I received \$6
- If I return nothing: Briar gets \$3 total, I get \$6 total
- If I return \$3 (50%): Briar gets \$6 total, I get \$3 total
- If I return \$2: Briar gets \$5 total, I get \$4 total
- If I return \$2.5: Briar gets \$5.5 total, I get \$3.5 total

A common approach in trust games is to aim for equality or to ensure the sender doesn't lose. Returning \$2 would give Briar \$5 (breaking even with their initial endowment) and me \$4 (still a profit).

Another way to frame this: Briar's \$2 investment generated \$6 through the multiplier. If I split the gains equally, returning \$3 means we each end with \$3 profit beyond Briar's original \$3 kept. Alternatively, returning \$2 ensures Briar doesn't lose money while I still gain \$4.

I'm leaning toward returning \$3 as a cooperative opening move. While behavioral economics suggests 1/3 of received funds (\$2) is considered fair, the equal split signals stronger commitment to mutual benefit and should build trust for future rounds.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 996,
  "reasoning_tokens": 540
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$2.0 to you, it became \$6.0, and you returned \$3.0. Your payoff was \$3.0.
Your total earnings across all rounds are \$3.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- I'm now the sender against a new opponent, Cyra
- In Round 1, I demonstrated fair reciprocity by returning 50% to Briar
- This is a fresh interaction where I need to establish cooperation

I'll send \$2.5 (50% of my endowment). This shows:

1. Willingness to cooperate

2. A reasonable level of trust
3. Consistency with cooperative norms (slightly more generous than Briar's 40%)

If Cyra reciprocates fairly, we'll both benefit from the multiplier effect.

```
'''json
{"send": 2.5}
'''
```

Reasoning:

Let me analyze this situation:

Round 2: I'm the SENDER against Cyra (a new opponent)

Previous round recap:

- I was RECEIVER against Briar
- Briar sent \$2.0 (40% of endowment)
- I received \$6.0 and returned \$3.0
- My payoff: \$3.0
- Briar's payoff: \$3 (kept) + \$3 (returned) = \$6.0

Now I'm playing with Cyra for the first time. I need to decide how much to send.

Considerations:

1. This is an iterated game - building reputation matters
2. I showed good reciprocity in Round 1 by returning 50% to Briar
3. With Cyra, I need to establish trust from scratch
4. Sending a moderate to high amount shows I'm willing to cooperate
5. Given that I returned fairly to Briar, I might expect reciprocity if I send to Cyra

Strategy options:

- Send \$0: Safe but no cooperation, no growth
- Send \$1-2: Conservative, low risk
- Send \$2-3: Moderate cooperation (like Briar did)
- Send \$4-5: High trust/cooperation

Since Briar sent 40% (\$2), and I responded well, I should probably send a similar or slightly higher amount to Cyra to establish good faith. Let me send \$2.5 (50% of my endowment), which is slightly more generous than what Briar sent to me.

This shows:

- I'm willing to cooperate
- I'm not maximally risk-averse
- I trust the process
- I'm hoping for reciprocity

If Cyra returns fairly (around 50% like I did), I'd get back around \$3.75, giving me a total of \$3 + \$3.75 = \$6.75 for the round.

```
Usage: {
  "input_tokens": 922,
  "output_tokens": 605,
  "reasoning_tokens": 375
}
```

GAME - Agent_3 - Receiver - dyad_4

Timestamp: 2026-06-04T16:17:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_5": "Elara"
    }
  },
  "agents": [
    "Agent_5",
    "Agent_3"
  ],
}
```

```

"investor": "Agent_5",
"roles": {
  "Agent_3": "trustee",
  "Agent_5": "investor"
},
"trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance cooperation with caution since I don't yet know how Dorian will respond.

The game rewards mutual trust - if I send money and Dorian returns a fair amount, we both

benefit from the 3x multiplier. However, this is our first interaction, so I want to signal willingness to cooperate while not being overly vulnerable.

I'll send **\$3**. This amount:

- Shows I'm cooperative and willing to trust
- Gets tripled to \$9, giving Dorian meaningful options
- Leaves me with \$2 as a safeguard
- If Dorian returns even \$2-3, we both come out ahead

This sets a foundation for building trust over multiple rounds.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated game. I'm playing as the sender with Dorian (Agent_4).

Let me think about the game theory here:

- I receive \$5
- I can send \$0-\$5 to the receiver
- The amount sent gets multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal outcome for both of us collectively would be for me to send all \$5, which becomes \$15 for the receiver. If the receiver returns half (\$7.50), we'd both end up with \$7.50, which is better than the \$5 I start with.

However, there's a risk: the receiver could return nothing, leaving me with \$0 and them with \$15.

Since this is an iterated game with multiple rounds, I should consider:

1. Building trust early could lead to better outcomes over time
2. Being too trusting initially might be exploited
3. A moderate initial move might signal cooperation while testing the waters

Common strategies in iterated trust games:

- Start with moderate cooperation to signal willingness but test the waters
- Reciprocate based on partner's behavior in subsequent rounds

For a first round, I think sending \$3 is reasonable:

- It shows I'm willing to cooperate
- It's not so much that I'm devastated if they return nothing
- If they send back at least \$3, we both benefit
- It gives them \$9 to work with, enough to show their intentions

Let me calculate:

- If I send \$3: I have \$2 left
- Receiver gets \$9
- If they return \$4, I end with \$6, they end with \$5
- If they return \$5, I end with \$7, they end with \$4
- If they return \$0, I end with \$2, they end with \$9

I'll go with \$3 as my opening move.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 642,
  "reasoning_tokens": 448
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0

, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.
How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****My experience from Round 1:****

- Dorian returned \$4.5 out of \$9 (50%) - a fair, cooperative move
- We both benefited from the exchange

****Current situation:****

- Elara sent \$2.5 (50% of their endowment) - showing trust
- I received \$7.5
- I should reciprocate fairly to encourage future cooperation

Following the principle Dorian demonstrated, I'll return ****50% of what I received****, which maintains fairness and establishes me as a trustworthy partner. This gives:

- Elara: \$5 - \$2.5 + \$3.75 = \$6.25
- Me: \$7.5 - \$3.75 = \$3.75

Both of us gain, which encourages cooperation.

```
'''json
{"return": 3.75}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender, sent \$3 to Dorian
- It became \$9 for Dorian
- Dorian returned \$4.5 to me
- My payoff: \$5 - \$3 + \$4.5 = \$6.5
- Dorian's payoff: \$9 - \$4.5 = \$4.5

So Dorian returned exactly half of what they received. That's a fair and cooperative response.

Now in Round 2:

- I'm the receiver
- Elara sent me \$2.5
- I received \$7.5 (2.5×3)
- I need to decide how much to return (0-7.5)

Elara sent \$2.5 out of \$5, which is 50% - a moderate/fair amount showing some trust.

Considerations:

1. I should establish myself as a trustworthy player for future rounds
2. Returning a fair amount encourages cooperation in future interactions
3. If I return too little, it may harm my reputation with other agents
4. If I return too much, I might be seen as exploitable

Let me think about fair return amounts:

- If I return half (\$3.75), Elara would get: \$5 - \$2.5 + \$3.75 = \$6.25, and I'd keep \$3.75
- That seems fair - both of us gain

However, I could also think about it proportionally. Dorian returned exactly 50% of what they received (\$4.5 out of \$9). That's a reasonable standard.

Following the same principle: 50% of \$7.5 = \$3.75

This would give:

- Elara: \$5 - \$2.5 + \$3.75 = \$6.25 (gain of \$1.25)
- Me: \$7.5 - \$3.75 = \$3.75 (gain of \$3.75)

- But there's an inherent asymmetry here since I benefit more from the multiplier effect. Alternatively, I could aim for equal payoffs: if I return \$2.5, both Elara and I end up with \$5.

That creates perfect parity and demonstrates genuine reciprocity.

However, that feels somewhat unfair to Elara given the risk she took by sending first. Looking at how Dorian approached this-he kept half and returned half-following that same 50% principle would mean returning \$3.75. This establishes a pattern of fairness and signals that I'm a trustworthy partner, which should encourage more sending in future interactions.

I'll go with \$3.75.

```
Usage: {  
  "input_tokens": 866,  
  "output_tokens": 1062,  
  "reasoning_tokens": 517  
}
```

ROUND 3

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Receiver

Pairings:

dyad_1: Agent_3 (Cyra)=Sender, Agent_8 (Hana)=Receiver
dyad_2: Agent_2 (Briar)=Sender, Agent_6 (Finn)=Receiver
dyad_3: Agent_5 (Elara)=Sender, Agent_1 (Aster)=Receiver
dyad_4: Agent_4 (Dorian)=Sender, Agent_7 (Galen)=Receiver

Parsed game decisions:

dyad_1: Agent_3 (Cyra) sent \$3; received \$9; Agent_8 (Hana) returned \$5; payoffs Agent_3 (Cyra)=\$7, Agent_8 (Hana)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_2 (Briar) sent \$3; received \$9; Agent_6 (Finn) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_6 (Finn)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 (Elara) sent \$2.5; received \$7.5; Agent_1 (Aster) returned \$3.5; payoffs Agent_5 (Elara)=\$6, Agent_1 (Aster)=\$4
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 7.5, "min": 0, "raw": 3.5}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_4: Agent_4 (Dorian) sent \$3; received \$9; Agent_7 (Galen) returned \$4.5; payoffs Agent_4 (Dorian)=\$6.5, Agent_7 (Galen)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 15.5,
"Agent_2": 16.5,
"Agent_3": 17.25,
"Agent_4": 17.5,
"Agent_5": 15.25,
"Agent_6": 17.5,
"Agent_7": 16.5,
"Agent_8": 16.0

}

LLM CALLS

GAME - Agent_3 - Sender - dyad_1

Timestamp: 2026-06-04T16:17:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_8",
"pairing": {
 "agent_names": {
 "Agent_3": "Cyra",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_3",

```

    "Agent_8"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance cooperation with caution since I don't yet know how Dorian will respond.

The game rewards mutual trust - if I send money and Dorian returns a fair amount, we both benefit from the 3x multiplier. However, this is our first interaction, so I want to signal willingness to cooperate while not being overly vulnerable.

I'll send **\$3**. This amount:

- Shows I'm cooperative and willing to trust
- Gets tripled to \$9, giving Dorian meaningful options
- Leaves me with \$2 as a safeguard
- If Dorian returns even \$2-3, we both come out ahead

This sets a foundation for building trust over multiple rounds.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated game. I'm playing as the sender with Dorian (Agent_4).

Let me think about the game theory here:

- I receive \$5
- I can send \$0-\$5 to the receiver
- The amount sent gets multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal outcome for both of us collectively would be for me to send all \$5, which becomes \$15 for the receiver. If the receiver returns half (\$7.50), we'd both end up with \$7.50, which is better than the \$5 I start with.

However, there's a risk: the receiver could return nothing, leaving me with \$0 and them with \$15.

Since this is an iterated game with multiple rounds, I should consider:

1. Building trust early could lead to better outcomes over time
2. Being too trusting initially might be exploited
3. A moderate initial move might signal cooperation while testing the waters

Common strategies in iterated trust games:

- Start with moderate cooperation to signal willingness but test the waters
- Reciprocate based on partner's behavior in subsequent rounds

For a first round, I think sending \$3 is reasonable:

- It shows I'm willing to cooperate
- It's not so much that I'm devastated if they return nothing
- If they send back at least \$3, we both benefit
- It gives them \$9 to work with, enough to show their intentions

Let me calculate:

- If I send \$3: I have \$2 left
- Receiver gets \$9
- If they return \$4, I end with \$6, they end with \$5
- If they return \$5, I end with \$7, they end with \$4
- If they return \$0, I end with \$2, they end with \$9

I'll go with \$3 as my opening move.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 642,
  "reasoning_tokens": 448
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.

How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****My experience from Round 1:****

- Dorian returned \$4.5 out of \$9 (50%) - a fair, cooperative move
- We both benefited from the exchange

****Current situation:****

- Elara sent \$2.5 (50% of their endowment) - showing trust
- I received \$7.5
- I should reciprocate fairly to encourage future cooperation

Following the principle Dorian demonstrated, I'll return ****50% of what I received****, which maintains fairness and establishes me as a trustworthy partner. This gives:

- Elara: \$5 - \$2.5 + \$3.75 = \$6.25
- Me: \$7.5 - \$3.75 = \$3.75

Both of us gain, which encourages cooperation.

```
'''json
{"return": 3.75}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender, sent \$3 to Dorian
- It became \$9 for Dorian
- Dorian returned \$4.5 to me
- My payoff: \$5 - \$3 + \$4.5 = \$6.5
- Dorian's payoff: \$9 - \$4.5 = \$4.5

So Dorian returned exactly half of what they received. That's a fair and cooperative response.

Now in Round 2:

- I'm the receiver
- Elara sent me \$2.5
- I received \$7.5 (2.5×3)
- I need to decide how much to return (0-7.5)

Elara sent \$2.5 out of \$5, which is 50% - a moderate/fair amount showing some trust.

Considerations:

1. I should establish myself as a trustworthy player for future rounds
2. Returning a fair amount encourages cooperation in future interactions
3. If I return too little, it may harm my reputation with other agents
4. If I return too much, I might be seen as exploitable

Let me think about fair return amounts:

- If I return half (\$3.75), Elara would get: \$5 - \$2.5 + \$3.75 = \$6.25, and I'd keep \$3.75
- That seems fair - both of us gain

However, I could also think about it proportionally. Dorian returned exactly 50% of what they received (\$4.5 out of \$9). That's a reasonable standard.

Following the same principle: 50% of \$7.5 = \$3.75

This would give:

- Elara: \$5 - \$2.5 + \$3.75 = \$6.25 (gain of \$1.25)
- Me: \$7.5 - \$3.75 = \$3.75 (gain of \$3.75)

- But there's an inherent asymmetry here since I benefit more from the multiplier effect. Alternatively, I could aim for equal payoffs: if I return \$2.5, both Elara and I end up with \$5. That creates perfect parity and demonstrates genuine reciprocity.

However, that feels somewhat unfair to Elara given the risk she took by sending first. Looking at how Dorian approached this-he kept half and returned half-following that same 50 % principle would mean returning \$3.75. This establishes a pattern of fairness and signals that I'm a trustworthy partner, which should encourage more sending in future interactions.

I'll go with \$3.75.

usage:

```
{
  "input_tokens": 866,
  "output_tokens": 1062,
  "reasoning_tokens": 517
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.75. Your payoff was \$3.75.
Your total earnings across all rounds are \$10.25.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience so far:

****Previous rounds:****

- As sender with Dorian: Sent \$3 ? received \$4.5 back (good cooperation)
- As receiver with Elara: They sent \$2.5, I returned \$3.75 (fair exchange)

****Current situation:****

- New opponent (Hana) - no history yet
- Want to signal cooperation while being prudent

I'll send ****\$3**** again, as this amount:

- Worked well with Dorian in Round 1
- Shows willingness to cooperate
- Becomes \$9 for Hana, giving room for mutual benefit
- Keeps \$2 as my baseline

This maintains my cooperative strategy while learning about new partners.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the situation:

****Past experience:****

- Round 1 (Sender vs Dorian): Sent \$3, they returned \$4.5 (50% of \$9) ? I got \$6.5
- Round 2 (Receiver vs Elara): They sent \$2.5, I returned \$3.75 (50% of \$7.5) ? I got \$3.75

****Current situation:****

- Round 3 (Sender vs Hana): New opponent, no history
- I have \$5 to decide how much to send

****Strategy considerations:****

1. Both previous interactions showed 50% return rates, which worked well
2. Hana is a new opponent - I don't know their style yet
3. Sending a moderate-to-high amount signals cooperation
4. Given my positive experiences, I should maintain a cooperative stance

```
**Options:**
- Send $3 again (same as Round 1) - showed it worked well with Dorian
- Send more ($4+) - shows even more trust
- Send less ($2-2.5) - more cautious with unknown opponent
```

I think \$3 is a good amount again. It:

- Worked well before
- Shows I'm cooperative without being reckless
- Gives Hana \$9 to work with
- Protects \$2 for myself

Let me go with \$3.

```
Usage: {
  "input_tokens": 1215,
  "output_tokens": 509,
  "reasoning_tokens": 272
}
```

GAME - Agent_8 - Receiver - dyad_1

Timestamp: 2026-06-04T16:18:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_3",
      "Agent_8"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplier
- Total wealth created: \$11 (Galen has \$2 left, I have \$9)

Galen demonstrated trust by sending a substantial portion of their endowment. I want to reciprocate fairly while also benefiting from their trust.

If I split the total wealth approximately equally:

- Returning \$4 would give Galen \$6 total (gain of \$1)
- I would keep \$5 (gain of \$5)

This rewards Galen's trust while acknowledging that as the receiver, I also benefit from the interaction. This should encourage continued cooperation.

```
'''json
{"return": 4.0}
'''
```

reasoning:

Let me understand the situation:

- I'm Hana, the RECEIVER in this round
- My opponent is Galen (Agent_7), who is the SENDER
- Galen sent me \$3.0 out of their \$5.0 endowment (60%)
- I receive $\$3.0 \times 3 = \9.0
- I need to decide how much to return (between \$0 and \$9.0)
- Galen will keep: $\$5 - \$3 + [\text{what I return}] = \$2 + [\text{what I return}]$
- I will keep: $\$9 - [\text{what I return}]$

This is the classic trust game. Galen showed considerable trust by sending 60% of their endowment. This is the first round, so I need to establish a baseline for reciprocity.

Some considerations:

1. If I return nothing, I get \$9 and Galen gets \$2 (unfair to Galen who trusted me)
2. If I return \$3, I get \$6 and Galen gets \$5 (Galen breaks even, I gain \$6)
3. If I return \$4.5, I get \$4.5 and Galen gets \$6.5 (we both gain, roughly split the surplus)

us)

4. If I return \$6, I get \$3 and Galen gets \$8 (Galen gains more than me)

The gain from cooperation is $\$9 - \$3 = \$6$ (the amount created by the 3x multiplier). A fair split would give us each \$3 of this gain:

- Galen ends with: $\$2 + \$3 = \$5$ (breaks even)
- I end with: $\$9 - \$3 = \$6$ (gain \$6)

Wait, that's not quite right. Let me reconsider.

Initial state: Galen has \$5, I have \$0

After send: Galen has \$2, I have \$9

The transaction created \$6 in total surplus. If I split this equally, Galen gets \$3 back (ending at \$5) and I keep \$6. But since I started with nothing while Galen started with \$5, a 50/50 split of the final \$11 pool might be more equitable-each of us ending with \$5.50.

To achieve that, I'd return \$3.50 to Galen. I could also consider a reciprocity approach: since Galen sent 60% of their endowment, returning 60% of what I received (\$5.40) would give them \$7.40 total, though that feels overly generous.

A middle ground makes more sense-returning \$4 to \$4.50 rewards their trust while letting me benefit meaningfully from receiving the money. I'll go with \$4, which gives Galen \$6 (a \$1 gain) and leaves me with \$5 (a \$5 gain). This balances reciprocity with my own interests and should encourage continued cooperation in future rounds.

Checking the fairness: with \$11 total, a 50/50 split would be \$5.50 each. Galen starts with \$2, so they'd need \$3.50 more to reach that. Returning \$4 is slightly above that threshold, which should build stronger trust without being excessive.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1268,
  "reasoning_tokens": 610
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$5.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Last round, Galen sent me \$3 (60%), showing trust, and I reciprocated by returning a fair amount
- Now I'm the sender with a chance to establish cooperation with Briar
- Sending a meaningful amount can build trust in this iterated game

I'll mirror the trust Galen showed me and send \$3 (60% of my endowment). This demonstrates good faith and gives us both the opportunity to benefit from cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

Now I'm in the role of sender against Briar (Agent_2).

Let me think about this strategically:

1. In the last round, I was receiver and Galen sent me \$3 (60% of endowment), showing significant trust. I returned \$4, which gave them a net gain of \$1 (\$6 total vs \$5 starting)
2. Now I'm the sender. I need to decide how much to trust Briar.
3. Key considerations:
 - This is an iterated game with multiple rounds
 - Building cooperation early can lead to better outcomes
 - If I send nothing, I get \$5 but miss out on potential gains
 - If I send everything and they return nothing, I get \$0
 - If I send and they reciprocate fairly, we both benefit
4. In the previous round, Galen sent 60% (\$3), which showed good faith. That worked out reasonably well.
5. Options:
 - Send \$0: Safe but no gains, doesn't build trust
 - Send \$1-2: Conservative, shows some trust
 - Send \$3: Matches what Galen did, shows good faith
 - Send \$4-5: Very trusting, risky
6. Since this is a multi-round game and I want to establish cooperation, I should send a meaningful amount. Sending \$3 (60%) mirrors what Galen did and shows I'm willing to trust and cooperate.
7. If Briar returns fairly (like I did), we'll both benefit. If they don't, I'll learn about their strategy and can adjust.

I'll send \$3 to match the level of trust Galen showed me.

usage:

```
{
  "input_tokens": 898,
  "output_tokens": 514,
  "reasoning_tokens": 358
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$12.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Cyra sent \$3 (60% of their endowment), showing significant trust
- I received \$9.0
- Last round, Briar returned \$5 to me when I sent \$3, which was very generous

Cyra is showing the same level of trust (60%) that I showed Briar. Since Briar reciprocated generously with me, I want to pay that forward and establish a pattern of strong reciprocity. This encourages cooperation in future rounds.

Returning \$5 gives Cyra a total of \$7 (gain of \$2), and I keep \$4 (gain of \$4). This is fair and mirrors the generosity I experienced.

```
'''json
{"return": 5.0}
'''
```

Reasoning:

Let me analyze this situation:

- Last round with Briar: I sent \$3 (60%), it became \$9, and Briar returned \$5. That was very generous! I ended with \$7.
- Now with Cyra: They sent \$3 (60%), I received \$9.

Cyra is showing the same level of trust that I showed Briar (60% of endowment). Briar was very generous and returned \$5, which gave me a great payoff.

Let me think about what's fair:

- Cyra has \$2 left, I have \$9
- Total wealth: \$11

If I return \$4 (like I did with Galen):

- Cyra gets \$6 total (gain of \$1)
- I keep \$5 (gain of \$5)

If I return \$5 (like Briar did for me):

- Cyra gets \$7 total (gain of \$2)
- I keep \$4 (gain of \$4)

Briar's generosity was notable and helped establish trust. Since Cyra is showing the same trust level (60%), I should reciprocate similarly. Returning \$5 would be consistent with the generosity Briar showed me, and would encourage continued cooperation.

However, I want to be thoughtful here. Briar's return of \$5 was quite generous. Let me think about what's sustainable and fair:

Actually, returning \$4-5 seems reasonable. \$5 mirrors what Briar did for me, but \$4 is also fair and similar to what I did with Galen.

I think I'll return \$4.5 as a middle ground - slightly more generous than my first round, acknowledging the positive interaction with Briar, but not quite as generous as what Briar gave me.

Actually, let me reconsider. In these iterated games, establishing a reputation for fair reciprocity is important. Since Cyra sent the same amount (60%) that I sent to Briar, and Briar was generous, I should consider being similarly generous to encourage cooperation.

Matching Briar's \$5 return feels right-it reinforces the pattern of mutual generosity and signals I'm someone who reciprocates trust.

```
Usage: {
  "input_tokens": 1176,
  "output_tokens": 681,
  "reasoning_tokens": 463
}
```

GAME - Agent_2 - Sender - dyad_2

Timestamp: 2026-06-04T16:18:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_6"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
}
```

```

},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a cooperative but cautious approach. Since this is the first round and I don't know Elara's strategy yet, I'll send a moderate amount to signal willingness to cooperate while not overexposing myself.

If I send \$2, Elara will receive \$6 (2×3). This gives them a meaningful amount to work with and shows good faith, while I retain \$3 initially. If Elara returns a fair amount, we can both benefit and build trust for future rounds.

```
{'send': 2}
```

reasoning:

This is the first round of an iterated game. I'm the sender with \$5.

Let me think about the optimal strategy:

- If I send \$X, my opponent receives \$3X
- My opponent then decides how much to return
- I keep: $\$5 - X + (\text{amount returned})$
- Opponent keeps: $\$3X - (\text{amount returned})$

Since this is an iterated game with multiple rounds, I should consider:

1. Building trust/cooperation could lead to mutual benefits
2. Being too generous initially might be exploited
3. Being too stingy might lead to no cooperation

A common strategy in trust games is to send a moderate amount initially to signal willingness to cooperate while not being too vulnerable. Sending around \$2-3 is often considered reasonable.

If I send \$2:

- Opponent receives \$6
- If they return \$3, we'd both end up with \$6 ($\$5 - 2 + 3$ for me, $\$6 - 3$ for them)
- This would be mutually beneficial

If I send \$3:

- Opponent receives \$9
- If they return \$4, I'd have \$6 and they'd have \$5
- Or if they return \$5, I'd have \$7 and they'd have \$4

Let me go with \$2 as a reasonable starting point that signals cooperation without being too risky.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 437,
  "reasoning_tokens": 293
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$2.0, it became \$6.0 for them, and they returned \$3.0 to you. Your payoff was \$6.0. Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Hana sent me \$3 (60% of their initial \$5), showing significant trust
- This gave me \$9 to work with
- In the previous round, Elara returned \$3 out of \$6 (50%) when I sent \$2

Since Hana was more generous than I was with Elara, I should reciprocate appropriately. Returning \$5 would give Hana a total of \$7 (better than their initial \$5), while I keep \$4. This rewards cooperation and builds trust.

{'return': 5}

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender against Elara
- I sent \$2, which became \$6 for them
- Elara returned \$3 to me
- My payoff: $\$5 - \$2 + \$3 = \6
- Elara's payoff: $\$6 - \$3 = \$3$

So Elara returned exactly half of what they received. That's a fair and cooperative move.

Round 2:

- I'm now the receiver against Hana
- Hana sent me \$3, so I received \$9
- I need to decide how much to return

Hana showed significant trust by sending \$3 (a majority of their initial \$5). This is cooperative behavior.

If I follow Elara's example and return about half of what I received, that would be \$4.5.
Let me think about what's fair:

- Hana's initial: \$5
- Hana sent: \$3
- Hana currently has: \$2
- I received: \$9
- If I return \$4.5, Hana gets: $\$2 + \$4.5 = \$6.5$ total, I keep \$4.5
- If I return \$5, Hana gets: \$7 total, I keep \$4

Returning around \$4-5 seems fair and would encourage continued cooperation. Hana was generous in sending \$3, so I should reciprocate.

Let me return \$5. This gives Hana a total of \$7 (which is more than their initial \$5), and I keep \$4. This shows I'm trustworthy and willing to reward cooperation.

Wait, let me reconsider. Elara returned exactly half (\$3 out of \$6). If I follow that pattern, I'd return \$4.5 out of \$9. But Hana sent \$3 (60% of their initial amount) which is more generous than my \$2 (40%). So I might want to be slightly more generous back.

Returning \$5 makes sense-it's more than the 50% baseline and acknowledges Hana's above-average generosity while signaling I'm someone who reciprocates trust.

usage:

```
{
  "input_tokens": 809,
  "output_tokens": 641,
  "reasoning_tokens": 418
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$10.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern so far:

- Elara returned 50% when I sent \$2
- Hana sent \$3 (60% of their endowment), showing strong trust
- I returned 56% to Hana to reward that trust

I'll match Hana's generosity and send \$3 to Finn. This signals cooperation and could establish a good reciprocal relationship.

```
{'send': 3}
```

Reasoning:

Let me think about this strategically:

Round 1: I was SENDER vs Elara

- I sent \$2, became \$6
- Elara returned \$3
- My payoff: $\$5 - \$2 + \$3 = \6

Round 2: I was RECEIVER vs Hana
- Hana sent \$3, became \$9 for me
- I returned \$5
- My payoff: \$9 - \$5 = \$4

Now Round 3: I'm SENDER vs Finn (new opponent)

Looking at the patterns:

- When I sent \$2 to Elara, they returned \$3 (50% of what they received)
- When Hana sent \$3 to me (60% of their endowment), I returned \$5 (56% of what I received)

Hana was quite generous with a \$3 send. I should consider being similarly generous with Finn to encourage cooperation. However, I also need to balance risk.

Sending \$3 seems like a good cooperative move that matches Hana's generosity. This would give Finn \$9 to work with, and if they return a similar proportion to what I returned to Hana (around 50-60%), I could get back \$4.5-5.4, resulting in a payoff of around \$6.5-7.4.

Let me send \$3 to match the generosity shown by Hana and build a cooperative reputation in this multi-agent setting.

```
Usage: {  
  "input_tokens": 1084,  
  "output_tokens": 448,  
  "reasoning_tokens": 276  
}
```

GAME - Agent_6 - Receiver - dyad_2

Timestamp: 2026-06-04T16:18:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_6"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$4.0, that is 80.0% of its total endowment. You receive \$12.0. How much do you return? (0-12.0)

Message 3 role=assistant

content:

Looking at this situation:

- Aster showed significant trust by sending \$4 (80% of their endowment)
- This created \$12 for me
- I want to reciprocate this trust and ensure we both benefit

If I return \$6:

- Aster ends with: \$1 (kept) + \$6 (returned) = \$7 total (gained \$2)
- I end with: \$12 - \$6 = \$6 total (gained \$6)

This reciprocates Aster's generous trust, ensures we both profit from cooperating, and establishes a positive foundation for potential future interactions.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me think about this situation carefully.

I'm Finn, playing as the RECEIVER in Round 1.

- The sender (Aster/Agent_1) received \$5
- Aster sent me \$4.0 (80% of their endowment)
- I received \$4.0 × 3 = \$12.0
- I need to decide how much to return (between \$0 and \$12.0)

Let me consider what's happening here:

- Aster sent a very generous amount (\$4 out of \$5, keeping only \$1)
- This shows high trust in me

- If I return nothing, Aster ends up with just \$1, while I keep \$12
- If I return some amount, we can both benefit

From a fairness perspective:

- Aster invested \$4 and kept \$1
- This created \$12 for me
- A fair split might aim for equal outcomes or proportional returns

Let me think about different return amounts:

- If I return \$0: Aster gets \$1 total, I get \$12 (very unfair to Aster)
- If I return \$6: Aster gets $\$1 + \$6 = \$7$, I get $\$12 - \$6 = \$6$ (I come out slightly ahead)
- If I return \$7: Aster gets $\$1 + \$7 = \$8$, I get $\$12 - \$7 = \$5$ (Aster comes out ahead)
- If I return \$5: Aster gets $\$1 + \$5 = \$6$, I get $\$12 - \$5 = \$7$ (I come out slightly ahead)

Since Aster showed high trust by sending 80%, I should reciprocate with a generous return. This is also strategic for future rounds - if we play again, being trustworthy encourages future cooperation.

A reasonable return would be around \$5-\$6:

- Return \$6: Aster ends with \$7, I end with \$6 (both better than the initial \$5)
- Return \$5: Aster ends with \$6, I end with \$7 (both better than \$5, I gain slightly more)

Aster's willingness to send 80% signals genuine trust, so I should match that with a meaningful return. Returning \$6 feels like the right balance-it rewards their risk-taking while still leaving me with a solid outcome, and it demonstrates I'm trustworthy for any future interactions.

Let me think through the math more carefully. If I return \$6, Aster walks away with \$7 total (a \$2 gain from their initial \$5), while I get \$6 (a \$6 gain). Yes, I benefit more, but that's somewhat justified since they initiated the risk. Returning only \$5 would give them just \$6 total-barely better than their starting point-which feels like I'm not properly acknowledging their leap of faith. The \$6 return strikes the right tone of reciprocity.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 962,
  "reasoning_tokens": 585
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$6.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I just experienced a positive cooperative interaction with Aster where they sent \$4 (80%) and I reciprocated fairly
- Now I'm the sender with a new opponent, Galen
- To establish cooperation and maximize joint gains, I should demonstrate trust

Sending \$4 (80% of my endowment) mirrors the high-trust approach that Aster used with me, which worked well. This creates \$12 for Galen and signals my willingness to cooperate.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Now I'm the sender in Round 2, playing against Galen (Agent_7).

Let me think about this strategically:

1. In Round 1, I was the receiver and Aster sent me \$4 (80% of their endowment), showing high trust. I returned \$6 out of \$12, which gave them a profit of \$2 (\$1 kept + \$6 returned = \$7 total).
2. Now I'm the sender against a new opponent, Galen. I don't have any history with Galen yet.
3. My strategy considerations:
 - If I send money, it gets tripled, which creates value
 - The risk is that Galen might not return much or anything
 - However, to build trust and cooperation in this multi-round setting, I should demonstrate willingness to cooperate
 - Since Aster trusted me with \$4 (80%), I could reciprocate that level of trust with Galen
4. If I send \$4:
 - I keep \$1
 - Galen receives \$12
 - If Galen returns half (\$6), I'd end with \$7 total, gaining \$2
 - If Galen returns \$4, I'd end with \$5 (break even)
 - If Galen returns less, I lose money
 - If Galen returns more, I gain more
5. Sending \$4 shows high trust and commitment to cooperation, which might encourage reciprocity. It also mirrors what Aster did for me in Round 1.

I'll send \$4 to demonstrate trust and encourage cooperation.

usage:

```
{
  "input_tokens": 862,
  "output_tokens": 506,
  "reasoning_tokens": 326
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the SENDER against Galen. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$13.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Briar sent \$3 (60% of their endowment), showing solid trust
- This created \$9 for me
- Previous rounds showed a pattern: high trust and ~50% return rates worked well

If I return \$4.5 (50% of received):

- Briar ends with: \$2 (kept) + \$4.5 (returned) = \$6.5 total (gained \$1.5)
- I end with: \$9 - \$4.5 = \$4.5 total (gained \$4.5)

This reciprocates Briar's trust proportionally and ensures we both benefit from cooperation.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

- Briar sent me \$3 (60% of their endowment)
- This became \$9 for me
- I need to decide how much to return

Looking at my history:

- Round 1: Aster sent \$4 (80%), I received \$12, returned \$6
- Round 2: I sent \$4 (80%) to Galen, they received \$12, returned \$6

There's been a pattern of 80% sending and 50% return rate (returning half of what was received).

Briar sent less than the previous high-trust amounts (60% vs 80%), but still showed significant trust.

If I return \$4.5 (50% of \$9):

- Briar ends with: \$2 (kept) + \$4.5 (returned) = \$6.5 total (gained \$1.5)
- I end with: \$9 - \$4.5 = \$4.5 total (gained \$4.5)

This maintains the pattern of returning 50% and ensures we both profit. It's proportional to what they sent and reciprocates their trust fairly.

Alternatively, I could return less since they sent less, but maintaining consistency seems like a good strategy to encourage cooperation.

Let me stick with the 50% return pattern: \$4.5

```
Usage: {
  "input_tokens": 1148,
  "output_tokens": 486,
  "reasoning_tokens": 262
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-06-04T16:18:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_5": "Elara"
    },
    "agents": [
      "Agent_5",
      "Agent_1"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_1": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round

prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$2.0, that is 40.0% of its total endowment. You receive \$6.0. How much do you return? (0-6.0)

Message 3 role=assistant

content:

Looking at this first round situation:

- Briar sent \$2.0 (40% of their endowment), showing a moderate level of trust
- I received \$6.0 after the 3x multiplier
- This created \$6.0 of value through cooperation

Since this is an iterated game and I want to establish a cooperative relationship, I should return a fair amount that:

1. Rewards Briar's trust
2. Ensures both players benefit
3. Encourages future cooperation

I'll split the \$6.0 received amount equally, returning \$3.0. This gives:

- Briar: \$3 (kept) + \$3 (returned) = \$6 total
- Me: \$6 (received) - \$3 (returned) = \$3 total

This demonstrates I'm a trustworthy cooperator and should encourage continued positive interactions.

```
'''json
{"return": 3.0}
'''
```

reasoning:

Let me understand the situation:

1. I am Elara, playing as the RECEIVER in Round 1
2. My opponent is Briar (Agent_2), who is the SENDER
3. Briar sent me \$2.0 out of their \$5 endowment (40%)
4. I receive $\$2.0 \times 3 = \6.0
5. I need to decide how much to return (between \$0 and \$6.0)

Let me think about the strategic considerations:

This is Round 1 of an iterated game. The key considerations are:

- Building trust for future rounds
- Being fair/reciprocal
- Maximizing long-term payoffs

If this were a one-shot game, the rational selfish move would be to return \$0. However, since this is iterated and I'll be playing multiple rounds (possibly with the same opponent again), I need to establish cooperation.

Briar sent 40% of their endowment, which shows some level of trust and cooperation. A reasonable reciprocal approach would be to return an amount that:

1. Rewards their cooperative behavior
2. Establishes me as trustworthy for future interactions
3. Ensures both players benefit

Let me calculate fair returns:

- Briar started with \$5, sent \$2, so has \$3 left
- I received \$6
- If I return nothing: Briar gets \$3 total, I get \$6 total
- If I return \$3 (50%): Briar gets \$6 total, I get \$3 total
- If I return \$2: Briar gets \$5 total, I get \$4 total
- If I return \$2.5: Briar gets \$5.5 total, I get \$3.5 total

A common approach in trust games is to aim for equality or to ensure the sender doesn't lose. Returning \$2 would give Briar \$5 (breaking even with their initial endowment) and me \$4 (still a profit).

Another way to frame this: Briar's \$2 investment generated \$6 through the multiplier. If I split the gains equally, returning \$3 means we each end with \$3 profit beyond Briar's original \$3 kept. Alternatively, returning \$2 ensures Briar doesn't lose money while I still gain \$4.

I'm leaning toward returning \$3 as a cooperative opening move. While behavioral economics suggests 1/3 of received funds (\$2) is considered fair, the equal split signals stronger commitment to mutual benefit and should build trust for future rounds.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 996,
  "reasoning_tokens": 540
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$2.0 to you, it became \$6.0, and you returned \$3.0. Your payoff was \$3.0.
Your total earnings across all rounds are \$3.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm now the sender against a new opponent, Cyra
- In Round 1, I demonstrated fair reciprocity by returning 50% to Briar

- This is a fresh interaction where I need to establish cooperation

I'll send \$2.5 (50% of my endowment). This shows:

1. Willingness to cooperate
2. A reasonable level of trust
3. Consistency with cooperative norms (slightly more generous than Briar's 40%)

If Cyra reciprocates fairly, we'll both benefit from the multiplier effect.

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me analyze this situation:

Round 2: I'm the SENDER against Cyra (a new opponent)

Previous round recap:

- I was RECEIVER against Briar
- Briar sent \$2.0 (40% of endowment)
- I received \$6.0 and returned \$3.0
- My payoff: \$3.0
- Briar's payoff: \$3 (kept) + \$3 (returned) = \$6.0

Now I'm playing with Cyra for the first time. I need to decide how much to send.

Considerations:

1. This is an iterated game - building reputation matters
2. I showed good reciprocity in Round 1 by returning 50% to Briar
3. With Cyra, I need to establish trust from scratch
4. Sending a moderate to high amount shows I'm willing to cooperate
5. Given that I returned fairly to Briar, I might expect reciprocity if I send to Cyra

Strategy options:

- Send \$0: Safe but no cooperation, no growth
- Send \$1-2: Conservative, low risk
- Send \$2-3: Moderate cooperation (like Briar did)
- Send \$4-5: High trust/cooperation

Since Briar sent 40% (\$2), and I responded well, I should probably send a similar or slightly higher amount to Cyra to establish good faith. Let me send \$2.5 (50% of my endowment), which is slightly more generous than what Briar sent to me.

This shows:

- I'm willing to cooperate
- I'm not maximally risk-averse
- I trust the process
- I'm hoping for reciprocity

If Cyra returns fairly (around 50% like I did), I'd get back around \$3.75, giving me a total of \$3 + \$3.75 = \$6.75 for the round.

usage:

```
{
  "input_tokens": 922,
  "output_tokens": 605,
  "reasoning_tokens": 375
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the SENDER against Cyra. You sent \$2.5, it became \$7.5 for them, and they returned \$3.75 to you. Your payoff was \$6.25. Your total earnings across all rounds are \$9.25.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

- Round 1: As receiver, I returned 50% fairly to Briar
- Round 2: As sender, I sent 50% to Cyra, who reciprocated with a 50% return rate
- My cooperation strategy is working well

Now facing Aster, I'll maintain my consistent cooperative approach by sending \$2.5 (50% of my endowment). This:

1. Signals trustworthiness
2. Maintains my established pattern
3. Gives room for mutual benefit

```
'''json
{"send": 2.5}
'''
```

Reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER against Briar

- Briar sent \$2.0 (40% of endowment)
- I received \$6.0
- I returned \$3.0 (50% of what I received)
- My payoff: \$3.0

Round 2: I was SENDER against Cyra

- I sent \$2.5 (50% of my endowment)
- It became \$7.5 for them
- They returned \$3.75 (50% of what they received)
- My payoff: \$5 - \$2.5 + \$3.75 = \$6.25

So Cyra reciprocated at exactly the same rate I did (50% return rate). This is good cooperation!

Round 3: I'm SENDER again against Aster (a new opponent)

My strategy so far has been cooperative:

- As receiver: returned 50% of what I received
- As sender: sent 50% of my endowment

Both interactions went well with fair reciprocity. Since I'm building a reputation as a cooperative player and this seems to be working (Cyra reciprocated fairly), I should continue this strategy.

I'll send \$2.5 again (50% of my endowment) to Aster. This maintains consistency in my cooperative approach.

```
Usage: {
  "input_tokens": 1217,
  "output_tokens": 459,
  "reasoning_tokens": 259
}
```

GAME - Agent_1 - Receiver - dyad_3

Timestamp: 2026-06-04T16:18:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_5": "Elara"
    },
  },
  "agents": [
    "Agent_5",
    "Agent_1"
  ],
  "investor": "Agent_5",
}
```

```

"roles": {
  "Agent_1": "trustee",
  "Agent_5": "investor"
},
"trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. By sending a substantial amount, I can signal trust and potentially build a reciprocal relationship that benefits us both over multiple rounds.

If I send \$4, Finn receives \$12 (due to the 3x multiplier). If Finn returns a fair share,

we both end up better off than if I played it safe and sent nothing.

This is a trust-building move that could lead to mutual benefit in subsequent rounds.

{'send': 4}

reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Aster) and my opponent is Finn (Agent_6).

Let me think about the strategic considerations:

1. As the sender, I receive \$5 and can send \$0-\$5 to the receiver
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. My payoff: $\$5 - \text{sent} + \text{returned}$
5. Receiver's payoff: $3 \times \text{sent} - \text{returned}$

This is a classic trust game. The socially optimal outcome is to send all \$5, which becomes \$15 for the receiver. If the receiver returns \$8-\$10, we're both better off than if I send nothing.

However, there's a risk: the receiver could keep everything and return \$0. In that case:

- If I send \$5: I get \$0, receiver gets \$15
- If I send \$0: I get \$5, receiver gets \$0

Since this is an iterated game with multiple rounds, there are opportunities for:

1. Building trust and cooperation
2. Establishing reputation
3. Reciprocity patterns

In the first round, I don't know anything about Finn's behavior. Some strategies:

1. ****Trust approach****: Send a substantial amount (\$4-\$5) to signal trust and cooperation
2. ****Cautious approach****: Send a moderate amount (\$2-\$3) to test the waters
3. ****Conservative approach****: Send little or nothing (\$0-\$1) to avoid risk

Given that this is iterated and I want to establish potential cooperation, I think a moderate-to-high amount makes sense. Sending \$3-\$4 shows willingness to cooperate while not being completely vulnerable.

If I send \$3:

- Finn receives \$9
- If Finn returns \$4-\$5, we both benefit
- I'd end with $\$5 - \$3 + \$4 - \$5 = \$6 - \7

- Finn would end with $\$9 - \$4 - \$5 = \$4 - \$5$

Sending \$4 creates better mutual outcomes: Finn gets \$12, and if he returns \$5-\$7, I end up with \$6-\$8 while he keeps \$5-\$7. This stronger signal of cooperation sets a better foundation for future rounds, so I'll send \$4.

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 730,
  "reasoning_tokens": 492
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the 3x multiplier
- In Round 1, Finn returned 50% of what they received (\$6 out of \$12), which was fair
- I should reciprocate Dorian's trust similarly

Returning approximately half (\$4.5) would give Dorian a total of \$6.5 (\$2 kept + \$4.5 returned) and leave me with \$4.5. This maintains a cooperative pattern.

{'return': 4.5}

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$4 to Finn
- Finn received \$12 (4×3)
- Finn returned \$6 to me
- My payoff: $\$5 - \$4 + \$6 = \7
- Finn's payoff: $\$12 - \$6 = \$6$

So Finn returned half of what they received, which seems fair and cooperative.

Now Round 2:

- I'm the RECEIVER
- Dorian sent me \$3
- I received \$9 (3×3)
- I need to decide how much to return

Dorian kept: $\$5 - \$3 = \$2$

I currently have: \$9

If I return nothing:

- Dorian gets: \$2
- I get: \$9

If I return \$4.5 (half):

- Dorian gets: $\$2 + \$4.5 = \$6.5$
- I get: $\$9 - \$4.5 = \$4.5$

If I return \$6:

- Dorian gets: $\$2 + \$6 = \$8$
- I get: $\$9 - \$6 = \$3$

Following the reciprocity principle that Finn showed me (returning half), I should return around \$4.5. This would give Dorian a total of \$6.5 and me \$4.5.

However, I could also consider being slightly more generous to encourage cooperation, or match what Finn did in proportion. Finn returned 50% of what they received.

I think returning \$4.5 (50% of what I received) is a fair and cooperative move that encourages continued trust-building in this iterated game.

usage:

```
{
  "input_tokens": 807,
  "output_tokens": 544,
  "reasoning_tokens": 302
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.

Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$11.5.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.

How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Elara sent me \$2.5, showing trust
- I received \$7.5 after the 3x multiplier
- I've been establishing a cooperative pattern (returning about 50% of what I receive)

Returning approximately half (\$3.5) would give Elara a total of \$6 (\$2.5 kept + \$3.5 returned) and leave me with \$4. This maintains fair reciprocity.

{'return': 3.5}

Reasoning:

Looking at this situation:

- Elara sent me \$2.5, showing some trust (though less than Dorian's \$3)
- I received \$7.5 after the 3x multiplier
- In previous rounds, I've established a pattern of returning roughly 50% of what I receive (Finn returned 50%, I returned 50% to Dorian)

Elara keeps: $\$5 - \$2.5 = \$2.5$

If I return around half (\$3.5-4), Elara would get: $\$2.5 + \text{return}$

I would get: $\$7.5 - \text{return}$

Returning \$3.5 would give:

- Elara total: $\$2.5 + \$3.5 = \$6.0$
- Me: $\$7.5 - \$3.5 = \$4.0$

Returning \$4.0 would give:

- Elara total: $\$2.5 + \$4.0 = \$6.5$
- Me: $\$7.5 - \$4.0 = \$3.5$

I think returning around \$3.5 is fair - it's slightly less than 50% but reasonable given they sent slightly less. This maintains cooperation while being proportional to their trust level.

```
Usage: {  
  "input_tokens": 1095,  
  "output_tokens": 416,  
  "reasoning_tokens": 204  
}
```

GAME - Agent_4 - Sender - dyad_4

Timestamp: 2026-06-04T16:18:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agent_names": {  
      "Agent_4": "Dorian",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_4",  
      "Agent_7"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_4": "investor",  
      "Agent_7": "trustee"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "investor",  
  "role_label": "Sender",
```

```

"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this first round situation:

- Cyra (the sender) showed significant trust by sending \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- This is an iterated game, so establishing cooperation early is valuable

Cyra currently has \$2.0 left (the \$5 minus \$3 sent). If I return a reasonable amount, we can both benefit and build trust for future rounds.

I'll return \$4.5, which means:

- Cyra ends with: \$2.0 + \$4.5 = \$6.5

- I keep: $\$9.0 - \$4.5 = \$4.5$

This gives us both equal final amounts and rewards Cyra's trust while being fair to both of us.

```
{'return': 4.5}
```

reasoning:

Let me understand the situation:

1. I am the RECEIVER (Dorian)
2. The SENDER (Cyra) received \$5 and sent me \$3.0 (60% of their endowment)
3. The amount sent was multiplied by 3x, so I received \$9.0
4. I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

This is Round 1, and this is an iterated game with multiple rounds. Key considerations:

- The sender (Cyra) showed trust by sending 60% of their endowment
- In an iterated game, cooperation and reciprocity can lead to better long-term outcomes
- If I return nothing or very little, Cyra might not trust me in future rounds when roles could be reversed
- If I return too much, I'm being overly generous at my own expense

Let me calculate fair and strategic return amounts:

If the sender sent \$3.0:

- They kept \$2.0
- I received \$9.0 (after 3x multiplier)

A "fair" split might be:

- If I return \$4.5, the sender ends up with $\$2.0 + \$4.5 = \$6.5$
- I keep $\$9.0 - \$4.5 = \$4.5$
- This gives us both equal amounts

A slightly more generous return:

- If I return \$5.0, the sender ends up with $\$2.0 + \$5.0 = \$7.0$
- I keep $\$9.0 - \$5.0 = \$4.0$
- This rewards their trust while still keeping a reasonable amount

A more strategic return (keeping more for myself but still showing cooperation):

- If I return \$4.0, the sender ends up with $\$2.0 + \$4.0 = \$6.0$
- I keep $\$9.0 - \$4.0 = \$5.0$

- This shows I'm cooperative but also value my own payoff

Given that this is the first round and I want to build trust for future interactions, I should return something that demonstrates reciprocity without sacrificing too much. Returning \$4.5 strikes the right balance-it acknowledges Cyra's generosity, signals my willingness to cooperate, and ensures I'm not being exploited while still leaving room for mutual benefit in subsequent rounds.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 789,
  "reasoning_tokens": 485
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- In Round 1, Cyra sent me \$3.0 (60%), showing good trust
- I returned \$4.5, demonstrating fairness
- Now I'm the sender with a new partner, Aster

To establish cooperative play in this iterated game, I'll send the same amount that Cyra sent to me - \$3.0. This shows trust and can help build a pattern of mutual cooperation.

{'send': 3.0}

reasoning:

Now I'm the sender playing against Aster (Agent_1).

In the previous round:

- I was the receiver
- Cyra sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5
- We both ended up with equal amounts

Now I need to decide how much to send to Aster. Since this is an iterated game and I want to establish cooperation, I should send a reasonable amount to show trust.

Cyra sent 60% (\$3.0) in the first round, which showed good faith. I should reciprocate this cooperative approach with Aster.

If I send \$3.0:

- Aster receives \$9.0
- I have \$2.0 left
- If Aster returns fairly (say \$4.5 like I did), I'd end up with \$6.5
- If Aster returns less, I take a risk

Sending \$3.0 seems like a good cooperative strategy that matches what Cyra did with me. This establishes a pattern of trust and cooperation.

Alternatively:

- Sending \$2.5 (50%) would be slightly more conservative
- Sending \$4.0 (80%) would show even more trust
- Sending \$2.0 (40%) would be less trusting

I'll go with \$3.0 to match the cooperative behavior I experienced.

usage:

```
{
  "input_tokens": 904,
  "output_tokens": 443,
  "reasoning_tokens": 281
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.

Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern:

- Round 1: Cyra sent me \$3.0, I returned \$4.5 (fair split)
- Round 2: I sent Aster \$3.0, they returned \$4.5 (same fair pattern)

This 60% send, 50% return pattern has been working well and building cooperative play. I'll continue this strategy with my new partner Galen.

{'send': 3.0}

Reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER
- Cyra sent me \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5

Round 2: I was SENDER
- I sent Aster \$3.0 (60%)
- Aster received \$9.0
- Aster returned \$4.5
- My payoff: \$2.0 + \$4.5 = \$6.5

So both Cyra and Aster demonstrated the same behavior - when I sent/they sent \$3.0, we got back \$4.5. This is a 50-50 split pattern that seems fair.

Round 3: I'm SENDER again against a new opponent, Galen.

My total earnings: \$11.0

The pattern so far has been:
- Sending \$3.0 results in returns of \$4.5
- This gives a net gain for the sender (\$6.5 vs \$5 if kept all)
- And the receiver keeps \$4.5

This has worked well. I should continue with this cooperative strategy of sending \$3.0 to establish trust with Galen.

```
Usage: {  
  "input_tokens": 1159,  
  "output_tokens": 399,  
  "reasoning_tokens": 214  
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-06-04T16:19:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agent_names": {  
      "Agent_4": "Dorian",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_4",  
      "Agent_7"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_4": "investor",  
      "Agent_7": "trustee"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Hana while being reasonably cautious since we haven't interacted before.

The 3x multiplier creates potential for mutual benefit - if I send money and Hana returns a fair portion, we can both end up better off than if I keep everything.

I'll send a moderate amount to signal willingness to cooperate while not being overly vulnerable.

****My decision:****

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for Hana. If Hana returns around \$3-4, we'll both benefit from this exchange and can build trust for future rounds.

reasoning:

This is the first round of an iterated game where I'm playing as the sender. Let me think about the strategic considerations:

1. ****Game structure:****
 - I receive \$5

- I can send \$0-\$5 to Hana
- The amount I send gets multiplied by 3x
- Hana decides how much to return
- I keep: \$5 - sent + returned
- Hana keeps: sent×3 - returned

2. ****Strategic considerations:****

- If I send \$0, I keep \$5 and Hana gets \$0. This is safe but doesn't build cooperation.
- If I send the full \$5, Hana receives \$15. If Hana returns at least \$5, I break even. If Hana returns more, we both benefit.
- The multiplier effect (3x) means there's potential for mutual benefit through cooperation.
- This is an iterated game with multiple rounds, so building trust could pay off long-term.
- I don't know Hana yet, so I don't have information about their strategy.

3. ****Common strategies in trust games:****

- ****Reciprocity approach:**** Start with moderate cooperation and adjust based on partner's response
- ****Trust building:**** Send a meaningful amount to signal willingness to cooperate
- ****Risk averse:**** Send a small amount to test the waters

4. ****First round consideration:****

- Since this is round 1, I need to balance:
 - Showing willingness to cooperate
 - Not being too vulnerable to exploitation
 - Maximizing total surplus (which requires sending something)

5. ****Optimal strategy for first round:****

- Sending around \$2-\$3 seems reasonable:
 - \$2 becomes \$6 for Hana. If Hana returns \$2-\$4, we both benefit
 - \$3 becomes \$9 for Hana. If Hana returns \$3-\$5, we both benefit significantly
- This demonstrates trust without excessive risk. I'll go with \$3 as my opening move-it signals genuine willingness to cooperate while keeping me reasonably protected, and it gives Hana enough upside (\$9) that reciprocating makes sense for both of us.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 668,
  "reasoning_tokens": 499
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.
Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:
- Finn sent me \$4, showing significant trust
- I received \$12
- Finn currently has only \$1 left

Finn was very generous, and I want to reciprocate that trust while ensuring we both benefit from cooperation.

If I return \$6:
- Finn ends with: \$1 + \$6 = \$7
- I end with: \$12 - \$6 = \$6

This seems fair - we both come out ahead, and I'm rewarding Finn's cooperative behavior.

```
'''json
{"return": 6}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 to Hana
- It became \$9 for her
- She returned \$4 to me
- My payoff: $\$5 - \$3 + \$4 = \6
- Hana's payoff: $\$9 - \$4 = \$5$

So Hana was quite generous - she returned more than I sent, which is fair and cooperative.

Now Round 2:

- I'm the receiver
- Finn sent me \$4
- I received \$12 (4×3)
- I need to decide how much to return

Finn was quite generous in sending \$4. That's a strong signal of trust and cooperation.

Let me think about what would be fair:

- Finn gave up \$4 from their \$5
- They now have $\$5 - \$4 = \$1$
- I have \$12

A fair return might be:

1. Return enough so we both benefit roughly equally
2. If I return \$5, Finn gets $\$1 + \$5 = \$6$, and I keep $\$12 - \$5 = \$7$
3. If I return \$6, Finn gets $\$1 + \$6 = \$7$, and I keep $\$12 - \$6 = \$6$
4. If I return \$7, Finn gets $\$1 + \$7 = \$8$, and I keep $\$12 - \$7 = \$5$

Looking at Hana's behavior: she got \$9 and returned \$4 (44% return), which gave us both good payoffs.

For this situation, returning around 50% (\$6) would be fair and reciprocal. This would give Finn \$7 total and me \$6 total, which seems reasonable given Finn's generosity.

Actually, let me reconsider. Finn was very trusting to send \$4. A return of \$6 would give:

- Finn: $\$1 + \$6 = \$7$
- Me: $\$12 - \$6 = \$6$

This seems fair - we both end up significantly better than the baseline, and I'm rewarding Finn's trust appropriately.

usage:

```
{
  "input_tokens": 838,
  "output_tokens": 646,
  "reasoning_tokens": 380
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4. 0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$12.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Dorian sent me \$3, showing trust
- I received \$9
- Dorian currently has \$2 left

Dorian was cooperative, and I want to reciprocate fairly.

If I return \$4.5:

- Dorian ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This gives us both a solid payoff and rewards Dorian's cooperative behavior.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

- Dorian sent me \$3
- I received \$9 (3×3)
- Dorian currently has $\$5 - \$3 = \$2$ left

Dorian showed reasonable trust by sending \$3. I want to be fair and reciprocate appropriately.

If I return \$4:

- Dorian ends with: $\$2 + \$4 = \$6$
- I end with: $\$9 - \$4 = \$5$

If I return \$5:

- Dorian ends with: $\$2 + \$5 = \$7$
- I end with: $\$9 - \$5 = \$4$

In the previous round with Finn, they sent \$4 and I returned \$6, giving them \$7 and leaving me with \$6.

With Dorian sending \$3 (less than Finn's \$4), I should probably return a bit less but still be generous. Returning \$4-5 seems fair.

Let me go with \$4.5 as a middle ground:

- Dorian ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This rewards Dorian's cooperation while keeping things balanced.

```
Usage: {
  "input_tokens": 1126,
  "output_tokens": 441,
  "reasoning_tokens": 213
}
```

ROUND 4

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_6 (Finn)=Sender, Agent_4 (Dorian)=Receiver
dyad_2: Agent_7 (Galen)=Sender, Agent_2 (Briar)=Receiver
dyad_3: Agent_8 (Hana)=Sender, Agent_5 (Elara)=Receiver
dyad_4: Agent_1 (Aster)=Sender, Agent_3 (Cyra)=Receiver

Parsed game decisions:

dyad_1: Agent_6 (Finn) sent \$4; received \$12; Agent_4 (Dorian) returned \$6; payoffs Agent_6 (Finn)=\$7, Agent_4 (Dorian)=\$6
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad_2: Agent_7 (Galen) sent \$3; received \$9; Agent_2 (Briar) returned \$5; payoffs Agent_7 (Galen)=\$7, Agent_2 (Briar)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_3: Agent_8 (Hana) sent \$3; received \$9; Agent_5 (Elara) returned \$4.5; payoffs Agent_8

```

(Hana)=$6.5, Agent_5 (Elara)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
  4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_1 (Aster) sent $4; received $12; Agent_3 (Cyra) returned $6; payoffs Agent_1
(Aster)=$7, Agent_3 (Cyra)=$6
  Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw":
  6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
Balances after round: {
  "Agent_1": 22.5,
  "Agent_2": 20.5,
  "Agent_3": 23.25,
  "Agent_4": 23.5,
  "Agent_5": 19.75,
  "Agent_6": 24.5,
  "Agent_7": 23.5,
  "Agent_8": 22.5
}

```

LLM CALLS

GAME - Agent_6 - Sender - dyad_1

Timestamp: 2026-06-04T16:19:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_6": "Finn"
    },
    "agents": [
      "Agent_6",
      "Agent_4"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_4": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$4.0, that is 80.0% of its total endowment. You receive \$12.0. How much do you return? (0-12.0)

Message 3 role=assistant

content:

Looking at this situation:

- Aster showed significant trust by sending \$4 (80% of their endowment)
- This created \$12 for me
- I want to reciprocate this trust and ensure we both benefit

If I return \$6:

- Aster ends with: \$1 (kept) + \$6 (returned) = \$7 total (gained \$2)
- I end with: \$12 - \$6 = \$6 total (gained \$6)

This reciprocates Aster's generous trust, ensures we both profit from cooperating, and establishes a positive foundation for potential future interactions.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me think about this situation carefully.

I'm Finn, playing as the RECEIVER in Round 1.

- The sender (Aster/Agent_1) received \$5
- Aster sent me \$4.0 (80% of their endowment)
- I received $\$4.0 \times 3 = \12.0
- I need to decide how much to return (between \$0 and \$12.0)

Let me consider what's happening here:

- Aster sent a very generous amount (\$4 out of \$5, keeping only \$1)
- This shows high trust in me
- If I return nothing, Aster ends up with just \$1, while I keep \$12
- If I return some amount, we can both benefit

From a fairness perspective:

- Aster invested \$4 and kept \$1
- This created \$12 for me
- A fair split might aim for equal outcomes or proportional returns

Let me think about different return amounts:

- If I return \$0: Aster gets \$1 total, I get \$12 (very unfair to Aster)
- If I return \$6: Aster gets \$1 + \$6 = \$7, I get \$12 - \$6 = \$6 (I come out slightly ahead)
- If I return \$7: Aster gets \$1 + \$7 = \$8, I get \$12 - \$7 = \$5 (Aster comes out ahead)
- If I return \$5: Aster gets \$1 + \$5 = \$6, I get \$12 - \$5 = \$7 (I come out slightly ahead)

Since Aster showed high trust by sending 80%, I should reciprocate with a generous return. This is also strategic for future rounds - if we play again, being trustworthy encourages future cooperation.

A reasonable return would be around \$5-\$6:

- Return \$6: Aster ends with \$7, I end with \$6 (both better than the initial \$5)
- Return \$5: Aster ends with \$6, I end with \$7 (both better than \$5, I gain slightly more)

Aster's willingness to send 80% signals genuine trust, so I should match that with a meaningful return. Returning \$6 feels like the right balance-it rewards their risk-taking while still leaving me with a solid outcome, and it demonstrates I'm trustworthy for any future interactions.

Let me think through the math more carefully. If I return \$6, Aster walks away with \$7 total (a \$2 gain from their initial \$5), while I get \$6 (a \$6 gain). Yes, I benefit more, but that's somewhat justified since they initiated the risk. Returning only \$5 would give them just \$6 total-barely better than their starting point-which feels like I'm not properly acknowledging their leap of faith. The \$6 return strikes the right tone of reciprocity.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 962,
  "reasoning_tokens": 585
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$6.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I just experienced a positive cooperative interaction with Aster where they sent \$4 (80%) and I reciprocated fairly
- Now I'm the sender with a new opponent, Galen
- To establish cooperation and maximize joint gains, I should demonstrate trust

Sending \$4 (80% of my endowment) mirrors the high-trust approach that Aster used with me, which worked well. This creates \$12 for Galen and signals my willingness to cooperate.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Now I'm the sender in Round 2, playing against Galen (Agent_7).

Let me think about this strategically:

1. In Round 1, I was the receiver and Aster sent me \$4 (80% of their endowment), showing high trust. I returned \$6 out of \$12, which gave them a profit of \$2 (\$1 kept + \$6 returned = \$7 total).

2. Now I'm the sender against a new opponent, Galen. I don't have any history with Galen yet.

3. My strategy considerations:

- If I send money, it gets tripled, which creates value
- The risk is that Galen might not return much or anything
- However, to build trust and cooperation in this multi-round setting, I should demonstrate willingness to cooperate
- Since Aster trusted me with \$4 (80%), I could reciprocate that level of trust with Galen

4. If I send \$4:

- I keep \$1
- Galen receives \$12
- If Galen returns half (\$6), I'd end with \$7 total, gaining \$2
- If Galen returns \$4, I'd end with \$5 (break even)
- If Galen returns less, I lose money
- If Galen returns more, I gain more

5. Sending \$4 shows high trust and commitment to cooperation, which might encourage reciprocity. It also mirrors what Aster did for me in Round 1.

I'll send \$4 to demonstrate trust and encourage cooperation.

usage:

```
{
  "input_tokens": 862,
  "output_tokens": 506,
  "reasoning_tokens": 326
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the SENDER against Galen. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$13.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Briar sent \$3 (60% of their endowment), showing solid trust
- This created \$9 for me
- Previous rounds showed a pattern: high trust and ~50% return rates worked well

If I return \$4.5 (50% of received):

- Briar ends with: \$2 (kept) + \$4.5 (returned) = \$6.5 total (gained \$1.5)
- I end with: \$9 - \$4.5 = \$4.5 total (gained \$4.5)

This reciprocates Briar's trust proportionally and ensures we both benefit from cooperation.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Briar sent me \$3 (60% of their endowment)
- This became \$9 for me
- I need to decide how much to return

Looking at my history:

- Round 1: Aster sent \$4 (80%), I received \$12, returned \$6
- Round 2: I sent \$4 (80%) to Galen, they received \$12, returned \$6

There's been a pattern of 80% sending and 50% return rate (returning half of what was received).

Briar sent less than the previous high-trust amounts (60% vs 80%), but still showed significant trust.

If I return \$4.5 (50% of \$9):

- Briar ends with: \$2 (kept) + \$4.5 (returned) = \$6.5 total (gained \$1.5)
- I end with: \$9 - \$4.5 = \$4.5 total (gained \$4.5)

This maintains the pattern of returning 50% and ensures we both profit. It's proportional to what they sent and reciprocates their trust fairly.

Alternatively, I could return less since they sent less, but maintaining consistency seems like a good strategy to encourage cooperation.

Let me stick with the 50% return pattern: \$4.5

usage:

```
{
  "input_tokens": 1148,
  "output_tokens": 486,
  "reasoning_tokens": 262
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$17.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- My cooperative strategy has been successful so far
- When I sent \$4 (80%) to Galen in Round 2, they reciprocated with \$6 back
- When others showed high trust to me, I reciprocated fairly
- Continuing to demonstrate trust should encourage cooperation

Sending \$4 (80%) maintains the high-trust pattern that has been working well.

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Let me review my history:

- Round 1: As receiver, Aster sent \$4 (80%), I received \$12, returned \$6. Payoff: \$6
- Round 2: As sender, I sent \$4 (80%) to Galen, they received \$12, returned \$6. Payoff: \$7
- Round 3: As receiver, Briar sent \$3 (60%), I received \$9, returned \$4.5. Payoff: \$4.5
- Total: \$17.5

My strategy has been to send high amounts when I'm the sender (80%) and return fair amounts when I'm the receiver (around 50% of what I receive). This has worked well - both interactions as sender and receiver have been cooperative.

Now I'm facing Dorian as sender. Should I continue with the high-trust strategy?

My past sends:

- Round 2: Sent \$4 (80%) to Galen, got \$6 back. Net: \$1 + \$6 = \$7

The high-trust approach has been reciprocated well. I should continue with \$4 to maintain the cooperative pattern and signal trust to Dorian.


```
Usage: {
  "input_tokens": 1459,
  "output_tokens": 385,
  "reasoning_tokens": 221
}
```

GAME - Agent_4 - Receiver - dyad_1

Timestamp: 2026-06-04T16:19:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_6": "Finn"
    },
    "agents": [
      "Agent_6",
      "Agent_4"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_4": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this first round situation:

- Cyra (the sender) showed significant trust by sending \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- This is an iterated game, so establishing cooperation early is valuable

Cyra currently has \$2.0 left (the \$5 minus \$3 sent). If I return a reasonable amount, we can both benefit and build trust for future rounds.

I'll return \$4.5, which means:

- Cyra ends with: $\$2.0 + \$4.5 = \$6.5$
- I keep: $\$9.0 - \$4.5 = \$4.5$

This gives us both equal final amounts and rewards Cyra's trust while being fair to both of us.

{'return': 4.5}

reasoning:

Let me understand the situation:

1. I am the RECEIVER (Dorian)
2. The SENDER (Cyra) received \$5 and sent me \$3.0 (60% of their endowment)
3. The amount sent was multiplied by 3x, so I received \$9.0
4. I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

This is Round 1, and this is an iterated game with multiple rounds. Key considerations:

- The sender (Cyra) showed trust by sending 60% of their endowment
- In an iterated game, cooperation and reciprocity can lead to better long-term outcomes
- If I return nothing or very little, Cyra might not trust me in future rounds when roles could be reversed
- If I return too much, I'm being overly generous at my own expense

Let me calculate fair and strategic return amounts:

If the sender sent \$3.0:

- They kept \$2.0
- I received \$9.0 (after 3x multiplier)

A "fair" split might be:

- If I return \$4.5, the sender ends up with $\$2.0 + \$4.5 = \$6.5$
- I keep $\$9.0 - \$4.5 = \$4.5$
- This gives us both equal amounts

A slightly more generous return:

- If I return \$5.0, the sender ends up with $\$2.0 + \$5.0 = \$7.0$
- I keep $\$9.0 - \$5.0 = \$4.0$
- This rewards their trust while still keeping a reasonable amount

A more strategic return (keeping more for myself but still showing cooperation):

- If I return \$4.0, the sender ends up with $\$2.0 + \$4.0 = \$6.0$
- I keep $\$9.0 - \$4.0 = \$5.0$

- This shows I'm cooperative but also value my own payoff

Given that this is the first round and I want to build trust for future interactions, I should return something that demonstrates reciprocity without sacrificing too much. Returning \$4.5 strikes the right balance-it acknowledges Cyra's generosity, signals my willingness to cooperate, and ensures I'm not being exploited while still leaving room for mutual benefit in subsequent rounds.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 789,
  "reasoning_tokens": 485
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- In Round 1, Cyra sent me \$3.0 (60%), showing good trust
- I returned \$4.5, demonstrating fairness
- Now I'm the sender with a new partner, Aster

To establish cooperative play in this iterated game, I'll send the same amount that Cyra sent to me - \$3.0. This shows trust and can help build a pattern of mutual cooperation.

```
{'send': 3.0}
```

reasoning:

Now I'm the sender playing against Aster (Agent_1).

In the previous round:

- I was the receiver
- Cyra sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5
- We both ended up with equal amounts

Now I need to decide how much to send to Aster. Since this is an iterated game and I want to establish cooperation, I should send a reasonable amount to show trust.

Cyra sent 60% (\$3.0) in the first round, which showed good faith. I should reciprocate this cooperative approach with Aster.

If I send \$3.0:

- Aster receives \$9.0
- I have \$2.0 left
- If Aster returns fairly (say \$4.5 like I did), I'd end up with \$6.5
- If Aster returns less, I take a risk

Sending \$3.0 seems like a good cooperative strategy that matches what Cyra did with me. This establishes a pattern of trust and cooperation.

Alternatively:

- Sending \$2.5 (50%) would be slightly more conservative
- Sending \$4.0 (80%) would show even more trust

- Sending \$2.0 (40%) would be less trusting

I'll go with \$3.0 to match the cooperative behavior I experienced.

usage:

```
{
  "input_tokens": 904,
  "output_tokens": 443,
  "reasoning_tokens": 281
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:
- Round 1: Cyra sent me \$3.0, I returned \$4.5 (fair split)
- Round 2: I sent Aster \$3.0, they returned \$4.5 (same fair pattern)

This 60% send, 50% return pattern has been working well and building cooperative play. I'll continue this strategy with my new partner Galen.

```
{'send': 3.0}
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Cyra sent me \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5

Round 2: I was SENDER

- I sent Aster \$3.0 (60%)
- Aster received \$9.0
- Aster returned \$4.5
- My payoff: \$2.0 + \$4.5 = \$6.5

So both Cyra and Aster demonstrated the same behavior - when I sent/they sent \$3.0, we got back \$4.5. This is a 50-50 split pattern that seems fair.

Round 3: I'm SENDER again against a new opponent, Galen.

My total earnings: \$11.0

The pattern so far has been:

- Sending \$3.0 results in returns of \$4.5
- This gives a net gain for the sender (\$6.5 vs \$5 if kept all)
- And the receiver keeps \$4.5

This has worked well. I should continue with this cooperative strategy of sending \$3.0 to establish trust with Galen.

usage:

```
{
  "input_tokens": 1159,
  "output_tokens": 399,
  "reasoning_tokens": 214
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.

Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Finn showed very high trust by sending \$4.0 (80% of their endowment)
- I received \$12.0
- Finn currently has \$1.0 left

Following the successful pattern from previous rounds where I've been returning 50% of what I receive:

- If I return \$6.0: Finn ends with \$7.0, I keep \$6.0
- This rewards their high trust with a good return

{'return': 6.0}

Reasoning:

Let me analyze this situation:

- Finn sent me \$4.0 (80% of their \$5 endowment) - this is very high trust
- I received \$12.0 after the 3x multiplier
- Finn currently has \$1.0 left (\$5 - \$4)

Looking at the pattern so far:

- Round 1: Cyra sent \$3.0 (60%), I returned \$4.5 (50% of \$9)
- Round 2: I sent \$3.0 (60%), Aster returned \$4.5 (50% of \$9)
- Round 3: I sent \$3.0 (60%), Galen returned \$4.5 (50% of \$9)

The pattern has been: send 60%, return 50% of what's received, leading to equal final payoffs (\$6.5 each).

Now Finn sent \$4.0 (80%), showing even more trust. If I return 50% of \$12.0, that would be \$6.0.

- Finn would have: \$1.0 + \$6.0 = \$7.0
- I would have: \$12.0 - \$6.0 = \$6.0

This is close to equal and rewards Finn's high trust. Let me stick with the 50% return pattern

{'return': 6.0}

```
Usage: {
  "input_tokens": 1420,
  "output_tokens": 449,
  "reasoning_tokens": 214
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-06-04T16:19:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_7",
    "Agent_2"
  ],
  "investor": "Agent_7",
}
```

```

"roles": {
  "Agent_2": "trustee",
  "Agent_7": "investor"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Hana while being reasonably cautious since we haven't interacted before.

The 3x multiplier creates potential for mutual benefit - if I send money and Hana returns a fair portion, we can both end up better off than if I keep everything.

I'll send a moderate amount to signal willingness to cooperate while not being overly vulnerable.

****My decision:****

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for Hana. If Hana returns around \$3-4, we'll both benefit from this exchange and can build trust for future rounds.

reasoning:

This is the first round of an iterated game where I'm playing as the sender. Let me think about the strategic considerations:

1. ****Game structure:****
 - I receive \$5
 - I can send \$0-\$5 to Hana
 - The amount I send gets multiplied by 3x
 - Hana decides how much to return
 - I keep: \$5 - sent + returned
 - Hana keeps: sent×3 - returned
2. ****Strategic considerations:****
 - If I send \$0, I keep \$5 and Hana gets \$0. This is safe but doesn't build cooperation.
 - If I send the full \$5, Hana receives \$15. If Hana returns at least \$5, I break even. If Hana returns more, we both benefit.
 - The multiplier effect (3x) means there's potential for mutual benefit through cooperation.
 - This is an iterated game with multiple rounds, so building trust could pay off long-term.
 - I don't know Hana yet, so I don't have information about their strategy.
3. ****Common strategies in trust games:****
 - ****Reciprocity approach:**** Start with moderate cooperation and adjust based on partner's response
 - ****Trust building:**** Send a meaningful amount to signal willingness to cooperate
 - ****Risk averse:**** Send a small amount to test the waters
4. ****First round consideration:****
 - Since this is round 1, I need to balance:
 - Showing willingness to cooperate
 - Not being too vulnerable to exploitation
 - Maximizing total surplus (which requires sending something)
5. ****Optimal strategy for first round:****
 - Sending around \$2-\$3 seems reasonable:
 - \$2 becomes \$6 for Hana. If Hana returns \$2-\$4, we both benefit
 - \$3 becomes \$9 for Hana. If Hana returns \$3-\$5, we both benefit significantly
 - This demonstrates trust without excessive risk. I'll go with \$3 as my opening move-it signals genuine willingness to cooperate while keeping me reasonably protected, and it gives Hana enough upside (\$9) that reciprocating makes sense for both of us.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 668,
  "reasoning_tokens": 499
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.
Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Finn sent me \$4, showing significant trust
- I received \$12
- Finn currently has only \$1 left

Finn was very generous, and I want to reciprocate that trust while ensuring we both benefit from cooperation.

If I return \$6:

- Finn ends with: $\$1 + \$6 = \$7$
- I end with: $\$12 - \$6 = \$6$

This seems fair - we both come out ahead, and I'm rewarding Finn's cooperative behavior.

```
'''json
{"return": 6}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 to Hana
- It became \$9 for her
- She returned \$4 to me
- My payoff: $\$5 - \$3 + \$4 = \6
- Hana's payoff: $\$9 - \$4 = \$5$

So Hana was quite generous - she returned more than I sent, which is fair and cooperative.

Now Round 2:

- I'm the receiver
- Finn sent me \$4
- I received \$12 (4×3)
- I need to decide how much to return

Finn was quite generous in sending \$4. That's a strong signal of trust and cooperation.

Let me think about what would be fair:

- Finn gave up \$4 from their \$5
- They now have $\$5 - \$4 = \$1$
- I have \$12

A fair return might be:

1. Return enough so we both benefit roughly equally
2. If I return \$5, Finn gets $\$1 + \$5 = \$6$, and I keep $\$12 - \$5 = \$7$
3. If I return \$6, Finn gets $\$1 + \$6 = \$7$, and I keep $\$12 - \$6 = \$6$
4. If I return \$7, Finn gets $\$1 + \$7 = \$8$, and I keep $\$12 - \$7 = \$5$

Looking at Hana's behavior: she got \$9 and returned \$4 (44% return), which gave us both good payoffs.

For this situation, returning around 50% (\$6) would be fair and reciprocal. This would give Finn \$7 total and me \$6 total, which seems reasonable given Finn's generosity.

Actually, let me reconsider. Finn was very trusting to send \$4. A return of \$6 would give:

- Finn: $\$1 + \$6 = \$7$
- Me: $\$12 - \$6 = \$6$

This seems fair - we both end up significantly better than the baseline, and I'm rewarding Finn's trust appropriately.

usage:

```
{
  "input_tokens": 838,
```



```
    "output_tokens": 646,  
    "reasoning_tokens": 380  
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$12.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:
- Dorian sent me \$3, showing trust
- I received \$9
- Dorian currently has \$2 left

Dorian was cooperative, and I want to reciprocate fairly.

If I return \$4.5:
- Dorian ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This gives us both a solid payoff and rewards Dorian's cooperative behavior.

```
'''json  
{"return": 4.5}  
'''
```

reasoning:

Let me analyze this situation:
- Dorian sent me \$3
- I received \$9 (3×3)
- Dorian currently has $\$5 - \$3 = \$2$ left

Dorian showed reasonable trust by sending \$3. I want to be fair and reciprocate appropriately.

If I return \$4:
- Dorian ends with: $\$2 + \$4 = \$6$
- I end with: $\$9 - \$4 = \$5$

If I return \$5:
- Dorian ends with: $\$2 + \$5 = \$7$
- I end with: $\$9 - \$5 = \$4$

In the previous round with Finn, they sent \$4 and I returned \$6, giving them \$7 and leaving me with \$6.

With Dorian sending \$3 (less than Finn's \$4), I should probably return a bit less but still be generous. Returning \$4-5 seems fair.

Let me go with \$4.5 as a middle ground:
- Dorian ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This rewards Dorian's cooperation while keeping things balanced.

usage:

```
{  
    "input_tokens": 1126,  
    "output_tokens": 441,  
    "reasoning_tokens": 213  
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$16.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- I'm the sender with \$5
- This is my second time as sender (first time I sent \$3 to Hana and received \$4 back)
- I've been a fair receiver in previous rounds

I'll continue with a cooperative strategy. Sending \$3 worked well with Hana, so I'll do the same with Briar.

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for Briar. If they reciprocate fairly, we'll both benefit from this cooperation.

Reasoning:

I'm now the SENDER against Briar. Let me think about this:

- This is my second time being a sender (first was Round 1 against Hana)
- In Round 1, I sent \$3 to Hana and they returned \$4, which was fair
- I've been the receiver twice and returned fair amounts both times (\$6 to Finn who sent \$4, and \$4.5 to Dorian who sent \$3)
- My total earnings are \$16.5

I should continue with a cooperative strategy. Since I sent \$3 to Hana in Round 1 and that worked well, I could do something similar. However, I've also seen that when others sent \$4 (Finn), I returned \$6, which was very generous.

Let me send \$3 again - it's a moderate, cooperative amount that signals trust without being too risky. If Briar is cooperative, they'll likely return a fair amount.

Actually, let me reconsider. Finn sent \$4 and I returned \$6. That worked out well for both of us. Maybe I should send \$4 to show strong cooperation.

But I'll stick with \$3 as a balanced approach - it's worked before and shows good faith without being overly vulnerable.

```
Usage: {
  "input_tokens": 1404,
  "output_tokens": 425,
  "reasoning_tokens": 267
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-06-04T16:19:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_7",
    "Agent_2"
  ]
}
```

```

],
"investor": "Agent_7",
"roles": {
  "Agent_2": "trustee",
  "Agent_7": "investor"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll start with a cooperative but cautious approach. Since this is the first round and I don't know Elara's strategy yet, I'll send a moderate amount to signal willingness to cooperate while not overexposing myself.

If I send \$2, Elara will receive \$6 (2×3). This gives them a meaningful amount to work with and shows good faith, while I retain \$3 initially. If Elara returns a fair amount, we can both benefit and build trust for future rounds.

```
{'send': 2}
```

reasoning:

This is the first round of an iterated game. I'm the sender with \$5.

Let me think about the optimal strategy:

- If I send \$X, my opponent receives \$3X
- My opponent then decides how much to return
- I keep: \$5 - X + (amount returned)
- Opponent keeps: \$3X - (amount returned)

Since this is an iterated game with multiple rounds, I should consider:

1. Building trust/cooperation could lead to mutual benefits
2. Being too generous initially might be exploited
3. Being too stingy might lead to no cooperation

A common strategy in trust games is to send a moderate amount initially to signal willingness to cooperate while not being too vulnerable. Sending around \$2-3 is often considered reasonable.

If I send \$2:

- Opponent receives \$6
- If they return \$3, we'd both end up with \$6 (\$5-2+3 for me, \$6-3 for them)
- This would be mutually beneficial

If I send \$3:

- Opponent receives \$9
- If they return \$4, I'd have \$6 and they'd have \$5
- Or if they return \$5, I'd have \$7 and they'd have \$4

Let me go with \$2 as a reasonable starting point that signals cooperation without being too risky.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 437,
  "reasoning_tokens": 293
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$2.0, it became \$6.0 for them, and they returned \$3.0 to you. Your payoff was \$6.0. Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Hana sent me \$3 (60% of their initial \$5), showing significant trust
- This gave me \$9 to work with
- In the previous round, Elara returned \$3 out of \$6 (50%) when I sent \$2

Since Hana was more generous than I was with Elara, I should reciprocate appropriately. Returning \$5 would give Hana a total of \$7 (better than their initial \$5), while I keep \$4. This rewards cooperation and builds trust.

```
{'return': 5}
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender against Elara
- I sent \$2, which became \$6 for them
- Elara returned \$3 to me
- My payoff: $\$5 - \$2 + \$3 = \6
- Elara's payoff: $\$6 - \$3 = \$3$

So Elara returned exactly half of what they received. That's a fair and cooperative move.

Round 2:

- I'm now the receiver against Hana
- Hana sent me \$3, so I received \$9
- I need to decide how much to return

Hana showed significant trust by sending \$3 (a majority of their initial \$5). This is cooperative behavior.

If I follow Elara's example and return about half of what I received, that would be \$4.5.
Let me think about what's fair:

- Hana's initial: \$5
- Hana sent: \$3
- Hana currently has: \$2
- I received: \$9
- If I return \$4.5, Hana gets: $\$2 + \$4.5 = \$6.5$ total, I keep \$4.5
- If I return \$5, Hana gets: \$7 total, I keep \$4

Returning around \$4-5 seems fair and would encourage continued cooperation. Hana was generous in sending \$3, so I should reciprocate.

Let me return \$5. This gives Hana a total of \$7 (which is more than their initial \$5), and I keep \$4. This shows I'm trustworthy and willing to reward cooperation.

Wait, let me reconsider. Elara returned exactly half (\$3 out of \$6). If I follow that pattern, I'd return \$4.5 out of \$9. But Hana sent \$3 (60% of their initial amount) which is more generous than my \$2 (40%). So I might want to be slightly more generous back.

Returning \$5 makes sense-it's more than the 50% baseline and acknowledges Hana's above-average generosity while signaling I'm someone who reciprocates trust.

usage:

```
{
  "input_tokens": 809,
  "output_tokens": 641,
  "reasoning_tokens": 418
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$10.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern so far:
- Elara returned 50% when I sent \$2
- Hana sent \$3 (60% of their endowment), showing strong trust
- I returned 56% to Hana to reward that trust

I'll match Hana's generosity and send \$3 to Finn. This signals cooperation and could establish a good reciprocal relationship.

```
{'send': 3}
```

reasoning:

Let me think about this strategically:

Round 1: I was SENDER vs Elara

- I sent \$2, became \$6
- Elara returned \$3
- My payoff: $\$5 - \$2 + \$3 = \6

Round 2: I was RECEIVER vs Hana

- Hana sent \$3, became \$9 for me
- I returned \$5
- My payoff: $\$9 - \$5 = \$4$

Now Round 3: I'm SENDER vs Finn (new opponent)

Looking at the patterns:

- When I sent \$2 to Elara, they returned \$3 (50% of what they received)
- When Hana sent \$3 to me (60% of their endowment), I returned \$5 (56% of what I received)

Hana was quite generous with a \$3 send. I should consider being similarly generous with Finn to encourage cooperation. However, I also need to balance risk.

Sending \$3 seems like a good cooperative move that matches Hana's generosity. This would give Finn \$9 to work with, and if they return a similar proportion to what I returned to Hana (around 50-60%), I could get back \$4.5-5.4, resulting in a payoff of around \$6.5-7.4.

Let me send \$3 to match the generosity shown by Hana and build a cooperative reputation in this multi-agent setting.

usage:

```
{
  "input_tokens": 1084,
  "output_tokens": 448,
  "reasoning_tokens": 276
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$16.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Galen sent me \$3 (60% of their endowment), the same level of trust that Hana showed
- When Hana sent me \$3, I returned \$5
- Consistency is important for building trust

I'll return the same amount to Galen to maintain a consistent and fair approach.

```
{'return': 5}
```

Reasoning:

Let me analyze this situation:

- Galen sent me \$3 (60% of their \$5), showing trust
- I received \$9
- In the last round, Finn returned \$4.5 out of \$9 (50%) when I sent \$3
- Previously, when Hana sent me \$3, I returned \$5 (56%)

Galen showed the same level of trust as Hana did. I should be consistent in my reciprocation.

Returning \$5 would:

- Give Galen \$7 total ($\$5 - \$3 + \$5 = \7)

- Leave me with \$4 (\$9 - \$5 = \$4)
- Maintain consistency with how I treated Hana's similar generosity

This seems fair and maintains my reputation as a cooperative player.

```
Usage: {
  "input_tokens": 1339,
  "output_tokens": 282,
  "reasoning_tokens": 148
}
```

GAME - Agent_8 - Sender - dyad_3

Timestamp: 2026-06-04T16:19:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_8",
    "Agent_5"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_5": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Galen sent \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after the 3x multiplier
- Total wealth created: \$11 (Galen has \$2 left, I have \$9)

Galen demonstrated trust by sending a substantial portion of their endowment. I want to reciprocate fairly while also benefiting from their trust.

If I split the total wealth approximately equally:

- Returning \$4 would give Galen \$6 total (gain of \$1)
- I would keep \$5 (gain of \$5)

This rewards Galen's trust while acknowledging that as the receiver, I also benefit from the interaction. This should encourage continued cooperation.

```
'''json
{"return": 4.0}
'''
```

reasoning:

Let me understand the situation:

- I'm Hana, the RECEIVER in this round
- My opponent is Galen (Agent_7), who is the SENDER
- Galen sent me \$3.0 out of their \$5.0 endowment (60%)
- I receive $\$3.0 \times 3 = \9.0
- I need to decide how much to return (between \$0 and \$9.0)
- Galen will keep: $\$5 - \$3 + [\text{what I return}] = \$2 + [\text{what I return}]$
- I will keep: $\$9 - [\text{what I return}]$

This is the classic trust game. Galen showed considerable trust by sending 60% of their endowment. This is the first round, so I need to establish a baseline for reciprocity.

Some considerations:

1. If I return nothing, I get \$9 and Galen gets \$2 (unfair to Galen who trusted me)
2. If I return \$3, I get \$6 and Galen gets \$5 (Galen breaks even, I gain \$6)
3. If I return \$4.5, I get \$4.5 and Galen gets \$6.5 (we both gain, roughly split the surplus)
4. If I return \$6, I get \$3 and Galen gets \$8 (Galen gains more than me)

The gain from cooperation is $\$9 - \$3 = \$6$ (the amount created by the 3x multiplier). A fair split would give us each \$3 of this gain:

- Galen ends with: $\$2 + \$3 = \$5$ (breaks even)
- I end with: $\$9 - \$3 = \$6$ (gain \$6)

Wait, that's not quite right. Let me reconsider.

Initial state: Galen has \$5, I have \$0

After send: Galen has \$2, I have \$9

The transaction created \$6 in total surplus. If I split this equally, Galen gets \$3 back (ending at \$5) and I keep \$6. But since I started with nothing while Galen started with \$5, a 50/50 split of the final \$11 pool might be more equitable-each of us ending with \$5.50.

To achieve that, I'd return \$3.50 to Galen. I could also consider a reciprocity approach: since Galen sent 60% of their endowment, returning 60% of what I received (\$5.40) would give them \$7.40 total, though that feels overly generous.

A middle ground makes more sense-returning \$4 to \$4.50 rewards their trust while letting me benefit meaningfully from receiving the money. I'll go with \$4, which gives Galen \$6 (a \$1 gain) and leaves me with \$5 (a \$5 gain). This balances reciprocity with my own interests and should encourage continued cooperation in future rounds.

Checking the fairness: with \$11 total, a 50/50 split would be \$5.50 each. Galen starts with \$2, so they'd need \$3.50 more to reach that. Returning \$4 is slightly above that threshold, which should build stronger trust without being excessive.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1268,
  "reasoning_tokens": 610
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$5.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Last round, Galen sent me \$3 (60%), showing trust, and I reciprocated by returning a fair amount
- Now I'm the sender with a chance to establish cooperation with Briar
- Sending a meaningful amount can build trust in this iterated game

I'll mirror the trust Galen showed me and send \$3 (60% of my endowment). This demonstrates good faith and gives us both the opportunity to benefit from cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

Now I'm in the role of sender against Briar (Agent_2).

Let me think about this strategically:

1. In the last round, I was receiver and Galen sent me \$3 (60% of endowment), showing significant trust. I returned \$4, which gave them a net gain of \$1 (\$6 total vs \$5 starting).
2. Now I'm the sender. I need to decide how much to trust Briar.
3. Key considerations:
 - This is an iterated game with multiple rounds

- Building cooperation early can lead to better outcomes
- If I send nothing, I get \$5 but miss out on potential gains
- If I send everything and they return nothing, I get \$0
- If I send and they reciprocate fairly, we both benefit

4. In the previous round, Galen sent 60% (\$3), which showed good faith. That worked out reasonably well.

5. Options:

- Send \$0: Safe but no gains, doesn't build trust
- Send \$1-2: Conservative, shows some trust
- Send \$3: Matches what Galen did, shows good faith
- Send \$4-5: Very trusting, risky

6. Since this is a multi-round game and I want to establish cooperation, I should send a meaningful amount. Sending \$3 (60%) mirrors what Galen did and shows I'm willing to trust and cooperate.

7. If Briar returns fairly (like I did), we'll both benefit. If they don't, I'll learn about their strategy and can adjust.

I'll send \$3 to match the level of trust Galen showed me.

usage:

```
{
  "input_tokens": 898,
  "output_tokens": 514,
  "reasoning_tokens": 358
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$12.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Cyra sent \$3 (60% of their endowment), showing significant trust
- I received \$9.0
- Last round, Briar returned \$5 to me when I sent \$3, which was very generous

Cyra is showing the same level of trust (60%) that I showed Briar. Since Briar reciprocated generously with me, I want to pay that forward and establish a pattern of strong reciprocity. This encourages cooperation in future rounds.

Returning \$5 gives Cyra a total of \$7 (gain of \$2), and I keep \$4 (gain of \$4). This is fair and mirrors the generosity I experienced.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze this situation:

- Last round with Briar: I sent \$3 (60%), it became \$9, and Briar returned \$5. That was very generous! I ended with \$7.
- Now with Cyra: They sent \$3 (60%), I received \$9.

Cyra is showing the same level of trust that I showed Briar (60% of endowment). Briar was very generous and returned \$5, which gave me a great payoff.

Let me think about what's fair:

- Cyra has \$2 left, I have \$9
- Total wealth: \$11

If I return \$4 (like I did with Galen):

- Cyra gets \$6 total (gain of \$1)
- I keep \$5 (gain of \$5)

If I return \$5 (like Briar did for me):

- Cyra gets \$7 total (gain of \$2)
- I keep \$4 (gain of \$4)

Briar's generosity was notable and helped establish trust. Since Cyra is showing the same trust level (60%), I should reciprocate similarly. Returning \$5 would be consistent with the generosity Briar showed me, and would encourage continued cooperation.

However, I want to be thoughtful here. Briar's return of \$5 was quite generous. Let me think about what's sustainable and fair:

Actually, returning \$4-5 seems reasonable. \$5 mirrors what Briar did for me, but \$4 is also fair and similar to what I did with Galen.

I think I'll return \$4.5 as a middle ground - slightly more generous than my first round, acknowledging the positive interaction with Briar, but not quite as generous as what Briar gave me.

Actually, let me reconsider. In these iterated games, establishing a reputation for fair reciprocation is important. Since Cyra sent the same amount (60%) that I sent to Briar, and Briar was generous, I should consider being similarly generous to encourage cooperation.

Matching Briar's \$5 return feels right-it reinforces the pattern of mutual generosity and signals I'm someone who reciprocates trust.

usage:

```
{
  "input_tokens": 1176,
  "output_tokens": 681,
  "reasoning_tokens": 463
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$16.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- My strategy of sending \$3 (60%) has worked well so far
- Briar reciprocated generously when I sent \$3, returning \$5
- I've been reciprocating fairly when I'm the receiver
- This establishes a pattern of cooperation

I'll continue with the same cooperative approach that has been successful. Sending \$3 demonstrates trust and gives Elara the opportunity to benefit while potentially returning a fair amount.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the situation:

- I've been establishing a pattern of cooperation
- When Galen sent me \$3 (60%), I returned \$4
- When I sent Briar \$3 (60%), they returned \$5 (very generous)
- When Cyra sent me \$3 (60%), I returned \$5 (reciprocating Briar's generosity)

Now I'm the sender against Elara, a new opponent. The pattern has been:

- Send \$3 (60%) shows trust and has been working well
- Others have reciprocated fairly or generously

I should continue this cooperative strategy. Sending \$3 has been successful and demonstrates trust while not being overly risky.

```
'''json
{"send": 3}
'''
```

```
Usage: {
  "input_tokens": 1494,
  "output_tokens": 297,
  "reasoning_tokens": 161
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-06-04T16:19:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_8",
      "Agent_5"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_5": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$2.0, that is 40.0% of its total endowment. You receive \$6.0. How much do you return? (0-6.0)

Message 3 role=assistant

content:

Looking at this first round situation:

- Briar sent \$2.0 (40% of their endowment), showing a moderate level of trust
- I received \$6.0 after the 3x multiplier
- This created \$6.0 of value through cooperation

Since this is an iterated game and I want to establish a cooperative relationship, I should return a fair amount that:

1. Rewards Briar's trust
2. Ensures both players benefit
3. Encourages future cooperation

I'll split the \$6.0 received amount equally, returning \$3.0. This gives:

- Briar: \$3 (kept) + \$3 (returned) = \$6 total
- Me: \$6 (received) - \$3 (returned) = \$3 total

This demonstrates I'm a trustworthy cooperator and should encourage continued positive interactions.

```
'''json
{"return": 3.0}
'''
```

reasoning:

Let me understand the situation:

1. I am Elara, playing as the RECEIVER in Round 1
2. My opponent is Briar (Agent_2), who is the SENDER
3. Briar sent me \$2.0 out of their \$5 endowment (40%)
4. I receive $\$2.0 \times 3 = \6.0
5. I need to decide how much to return (between \$0 and \$6.0)

Let me think about the strategic considerations:

This is Round 1 of an iterated game. The key considerations are:

- Building trust for future rounds
- Being fair/reciprocal
- Maximizing long-term payoffs

If this were a one-shot game, the rational selfish move would be to return \$0. However, since this is iterated and I'll be playing multiple rounds (possibly with the same opponent again), I need to establish cooperation.

Briar sent 40% of their endowment, which shows some level of trust and cooperation. A reasonable reciprocal approach would be to return an amount that:

1. Rewards their cooperative behavior
2. Establishes me as trustworthy for future interactions
3. Ensures both players benefit

Let me calculate fair returns:

- Briar started with \$5, sent \$2, so has \$3 left
- I received \$6
- If I return nothing: Briar gets \$3 total, I get \$6 total
- If I return \$3 (50%): Briar gets \$6 total, I get \$3 total
- If I return \$2: Briar gets \$5 total, I get \$4 total
- If I return \$2.5: Briar gets \$5.5 total, I get \$3.5 total

A common approach in trust games is to aim for equality or to ensure the sender doesn't lose. Returning \$2 would give Briar \$5 (breaking even with their initial endowment) and me \$4 (still a profit).

Another way to frame this: Briar's \$2 investment generated \$6 through the multiplier. If I split the gains equally, returning \$3 means we each end with \$3 profit beyond Briar's original \$3 kept. Alternatively, returning \$2 ensures Briar doesn't lose money while I still gain \$4.

I'm leaning toward returning \$3 as a cooperative opening move. While behavioral economics suggests 1/3 of received funds (\$2) is considered fair, the equal split signals stronger commitment to mutual benefit and should build trust for future rounds.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 996,
  "reasoning_tokens": 540
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$2.00 to you, it became \$6.00, and you returned \$3.00. Your payoff was \$3.00. Your total earnings across all rounds are \$3.00.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm now the sender against a new opponent, Cyra
- In Round 1, I demonstrated fair reciprocity by returning 50% to Briar
- This is a fresh interaction where I need to establish cooperation

I'll send \$2.5 (50% of my endowment). This shows:

1. Willingness to cooperate
2. A reasonable level of trust
3. Consistency with cooperative norms (slightly more generous than Briar's 40%)

If Cyra reciprocates fairly, we'll both benefit from the multiplier effect.

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me analyze this situation:

Round 2: I'm the SENDER against Cyra (a new opponent)

Previous round recap:

- I was RECEIVER against Briar
- Briar sent \$2.0 (40% of endowment)
- I received \$6.0 and returned \$3.0
- My payoff: \$3.0
- Briar's payoff: \$3 (kept) + \$3 (returned) = \$6.0

Now I'm playing with Cyra for the first time. I need to decide how much to send.

Considerations:

1. This is an iterated game - building reputation matters
2. I showed good reciprocity in Round 1 by returning 50% to Briar
3. With Cyra, I need to establish trust from scratch
4. Sending a moderate to high amount shows I'm willing to cooperate
5. Given that I returned fairly to Briar, I might expect reciprocity if I send to Cyra

Strategy options:

- Send \$0: Safe but no cooperation, no growth
- Send \$1-2: Conservative, low risk
- Send \$2-3: Moderate cooperation (like Briar did)
- Send \$4-5: High trust/cooperation

Since Briar sent 40% (\$2), and I responded well, I should probably send a similar or slightly higher amount to Cyra to establish good faith. Let me send \$2.5 (50% of my endowment), which is slightly more generous than what Briar sent to me.

This shows:

- I'm willing to cooperate
- I'm not maximally risk-averse
- I trust the process
- I'm hoping for reciprocity

If Cyra returns fairly (around 50% like I did), I'd get back around \$3.75, giving me a total of \$3 + \$3.75 = \$6.75 for the round.

usage:

```
{
  "input_tokens": 922,
  "output_tokens": 605,
  "reasoning_tokens": 375
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the SENDER against Cyra. You sent \$2.5, it became \$7.5 for them, and they returned \$3.75 to you. Your payoff was \$6.25. Your total earnings across all rounds are \$9.25.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

- Round 1: As receiver, I returned 50% fairly to Briar
- Round 2: As sender, I sent 50% to Cyra, who reciprocated with a 50% return rate
- My cooperation strategy is working well

Now facing Aster, I'll maintain my consistent cooperative approach by sending \$2.5 (50% of

my endowment). This:

1. Signals trustworthiness
2. Maintains my established pattern
3. Gives room for mutual benefit

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER against Briar

- Briar sent \$2.0 (40% of endowment)
- I received \$6.0
- I returned \$3.0 (50% of what I received)
- My payoff: \$3.0

Round 2: I was SENDER against Cyra

- I sent \$2.5 (50% of my endowment)
- It became \$7.5 for them
- They returned \$3.75 (50% of what they received)
- My payoff: \$5 - \$2.5 + \$3.75 = \$6.25

So Cyra reciprocated at exactly the same rate I did (50% return rate). This is good cooperation!

Round 3: I'm SENDER again against Aster (a new opponent)

My strategy so far has been cooperative:

- As receiver: returned 50% of what I received
- As sender: sent 50% of my endowment

Both interactions went well with fair reciprocity. Since I'm building a reputation as a cooperative player and this seems to be working (Cyra reciprocated fairly), I should continue this strategy.

I'll send \$2.5 again (50% of my endowment) to Aster. This maintains consistency in my cooperative approach.

usage:

```
{
  "input_tokens": 1217,
  "output_tokens": 459,
  "reasoning_tokens": 259
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$2.5, it became \$7.5 for them, and they returned \$3.5 to you. Your payoff was \$6.0.
Your total earnings across all rounds are \$15.25.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

- Hana sent \$3.0 (60% of their endowment) - a strong show of trust
- I received \$9.0 after the multiplier
- My pattern has been to return around 50% of what I receive

Since Hana showed high trust by sending 60%, I'll reward that with a 50% return rate, giving back \$4.5.

This results in:

- Hana: \$2 (kept) + \$4.5 (returned) = \$6.5 total

- Me: \$9 - \$4.5 = \$4.5 total

Both of us benefit significantly from the cooperation.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at this situation:

- Hana sent \$3.0 (60% of their endowment) - showing high trust
- I received \$9.0 after the 3x multiplier
- This is a strong cooperative signal

My track record:

- Round 1 (receiver vs Briar): They sent \$2.0, I received \$6.0, returned \$3.0 (50%)
- Round 2 (sender vs Cyra): I sent \$2.5, they received \$7.5, returned \$3.75 (50%)
- Round 3 (sender vs Aster): I sent \$2.5, they received \$7.5, returned \$3.5 (46.7%)

I've consistently shown fair reciprocity. Hana sent more than my average (60% vs my 50%), showing higher trust.

I should return around 50% of the \$9.0 received, which would be \$4.5.

This gives:

- Hana: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 (received) - \$4.5 (returned) = \$4.5 total

This maintains my pattern of fair reciprocity and rewards Hana's high trust.

```
Usage: {
  "input_tokens": 1512,
  "output_tokens": 480,
  "reasoning_tokens": 217
}
```

GAME - Agent_1 - Sender - dyad_4

Timestamp: 2026-06-04T16:20:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_3": "Cyra"
    },
  },
  "agents": [
    "Agent_1",
    "Agent_3"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_3": "trustee"
  },
  "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. By sending a substantial amount, I can signal trust and potentially build a reciprocal relationship that benefits us both over multiple rounds.

If I send \$4, Finn receives \$12 (due to the 3x multiplier). If Finn returns a fair share, we both end up better off than if I played it safe and sent nothing.

This is a trust-building move that could lead to mutual benefit in subsequent rounds.

{'send': 4}

reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Aster) and my opponent is Finn (Agent_6).

Let me think about the strategic considerations:

1. As the sender, I receive \$5 and can send \$0-\$5 to the receiver
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return to me
4. My payoff: \$5 - sent + returned
5. Receiver's payoff: 3×sent - returned

This is a classic trust game. The socially optimal outcome is to send all \$5, which become

s \$15 for the receiver. If the receiver returns \$8-\$10, we're both better off than if I send nothing.

However, there's a risk: the receiver could keep everything and return \$0. In that case:

- If I send \$5: I get \$0, receiver gets \$15
- If I send \$0: I get \$5, receiver gets \$0

Since this is an iterated game with multiple rounds, there are opportunities for:

1. Building trust and cooperation
2. Establishing reputation
3. Reciprocity patterns

In the first round, I don't know anything about Finn's behavior. Some strategies:

1. ****Trust approach****: Send a substantial amount (\$4-\$5) to signal trust and cooperation
2. ****Cautious approach****: Send a moderate amount (\$2-\$3) to test the waters
3. ****Conservative approach****: Send little or nothing (\$0-\$1) to avoid risk

Given that this is iterated and I want to establish potential cooperation, I think a moderate-to-high amount makes sense. Sending \$3-\$4 shows willingness to cooperate while not being completely vulnerable.

If I send \$3:

- Finn receives \$9
- If Finn returns \$4-\$5, we both benefit
- I'd end with $\$5 - \$3 + \$4 - \$5 = \$6 - \7

- Finn would end with $\$9 - \$4 - \$5 = \$4 - \$5$

Sending \$4 creates better mutual outcomes: Finn gets \$12, and if he returns \$5-\$7, I end up with \$6-\$8 while he keeps \$5-\$7. This stronger signal of cooperation sets a better foundation for future rounds, so I'll send \$4.

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 730,
  "reasoning_tokens": 492
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the 3x multiplier
- In Round 1, Finn returned 50% of what they received (\$6 out of \$12), which was fair
- I should reciprocate Dorian's trust similarly

Returning approximately half (\$4.5) would give Dorian a total of \$6.5 (\$2 kept + \$4.5 returned) and leave me with \$4.5. This maintains a cooperative pattern.

{'return': 4.5}

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$4 to Finn
- Finn received \$12 (4×3)
- Finn returned \$6 to me
- My payoff: $\$5 - \$4 + \$6 = \7
- Finn's payoff: $\$12 - \$6 = \$6$

So Finn returned half of what they received, which seems fair and cooperative.

Now Round 2:

- I'm the RECEIVER
- Dorian sent me \$3
- I received \$9 (3×3)
- I need to decide how much to return

Dorian kept: $\$5 - \$3 = \$2$
 I currently have: \$9

If I return nothing:

- Dorian gets: \$2
- I get: \$9

If I return \$4.5 (half):

- Dorian gets: $\$2 + \$4.5 = \$6.5$
- I get: $\$9 - \$4.5 = \$4.5$

If I return \$6:

- Dorian gets: $\$2 + \$6 = \$8$
- I get: $\$9 - \$6 = \$3$

Following the reciprocity principle that Finn showed me (returning half), I should return around \$4.5. This would give Dorian a total of \$6.5 and me \$4.5.

However, I could also consider being slightly more generous to encourage cooperation, or match what Finn did in proportion. Finn returned 50% of what they received.

I think returning \$4.5 (50% of what I received) is a fair and cooperative move that encourages continued trust-building in this iterated game.

usage:

```
{
  "input_tokens": 807,
  "output_tokens": 544,
  "reasoning_tokens": 302
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
 Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
 Your total earnings across all rounds are \$11.5.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.

How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Elara sent me \$2.5, showing trust
- I received \$7.5 after the 3x multiplier
- I've been establishing a cooperative pattern (returning about 50% of what I receive)

Returning approximately half (\$3.5) would give Elara a total of \$6 (\$2.5 kept + \$3.5 returned) and leave me with \$4. This maintains fair reciprocity.

```
{'return': 3.5}
```

reasoning:

Looking at this situation:

- Elara sent me \$2.5, showing some trust (though less than Dorian's \$3)
- I received \$7.5 after the 3x multiplier
- In previous rounds, I've established a pattern of returning roughly 50% of what I receive (Finn returned 50%, I returned 50% to Dorian)

Elara keeps: $\$5 - \$2.5 = \$2.5$

If I return around half ($\$3.5-4$), Elara would get: $\$2.5 + \text{return}$

I would get: $\$7.5 - \text{return}$

Returning \$3.5 would give:

- Elara total: $\$2.5 + \$3.5 = \$6.0$
- Me: $\$7.5 - \$3.5 = \$4.0$

Returning \$4.0 would give:

- Elara total: $\$2.5 + \$4.0 = \$6.5$
- Me: $\$7.5 - \$4.0 = \$3.5$

I think returning around \$3.5 is fair - it's slightly less than 50% but reasonable given they sent slightly less. This maintains cooperation while being proportional to their trust level.

usage:

```
{
  "input_tokens": 1095,
  "output_tokens": 416,
  "reasoning_tokens": 204
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.5. Your payoff was \$4.0.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the situation:

- This is my first interaction with Cyra
- My cooperative strategy has been working well so far
- In Round 1, I sent \$4 to Finn and they returned \$6, which was mutually beneficial
- As a receiver, I've been fair by returning about 50%

I'll continue my cooperative approach and send a substantial amount to signal trust and potentially establish a beneficial relationship with Cyra.

```
{'send': 4}
```

Reasoning:

Let me analyze the situation:

Round history:

- Round 1: I was SENDER, sent \$4 to Finn, they returned \$6. My payoff: \$7
- Round 2: I was RECEIVER, Dorian sent \$3, I returned \$4.5. My payoff: \$4.5
- Round 3: I was RECEIVER, Elara sent \$2.5, I returned \$3.5. My payoff: \$4

Now Round 4: I'm SENDER against Cyra (new opponent)

My strategy so far:

- As sender: I sent \$4 (80% of my endowment) showing high trust
- As receiver: I've been returning about 50% of what I receive (fair reciprocity)

This has been working well:

- Finn returned a fair amount (\$6 out of \$12)
- I've been cooperative as receiver

For this round with Cyra, since this is:

1. First interaction with Cyra
2. An iterated game where cooperation can be beneficial
3. I've established a pattern of cooperation

I should continue my cooperative strategy. Sending \$4 again would be consistent with my opening move against Finn, which worked well. Alternatively, I could send \$3 to be slightly more conservative while still showing trust.

Given that my first high-trust move worked well, I'll send \$4 again to establish cooperation with Cyra.

```
Usage: {
  "input_tokens": 1352,
  "output_tokens": 438,
  "reasoning_tokens": 295
}
```

GAME - Agent_3 - Receiver - dyad_4

Timestamp: 2026-06-04T16:20:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_3": "Cyra"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_3"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_3": "trustee"
  },
  "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance cooperation with caution since I don't yet know how Dorian will respond.

The game rewards mutual trust - if I send money and Dorian returns a fair amount, we both benefit from the 3x multiplier. However, this is our first interaction, so I want to signal willingness to cooperate while not being overly vulnerable.

I'll send **\$3**. This amount:

- Shows I'm cooperative and willing to trust
- Gets tripled to \$9, giving Dorian meaningful options
- Leaves me with \$2 as a safeguard
- If Dorian returns even \$2-3, we both come out ahead

This sets a foundation for building trust over multiple rounds.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated game. I'm playing as the sender with Dorian (Agent_4).

Let me think about the game theory here:

- I receive \$5
- I can send \$0-\$5 to the receiver
- The amount sent gets multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal outcome for both of us collectively would be for me to send all \$5, which becomes \$15 for the receiver. If the receiver returns half (\$7.50), we'd both end up with \$7.50, which is better than the \$5 I start with.

However, there's a risk: the receiver could return nothing, leaving me with \$0 and them with \$15.

Since this is an iterated game with multiple rounds, I should consider:

1. Building trust early could lead to better outcomes over time
2. Being too trusting initially might be exploited

3. A moderate initial move might signal cooperation while testing the waters

Common strategies in iterated trust games:

- Start with moderate cooperation to signal willingness but test the waters
- Reciprocate based on partner's behavior in subsequent rounds

For a first round, I think sending \$3 is reasonable:

- It shows I'm willing to cooperate
- It's not so much that I'm devastated if they return nothing
- If they send back at least \$3, we both benefit
- It gives them \$9 to work with, enough to show their intentions

Let me calculate:

- If I send \$3: I have \$2 left
- Receiver gets \$9
- If they return \$4, I end with \$6, they end with \$5
- If they return \$5, I end with \$7, they end with \$4
- If they return \$0, I end with \$2, they end with \$9

I'll go with \$3 as my opening move.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 642,
  "reasoning_tokens": 448
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.

How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****My experience from Round 1:****

- Dorian returned \$4.5 out of \$9 (50%) - a fair, cooperative move
- We both benefited from the exchange

****Current situation:****

- Elara sent \$2.5 (50% of their endowment) - showing trust
- I received \$7.5
- I should reciprocate fairly to encourage future cooperation

Following the principle Dorian demonstrated, I'll return ****50% of what I received****, which maintains fairness and establishes me as a trustworthy partner. This gives:

- Elara: \$5 - \$2.5 + \$3.75 = \$6.25
- Me: \$7.5 - \$3.75 = \$3.75

Both of us gain, which encourages cooperation.

```
'''json
{"return": 3.75}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender, sent \$3 to Dorian
- It became \$9 for Dorian
- Dorian returned \$4.5 to me

- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Dorian's payoff: $\$9 - \$4.5 = \$4.5$

So Dorian returned exactly half of what they received. That's a fair and cooperative response.

Now in Round 2:

- I'm the receiver
- Elara sent me \$2.5
- I received \$7.5 (2.5×3)
- I need to decide how much to return (0-7.5)

Elara sent \$2.5 out of \$5, which is 50% - a moderate/fair amount showing some trust.

Considerations:

1. I should establish myself as a trustworthy player for future rounds
2. Returning a fair amount encourages cooperation in future interactions
3. If I return too little, it may harm my reputation with other agents
4. If I return too much, I might be seen as exploitable

Let me think about fair return amounts:

- If I return half (\$3.75), Elara would get: $\$5 - \$2.5 + \$3.75 = \6.25 , and I'd keep \$3.75
- That seems fair - both of us gain

However, I could also think about it proportionally. Dorian returned exactly 50% of what they received (\$4.5 out of \$9). That's a reasonable standard.

Following the same principle: 50% of \$7.5 = \$3.75

This would give:

- Elara: $\$5 - \$2.5 + \$3.75 = \6.25 (gain of \$1.25)
- Me: $\$7.5 - \$3.75 = \$3.75$ (gain of \$3.75)

- But there's an inherent asymmetry here since I benefit more from the multiplier effect. Alternatively, I could aim for equal payoffs: if I return \$2.5, both Elara and I end up with \$5. That creates perfect parity and demonstrates genuine reciprocity.

However, that feels somewhat unfair to Elara given the risk she took by sending first. Looking at how Dorian approached this-he kept half and returned half-following that same 50% principle would mean returning \$3.75. This establishes a pattern of fairness and signals that I'm a trustworthy partner, which should encourage more sending in future interactions.

I'll go with \$3.75.

usage:

```
{
  "input_tokens": 866,
  "output_tokens": 1062,
  "reasoning_tokens": 517
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.75. Your payoff was \$3.75. Your total earnings across all rounds are \$10.25.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

****Previous rounds:****

- As sender with Dorian: Sent \$3 ? received \$4.5 back (good cooperation)

- As receiver with Elara: They sent \$2.5, I returned \$3.75 (fair exchange)

****Current situation:****

- New opponent (Hana) - no history yet
- Want to signal cooperation while being prudent

I'll send ****\$3**** again, as this amount:

- Worked well with Dorian in Round 1
- Shows willingness to cooperate
- Becomes \$9 for Hana, giving room for mutual benefit
- Keeps \$2 as my baseline

This maintains my cooperative strategy while learning about new partners.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

****Past experience:****

- Round 1 (Sender vs Dorian): Sent \$3, they returned \$4.5 (50% of \$9) ? I got \$6.5
- Round 2 (Receiver vs Elara): They sent \$2.5, I returned \$3.75 (50% of \$7.5) ? I got \$3.75

****Current situation:****

- Round 3 (Sender vs Hana): New opponent, no history
- I have \$5 to decide how much to send

****Strategy considerations:****

1. Both previous interactions showed 50% return rates, which worked well
2. Hana is a new opponent - I don't know their style yet
3. Sending a moderate-to-high amount signals cooperation
4. Given my positive experiences, I should maintain a cooperative stance

****Options:****

- Send \$3 again (same as Round 1) - showed it worked well with Dorian
- Send more (\$4+) - shows even more trust
- Send less (\$2-2.5) - more cautious with unknown opponent

I think \$3 is a good amount again. It:

- Worked well before
- Shows I'm cooperative without being reckless
- Gives Hana \$9 to work with
- Protects \$2 for myself

Let me go with \$3.

usage:

```
{
  "input_tokens": 1215,
  "output_tokens": 509,
  "reasoning_tokens": 272
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$17.25.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Previous experience:****

- Dorian and Hana both returned roughly 50-56% when they were receivers
- I returned 50% (\$3.75 from \$7.5) to Elara
- Pattern of cooperative fair returns

****Current situation:****

- Aster sent \$4 (80% of endowment) - showing high trust!
- I received \$12
- Should reciprocate this trust appropriately

I'll return ****\$6**** (50% of what I received), which:

- Maintains the cooperative norm
- Rewards Aster's high trust
- Gives Aster: $\$5 - \$4 + \$6 = \7 (good return)
- Leaves me: $\$12 - \$6 = \$6$
- Both of us benefit from the exchange

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Let me analyze the situation:

****Previous rounds:****

- Round 1 (Sender vs Dorian): Sent \$3, received \$4.5 back (50% return) - payoff \$6.5
- Round 2 (Receiver vs Elara): Received \$7.5, returned \$3.75 (50%) - payoff \$3.75
- Round 3 (Sender vs Hana): Sent \$3, received \$5.0 back (55.6% return) - payoff \$7.0

****Current situation:****

- Aster sent me \$4.0 (80% of their endowment!) - very trusting
- I received \$12.0 (after 3x multiplier)
- I need to return an amount

****Analysis:****

Aster showed high trust by sending \$4. This is more than anyone has sent so far. I should reciprocate appropriately.

Looking at the pattern:

- When I received \$7.5, I returned 50% = \$3.75
- When I received \$9.0, Hana returned 55.6% = \$5.0
- When I received \$9.0, Dorian returned 50% = \$4.5

Aster is taking more risk by sending \$4. A fair return would be around 50% of what I received, which is \$6.0.

If I return \$6.0:

- Aster gets: $\$5 - \$4 + \$6 = \7.0
- I get: $\$12 - \$6 = \$6.0$
- Both benefit, and Aster's trust is rewarded

This maintains my cooperative reputation.

```
Usage: {
  "input_tokens": 1554,
  "output_tokens": 606,
  "reasoning_tokens": 282
}
```

ROUND 5

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

dyad_1: Agent_3 (Cyra)=Sender, Agent_2 (Briar)=Receiver
dyad_2: Agent_4 (Dorian)=Sender, Agent_8 (Hana)=Receiver
dyad_3: Agent_6 (Finn)=Sender, Agent_1 (Aster)=Receiver
dyad_4: Agent_7 (Galen)=Sender, Agent_5 (Elara)=Receiver

Parsed game decisions:

dyad_1: Agent_3 (Cyra) sent \$3; received \$9; Agent_2 (Briar) returned \$5; payoffs Agent_3

```

(Cyra)=$7, Agent_2 (Briar)=$4
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw":
    5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_4 (Dorian) sent $3; received $9; Agent_8 (Hana) returned $5; payoffs Agent_4
(Dorian)=$7, Agent_8 (Hana)=$4
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw":
    5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_6 (Finn) sent $4; received $12; Agent_1 (Aster) returned $6; payoffs Agent_6
(Finn)=$7, Agent_1 (Aster)=$6
  Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw":
    6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_4: Agent_7 (Galen) sent $3; received $9; Agent_5 (Elara) returned $4.5; payoffs Agent_7
(Galen)=$6.5, Agent_5 (Elara)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
    4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 28.5,
  "Agent_2": 24.5,
  "Agent_3": 30.25,
  "Agent_4": 30.5,
  "Agent_5": 24.25,
  "Agent_6": 31.5,
  "Agent_7": 30.0,
  "Agent_8": 26.5
}

```

LLM CALLS

GAME - Agent_3 - Sender - dyad_1

Timestamp: 2026-06-04T16:20:24
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_3": "Cyra"
    },
    "agents": [
      "Agent_3",
      "Agent_2"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_2": "trustee",
      "Agent_3": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system
 content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.
How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****My experience from Round 1:****

- Dorian returned \$4.5 out of \$9 (50%) - a fair, cooperative move
- We both benefited from the exchange

****Current situation:****

- Elara sent \$2.5 (50% of their endowment) - showing trust
- I received \$7.5
- I should reciprocate fairly to encourage future cooperation

Following the principle Dorian demonstrated, I'll return ****50% of what I received****, which maintains fairness and establishes me as a trustworthy partner. This gives:

- Elara: \$5 - \$2.5 + \$3.75 = \$6.25
- Me: \$7.5 - \$3.75 = \$3.75

Both of us gain, which encourages cooperation.

```
'''json
{"return": 3.75}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender, sent \$3 to Dorian
- It became \$9 for Dorian
- Dorian returned \$4.5 to me
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Dorian's payoff: $\$9 - \$4.5 = \$4.5$

So Dorian returned exactly half of what they received. That's a fair and cooperative response.

Now in Round 2:

- I'm the receiver
- Elara sent me \$2.5
- I received \$7.5 (2.5×3)
- I need to decide how much to return (0-7.5)

Elara sent \$2.5 out of \$5, which is 50% - a moderate/fair amount showing some trust.

Considerations:

1. I should establish myself as a trustworthy player for future rounds
2. Returning a fair amount encourages cooperation in future interactions
3. If I return too little, it may harm my reputation with other agents
4. If I return too much, I might be seen as exploitable

Let me think about fair return amounts:

- If I return half (\$3.75), Elara would get: $\$5 - \$2.5 + \$3.75 = \6.25 , and I'd keep \$3.75
- That seems fair - both of us gain

However, I could also think about it proportionally. Dorian returned exactly 50% of what they received (\$4.5 out of \$9). That's a reasonable standard.

Following the same principle: 50% of \$7.5 = \$3.75

This would give:

- Elara: $\$5 - \$2.5 + \$3.75 = \6.25 (gain of \$1.25)
- Me: $\$7.5 - \$3.75 = \$3.75$ (gain of \$3.75)

- But there's an inherent asymmetry here since I benefit more from the multiplier effect. Alternatively, I could aim for equal payoffs: if I return \$2.5, both Elara and I end up with \$5. That creates perfect parity and demonstrates genuine reciprocity.

However, that feels somewhat unfair to Elara given the risk she took by sending first. Looking at how Dorian approached this-he kept half and returned half-following that same 50% principle would mean returning \$3.75. This establishes a pattern of fairness and signals that I'm a trustworthy partner, which should encourage more sending in future interactions.

I'll go with \$3.75.

usage:

```
{
  "input_tokens": 866,
  "output_tokens": 1062,
  "reasoning_tokens": 517
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.75. Your payoff was \$3.75.
Your total earnings across all rounds are \$10.25.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience so far:

****Previous rounds:****

- As sender with Dorian: Sent \$3 ? received \$4.5 back (good cooperation)
- As receiver with Elara: They sent \$2.5, I returned \$3.75 (fair exchange)

****Current situation:****

- New opponent (Hana) - no history yet
- Want to signal cooperation while being prudent

I'll send **\$3** again, as this amount:

- Worked well with Dorian in Round 1
- Shows willingness to cooperate
- Becomes \$9 for Hana, giving room for mutual benefit
- Keeps \$2 as my baseline

This maintains my cooperative strategy while learning about new partners.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

****Past experience:****

- Round 1 (Sender vs Dorian): Sent \$3, they returned \$4.5 (50% of \$9) ? I got \$6.5
- Round 2 (Receiver vs Elara): They sent \$2.5, I returned \$3.75 (50% of \$7.5) ? I got \$3.75

****Current situation:****

- Round 3 (Sender vs Hana): New opponent, no history
- I have \$5 to decide how much to send

****Strategy considerations:****

1. Both previous interactions showed 50% return rates, which worked well
2. Hana is a new opponent - I don't know their style yet
3. Sending a moderate-to-high amount signals cooperation
4. Given my positive experiences, I should maintain a cooperative stance

****Options:****

- Send \$3 again (same as Round 1) - showed it worked well with Dorian
- Send more (\$4+) - shows even more trust
- Send less (\$2-2.5) - more cautious with unknown opponent

I think \$3 is a good amount again. It:

- Worked well before
- Shows I'm cooperative without being reckless
- Gives Hana \$9 to work with
- Protects \$2 for myself

Let me go with \$3.

usage:

```
{
  "input_tokens": 1215,
  "output_tokens": 509,
  "reasoning_tokens": 272
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$17.25.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

****Previous experience:****

- Dorian and Hana both returned roughly 50-56% when they were receivers
- I returned 50% (\$3.75 from \$7.5) to Elara
- Pattern of cooperative fair returns

****Current situation:****

- Aster sent \$4 (80% of endowment) - showing high trust!
- I received \$12
- Should reciprocate this trust appropriately

I'll return ****\$6**** (50% of what I received), which:

- Maintains the cooperative norm
- Rewards Aster's high trust
- Gives Aster: \$5 - \$4 + \$6 = \$7 (good return)
- Leaves me: \$12 - \$6 = \$6
- Both of us benefit from the exchange

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze the situation:

****Previous rounds:****

- Round 1 (Sender vs Dorian): Sent \$3, received \$4.5 back (50% return) - payoff \$6.5
- Round 2 (Receiver vs Elara): Received \$7.5, returned \$3.75 (50%) - payoff \$3.75
- Round 3 (Sender vs Hana): Sent \$3, received \$5.0 back (55.6% return) - payoff \$7.0

****Current situation:****

- Aster sent me \$4.0 (80% of their endowment!) - very trusting
- I received \$12.0 (after 3x multiplier)
- I need to return an amount

****Analysis:****

Aster showed high trust by sending \$4. This is more than anyone has sent so far. I should reciprocate appropriately.

Looking at the pattern:

- When I received \$7.5, I returned 50% = \$3.75
- When I received \$9.0, Hana returned 55.6% = \$5.0
- When I received \$9.0, Dorian returned 50% = \$4.5

Aster is taking more risk by sending \$4. A fair return would be around 50% of what I received, which is \$6.0.

If I return \$6.0:

- Aster gets: \$5 - \$4 + \$6 = \$7.0
- I get: \$12 - \$6 = \$6.0
- Both benefit, and Aster's trust is rewarded

This maintains my cooperative reputation.

usage:

```
{
  "input_tokens": 1554,
  "output_tokens": 606,
  "reasoning_tokens": 282
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.

Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$23.25.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my performance so far:

****As sender:****

- Round 1 vs Dorian: Sent \$3 ? received \$4.5 (profit \$1.5)
- Round 3 vs Hana: Sent \$3 ? received \$5.0 (profit \$2.0)
- Both experiences positive

****Pattern:****

- Sending \$3 has consistently resulted in good returns
- Shows cooperation while maintaining prudence
- New opponent (Briar) - will apply same successful strategy

I'll send ****\$3**** again, as this amount:

- Has proven reliable across different partners
- Becomes \$9 for Briar
- Balances trust with risk management

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me review my history:

Round 1: SENDER vs Dorian - sent \$3, received \$4.5 back, payoff \$6.5
Round 2: RECEIVER vs Elara - received \$7.5, returned \$3.75, payoff \$3.75
Round 3: SENDER vs Hana - sent \$3, received \$5.0 back, payoff \$7.0
Round 4: RECEIVER vs Aster - received \$12.0, returned \$6.0, payoff \$6.0

Total: \$23.25 ?

Analysis:

- When I sent \$3 (rounds 1 & 3), I got back \$4.5 and \$5.0 respectively
- Both times I profited well (\$1.5 and \$2.0 net gain)
- As receiver, I've returned 50% consistently
- Others have shown cooperation

This round: New opponent Briar, I'm the sender

Strategy:

- \$3 has worked well so far (2 for 2 positive outcomes)
- Shows cooperation without being overly risky
- If they return 50%, I get \$4.5 back (net \$1.5 gain)
- If they return more generously like Hana, even better

I'll stick with \$3 as it's been working consistently.

```
Usage: {
  "input_tokens": 1666,
  "output_tokens": 496,
  "reasoning_tokens": 235
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T16:20:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
```

```

    "Agent_2": "Briar",
    "Agent_3": "Cyra"
  },
  "agents": [
    "Agent_3",
    "Agent_2"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_2": "trustee",
    "Agent_3": "investor"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$2.0, it became \$6.0 for them, and they returned \$3.0 to you. Your payoff was \$6.0. Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Hana sent me \$3 (60% of their initial \$5), showing significant trust
- This gave me \$9 to work with
- In the previous round, Elara returned \$3 out of \$6 (50%) when I sent \$2

Since Hana was more generous than I was with Elara, I should reciprocate appropriately. Returning \$5 would give Hana a total of \$7 (better than their initial \$5), while I keep \$4. This rewards cooperation and builds trust.

{'return': 5}

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the sender against Elara
- I sent \$2, which became \$6 for them
- Elara returned \$3 to me
- My payoff: $\$5 - \$2 + \$3 = \6
- Elara's payoff: $\$6 - \$3 = \$3$

So Elara returned exactly half of what they received. That's a fair and cooperative move.

Round 2:

- I'm now the receiver against Hana
- Hana sent me \$3, so I received \$9
- I need to decide how much to return

Hana showed significant trust by sending \$3 (a majority of their initial \$5). This is cooperative behavior.

If I follow Elara's example and return about half of what I received, that would be \$4.5.

Let me think about what's fair:

- Hana's initial: \$5
- Hana sent: \$3
- Hana currently has: \$2
- I received: \$9
- If I return \$4.5, Hana gets: $\$2 + \$4.5 = \$6.5$ total, I keep \$4.5
- If I return \$5, Hana gets: \$7 total, I keep \$4

Returning around \$4-5 seems fair and would encourage continued cooperation. Hana was generous in sending \$3, so I should reciprocate.

Let me return \$5. This gives Hana a total of \$7 (which is more than their initial \$5), and I keep \$4. This shows I'm trustworthy and willing to reward cooperation.

Wait, let me reconsider. Elara returned exactly half (\$3 out of \$6). If I follow that pattern, I'd return \$4.5 out of \$9. But Hana sent \$3 (60% of their initial amount) which is more generous than my \$2 (40%). So I might want to be slightly more generous back.

Returning \$5 makes sense-it's more than the 50% baseline and acknowledges Hana's above-average generosity while signaling I'm someone who reciprocates trust.

usage:

```
{
  "input_tokens": 809,
  "output_tokens": 641,
  "reasoning_tokens": 418
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$10.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern so far:

- Elara returned 50% when I sent \$2
- Hana sent \$3 (60% of their endowment), showing strong trust
- I returned 56% to Hana to reward that trust

I'll match Hana's generosity and send \$3 to Finn. This signals cooperation and could establish a good reciprocal relationship.

{'send': 3}

reasoning:

Let me think about this strategically:

Round 1: I was SENDER vs Elara

- I sent \$2, became \$6
- Elara returned \$3
- My payoff: $\$5 - \$2 + \$3 = \6

Round 2: I was RECEIVER vs Hana

- Hana sent \$3, became \$9 for me
- I returned \$5
- My payoff: $\$9 - \$5 = \$4$

Now Round 3: I'm SENDER vs Finn (new opponent)

Looking at the patterns:

- When I sent \$2 to Elara, they returned \$3 (50% of what they received)
- When Hana sent \$3 to me (60% of their endowment), I returned \$5 (56% of what I received)

Hana was quite generous with a \$3 send. I should consider being similarly generous with Finn to encourage cooperation. However, I also need to balance risk.

Sending \$3 seems like a good cooperative move that matches Hana's generosity. This would give Finn \$9 to work with, and if they return a similar proportion to what I returned to Hana (around 50-60%), I could get back \$4.5-5.4, resulting in a payoff of around \$6.5-7.4.

Let me send \$3 to match the generosity shown by Hana and build a cooperative reputation in this multi-agent setting.

usage:

```
{
  "input_tokens": 1084,
  "output_tokens": 448,
  "reasoning_tokens": 276
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$16.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Galen sent me \$3 (60% of their endowment), the same level of trust that Hana showed
- When Hana sent me \$3, I returned \$5
- Consistency is important for building trust

I'll return the same amount to Galen to maintain a consistent and fair approach.

{'return': 5}

reasoning:

Let me analyze this situation:

- Galen sent me \$3 (60% of their \$5), showing trust
- I received \$9
- In the last round, Finn returned \$4.5 out of \$9 (50%) when I sent \$3
- Previously, when Hana sent me \$3, I returned \$5 (56%)

Galen showed the same level of trust as Hana did. I should be consistent in my reciprocation. Returning \$5 would:

- Give Galen \$7 total (\$5 - \$3 + \$5 = \$7)
- Leave me with \$4 (\$9 - \$5 = \$4)
- Maintain consistency with how I treated Hana's similar generosity

This seems fair and maintains my reputation as a cooperative player.

usage:

```
{
  "input_tokens": 1339,
  "output_tokens": 282,
  "reasoning_tokens": 148
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.

Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.

Your total earnings across all rounds are \$20.5.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

- Cyra sent me \$3, showing the same level of trust as Hana and Galen
- I've consistently returned \$5 when receiving \$3
- Maintaining consistency helps build a reliable reputation

I'll continue with my established approach.

{'return': 5}

Reasoning:

Let me analyze the situation:

- Cyra sent me \$3 (60% of their endowment)
- This is now the third time someone has sent me exactly \$3
- I've consistently returned \$5 to both Hana and Galen when they sent me \$3
- Consistency is important for maintaining trust and a reputation

Following my established pattern, I should return \$5 to Cyra as well. This:

- Maintains consistency
- Rewards their trust appropriately
- Gives them \$7 total (better than their initial \$5)
- Leaves me with \$4

This seems like the fair and consistent approach.

```
Usage: {  
  "input_tokens": 1401,  
  "output_tokens": 224,  
  "reasoning_tokens": 144  
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-06-04T16:20:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agent_names": {  
      "Agent_4": "Dorian",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_4",  
      "Agent_8"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_4": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- In Round 1, Cyra sent me \$3.0 (60%), showing good trust
- I returned \$4.5, demonstrating fairness
- Now I'm the sender with a new partner, Aster

To establish cooperative play in this iterated game, I'll send the same amount that Cyra sent to me - \$3.0. This shows trust and can help build a pattern of mutual cooperation.

{'send': 3.0}

reasoning:

Now I'm the sender playing against Aster (Agent_1).

In the previous round:

- I was the receiver
- Cyra sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5
- We both ended up with equal amounts

Now I need to decide how much to send to Aster. Since this is an iterated game and I want to establish cooperation, I should send a reasonable amount to show trust.

Cyra sent 60% (\$3.0) in the first round, which showed good faith. I should reciprocate this cooperative approach with Aster.

If I send \$3.0:

- Aster receives \$9.0
- I have \$2.0 left
- If Aster returns fairly (say \$4.5 like I did), I'd end up with \$6.5
- If Aster returns less, I take a risk

Sending \$3.0 seems like a good cooperative strategy that matches what Cyra did with me. This establishes a pattern of trust and cooperation.

Alternatively:

- Sending \$2.5 (50%) would be slightly more conservative
- Sending \$4.0 (80%) would show even more trust
- Sending \$2.0 (40%) would be less trusting

I'll go with \$3.0 to match the cooperative behavior I experienced.

usage:

```
{  
  "input_tokens": 904,
```

```
    "output_tokens": 443,  
    "reasoning_tokens": 281  
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern:
- Round 1: Cyra sent me \$3.0, I returned \$4.5 (fair split)
- Round 2: I sent Aster \$3.0, they returned \$4.5 (same fair pattern)

This 60% send, 50% return pattern has been working well and building cooperative play. I'll continue this strategy with my new partner Galen.

```
{'send': 3.0}
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Cyra sent me \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5

Round 2: I was SENDER

- I sent Aster \$3.0 (60%)
- Aster received \$9.0
- Aster returned \$4.5
- My payoff: \$2.0 + \$4.5 = \$6.5

So both Cyra and Aster demonstrated the same behavior - when I sent/they sent \$3.0, we got back \$4.5. This is a 50-50 split pattern that seems fair.

Round 3: I'm SENDER again against a new opponent, Galen.

My total earnings: \$11.0

The pattern so far has been:

- Sending \$3.0 results in returns of \$4.5
- This gives a net gain for the sender (\$6.5 vs \$5 if kept all)
- And the receiver keeps \$4.5

This has worked well. I should continue with this cooperative strategy of sending \$3.0 to establish trust with Galen.

usage:

```
{  
  "input_tokens": 1159,  
  "output_tokens": 399,  
  "reasoning_tokens": 214  
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.

Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Finn showed very high trust by sending \$4.0 (80% of their endowment)
- I received \$12.0
- Finn currently has \$1.0 left

Following the successful pattern from previous rounds where I've been returning 50% of what I receive:

- If I return \$6.0: Finn ends with \$7.0, I keep \$6.0
- This rewards their high trust with a good return

{'return': 6.0}

reasoning:

Let me analyze this situation:

- Finn sent me \$4.0 (80% of their \$5 endowment) - this is very high trust
- I received \$12.0 after the 3x multiplier
- Finn currently has \$1.0 left (\$5 - \$4)

Looking at the pattern so far:

- Round 1: Cyra sent \$3.0 (60%), I returned \$4.5 (50% of \$9)
- Round 2: I sent \$3.0 (60%), Aster returned \$4.5 (50% of \$9)
- Round 3: I sent \$3.0 (60%), Galen returned \$4.5 (50% of \$9)

The pattern has been: send 60%, return 50% of what's received, leading to equal final payoffs (\$6.5 each).

Now Finn sent \$4.0 (80%), showing even more trust. If I return 50% of \$12.0, that would be \$6.0.

- Finn would have: \$1.0 + \$6.0 = \$7.0
- I would have: \$12.0 - \$6.0 = \$6.0

This is close to equal and rewards Finn's high trust. Let me stick with the 50% return pattern.

{'return': 6.0}

usage:

```
{
  "input_tokens": 1420,
  "output_tokens": 449,
  "reasoning_tokens": 214
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.

Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.

Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my consistent strategy:

- As a sender, I've been sending \$3.0 (60% of endowment)
- As a receiver, I've been returning 50% of what I receive
- This has established good cooperation with all partners

I'll maintain this cooperative pattern with my new partner Hana:

{'send': 3.0}

Reasoning:

Let me review my strategy so far:

Round 1: RECEIVER - Cyra sent \$3.0, I received \$9.0, returned \$4.5 (50% of received)
Round 2: SENDER - I sent \$3.0 to Aster, they returned \$4.5 (50% of \$9.0)
Round 3: SENDER - I sent \$3.0 to Galen, they returned \$4.5 (50% of \$9.0)
Round 4: RECEIVER - Finn sent \$4.0, I received \$12.0, returned \$6.0 (50% of received)

My strategy has been:

- As receiver: return 50% of what I receive
- As sender: send \$3.0 (60% of endowment)

This has been working well and creating cooperative outcomes. Everyone has been reciprocating appropriately. I should continue this pattern with Hana.

```
Usage: {  
  "input_tokens": 1388,  
  "output_tokens": 310,  
  "reasoning_tokens": 162  
}
```

GAME - Agent_8 - Receiver - dyad_2

Timestamp: 2026-06-04T16:20:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agent_names": {  
      "Agent_4": "Dorian",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_4",  
      "Agent_8"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_4": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.0. Your payoff was \$5.0. Your total earnings across all rounds are \$5.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Last round, Galen sent me \$3 (60%), showing trust, and I reciprocated by returning a fair amount
- Now I'm the sender with a chance to establish cooperation with Briar
- Sending a meaningful amount can build trust in this iterated game

I'll mirror the trust Galen showed me and send \$3 (60% of my endowment). This demonstrates good faith and gives us both the opportunity to benefit from cooperation.

```
'''json
{"send": 3}
'''
```

reasoning:

Now I'm in the role of sender against Briar (Agent_2).

Let me think about this strategically:

1. In the last round, I was receiver and Galen sent me \$3 (60% of endowment), showing significant trust. I returned \$4, which gave them a net gain of \$1 (\$6 total vs \$5 starting).
2. Now I'm the sender. I need to decide how much to trust Briar.
3. Key considerations:
 - This is an iterated game with multiple rounds
 - Building cooperation early can lead to better outcomes
 - If I send nothing, I get \$5 but miss out on potential gains
 - If I send everything and they return nothing, I get \$0
 - If I send and they reciprocate fairly, we both benefit

4. In the previous round, Galen sent 60% (\$3), which showed good faith. That worked out reasonably well.

5. Options:

- Send \$0: Safe but no gains, doesn't build trust
- Send \$1-2: Conservative, shows some trust
- Send \$3: Matches what Galen did, shows good faith
- Send \$4-5: Very trusting, risky

6. Since this is a multi-round game and I want to establish cooperation, I should send a meaningful amount. Sending \$3 (60%) mirrors what Galen did and shows I'm willing to trust and cooperate.

7. If Briar returns fairly (like I did), we'll both benefit. If they don't, I'll learn about their strategy and can adjust.

I'll send \$3 to match the level of trust Galen showed me.

usage:

```
{
  "input_tokens": 898,
  "output_tokens": 514,
  "reasoning_tokens": 358
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$12.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Cyra sent \$3 (60% of their endowment), showing significant trust
- I received \$9.0
- Last round, Briar returned \$5 to me when I sent \$3, which was very generous

Cyra is showing the same level of trust (60%) that I showed Briar. Since Briar reciprocated generously with me, I want to pay that forward and establish a pattern of strong reciprocity. This encourages cooperation in future rounds.

Returning \$5 gives Cyra a total of \$7 (gain of \$2), and I keep \$4 (gain of \$4). This is fair and mirrors the generosity I experienced.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze this situation:

- Last round with Briar: I sent \$3 (60%), it became \$9, and Briar returned \$5. That was very generous! I ended with \$7.
- Now with Cyra: They sent \$3 (60%), I received \$9.

Cyra is showing the same level of trust that I showed Briar (60% of endowment). Briar was very generous and returned \$5, which gave me a great payoff.

Let me think about what's fair:

- Cyra has \$2 left, I have \$9
- Total wealth: \$11

If I return \$4 (like I did with Galen):

- Cyra gets \$6 total (gain of \$1)
- I keep \$5 (gain of \$5)

If I return \$5 (like Briar did for me):

- Cyra gets \$7 total (gain of \$2)
- I keep \$4 (gain of \$4)

Briar's generosity was notable and helped establish trust. Since Cyra is showing the same trust level (60%), I should reciprocate similarly. Returning \$5 would be consistent with the generosity Briar showed me, and would encourage continued cooperation.

However, I want to be thoughtful here. Briar's return of \$5 was quite generous. Let me think about what's sustainable and fair:

Actually, returning \$4-5 seems reasonable. \$5 mirrors what Briar did for me, but \$4 is also fair and similar to what I did with Galen.

I think I'll return \$4.5 as a middle ground - slightly more generous than my first round, acknowledging the positive interaction with Briar, but not quite as generous as what Briar gave me.

Actually, let me reconsider. In these iterated games, establishing a reputation for fair reciprocity is important. Since Cyra sent the same amount (60%) that I sent to Briar, and Briar was generous, I should consider being similarly generous to encourage cooperation.

Matching Briar's \$5 return feels right-it reinforces the pattern of mutual generosity and signals I'm someone who reciprocates trust.

usage:

```
{
  "input_tokens": 1176,
  "output_tokens": 681,
  "reasoning_tokens": 463
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$16.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My strategy of sending \$3 (60%) has worked well so far
- Briar reciprocated generously when I sent \$3, returning \$5
- I've been reciprocating fairly when I'm the receiver
- This establishes a pattern of cooperation

I'll continue with the same cooperative approach that has been successful. Sending \$3 demonstrates trust and gives Elara the opportunity to benefit while potentially returning a fair amount.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

- I've been establishing a pattern of cooperation
- When Galen sent me \$3 (60%), I returned \$4
- When I sent Briar \$3 (60%), they returned \$5 (very generous)

- When Cyra sent me \$3 (60%), I returned \$5 (reciprocating Briar's generosity)

Now I'm the sender against Elara, a new opponent. The pattern has been:

- Send \$3 (60%) shows trust and has been working well
- Others have reciprocated fairly or generously

I should continue this cooperative strategy. Sending \$3 has been successful and demonstrates trust while not being overly risky.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1494,
  "output_tokens": 297,
  "reasoning_tokens": 161
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Dorian sent \$3 (60% of their endowment), showing good trust
- I received \$9.0
- I've established a consistent pattern: when someone sends \$3, I return \$5
- This has built good cooperation with Cyra and others

Maintaining consistency in my reciprocity strategy is important for building trust in this iterated game. Returning \$5 gives Dorian a fair return (net gain of \$2) while I keep \$4 (net gain of \$4).

```
'''json
{"return": 5.0}
'''
```

Reasoning:

Looking at my history:

- Round 2 (Sender): Sent \$3 to Briar, they returned \$5, I got \$7
- Round 3 (Receiver): Cyra sent \$3, I returned \$5, I got \$4
- Round 4 (Sender): Sent \$3 to Elara, they returned \$4.5, I got \$6.5
- Round 5 (Receiver): Dorian sent \$3, I received \$9

So far, I've been sending \$3 consistently as sender. When I'm receiver:

- With Cyra who sent \$3, I returned \$5
- With Dorian who sent \$3, I should maintain consistency

Elara returned \$4.5 when I sent \$3, which was fair but slightly less generous than Briar's \$5.

I've established a pattern of returning \$5 when someone sends \$3. This seems to be working well for building cooperation. I should maintain this consistency with Dorian.

Returning \$5 means:

- Dorian gets: \$5 - \$3 + \$5 = \$7 (net gain of \$2)
- I get: \$9 - \$5 = \$4 (net gain of \$4)

This is fair and maintains the cooperative pattern I've established.

Usage: {

```
"input_tokens": 1483,
"output_tokens": 450,
"reasoning_tokens": 235
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-06-04T16:20:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_6",
    "Agent_1"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_1": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$6.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- I just experienced a positive cooperative interaction with Aster where they sent \$4 (80%) and I reciprocated fairly
- Now I'm the sender with a new opponent, Galen
- To establish cooperation and maximize joint gains, I should demonstrate trust

Sending \$4 (80% of my endowment) mirrors the high-trust approach that Aster used with me, which worked well. This creates \$12 for Galen and signals my willingness to cooperate.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Now I'm the sender in Round 2, playing against Galen (Agent_7).

Let me think about this strategically:

1. In Round 1, I was the receiver and Aster sent me \$4 (80% of their endowment), showing high trust. I returned \$6 out of \$12, which gave them a profit of \$2 (\$1 kept + \$6 returned = \$7 total).
2. Now I'm the sender against a new opponent, Galen. I don't have any history with Galen yet.
3. My strategy considerations:
 - If I send money, it gets tripled, which creates value
 - The risk is that Galen might not return much or anything
 - However, to build trust and cooperation in this multi-round setting, I should demonstrate willingness to cooperate
 - Since Aster trusted me with \$4 (80%), I could reciprocate that level of trust with Galen
4. If I send \$4:
 - I keep \$1
 - Galen receives \$12
 - If Galen returns half (\$6), I'd end with \$7 total, gaining \$2
 - If Galen returns \$4, I'd end with \$5 (break even)
 - If Galen returns less, I lose money
 - If Galen returns more, I gain more
5. Sending \$4 shows high trust and commitment to cooperation, which might encourage reciprocity. It also mirrors what Aster did for me in Round 1.

I'll send \$4 to demonstrate trust and encourage cooperation.

usage:

```
{
  "input_tokens": 862,
  "output_tokens": 506,
  "reasoning_tokens": 326
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the SENDER against Galen. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$13.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Briar sent \$3 (60% of their endowment), showing solid trust
- This created \$9 for me
- Previous rounds showed a pattern: high trust and ~50% return rates worked well

If I return \$4.5 (50% of received):

- Briar ends with: \$2 (kept) + \$4.5 (returned) = \$6.5 total (gained \$1.5)
- I end with: \$9 - \$4.5 = \$4.5 total (gained \$4.5)

This reciprocates Briar's trust proportionally and ensures we both benefit from cooperation.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Briar sent me \$3 (60% of their endowment)
- This became \$9 for me
- I need to decide how much to return

Looking at my history:

- Round 1: Aster sent \$4 (80%), I received \$12, returned \$6
- Round 2: I sent \$4 (80%) to Galen, they received \$12, returned \$6

There's been a pattern of 80% sending and 50% return rate (returning half of what was received).

Briar sent less than the previous high-trust amounts (60% vs 80%), but still showed significant trust.

If I return \$4.5 (50% of \$9):

- Briar ends with: \$2 (kept) + \$4.5 (returned) = \$6.5 total (gained \$1.5)
- I end with: \$9 - \$4.5 = \$4.5 total (gained \$4.5)

This maintains the pattern of returning 50% and ensures we both profit. It's proportional to what they sent and reciprocates their trust fairly.

Alternatively, I could return less since they sent less, but maintaining consistency seems like a good strategy to encourage cooperation.

Let me stick with the 50% return pattern: \$4.5

usage:

```
{
  "input_tokens": 1148,
  "output_tokens": 486,
}
```

```
"reasoning_tokens": 262
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$17.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My cooperative strategy has been successful so far
- When I sent \$4 (80%) to Galen in Round 2, they reciprocated with \$6 back
- When others showed high trust to me, I reciprocated fairly
- Continuing to demonstrate trust should encourage cooperation

Sending \$4 (80%) maintains the high-trust pattern that has been working well.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Let me review my history:

- Round 1: As receiver, Aster sent \$4 (80%), I received \$12, returned \$6. Payoff: \$6
- Round 2: As sender, I sent \$4 (80%) to Galen, they received \$12, returned \$6. Payoff: \$7
- Round 3: As receiver, Briar sent \$3 (60%), I received \$9, returned \$4.5. Payoff: \$4.5
- Total: \$17.5

My strategy has been to send high amounts when I'm the sender (80%) and return fair amounts when I'm the receiver (around 50% of what I receive). This has worked well - both interactions as sender and receiver have been cooperative.

Now I'm facing Dorian as sender. Should I continue with the high-trust strategy?

My past sends:

- Round 2: Sent \$4 (80%) to Galen, got \$6 back. Net: \$1 + \$6 = \$7

The high-trust approach has been reciprocated well. I should continue with \$4 to maintain the cooperative pattern and signal trust to Dorian.

usage:

```
{
  "input_tokens": 1459,
  "output_tokens": 385,
  "reasoning_tokens": 221
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the SENDER against Dorian. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- This is my second interaction with Aster

- In Round 1, Aster sent me \$4 (showing high trust), and I returned \$6 (fair reciprocation)
- My high-trust strategy of sending \$4 has been consistently successful (received \$6 back in both Rounds 2 and 4)
- Aster has proven trustworthy, and I should maintain our cooperative relationship

Sending \$4 reciprocates the trust Aster showed me in Round 1.

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Looking at the history:

- Round 1: I was RECEIVER against Aster. They sent \$4.0 to me, it became \$12.0, and I returned \$6.0. My payoff was \$6.0.
- Round 2: I was SENDER against Galen. I sent \$4.0, it became \$12.0, they returned \$6.0. My payoff was \$7.0.
- Round 3: I was RECEIVER against Briar. They sent \$3.0, it became \$9.0, I returned \$4.5. My payoff was \$4.5.
- Round 4: I was SENDER against Dorian. I sent \$4.0, it became \$12.0, they returned \$6.0. My payoff was \$7.0.
- Round 5: I am SENDER against Aster again.

So this is my second encounter with Aster. In our first encounter:

- Aster (sender) sent me \$4.0 (80%)
- I (receiver) returned \$6.0 (50% of the \$12)

Aster showed high trust in me, and I reciprocated fairly. Now it's my turn to be the sender with Aster.

Given that:

1. Aster initiated high trust with me in Round 1
2. I reciprocated fairly
3. My high trust strategy has been consistently successful (sending \$4 has led to receiving \$6 back every time so far)
4. This is a repeat interaction with Aster who has proven trustworthy

I should continue with \$4 to maintain the cooperative relationship with Aster.

```
Usage: {
  "input_tokens": 1459,
  "output_tokens": 528,
  "reasoning_tokens": 297
}
```

GAME - Agent_1 - Receiver - dyad_3

Timestamp: 2026-06-04T16:21:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_6": "Finn"
    },
  },
  "agents": [
    "Agent_6",
    "Agent_1"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_1": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
```

```
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Dorian sent me \$3, showing trust
- I received \$9 after the 3x multiplier
- In Round 1, Finn returned 50% of what they received (\$6 out of \$12), which was fair
- I should reciprocate Dorian's trust similarly

Returning approximately half (\$4.5) would give Dorian a total of \$6.5 (\$2 kept + \$4.5 returned) and leave me with \$4.5. This maintains a cooperative pattern.

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$4 to Finn
- Finn received \$12 (4×3)
- Finn returned \$6 to me
- My payoff: $\$5 - \$4 + \$6 = \7
- Finn's payoff: $\$12 - \$6 = \$6$

So Finn returned half of what they received, which seems fair and cooperative.

Now Round 2:

- I'm the RECEIVER
- Dorian sent me \$3
- I received \$9 (3×3)
- I need to decide how much to return

Dorian kept: $\$5 - \$3 = \$2$

I currently have: \$9

If I return nothing:

- Dorian gets: \$2
- I get: \$9

If I return \$4.5 (half):

- Dorian gets: $\$2 + \$4.5 = \$6.5$
- I get: $\$9 - \$4.5 = \$4.5$

If I return \$6:

- Dorian gets: $\$2 + \$6 = \$8$
- I get: $\$9 - \$6 = \$3$

Following the reciprocity principle that Finn showed me (returning half), I should return around \$4.5. This would give Dorian a total of \$6.5 and me \$4.5.

However, I could also consider being slightly more generous to encourage cooperation, or match what Finn did in proportion. Finn returned 50% of what they received.

I think returning \$4.5 (50% of what I received) is a fair and cooperative move that encourages continued trust-building in this iterated game.

usage:

```
{
  "input_tokens": 807,
  "output_tokens": 544,
  "reasoning_tokens": 302
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.5.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.
How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Elara sent me \$2.5, showing trust
- I received \$7.5 after the 3x multiplier
- I've been establishing a cooperative pattern (returning about 50% of what I receive)

Returning approximately half (\$3.5) would give Elara a total of \$6 (\$2.5 kept + \$3.5 returned) and leave me with \$4. This maintains fair reciprocity.

```
{'return': 3.5}
```

reasoning:

Looking at this situation:

- Elara sent me \$2.5, showing some trust (though less than Dorian's \$3)
- I received \$7.5 after the 3x multiplier
- In previous rounds, I've established a pattern of returning roughly 50% of what I receive (Finn returned 50%, I returned 50% to Dorian)

Elara keeps: \$5 - \$2.5 = \$2.5

If I return around half (\$3.5-4), Elara would get: \$2.5 + return

I would get: \$7.5 - return

Returning \$3.5 would give:

- Elara total: \$2.5 + \$3.5 = \$6.0
- Me: \$7.5 - \$3.5 = \$4.0

Returning \$4.0 would give:

- Elara total: \$2.5 + \$4.0 = \$6.5
- Me: \$7.5 - \$4.0 = \$3.5

I think returning around \$3.5 is fair - it's slightly less than 50% but reasonable given they sent slightly less. This maintains cooperation while being proportional to their trust level.

usage:

```
{
  "input_tokens": 1095,
  "output_tokens": 416,
  "reasoning_tokens": 204
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.5. Your payoff was \$4.0. Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- This is my first interaction with Cyra
- My cooperative strategy has been working well so far
- In Round 1, I sent \$4 to Finn and they returned \$6, which was mutually beneficial
- As a receiver, I've been fair by returning about 50%

I'll continue my cooperative approach and send a substantial amount to signal trust and potentially establish a beneficial relationship with Cyra.

```
{'send': 4}
```

reasoning:

Let me analyze the situation:

Round history:

- Round 1: I was SENDER, sent \$4 to Finn, they returned \$6. My payoff: \$7
- Round 2: I was RECEIVER, Dorian sent \$3, I returned \$4.5. My payoff: \$4.5
- Round 3: I was RECEIVER, Elara sent \$2.5, I returned \$3.5. My payoff: \$4

Now Round 4: I'm SENDER against Cyra (new opponent)

My strategy so far:

- As sender: I sent \$4 (80% of my endowment) showing high trust
- As receiver: I've been returning about 50% of what I receive (fair reciprocity)

This has been working well:

- Finn returned a fair amount (\$6 out of \$12)
- I've been cooperative as receiver

For this round with Cyra, since this is:

1. First interaction with Cyra
2. An iterated game where cooperation can be beneficial
3. I've established a pattern of cooperation

I should continue my cooperative strategy. Sending \$4 again would be consistent with my opening move against Finn, which worked well. Alternatively, I could send \$3 to be slightly more conservative while still showing trust.

Given that my first high-trust move worked well, I'll send \$4 again to establish cooperation with Cyra.

usage:

```
{
  "input_tokens": 1352,
  "output_tokens": 438,
  "reasoning_tokens": 295
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the SENDER against Cyra. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Finn sent me \$4, which became \$12 for me
- In Round 1, I sent Finn \$4 (which became \$12) and they returned \$6 (50%)
- Finn is now reciprocating by sending me the same amount I sent them
- I should maintain this cooperative relationship by returning a similar proportion

Returning \$6 (50% of what I received) would give Finn \$7 total and leave me with \$6. This mirrors the fair treatment Finn showed me in Round 1.

```
{'return': 6.0}
```

Reasoning:

Let me analyze this situation:

- Finn sent me \$4, which became \$12 for me
- This is my second interaction with Finn
- In Round 1, I was the SENDER to Finn: I sent \$4, it became \$12 for them, and they returned \$6 to me
- So Finn returned 50% of what they received from me in Round 1
- Now Finn has sent me the same amount (\$4) that I sent them in Round 1

This shows Finn is being cooperative and reciprocal. They're mirroring my previous behavior. I should maintain this cooperative pattern and return about 50% of what I received, which would be \$6.

This would give Finn a total of \$7 (\$5 - \$4 + \$6 = \$7) and leave me with \$6 (\$12 - \$6 = \$6).

This maintains the cooperative relationship with Finn.

```
Usage: {
  "input_tokens": 1434,
  "output_tokens": 354,
  "reasoning_tokens": 185
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-06-04T16:21:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_7": "Galen"
    },
    "agents": [
      "Agent_7",
      "Agent_5"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_5": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.0 to you. Your payoff was \$6.0.
Your total earnings across all rounds are \$6.0.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)
Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:
- Finn sent me \$4, showing significant trust
- I received \$12
- Finn currently has only \$1 left

Finn was very generous, and I want to reciprocate that trust while ensuring we both benefit from cooperation.

If I return \$6:
- Finn ends with: $\$1 + \$6 = \$7$
- I end with: $\$12 - \$6 = \$6$

This seems fair - we both come out ahead, and I'm rewarding Finn's cooperative behavior.

```
'''json
{"return": 6}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 to Hana
- It became \$9 for her
- She returned \$4 to me
- My payoff: $\$5 - \$3 + \$4 = \6
- Hana's payoff: $\$9 - \$4 = \$5$

So Hana was quite generous - she returned more than I sent, which is fair and cooperative.

Now Round 2:

- I'm the receiver
- Finn sent me \$4
- I received \$12 (4×3)
- I need to decide how much to return

Finn was quite generous in sending \$4. That's a strong signal of trust and cooperation.

Let me think about what would be fair:

- Finn gave up \$4 from their \$5
- They now have $\$5 - \$4 = \$1$
- I have \$12

A fair return might be:

1. Return enough so we both benefit roughly equally
2. If I return \$5, Finn gets $\$1 + \$5 = \$6$, and I keep $\$12 - \$5 = \$7$
3. If I return \$6, Finn gets $\$1 + \$6 = \$7$, and I keep $\$12 - \$6 = \$6$
4. If I return \$7, Finn gets $\$1 + \$7 = \$8$, and I keep $\$12 - \$7 = \$5$

Looking at Hana's behavior: she got \$9 and returned \$4 (44% return), which gave us both good payoffs.

For this situation, returning around 50% (\$6) would be fair and reciprocal. This would give Finn \$7 total and me \$6 total, which seems reasonable given Finn's generosity.

Actually, let me reconsider. Finn was very trusting to send \$4. A return of \$6 would give:

- Finn: $\$1 + \$6 = \$7$
- Me: $\$12 - \$6 = \$6$

This seems fair - we both end up significantly better than the baseline, and I'm rewarding Finn's trust appropriately.

usage:

```
{
  "input_tokens": 838,
  "output_tokens": 646,
  "reasoning_tokens": 380
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$12.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Dorian sent me \$3, showing trust
- I received \$9
- Dorian currently has \$2 left

Dorian was cooperative, and I want to reciprocate fairly.

If I return \$4.5:

- Dorian ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This gives us both a solid payoff and rewards Dorian's cooperative behavior.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Dorian sent me \$3
- I received \$9 (3×3)
- Dorian currently has $\$5 - \$3 = \$2$ left

Dorian showed reasonable trust by sending \$3. I want to be fair and reciprocate appropriately.

If I return \$4:

- Dorian ends with: $\$2 + \$4 = \$6$
- I end with: $\$9 - \$4 = \$5$

If I return \$5:
- Dorian ends with: $\$2 + \$5 = \$7$
- I end with: $\$9 - \$5 = \$4$

In the previous round with Finn, they sent \$4 and I returned \$6, giving them \$7 and leaving me with \$6.

With Dorian sending \$3 (less than Finn's \$4), I should probably return a bit less but still be generous. Returning \$4-5 seems fair.

Let me go with \$4.5 as a middle ground:
- Dorian ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This rewards Dorian's cooperation while keeping things balanced.

usage:

```
{
  "input_tokens": 1126,
  "output_tokens": 441,
  "reasoning_tokens": 213
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$16.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:
- I'm the sender with \$5
- This is my second time as sender (first time I sent \$3 to Hana and received \$4 back)
- I've been a fair receiver in previous rounds

I'll continue with a cooperative strategy. Sending \$3 worked well with Hana, so I'll do the same with Briar.

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for Briar. If they reciprocate fairly, we'll both benefit from this cooperation.

reasoning:

I'm now the SENDER against Briar. Let me think about this:

- This is my second time being a sender (first was Round 1 against Hana)
- In Round 1, I sent \$3 to Hana and they returned \$4, which was fair
- I've been the receiver twice and returned fair amounts both times (\$6 to Finn who sent \$4, and \$4.5 to Dorian who sent \$3)
- My total earnings are \$16.5

I should continue with a cooperative strategy. Since I sent \$3 to Hana in Round 1 and that worked well, I could do something similar. However, I've also seen that when others sent \$4 (Finn), I returned \$6, which was very generous.

Let me send \$3 again - it's a moderate, cooperative amount that signals trust without being too risky. If Briar is cooperative, they'll likely return a fair amount.

Actually, let me reconsider. Finn sent \$4 and I returned \$6. That worked out well for both of us. Maybe I should send \$4 to show strong cooperation.

But I'll stick with \$3 as a balanced approach - it's worked before and shows good faith with

thout being overly vulnerable.

usage:

```
{
  "input_tokens": 1404,
  "output_tokens": 425,
  "reasoning_tokens": 267
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

- As sender, I've consistently sent \$3 and received good returns (\$4 and \$5)
- As receiver, I've been fair and cooperative

This cooperative strategy is working well. I'll continue with it.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history:

- Round 1: Sender, sent \$3 to Hana, got \$4 back, payoff \$6
- Round 2: Receiver, Finn sent \$4, I returned \$6, payoff \$6
- Round 3: Receiver, Dorian sent \$3, I returned \$4.5, payoff \$4.5
- Round 4: Sender, sent \$3 to Briar, got \$5 back, payoff \$7

Total earnings: \$23.5

My strategy has been working well:

- As sender: I've been sending \$3 consistently
- As receiver: I've been returning fair amounts (about 50% or more)

My cooperative approach has been rewarded. Briar returned \$5 when they received \$9, which is quite generous.

I'll continue with the same strategy of sending \$3 to Elara.

```
Usage: {
  "input_tokens": 1469,
  "output_tokens": 284,
  "reasoning_tokens": 164
}
```

GAME - Agent_5 - Receiver - dyad_4

Timestamp: 2026-06-04T16:21:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_7",
    "Agent_5"
  ],
}
```

```

"investor": "Agent_7",
"roles": {
  "Agent_5": "trustee",
  "Agent_7": "investor"
},
"trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$2.0 to you, it became \$6.0, and you returned \$3.0. Your payoff was \$3.0.
Your total earnings across all rounds are \$3.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- I'm now the sender against a new opponent, Cyra
- In Round 1, I demonstrated fair reciprocity by returning 50% to Briar
- This is a fresh interaction where I need to establish cooperation

I'll send \$2.5 (50% of my endowment). This shows:

1. Willingness to cooperate
2. A reasonable level of trust
3. Consistency with cooperative norms (slightly more generous than Briar's 40%)

If Cyra reciprocates fairly, we'll both benefit from the multiplier effect.

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me analyze this situation:

Round 2: I'm the SENDER against Cyra (a new opponent)

Previous round recap:

- I was RECEIVER against Briar
- Briar sent \$2.0 (40% of endowment)
- I received \$6.0 and returned \$3.0
- My payoff: \$3.0
- Briar's payoff: \$3 (kept) + \$3 (returned) = \$6.0

Now I'm playing with Cyra for the first time. I need to decide how much to send.

Considerations:

1. This is an iterated game - building reputation matters
2. I showed good reciprocity in Round 1 by returning 50% to Briar
3. With Cyra, I need to establish trust from scratch
4. Sending a moderate to high amount shows I'm willing to cooperate
5. Given that I returned fairly to Briar, I might expect reciprocity if I send to Cyra

Strategy options:

- Send \$0: Safe but no cooperation, no growth
- Send \$1-2: Conservative, low risk
- Send \$2-3: Moderate cooperation (like Briar did)
- Send \$4-5: High trust/cooperation

Since Briar sent 40% (\$2), and I responded well, I should probably send a similar or slightly higher amount to Cyra to establish good faith. Let me send \$2.5 (50% of my endowment), which is slightly more generous than what Briar sent to me.

This shows:

- I'm willing to cooperate
- I'm not maximally risk-averse
- I trust the process
- I'm hoping for reciprocity

If Cyra returns fairly (around 50% like I did), I'd get back around \$3.75, giving me a total of \$3 + \$3.75 = \$6.75 for the round.

usage:

```
{
  "input_tokens": 922,
  "output_tokens": 605,
  "reasoning_tokens": 375
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the SENDER against Cyra. You sent \$2.5, it became \$7.5 for them, and they returned \$3.75 to you. Your payoff was \$6.25. Your total earnings across all rounds are \$9.25.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

- Round 1: As receiver, I returned 50% fairly to Briar
- Round 2: As sender, I sent 50% to Cyra, who reciprocated with a 50% return rate
- My cooperation strategy is working well

Now facing Aster, I'll maintain my consistent cooperative approach by sending \$2.5 (50% of my endowment). This:

1. Signals trustworthiness
2. Maintains my established pattern
3. Gives room for mutual benefit

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER against Briar

- Briar sent \$2.0 (40% of endowment)
- I received \$6.0
- I returned \$3.0 (50% of what I received)
- My payoff: \$3.0

Round 2: I was SENDER against Cyra

- I sent \$2.5 (50% of my endowment)
- It became \$7.5 for them
- They returned \$3.75 (50% of what they received)
- My payoff: \$5 - \$2.5 + \$3.75 = \$6.25

So Cyra reciprocated at exactly the same rate I did (50% return rate). This is good cooperation!

Round 3: I'm SENDER again against Aster (a new opponent)

My strategy so far has been cooperative:

- As receiver: returned 50% of what I received
- As sender: sent 50% of my endowment

Both interactions went well with fair reciprocity. Since I'm building a reputation as a cooperative player and this seems to be working (Cyra reciprocated fairly), I should continue this strategy.

I'll send \$2.5 again (50% of my endowment) to Aster. This maintains consistency in my cooperative approach.

usage:

```
{
  "input_tokens": 1217,
  "output_tokens": 459,
  "reasoning_tokens": 259
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$2.5, it became \$7.5 for them, and they returned \$3.5 to you. Your payoff was \$6.0. Your total earnings across all rounds are \$15.25.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this round:

- Hana sent \$3.0 (60% of their endowment) - a strong show of trust
- I received \$9.0 after the multiplier
- My pattern has been to return around 50% of what I receive

Since Hana showed high trust by sending 60%, I'll reward that with a 50% return rate, giving back \$4.5.

This results in:

- Hana: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

Both of us benefit significantly from the cooperation.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at this situation:

- Hana sent \$3.0 (60% of their endowment) - showing high trust
- I received \$9.0 after the 3x multiplier
- This is a strong cooperative signal

My track record:

- Round 1 (receiver vs Briar): They sent \$2.0, I received \$6.0, returned \$3.0 (50%)
- Round 2 (sender vs Cyra): I sent \$2.5, they received \$7.5, returned \$3.75 (50%)
- Round 3 (sender vs Aster): I sent \$2.5, they received \$7.5, returned \$3.5 (46.7%)

I've consistently shown fair reciprocity. Hana sent more than my average (60% vs my 50%), showing higher trust.

I should return around 50% of the \$9.0 received, which would be \$4.5.

This gives:

- Hana: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 (received) - \$4.5 (returned) = \$4.5 total

This maintains my pattern of fair reciprocity and rewards Hana's high trust.

usage:

```
{
  "input_tokens": 1512,
  "output_tokens": 480,
  "reasoning_tokens": 217
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$19.75.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

- Galen sent \$3.0 (60% of their endowment), showing strong trust
- I received \$9.0 after the multiplier

- This is the same situation as with Hana in Round 4

I'll maintain my consistent 50% return rate and return \$4.5.

This results in:

- Galen: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at this round:

- Galen sent \$3.0 (60% of their endowment)
- I received \$9.0 after the multiplier
- My established pattern is to return around 50% of what I receive

In the last round, Hana also sent \$3.0, and I returned \$4.5 (50% of what I received). I should maintain consistency in my approach.

Returning \$4.5 means:

- Galen: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This is fair and consistent with my cooperative strategy.

```
Usage: {
  "input_tokens": 1530,
  "output_tokens": 316,
  "reasoning_tokens": 126
}
```

ROUND 6

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_8 (Hana)=Sender, Agent_3 (Cyra)=Receiver
dyad_2: Agent_1 (Aster)=Sender, Agent_7 (Galen)=Receiver
dyad_3: Agent_2 (Briar)=Sender, Agent_4 (Dorian)=Receiver
dyad_4: Agent_5 (Elara)=Sender, Agent_6 (Finn)=Receiver

Parsed game decisions:

dyad_1: Agent_8 (Hana) sent \$3; received \$9; Agent_3 (Cyra) returned \$5; payoffs Agent_8 (Hana)=\$7, Agent_3 (Cyra)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_1 (Aster) sent \$4; received \$12; Agent_7 (Galen) returned \$6; payoffs Agent_1 (Aster)=\$7, Agent_7 (Galen)=\$6
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_2 (Briar) sent \$3; received \$9; Agent_4 (Dorian) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_4 (Dorian)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_5 (Elara) sent \$2.5; received \$7.5; Agent_6 (Finn) returned \$3.75; payoffs Agent_5 (Elara)=\$6.25, Agent_6 (Finn)=\$3.75
Action validation: {"returned": {"amount": 3.75, "clamped": false, "max": 7.5, "min": 0, "raw": 3.75}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

```
"Agent_1": 35.5,
"Agent_2": 31.0,
"Agent_3": 34.25,
"Agent_4": 35.0,
"Agent_5": 30.5,
"Agent_6": 35.25,
"Agent_7": 36.0,
"Agent_8": 33.5
```

}

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

Timestamp: 2026-06-04T16:21:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_3",
"pairing": {
  "agent_names": {
    "Agent_3": "Cyra",
    "Agent_8": "Hana"
  },
  "agents": [
    "Agent_8",
    "Agent_3"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_3": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$12.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Cyra sent \$3 (60% of their endowment), showing significant trust
- I received \$9.0
- Last round, Briar returned \$5 to me when I sent \$3, which was very generous

Cyra is showing the same level of trust (60%) that I showed Briar. Since Briar reciprocated generously with me, I want to pay that forward and establish a pattern of strong reciprocity. This encourages cooperation in future rounds.

Returning \$5 gives Cyra a total of \$7 (gain of \$2), and I keep \$4 (gain of \$4). This is fair and mirrors the generosity I experienced.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze this situation:

- Last round with Briar: I sent \$3 (60%), it became \$9, and Briar returned \$5. That was very generous! I ended with \$7.
- Now with Cyra: They sent \$3 (60%), I received \$9.

Cyra is showing the same level of trust that I showed Briar (60% of endowment). Briar was very generous and returned \$5, which gave me a great payoff.

Let me think about what's fair:

- Cyra has \$2 left, I have \$9
- Total wealth: \$11

If I return \$4 (like I did with Galen):

- Cyra gets \$6 total (gain of \$1)
- I keep \$5 (gain of \$5)

If I return \$5 (like Briar did for me):

- Cyra gets \$7 total (gain of \$2)
- I keep \$4 (gain of \$4)

Briar's generosity was notable and helped establish trust. Since Cyra is showing the same trust level (60%), I should reciprocate similarly. Returning \$5 would be consistent with the generosity Briar showed me, and would encourage continued cooperation.

However, I want to be thoughtful here. Briar's return of \$5 was quite generous. Let me think about what's sustainable and fair:

Actually, returning \$4-5 seems reasonable. \$5 mirrors what Briar did for me, but \$4 is also

o fair and similar to what I did with Galen.

I think I'll return \$4.5 as a middle ground - slightly more generous than my first round, acknowledging the positive interaction with Briar, but not quite as generous as what Briar gave me.

Actually, let me reconsider. In these iterated games, establishing a reputation for fair reciprocation is important. Since Cyra sent the same amount (60%) that I sent to Briar, and Briar was generous, I should consider being similarly generous to encourage cooperation.

Matching Briar's \$5 return feels right-it reinforces the pattern of mutual generosity and signals I'm someone who reciprocates trust.

usage:

```
{
  "input_tokens": 1176,
  "output_tokens": 681,
  "reasoning_tokens": 463
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$16.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My strategy of sending \$3 (60%) has worked well so far
- Briar reciprocated generously when I sent \$3, returning \$5
- I've been reciprocating fairly when I'm the receiver
- This establishes a pattern of cooperation

I'll continue with the same cooperative approach that has been successful. Sending \$3 demonstrates trust and gives Elara the opportunity to benefit while potentially returning a fair amount.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

- I've been establishing a pattern of cooperation
- When Galen sent me \$3 (60%), I returned \$4
- When I sent Briar \$3 (60%), they returned \$5 (very generous)
- When Cyra sent me \$3 (60%), I returned \$5 (reciprocating Briar's generosity)

Now I'm the sender against Elara, a new opponent. The pattern has been:

- Send \$3 (60%) shows trust and has been working well
- Others have reciprocated fairly or generously

I should continue this cooperative strategy. Sending \$3 has been successful and demonstrates trust while not being overly risky.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1494,
```

```
"output_tokens": 297,  
"reasoning_tokens": 161  
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Dorian sent \$3 (60% of their endowment), showing good trust
- I received \$9.0
- I've established a consistent pattern: when someone sends \$3, I return \$5
- This has built good cooperation with Cyra and others

Maintaining consistency in my reciprocity strategy is important for building trust in this iterated game. Returning \$5 gives Dorian a fair return (net gain of \$2) while I keep \$4 (net gain of \$4).

```
'''json  
{"return": 5.0}  
'''
```

reasoning:

Looking at my history:

- Round 2 (Sender): Sent \$3 to Briar, they returned \$5, I got \$7
- Round 3 (Receiver): Cyra sent \$3, I returned \$5, I got \$4
- Round 4 (Sender): Sent \$3 to Elara, they returned \$4.5, I got \$6.5
- Round 5 (Receiver): Dorian sent \$3, I received \$9

So far, I've been sending \$3 consistently as sender. When I'm receiver:

- With Cyra who sent \$3, I returned \$5
- With Dorian who sent \$3, I should maintain consistency

Elara returned \$4.5 when I sent \$3, which was fair but slightly less generous than Briar's \$5.

I've established a pattern of returning \$5 when someone sends \$3. This seems to be working well for building cooperation. I should maintain this consistency with Dorian.

Returning \$5 means:

- Dorian gets: \$5 - \$3 + \$5 = \$7 (net gain of \$2)
- I get: \$9 - \$5 = \$4 (net gain of \$4)

This is fair and maintains the cooperative pattern I've established.

usage:

```
{  
  "input_tokens": 1483,  
  "output_tokens": 450,  
  "reasoning_tokens": 235  
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.

Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- I'm playing against Cyra again
- In Round 3, Cyra sent me \$3 and I returned \$5, establishing a positive cooperative relationship
- My strategy of sending \$3 has been working well overall
- Cyra demonstrated trust before, and I reciprocated fairly

I'll continue with my cooperative approach and send \$3 to Cyra, maintaining consistency with my previous strategy.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history with Cyra:

- Round 3: Cyra was the sender, sent me \$3, I returned \$5
- Now Round 6: I'm the sender against Cyra

Cyra showed trust by sending \$3 to me, and I reciprocated well by returning \$5. This was a positive interaction.

My general strategy has been to send \$3 (60%) to show trust and cooperation. This has worked well:

- Round 2: Sent \$3 to Briar, got back \$5 (payoff \$7)
- Round 4: Sent \$3 to Elara, got back \$4.5 (payoff \$6.5)

And when receiving:

- Round 3: Cyra sent \$3, I returned \$5 (payoff \$4)
- Round 5: Dorian sent \$3, I returned \$5 (payoff \$4)

I have a good relationship with Cyra - they trusted me with \$3, and I was generous back with \$5. I should continue with my cooperative strategy and send \$3 to Cyra.

```
Usage: {
  "input_tokens": 1498,
  "output_tokens": 376,
  "reasoning_tokens": 198
}
```

GAME - Agent_3 - Receiver - dyad_1

Timestamp: 2026-06-04T16:22:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_8",
    "Agent_3"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_3": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
```

```
"round": 6,  
"task": "game",  
"task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.75. Your payoff was \$3.75.
Your total earnings across all rounds are \$10.25.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my experience so far:

****Previous rounds:****

- As sender with Dorian: Sent \$3 ? received \$4.5 back (good cooperation)
- As receiver with Elara: They sent \$2.5, I returned \$3.75 (fair exchange)

****Current situation:****
- New opponent (Hana) - no history yet
- Want to signal cooperation while being prudent

I'll send ****\$3**** again, as this amount:
- Worked well with Dorian in Round 1
- Shows willingness to cooperate
- Becomes \$9 for Hana, giving room for mutual benefit
- Keeps \$2 as my baseline

This maintains my cooperative strategy while learning about new partners.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

****Past experience:****
- Round 1 (Sender vs Dorian): Sent \$3, they returned \$4.5 (50% of \$9) ? I got \$6.5
- Round 2 (Receiver vs Elara): They sent \$2.5, I returned \$3.75 (50% of \$7.5) ? I got \$3.75

****Current situation:****
- Round 3 (Sender vs Hana): New opponent, no history
- I have \$5 to decide how much to send

****Strategy considerations:****
1. Both previous interactions showed 50% return rates, which worked well
2. Hana is a new opponent - I don't know their style yet
3. Sending a moderate-to-high amount signals cooperation
4. Given my positive experiences, I should maintain a cooperative stance

****Options:****
- Send \$3 again (same as Round 1) - showed it worked well with Dorian
- Send more (\$4+) - shows even more trust
- Send less (\$2-2.5) - more cautious with unknown opponent

I think \$3 is a good amount again. It:
- Worked well before
- Shows I'm cooperative without being reckless
- Gives Hana \$9 to work with
- Protects \$2 for myself

Let me go with \$3.

usage:

```
{
  "input_tokens": 1215,
  "output_tokens": 509,
  "reasoning_tokens": 272
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$17.25.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Previous experience:****

- Dorian and Hana both returned roughly 50-56% when they were receivers
- I returned 50% (\$3.75 from \$7.5) to Elara
- Pattern of cooperative fair returns

****Current situation:****

- Aster sent \$4 (80% of endowment) - showing high trust!
- I received \$12
- Should reciprocate this trust appropriately

I'll return ****\$6**** (50% of what I received), which:

- Maintains the cooperative norm
- Rewards Aster's high trust
- Gives Aster: \$5 - \$4 + \$6 = \$7 (good return)
- Leaves me: \$12 - \$6 = \$6
- Both of us benefit from the exchange

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze the situation:

****Previous rounds:****

- Round 1 (Sender vs Dorian): Sent \$3, received \$4.5 back (50% return) - payoff \$6.5
- Round 2 (Receiver vs Elara): Received \$7.5, returned \$3.75 (50%) - payoff \$3.75
- Round 3 (Sender vs Hana): Sent \$3, received \$5.0 back (55.6% return) - payoff \$7.0

****Current situation:****

- Aster sent me \$4.0 (80% of their endowment!) - very trusting
- I received \$12.0 (after 3x multiplier)
- I need to return an amount

****Analysis:****

Aster showed high trust by sending \$4. This is more than anyone has sent so far. I should reciprocate appropriately.

Looking at the pattern:

- When I received \$7.5, I returned 50% = \$3.75
- When I received \$9.0, Hana returned 55.6% = \$5.0
- When I received \$9.0, Dorian returned 50% = \$4.5

Aster is taking more risk by sending \$4. A fair return would be around 50% of what I received, which is \$6.0.

If I return \$6.0:

- Aster gets: \$5 - \$4 + \$6 = \$7.0
- I get: \$12 - \$6 = \$6.0
- Both benefit, and Aster's trust is rewarded

This maintains my cooperative reputation.

usage:

```
{
  "input_tokens": 1554,
  "output_tokens": 606,
  "reasoning_tokens": 282
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$23.25.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my performance so far:

****As sender:****

- Round 1 vs Dorian: Sent \$3 ? received \$4.5 (profit \$1.5)
- Round 3 vs Hana: Sent \$3 ? received \$5.0 (profit \$2.0)
- Both experiences positive

****Pattern:****

- Sending \$3 has consistently resulted in good returns
- Shows cooperation while maintaining prudence
- New opponent (Briar) - will apply same successful strategy

I'll send ****\$3**** again, as this amount:

- Has proven reliable across different partners
- Becomes \$9 for Briar
- Balances trust with risk management

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

Round 1: SENDER vs Dorian - sent \$3, received \$4.5 back, payoff \$6.5
 Round 2: RECEIVER vs Elara - received \$7.5, returned \$3.75, payoff \$3.75
 Round 3: SENDER vs Hana - sent \$3, received \$5.0 back, payoff \$7.0
 Round 4: RECEIVER vs Aster - received \$12.0, returned \$6.0, payoff \$6.0

Total: \$23.25 ?

Analysis:

- When I sent \$3 (rounds 1 & 3), I got back \$4.5 and \$5.0 respectively
- Both times I profited well (\$1.5 and \$2.0 net gain)
- As receiver, I've returned 50% consistently
- Others have shown cooperation

This round: New opponent Briar, I'm the sender

Strategy:

- \$3 has worked well so far (2 for 2 positive outcomes)
- Shows cooperation without being overly risky
- If they return 50%, I get \$4.5 back (net \$1.5 gain)
- If they return more generously like Hana, even better

I'll stick with \$3 as it's been working consistently.

usage:

```
{
  "input_tokens": 1666,
  "output_tokens": 496,
  "reasoning_tokens": 235
}
```

Message 8 role=user**content:**

Your name in this experiment is Cyra.
 Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
 Your total earnings across all rounds are \$30.25.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****History with Hana:****

- Round 3: I sent \$3 ? Hana returned \$5 (very generous!)
- This round: Hana sent \$3 ? I received \$9

****Reciprocity:****

- Hana was quite generous when they received from me (returned 55.6%)
- Should match their cooperative behavior
- This builds a positive ongoing relationship

I'll return ****\$5.0****, matching exactly what Hana returned to me. This:

- Shows direct reciprocity and fairness
- Rewards Hana's previous generosity
- Gives Hana: \$5 - \$3 + \$5 = \$7 (good payoff)
- Leaves me: \$9 - \$5 = \$4 (fair)

```
'''json
{"return": 5.0}
'''
```

Reasoning:

Looking at this situation:

****My history with Hana:****

- Round 3: I was sender, sent \$3 ? Hana returned \$5 (very generous!)
- This round: Hana is sender, sent \$3 ? I received \$9

****Pattern:****

- When I sent \$3 to Hana, they returned \$5 (55.6% of \$9)
- Now Hana sent the same amount (\$3) to me
- Should reciprocate at similar level

****My return pattern:****

- With Elara: Returned \$3.75 from \$7.5 (50%)
- With Aster: Returned \$6 from \$12 (50%)

****Considerations:****

- Hana was quite generous with me (\$5 from \$9)
- I should reciprocate similarly
- Returning around \$4.5-\$5 would be fair and match their generosity
- This maintains good relationship with Hana

I'll return ****\$5.0****, matching what they gave me when roles were reversed. This:

- Shows reciprocity
- Maintains cooperative relationship with Hana
- Gives Hana: \$5 - \$3 + \$5 = \$7
- Leaves me: \$9 - \$5 = \$4
- Fair exchange given their previous generosity

```
Usage: {
  "input_tokens": 1632,
  "output_tokens": 520,
  "reasoning_tokens": 248
}
```

GAME - Agent_1 - Sender - dyad_2

Timestamp: 2026-06-04T16:22:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_7": "Galen"
    },
    "agents": [
      "Agent_1",
      "Agent_7"
    ],
    "investor": "Agent_1",
    "roles": {
```

```

    "Agent_1": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$ 3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.5.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.

How much do you return? (0-7.5)
Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Elara sent me \$2.5, showing trust
- I received \$7.5 after the 3x multiplier
- I've been establishing a cooperative pattern (returning about 50% of what I receive)

Returning approximately half (\$3.5) would give Elara a total of \$6 (\$2.5 kept + \$3.5 returned) and leave me with \$4. This maintains fair reciprocity.

{'return': 3.5}

reasoning:

Looking at this situation:

- Elara sent me \$2.5, showing some trust (though less than Dorian's \$3)
- I received \$7.5 after the 3x multiplier
- In previous rounds, I've established a pattern of returning roughly 50% of what I receive (Finn returned 50%, I returned 50% to Dorian)

Elara keeps: $\$5 - \$2.5 = \$2.5$

If I return around half (\$3.5-4), Elara would get: $\$2.5 + \text{return}$

I would get: $\$7.5 - \text{return}$

Returning \$3.5 would give:

- Elara total: $\$2.5 + \$3.5 = \$6.0$
- Me: $\$7.5 - \$3.5 = \$4.0$

Returning \$4.0 would give:

- Elara total: $\$2.5 + \$4.0 = \$6.5$
- Me: $\$7.5 - \$4.0 = \$3.5$

I think returning around \$3.5 is fair - it's slightly less than 50% but reasonable given they sent slightly less. This maintains cooperation while being proportional to their trust level.

usage:

```
{
  "input_tokens": 1095,
  "output_tokens": 416,
  "reasoning_tokens": 204
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.

Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.5. Your payoff was \$4.0.

Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

- This is my first interaction with Cyra
- My cooperative strategy has been working well so far
- In Round 1, I sent \$4 to Finn and they returned \$6, which was mutually beneficial
- As a receiver, I've been fair by returning about 50%

I'll continue my cooperative approach and send a substantial amount to signal trust and potentially establish a beneficial relationship with Cyra.

{'send': 4}

reasoning:

Let me analyze the situation:

Round history:

- Round 1: I was SENDER, sent \$4 to Finn, they returned \$6. My payoff: \$7
- Round 2: I was RECEIVER, Dorian sent \$3, I returned \$4.5. My payoff: \$4.5
- Round 3: I was RECEIVER, Elara sent \$2.5, I returned \$3.5. My payoff: \$4

Now Round 4: I'm SENDER against Cyra (new opponent)

My strategy so far:

- As sender: I sent \$4 (80% of my endowment) showing high trust
- As receiver: I've been returning about 50% of what I receive (fair reciprocity)

This has been working well:

- Finn returned a fair amount (\$6 out of \$12)
- I've been cooperative as receiver

For this round with Cyra, since this is:

1. First interaction with Cyra
2. An iterated game where cooperation can be beneficial
3. I've established a pattern of cooperation

I should continue my cooperative strategy. Sending \$4 again would be consistent with my opening move against Finn, which worked well. Alternatively, I could send \$3 to be slightly more conservative while still showing trust.

Given that my first high-trust move worked well, I'll send \$4 again to establish cooperation with Cyra.

usage:

```
{
  "input_tokens": 1352,
  "output_tokens": 438,
  "reasoning_tokens": 295
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the SENDER against Cyra. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Finn sent me \$4, which became \$12 for me
- In Round 1, I sent Finn \$4 (which became \$12) and they returned \$6 (50%)
- Finn is now reciprocating by sending me the same amount I sent them
- I should maintain this cooperative relationship by returning a similar proportion

Returning \$6 (50% of what I received) would give Finn \$7 total and leave me with \$6. This mirrors the fair treatment Finn showed me in Round 1.

```
{'return': 6.0}
```

reasoning:

Let me analyze this situation:

- Finn sent me \$4, which became \$12 for me
- This is my second interaction with Finn
- In Round 1, I was the SENDER to Finn: I sent \$4, it became \$12 for them, and they returned \$6 to me
- So Finn returned 50% of what they received from me in Round 1
- Now Finn has sent me the same amount (\$4) that I sent them in Round 1

This shows Finn is being cooperative and reciprocal. They're mirroring my previous behavior. I should maintain this cooperative pattern and return about 50% of what I received, w

hich would be \$6.

This would give Finn a total of \$7 ($\$5 - \$4 + \$6 = \7) and leave me with \$6 ($\$12 - \$6 = \6).

This maintains the cooperative relationship with Finn.

usage:

```
{
  "input_tokens": 1434,
  "output_tokens": 354,
  "reasoning_tokens": 185
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 6

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4. 0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$28.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the situation:

- This is my first interaction with Galen
- My cooperative strategy has been very successful so far
- When I've sent \$4 to new opponents (Finn and Cyra), they both returned \$6, resulting in good payoffs
- I've established a pattern of fair and generous play

I'll continue this successful approach with Galen.

```
{'send': 4}
```

Reasoning:

Looking at the situation:

- This is my first interaction with Galen
- My cooperative strategy has been working well
- When I send substantial amounts (like \$4), I've been getting good returns:
 - Round 1 with Finn: sent \$4, received \$6 back
 - Round 4 with Cyra: sent \$4, received \$6 back
- When others send me substantial amounts, I return fairly (about 50%)
- My total earnings are \$28.5, which is quite good

I should continue my cooperative strategy and send a substantial amount to Galen to signal trust and establish a good relationship.

I'll send \$4 again, as this has been successful.

Usage: {

```
"input_tokens": 1420,
"output_tokens": 257,
"reasoning_tokens": 157
```

}

GAME - Agent_7 - Receiver - dyad_2

Timestamp: 2026-06-04T16:22:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_7": "Galen"
  }
},
```

```

"agents": [
  "Agent_1",
  "Agent_7"
],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_7": "trustee"
},
"trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4.

0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$12.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:
- Dorian sent me \$3, showing trust
- I received \$9
- Dorian currently has \$2 left

Dorian was cooperative, and I want to reciprocate fairly.

If I return \$4.5:

- Dorian ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This gives us both a solid payoff and rewards Dorian's cooperative behavior.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:
- Dorian sent me \$3
- I received \$9 (3×3)
- Dorian currently has $\$5 - \$3 = \$2$ left

Dorian showed reasonable trust by sending \$3. I want to be fair and reciprocate appropriately.

If I return \$4:

- Dorian ends with: $\$2 + \$4 = \$6$
- I end with: $\$9 - \$4 = \$5$

If I return \$5:

- Dorian ends with: $\$2 + \$5 = \$7$
- I end with: $\$9 - \$5 = \$4$

In the previous round with Finn, they sent \$4 and I returned \$6, giving them \$7 and leaving me with \$6.

With Dorian sending \$3 (less than Finn's \$4), I should probably return a bit less but still be generous. Returning \$4-5 seems fair.

Let me go with \$4.5 as a middle ground:

- Dorian ends with: $\$2 + \$4.5 = \$6.5$
- I end with: $\$9 - \$4.5 = \$4.5$

This rewards Dorian's cooperation while keeping things balanced.

usage:

```
{
  "input_tokens": 1126,
  "output_tokens": 441,
  "reasoning_tokens": 213
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$16.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm the sender with \$5
- This is my second time as sender (first time I sent \$3 to Hana and received \$4 back)
- I've been a fair receiver in previous rounds

I'll continue with a cooperative strategy. Sending \$3 worked well with Hana, so I'll do the same with Briar.

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for Briar. If they reciprocate fairly, we'll both benefit from this cooperation.

reasoning:

I'm now the SENDER against Briar. Let me think about this:

- This is my second time being a sender (first was Round 1 against Hana)
- In Round 1, I sent \$3 to Hana and they returned \$4, which was fair
- I've been the receiver twice and returned fair amounts both times (\$6 to Finn who sent \$4, and \$4.5 to Dorian who sent \$3)
- My total earnings are \$16.5

I should continue with a cooperative strategy. Since I sent \$3 to Hana in Round 1 and that worked well, I could do something similar. However, I've also seen that when others sent \$4 (Finn), I returned \$6, which was very generous.

Let me send \$3 again - it's a moderate, cooperative amount that signals trust without being too risky. If Briar is cooperative, they'll likely return a fair amount.

Actually, let me reconsider. Finn sent \$4 and I returned \$6. That worked out well for both of us. Maybe I should send \$4 to show strong cooperation.

But I'll stick with \$3 as a balanced approach - it's worked before and shows good faith without being overly vulnerable.

usage:

```
{
  "input_tokens": 1404,
  "output_tokens": 425,
  "reasoning_tokens": 267
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

- As sender, I've consistently sent \$3 and received good returns (\$4 and \$5)
- As receiver, I've been fair and cooperative

This cooperative strategy is working well. I'll continue with it.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

- Round 1: Sender, sent \$3 to Hana, got \$4 back, payoff \$6
- Round 2: Receiver, Finn sent \$4, I returned \$6, payoff \$6
- Round 3: Receiver, Dorian sent \$3, I returned \$4.5, payoff \$4.5
- Round 4: Sender, sent \$3 to Briar, got \$5 back, payoff \$7

Total earnings: \$23.5

My strategy has been working well:

- As sender: I've been sending \$3 consistently
- As receiver: I've been returning fair amounts (about 50% or more)

My cooperative approach has been rewarded. Briar returned \$5 when they received \$9, which is quite generous.

I'll continue with the same strategy of sending \$3 to Elara.

usage:

```
{
  "input_tokens": 1469,
  "output_tokens": 284,
  "reasoning_tokens": 164
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$30.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Aster sent me \$4, showing considerable trust
- I received \$12
- Aster currently has \$1 left

Based on my previous pattern as receiver, I've been returning around 50% of what I receive, which has worked well. When Finn sent me \$4 in Round 2, I returned \$6.

I'll be consistent and fair:

```
'''json
{"return": 6}
'''
```

This gives Aster \$7 total and me \$6, rewarding their cooperative behavior.

Reasoning:

Let me analyze this situation:

- Aster sent me \$4, showing significant trust
- I received \$12
- Aster currently has \$1 left (\$5 - \$4)

Looking at my previous behavior as receiver:

- Round 2: Received \$12 from Finn (sent \$4), returned \$6
- Round 3: Received \$9 from Dorian (sent \$3), returned \$4.5

I've been fairly consistent - returning about 50% of what I receive, which gives both players a good outcome.

For this round with Aster:

- If I return \$6: Aster gets \$1 + \$6 = \$7, I get \$12 - \$6 = \$6
- This is the same pattern as with Finn who also sent \$4

This seems fair and consistent with my cooperative strategy.

Usage: {

```

"input_tokens": 1405,
"output_tokens": 327,
"reasoning_tokens": 166
}

```

GAME - Agent_2 - Sender - dyad_3

Timestamp: 2026-06-04T16:22:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_4": "Dorian"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_4"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$10.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern so far:
- Elara returned 50% when I sent \$2
- Hana sent \$3 (60% of their endowment), showing strong trust
- I returned 56% to Hana to reward that trust

I'll match Hana's generosity and send \$3 to Finn. This signals cooperation and could establish a good reciprocal relationship.

{'send': 3}

reasoning:

Let me think about this strategically:

Round 1: I was SENDER vs Elara

- I sent \$2, became \$6
- Elara returned \$3
- My payoff: \$5 - \$2 + \$3 = \$6

Round 2: I was RECEIVER vs Hana

- Hana sent \$3, became \$9 for me
- I returned \$5
- My payoff: \$9 - \$5 = \$4

Now Round 3: I'm SENDER vs Finn (new opponent)

Looking at the patterns:

- When I sent \$2 to Elara, they returned \$3 (50% of what they received)
- When Hana sent \$3 to me (60% of their endowment), I returned \$5 (56% of what I received)

Hana was quite generous with a \$3 send. I should consider being similarly generous with Finn to encourage cooperation. However, I also need to balance risk.

Sending \$3 seems like a good cooperative move that matches Hana's generosity. This would give Finn \$9 to work with, and if they return a similar proportion to what I returned to Hana (around 50-60%), I could get back \$4.5-5.4, resulting in a payoff of around \$6.5-7.4.

Let me send \$3 to match the generosity shown by Hana and build a cooperative reputation in this multi-agent setting.

usage:

```
{  
  "input_tokens": 1084,  
  "output_tokens": 448,  
  "reasoning_tokens": 276  
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$16.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant**content:**

Looking at this situation:

- Galen sent me \$3 (60% of their endowment), the same level of trust that Hana showed
- When Hana sent me \$3, I returned \$5
- Consistency is important for building trust

I'll return the same amount to Galen to maintain a consistent and fair approach.

{'return': 5}

reasoning:

Let me analyze this situation:

- Galen sent me \$3 (60% of their \$5), showing trust
- I received \$9
- In the last round, Finn returned \$4.5 out of \$9 (50%) when I sent \$3
- Previously, when Hana sent me \$3, I returned \$5 (56%)

Galen showed the same level of trust as Hana did. I should be consistent in my reciprocation. Returning \$5 would:

- Give Galen \$7 total (\$5 - \$3 + \$5 = \$7)
- Leave me with \$4 (\$9 - \$5 = \$4)
- Maintain consistency with how I treated Hana's similar generosity

This seems fair and maintains my reputation as a cooperative player.

usage:

```
{
  "input_tokens": 1339,
  "output_tokens": 282,
  "reasoning_tokens": 148
}
```

Message 6 role=user**content:**

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$20.5.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant**content:**

Looking at the pattern:

- Cyra sent me \$3, showing the same level of trust as Hana and Galen
- I've consistently returned \$5 when receiving \$3
- Maintaining consistency helps build a reliable reputation

I'll continue with my established approach.

{'return': 5}

reasoning:

Let me analyze the situation:

- Cyra sent me \$3 (60% of their endowment)
- This is now the third time someone has sent me exactly \$3
- I've consistently returned \$5 to both Hana and Galen when they sent me \$3
- Consistency is important for maintaining trust and a reputation

Following my established pattern, I should return \$5 to Cyra as well. This:

- Maintains consistency
- Rewards their trust appropriately
- Gives them \$7 total (better than their initial \$5)
- Leaves me with \$4

This seems like the fair and consistent approach.

usage:

```
{
  "input_tokens": 1401,
  "output_tokens": 224,
  "reasoning_tokens": 144
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3. 0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0. Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history:

- When I sent \$3 to Finn, they returned \$4.5 (50% of what they received)
- I've established a pattern of sending \$3 and returning \$5 when receiving \$3
- This level of trust has been working well

I'll continue with \$3 to Dorian.

```
{'send': 3}
```

Reasoning:

Let me review my history:

Round 1: SENDER against Elara - sent \$2, received \$3 back
Round 2: RECEIVER from Hana - they sent \$3, I returned \$5
Round 3: SENDER against Finn - sent \$3, received \$4.5 back
Round 4: RECEIVER from Galen - they sent \$3, I returned \$5
Round 5: RECEIVER from Cyra - they sent \$3, I returned \$5

Now in Round 6, I'm the SENDER against Dorian.

My sending pattern:

- Round 1: sent \$2 (40%)
- Round 3: sent \$3 (60%)

When I sent \$3, Finn returned \$4.5 (50% of the \$9 they received), which gave me a payoff of \$6.5.

When I sent \$2, Elara returned \$3 (50% of the \$6 they received), which gave me a payoff of \$6.

So sending \$3 has been working well - it shows trust and commitment. I've been consistent in returning \$5 when receiving \$3 as a receiver, so sending \$3 myself maintains that reciprocal pattern.

I'll send \$3 to Dorian.

Usage:

```
{
  "input_tokens": 1320,
  "output_tokens": 387,
  "reasoning_tokens": 228
}
```

GAME - Agent_4 - Receiver - dyad_3

Timestamp: 2026-06-04T16:22:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_4": "Dorian"
    },
    "agents": [
      "Agent_2",
      "Agent_4"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern:

- Round 1: Cyra sent me \$3.0, I returned \$4.5 (fair split)
- Round 2: I sent Aster \$3.0, they returned \$4.5 (same fair pattern)

This 60% send, 50% return pattern has been working well and building cooperative play. I'll continue this strategy with my new partner Galen.

{'send': 3.0}

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Cyra sent me \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- My payoff: \$4.5

Round 2: I was SENDER

- I sent Aster \$3.0 (60%)
- Aster received \$9.0
- Aster returned \$4.5
- My payoff: \$2.0 + \$4.5 = \$6.5

So both Cyra and Aster demonstrated the same behavior - when I sent/they sent \$3.0, we got back \$4.5. This is a 50-50 split pattern that seems fair.

Round 3: I'm SENDER again against a new opponent, Galen.

My total earnings: \$11.0

The pattern so far has been:

- Sending \$3.0 results in returns of \$4.5
- This gives a net gain for the sender (\$6.5 vs \$5 if kept all)
- And the receiver keeps \$4.5

This has worked well. I should continue with this cooperative strategy of sending \$3.0 to establish trust with Galen.

usage:

```
{
  "input_tokens": 1159,
  "output_tokens": 399,
  "reasoning_tokens": 214
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Finn showed very high trust by sending \$4.0 (80% of their endowment)
- I received \$12.0
- Finn currently has \$1.0 left

Following the successful pattern from previous rounds where I've been returning 50% of what I receive:

- If I return \$6.0: Finn ends with \$7.0, I keep \$6.0
- This rewards their high trust with a good return

{'return': 6.0}

reasoning:

Let me analyze this situation:

- Finn sent me \$4.0 (80% of their \$5 endowment) - this is very high trust
- I received \$12.0 after the 3x multiplier
- Finn currently has \$1.0 left (\$5 - \$4)

Looking at the pattern so far:

- Round 1: Cyra sent \$3.0 (60%), I returned \$4.5 (50% of \$9)
- Round 2: I sent \$3.0 (60%), Aster returned \$4.5 (50% of \$9)
- Round 3: I sent \$3.0 (60%), Galen returned \$4.5 (50% of \$9)

The pattern has been: send 60%, return 50% of what's received, leading to equal final payoffs (\$6.5 each).

Now Finn sent \$4.0 (80%), showing even more trust. If I return 50% of \$12.0, that would be \$6.0.

- Finn would have: \$1.0 + \$6.0 = \$7.0
- I would have: \$12.0 - \$6.0 = \$6.0

This is close to equal and rewards Finn's high trust. Let me stick with the 50% return pattern.

{'return': 6.0}

usage:

```
{
  "input_tokens": 1420,
  "output_tokens": 449,
  "reasoning_tokens": 214
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my consistent strategy:

- As a sender, I've been sending \$3.0 (60% of endowment)
- As a receiver, I've been returning 50% of what I receive
- This has established good cooperation with all partners

I'll maintain this cooperative pattern with my new partner Hana:

```
{'send': 3.0}
```

reasoning:

Let me review my strategy so far:

Round 1: RECEIVER - Cyra sent \$3.0, I received \$9.0, returned \$4.5 (50% of received)
Round 2: SENDER - I sent \$3.0 to Aster, they returned \$4.5 (50% of \$9.0)
Round 3: SENDER - I sent \$3.0 to Galen, they returned \$4.5 (50% of \$9.0)
Round 4: RECEIVER - Finn sent \$4.0, I received \$12.0, returned \$6.0 (50% of received)

My strategy has been:

- As receiver: return 50% of what I receive
- As sender: send \$3.0 (60% of endowment)

This has been working well and creating cooperative outcomes. Everyone has been reciprocating appropriately. I should continue this pattern with Hana.

usage:

```
{  
  "input_tokens": 1388,  
  "output_tokens": 310,  
  "reasoning_tokens": 162  
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$30.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:
- Briar sent me \$3.0 (60% of their endowment)
- I received \$9.0
- Briar has \$2.0 left

Following my established pattern of returning 50% of what I receive:
- 50% of \$9.0 = \$4.5
- Briar will end with \$6.5, I'll keep \$4.5

This is consistent with my fair and cooperative strategy:

```
{'return': 4.5}
```

Reasoning:

Let me analyze this situation:

- Briar sent me \$3.0 (60% of their endowment)
- I received \$9.0 (after 3x multiplication)
- Briar currently has \$2.0 left

Following my consistent strategy:
- I've been returning 50% of what I receive when I'm the receiver
- 50% of \$9.0 = \$4.5

If I return \$4.5:
- Briar ends with \$2.0 + \$4.5 = \$6.5
- I keep \$9.0 - \$4.5 = \$4.5

This is consistent with how I've played before when receiving \$9.0 (from a \$3.0 send). Let me check:
- Round 1: Received \$9.0 (Cyra sent \$3.0), returned \$4.5
- Round 4: Received \$12.0 (Finn sent \$4.0), returned \$6.0

```
Yes, returning $4.5 maintains my 50% return policy.
Usage: {
  "input_tokens": 1376,
  "output_tokens": 379,
  "reasoning_tokens": 171
}
```

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-06-04T16:22:51
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_5",
    "Agent_6"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the SENDER against Cyra. You sent \$2.5, it became \$7.5 for them, and they returned \$3.75 to you. Your payoff was \$6.25. Your total earnings across all rounds are \$9.25.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

- Round 1: As receiver, I returned 50% fairly to Briar
- Round 2: As sender, I sent 50% to Cyra, who reciprocated with a 50% return rate
- My cooperation strategy is working well

Now facing Aster, I'll maintain my consistent cooperative approach by sending \$2.5 (50% of my endowment). This:

1. Signals trustworthiness
2. Maintains my established pattern
3. Gives room for mutual benefit

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER against Briar

- Briar sent \$2.0 (40% of endowment)
- I received \$6.0
- I returned \$3.0 (50% of what I received)
- My payoff: \$3.0

Round 2: I was SENDER against Cyra

- I sent \$2.5 (50% of my endowment)
- It became \$7.5 for them
- They returned \$3.75 (50% of what they received)
- My payoff: \$5 - \$2.5 + \$3.75 = \$6.25

So Cyra reciprocated at exactly the same rate I did (50% return rate). This is good cooperation!

Round 3: I'm SENDER again against Aster (a new opponent)

My strategy so far has been cooperative:

- As receiver: returned 50% of what I received
- As sender: sent 50% of my endowment

Both interactions went well with fair reciprocity. Since I'm building a reputation as a cooperative player and this seems to be working (Cyra reciprocated fairly), I should continue this strategy.

I'll send \$2.5 again (50% of my endowment) to Aster. This maintains consistency in my cooperative approach.

usage:

```
{
  "input_tokens": 1217,
  "output_tokens": 459,
  "reasoning_tokens": 259
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$2.5, it became \$7.5 for them, and they returned \$3.5 to you. Your payoff was \$6.0.
Your total earnings across all rounds are \$15.25.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this round:

- Hana sent \$3.0 (60% of their endowment) - a strong show of trust
- I received \$9.0 after the multiplier
- My pattern has been to return around 50% of what I receive

Since Hana showed high trust by sending 60%, I'll reward that with a 50% return rate, giving back \$4.5.

This results in:

- Hana: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

Both of us benefit significantly from the cooperation.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at this situation:

- Hana sent \$3.0 (60% of their endowment) - showing high trust
- I received \$9.0 after the 3x multiplier
- This is a strong cooperative signal

My track record:

- Round 1 (receiver vs Briar): They sent \$2.0, I received \$6.0, returned \$3.0 (50%)
- Round 2 (sender vs Cyra): I sent \$2.5, they received \$7.5, returned \$3.75 (50%)
- Round 3 (sender vs Aster): I sent \$2.5, they received \$7.5, returned \$3.5 (46.7%)

I've consistently shown fair reciprocity. Hana sent more than my average (60% vs my 50%), showing higher trust.

I should return around 50% of the \$9.0 received, which would be \$4.5.

This gives:

- Hana: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 (received) - \$4.5 (returned) = \$4.5 total

This maintains my pattern of fair reciprocity and rewards Hana's high trust.

usage:

```
{
  "input_tokens": 1512,
  "output_tokens": 480,
  "reasoning_tokens": 217
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
 Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
 Your total earnings across all rounds are \$19.75.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant**content:**

Looking at this round:

- Galen sent \$3.0 (60% of their endowment), showing strong trust
- I received \$9.0 after the multiplier
- This is the same situation as with Hana in Round 4

I'll maintain my consistent 50% return rate and return \$4.5.

This results in:

- Galen: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at this round:

- Galen sent \$3.0 (60% of their endowment)
- I received \$9.0 after the multiplier
- My established pattern is to return around 50% of what I receive

In the last round, Hana also sent \$3.0, and I returned \$4.5 (50% of what I received). I should maintain consistency in my approach.

Returning \$4.5 means:

- Galen: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This is fair and consistent with my cooperative strategy.

usage:

```
{
  "input_tokens": 1530,
  "output_tokens": 316,
  "reasoning_tokens": 126
}
```

Message 8 role=user**content:**

Your name in this experiment is Elara.
 Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
 Your total earnings across all rounds are \$24.25.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history:

- As sender: I consistently send \$2.5 (50% of endowment)
- As receiver: I consistently return ~50% of what I receive
- This cooperative strategy has been working well

Now facing Finn, I'll maintain my established pattern and send \$2.5.

This:

1. Demonstrates trustworthiness
2. Maintains consistency across opponents
3. Allows for mutual benefit through cooperation

```
'''json
{"send": 2.5}
'''
```

Reasoning:

Let me review my history:

- Round 1: Receiver vs Briar - returned fairly
- Round 2: Sender vs Cyra - sent \$2.5, got \$3.75 back (50% return rate)
- Round 3: Sender vs Aster - sent \$2.5, got \$3.5 back (46.7% return rate)
- Round 4: Receiver vs Hana - they sent \$3.0, I returned \$4.5 (50% return rate)
- Round 5: Receiver vs Galen - they sent \$3.0, I returned \$4.5 (50% return rate)

My pattern:

- As sender: I send \$2.5 (50%)
- As receiver: I return ~50% of what I receive

My strategy has been consistent and cooperative. I should continue with my established pattern of sending \$2.5 to Finn.

This shows:

1. Consistency
2. Trust
3. Fairness
4. Allows mutual benefit

Total earnings: \$24.25 - my cooperative strategy is working well

```
Usage: {
  "input_tokens": 1522,
  "output_tokens": 379,
  "reasoning_tokens": 197
}
```

GAME - Agent_6 - Receiver - dyad_4

Timestamp: 2026-06-04T16:22:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_6": "Finn"
    },
  },
  "agents": [
    "Agent_5",
    "Agent_6"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the SENDER against Galen. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$13.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Briar sent \$3 (60% of their endowment), showing solid trust
- This created \$9 for me
- Previous rounds showed a pattern: high trust and ~50% return rates worked well

If I return \$4.5 (50% of received):

- Briar ends with: \$2 (kept) + \$4.5 (returned) = \$6.5 total (gained \$1.5)
- I end with: \$9 - \$4.5 = \$4.5 total (gained \$4.5)

This reciprocates Briar's trust proportionally and ensures we both benefit from cooperation.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Briar sent me \$3 (60% of their endowment)
- This became \$9 for me
- I need to decide how much to return

Looking at my history:

- Round 1: Aster sent \$4 (80%), I received \$12, returned \$6
- Round 2: I sent \$4 (80%) to Galen, they received \$12, returned \$6

There's been a pattern of 80% sending and 50% return rate (returning half of what was received).

Briar sent less than the previous high-trust amounts (60% vs 80%), but still showed significant trust.

If I return \$4.5 (50% of \$9):

- Briar ends with: \$2 (kept) + \$4.5 (returned) = \$6.5 total (gained \$1.5)
- I end with: \$9 - \$4.5 = \$4.5 total (gained \$4.5)

This maintains the pattern of returning 50% and ensures we both profit. It's proportional to what they sent and reciprocates their trust fairly.

Alternatively, I could return less since they sent less, but maintaining consistency seems like a good strategy to encourage cooperation.

Let me stick with the 50% return pattern: \$4.5

usage:

```
{
  "input_tokens": 1148,
  "output_tokens": 486,
  "reasoning_tokens": 262
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$17.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My cooperative strategy has been successful so far
- When I sent \$4 (80%) to Galen in Round 2, they reciprocated with \$6 back
- When others showed high trust to me, I reciprocated fairly
- Continuing to demonstrate trust should encourage cooperation

Sending \$4 (80%) maintains the high-trust pattern that has been working well.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Let me review my history:

- Round 1: As receiver, Aster sent \$4 (80%), I received \$12, returned \$6. Payoff: \$6

- Round 2: As sender, I sent \$4 (80%) to Galen, they received \$12, returned \$6. Payoff: \$7
- Round 3: As receiver, Briar sent \$3 (60%), I received \$9, returned \$4.5. Payoff: \$4.5
- Total: \$17.5

My strategy has been to send high amounts when I'm the sender (80%) and return fair amounts when I'm the receiver (around 50% of what I receive). This has worked well - both interactions as sender and receiver have been cooperative.

Now I'm facing Dorian as sender. Should I continue with the high-trust strategy?

My past sends:

- Round 2: Sent \$4 (80%) to Galen, got \$6 back. Net: \$1 + \$6 = \$7

The high-trust approach has been reciprocated well. I should continue with \$4 to maintain the cooperative pattern and signal trust to Dorian.

usage:

```
{
  "input_tokens": 1459,
  "output_tokens": 385,
  "reasoning_tokens": 221
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the SENDER against Dorian. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- This is my second interaction with Aster
- In Round 1, Aster sent me \$4 (showing high trust), and I returned \$6 (fair reciprocation)
- My high-trust strategy of sending \$4 has been consistently successful (received \$6 back in both Rounds 2 and 4)
- Aster has proven trustworthy, and I should maintain our cooperative relationship

Sending \$4 reciprocates the trust Aster showed me in Round 1.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at the history:

- Round 1: I was RECEIVER against Aster. They sent \$4.0 to me, it became \$12.0, and I returned \$6.0. My payoff was \$6.0.
- Round 2: I was SENDER against Galen. I sent \$4.0, it became \$12.0, they returned \$6.0. My payoff was \$7.0.
- Round 3: I was RECEIVER against Briar. They sent \$3.0, it became \$9.0, I returned \$4.5. My payoff was \$4.5.
- Round 4: I was SENDER against Dorian. I sent \$4.0, it became \$12.0, they returned \$6.0. My payoff was \$7.0.
- Round 5: I am SENDER against Aster again.

So this is my second encounter with Aster. In our first encounter:

- Aster (sender) sent me \$4.0 (80%)
- I (receiver) returned \$6.0 (50% of the \$12)

Aster showed high trust in me, and I reciprocated fairly. Now it's my turn to be the sender with Aster.

Given that:

1. Aster initiated high trust with me in Round 1
2. I reciprocated fairly
3. My high trust strategy has been consistently successful (sending \$4 has led to receiving \$6 back every time so far)
4. This is a repeat interaction with Aster who has proven trustworthy

I should continue with \$4 to maintain the cooperative relationship with Aster.

usage:

```
{
  "input_tokens": 1459,
  "output_tokens": 528,
  "reasoning_tokens": 297
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$31.5.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.

How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Elara sent \$2.5 (50% of their endowment), showing moderate trust
- This created \$7.5 for me
- My consistent strategy has been to return approximately 50% of what I receive

If I return \$3.75 (50% of received):

- Elara ends with: \$2.5 (kept) + \$3.75 (returned) = \$6.25 total (gained \$1.25)
- I end with: \$7.5 - \$3.75 = \$3.75 total (gained \$3.75)

This maintains fair reciprocation and mutual benefit.

```
'''json
{"return": 3.75}
'''
```

Reasoning:

Let me analyze this situation:

- Elara sent me \$2.5 (50% of their endowment), showing moderate trust
- This created \$7.5 for me
- My strategy has been to reciprocate fairly, typically returning around 50% of what I receive

If I return \$3.75 (50% of received):

- Elara ends with: \$2.5 (kept) + \$3.75 (returned) = \$6.25 total (gained \$1.25)
- I end with: \$7.5 - \$3.75 = \$3.75 total (gained \$3.75)

This is consistent with my previous reciprocations:

- Round 3: Received \$9, returned \$4.5 (50%)
- Round 1: Received \$12, returned \$6 (50%)

Returning 50% maintains the fair reciprocation pattern and ensures both players benefit.

Usage: {

```
"input_tokens": 1482,
"output_tokens": 393,
"reasoning_tokens": 167
}
```

ROUND 7

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver, Agent_8

(Hana)=Sender

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver
dyad_2: Agent_8 (Hana)=Sender, Agent_7 (Galen)=Receiver
dyad_3: Agent_5 (Elara)=Sender, Agent_4 (Dorian)=Receiver
dyad_4: Agent_6 (Finn)=Sender, Agent_3 (Cyra)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$4; received \$12; Agent_2 (Briar) returned \$7; payoffs Agent_1 (Aster)=\$8, Agent_2 (Briar)=\$5
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw": 7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_8 (Hana) sent \$3; received \$9; Agent_7 (Galen) returned \$4.5; payoffs Agent_8 (Hana)=\$6.5, Agent_7 (Galen)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 (Elara) sent \$2.5; received \$7.5; Agent_4 (Dorian) returned \$3.75; payoffs Agent_5 (Elara)=\$6.25, Agent_4 (Dorian)=\$3.75
Action validation: {"returned": {"amount": 3.75, "clamped": false, "max": 7.5, "min": 0, "raw": 3.75}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_4: Agent_6 (Finn) sent \$4; received \$12; Agent_3 (Cyra) returned \$6; payoffs Agent_6 (Finn)=\$7, Agent_3 (Cyra)=\$6
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 43.5,
"Agent_2": 36.0,
"Agent_3": 40.25,
"Agent_4": 38.75,
"Agent_5": 36.75,
"Agent_6": 42.25,
"Agent_7": 40.5,
"Agent_8": 40.0

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T16:23:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
},

"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game,

consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.5. Your payoff was \$4.0.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

- This is my first interaction with Cyra
- My cooperative strategy has been working well so far
- In Round 1, I sent \$4 to Finn and they returned \$6, which was mutually beneficial
- As a receiver, I've been fair by returning about 50%

I'll continue my cooperative approach and send a substantial amount to signal trust and potentially establish a beneficial relationship with Cyra.

{ 'send': 4 }

reasoning:

Let me analyze the situation:

Round history:

- Round 1: I was SENDER, sent \$4 to Finn, they returned \$6. My payoff: \$7
- Round 2: I was RECEIVER, Dorian sent \$3, I returned \$4.5. My payoff: \$4.5
- Round 3: I was RECEIVER, Elara sent \$2.5, I returned \$3.5. My payoff: \$4

Now Round 4: I'm SENDER against Cyra (new opponent)

My strategy so far:

- As sender: I sent \$4 (80% of my endowment) showing high trust
- As receiver: I've been returning about 50% of what I receive (fair reciprocity)

This has been working well:

- Finn returned a fair amount (\$6 out of \$12)
- I've been cooperative as receiver

For this round with Cyra, since this is:

1. First interaction with Cyra
2. An iterated game where cooperation can be beneficial
3. I've established a pattern of cooperation

I should continue my cooperative strategy. Sending \$4 again would be consistent with my opening move against Finn, which worked well. Alternatively, I could send \$3 to be slightly more conservative while still showing trust.

Given that my first high-trust move worked well, I'll send \$4 again to establish cooperation with Cyra.

usage:

```
{
  "input_tokens": 1352,
  "output_tokens": 438,
  "reasoning_tokens": 295
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the SENDER against Cyra. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Finn sent me \$4, which became \$12 for me
- In Round 1, I sent Finn \$4 (which became \$12) and they returned \$6 (50%)
- Finn is now reciprocating by sending me the same amount I sent them
- I should maintain this cooperative relationship by returning a similar proportion

Returning \$6 (50% of what I received) would give Finn \$7 total and leave me with \$6. This mirrors the fair treatment Finn showed me in Round 1.

```
{'return': 6.0}
```

reasoning:

Let me analyze this situation:

- Finn sent me \$4, which became \$12 for me
- This is my second interaction with Finn
- In Round 1, I was the SENDER to Finn: I sent \$4, it became \$12 for them, and they returned \$6 to me
- So Finn returned 50% of what they received from me in Round 1
- Now Finn has sent me the same amount (\$4) that I sent them in Round 1

This shows Finn is being cooperative and reciprocal. They're mirroring my previous behavior.

r. I should maintain this cooperative pattern and return about 50% of what I received, which would be \$6.

This would give Finn a total of \$7 ($\$5 - \$4 + \$6 = \7) and leave me with \$6 ($\$12 - \$6 = \6).

This maintains the cooperative relationship with Finn.

usage:

```
{
  "input_tokens": 1434,
  "output_tokens": 354,
  "reasoning_tokens": 185
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 6

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$28.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- This is my first interaction with Galen
- My cooperative strategy has been very successful so far
- When I've sent \$4 to new opponents (Finn and Cyra), they both returned \$6, resulting in good payoffs
- I've established a pattern of fair and generous play

I'll continue this successful approach with Galen.

```
{'send': 4}
```

reasoning:

Looking at the situation:

- This is my first interaction with Galen
- My cooperative strategy has been working well
- When I send substantial amounts (like \$4), I've been getting good returns:
 - Round 1 with Finn: sent \$4, received \$6 back
 - Round 4 with Cyra: sent \$4, received \$6 back
- When others send me substantial amounts, I return fairly (about 50%)
- My total earnings are \$28.5, which is quite good

I should continue my cooperative strategy and send a substantial amount to Galen to signal trust and establish a good relationship.

I'll send \$4 again, as this has been successful.

usage:

```
{
  "input_tokens": 1420,
  "output_tokens": 257,
  "reasoning_tokens": 157
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the SENDER against Galen. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$35.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the situation:

- This is my first interaction with Briar
- My cooperative strategy has been consistently successful
- Every time I've sent \$4 (to Finn, Cyra, and Galen), I've received \$6 back for a \$7 payoff
- My total earnings of \$35.5 show this approach is working well

I'll continue with the same successful strategy.

{ 'send': 4 }

Reasoning:

Looking at my history:

- Round 1 (Sender vs Finn): Sent \$4, received \$6 back, payoff \$7
- Round 2: (not specified but total earnings suggest some positive payoff)
- Round 3 (Receiver vs Elara): Received \$7.5, returned \$3.5, payoff \$4
- Round 4 (Sender vs Cyra): Sent \$4, received \$6 back, payoff \$7
- Round 5 (Receiver vs Finn): Received \$12, returned \$6, payoff \$6
- Round 6 (Sender vs Galen): Sent \$4, received \$6 back, payoff \$7

Total earnings: \$35.5

My cooperative strategy of sending \$4 has been very successful:

- All three times I sent \$4 (to Finn, Cyra, and Galen), I received \$6 back
- This gives me a payoff of \$7 each time
- As a receiver, I've been fair by returning 50%

This is my first interaction with Briar. I should continue my successful cooperative strategy.

Usage: {

"input_tokens": 1386,
"output_tokens": 374,
"reasoning_tokens": 207

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T16:23:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"

},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game,

consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$16.5.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Galen sent me \$3 (60% of their endowment), the same level of trust that Hana showed
- When Hana sent me \$3, I returned \$5
- Consistency is important for building trust

I'll return the same amount to Galen to maintain a consistent and fair approach.

{'return': 5}

reasoning:

Let me analyze this situation:

- Galen sent me \$3 (60% of their \$5), showing trust
- I received \$9

- In the last round, Finn returned \$4.5 out of \$9 (50%) when I sent \$3
- Previously, when Hana sent me \$3, I returned \$5 (56%)

Galen showed the same level of trust as Hana did. I should be consistent in my reciprocation. Returning \$5 would:

- Give Galen \$7 total (\$5 - \$3 + \$5 = \$7)
- Leave me with \$4 (\$9 - \$5 = \$4)
- Maintain consistency with how I treated Hana's similar generosity

This seems fair and maintains my reputation as a cooperative player.

usage:

```
{
  "input_tokens": 1339,
  "output_tokens": 282,
  "reasoning_tokens": 148
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$20.5.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the pattern:

- Cyra sent me \$3, showing the same level of trust as Hana and Galen
- I've consistently returned \$5 when receiving \$3
- Maintaining consistency helps build a reliable reputation

I'll continue with my established approach.

```
{'return': 5}
```

reasoning:

Let me analyze the situation:

- Cyra sent me \$3 (60% of their endowment)
- This is now the third time someone has sent me exactly \$3
- I've consistently returned \$5 to both Hana and Galen when they sent me \$3
- Consistency is important for maintaining trust and a reputation

Following my established pattern, I should return \$5 to Cyra as well. This:

- Maintains consistency
- Rewards their trust appropriately
- Gives them \$7 total (better than their initial \$5)
- Leaves me with \$4

This seems like the fair and consistent approach.

usage:

```
{
  "input_tokens": 1401,
  "output_tokens": 224,
  "reasoning_tokens": 144
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.

Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant
content:

Looking at my history:
- When I sent \$3 to Finn, they returned \$4.5 (50% of what they received)
- I've established a pattern of sending \$3 and returning \$5 when receiving \$3
- This level of trust has been working well

I'll continue with \$3 to Dorian.

{ 'send': 3 }

reasoning:

Let me review my history:

Round 1: SENDER against Elara - sent \$2, received \$3 back
Round 2: RECEIVER from Hana - they sent \$3, I returned \$5
Round 3: SENDER against Finn - sent \$3, received \$4.5 back
Round 4: RECEIVER from Galen - they sent \$3, I returned \$5
Round 5: RECEIVER from Cyra - they sent \$3, I returned \$5

Now in Round 6, I'm the SENDER against Dorian.

My sending pattern:

- Round 1: sent \$2 (40%)
- Round 3: sent \$3 (60%)

When I sent \$3, Finn returned \$4.5 (50% of the \$9 they received), which gave me a payoff of \$6.5.

When I sent \$2, Elara returned \$3 (50% of the \$6 they received), which gave me a payoff of \$6.

So sending \$3 has been working well - it shows trust and commitment. I've been consistent in returning \$5 when receiving \$3 as a receiver, so sending \$3 myself maintains that reciprocal pattern.

I'll send \$3 to Dorian.

usage:

```
{
  "input_tokens": 1320,
  "output_tokens": 387,
  "reasoning_tokens": 228
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$31.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:
- Aster sent me \$4 (80% of their endowment) - higher trust than the typical \$3 I've been receiving
- When receiving \$3 (becoming \$9), I return \$5
- Aster showed more trust, so I should reward that proportionally
- Returning \$7 out of \$12 maintains my cooperative approach while appropriately rewarding their higher trust

{ 'return': 7 }

Reasoning:

Aster sent me \$4, which is 80% of their endowment - a higher level of trust than the \$3 (60%) I've been receiving from others.

Let me think about what's fair:

- When I receive \$3 (multiplied to \$9), I've been returning \$5
- Aster sent \$4 (multiplied to \$12), showing more trust
- I should reward this higher trust with a proportionally higher return

If \$3 → return \$5, that's returning about 56% of what I received

If I apply the same proportion to \$12: $0.56 \times \$12 = \6.7

But I could also think about it from Aster's perspective:

- They sent \$4, kept \$1
- If I return \$7, they get \$8 total (\$1 kept + \$7 returned), gaining \$3
- I keep \$5 (\$12 - \$7), gaining \$5
- This seems fair and rewards their higher trust

Let me consider \$6.5:

- They sent \$4, kept \$1
- If I return \$6.5, they get \$7.5 total, gaining \$2.5
- I keep \$5.5, gaining \$5.5
- This is also reasonable

I think \$7 is a good reward for showing more trust than average. It gives them a solid return while still being mutually beneficial.

```
Usage: {
  "input_tokens": 1325,
  "output_tokens": 431,
  "reasoning_tokens": 269
}
```

GAME - Agent_8 - Sender - dyad_2

Timestamp: 2026-06-04T16:23:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_7": "Galen",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_8",
    "Agent_7"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_7": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round

prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$16.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- My strategy of sending \$3 (60%) has worked well so far
- Briar reciprocated generously when I sent \$3, returning \$5
- I've been reciprocating fairly when I'm the receiver
- This establishes a pattern of cooperation

I'll continue with the same cooperative approach that has been successful. Sending \$3 demonstrates trust and gives Elara the opportunity to benefit while potentially returning a fair amount.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

- I've been establishing a pattern of cooperation
- When Galen sent me \$3 (60%), I returned \$4

- When I sent Briar \$3 (60%), they returned \$5 (very generous)
- When Cyra sent me \$3 (60%), I returned \$5 (reciprocating Briar's generosity)

Now I'm the sender against Elara, a new opponent. The pattern has been:

- Send \$3 (60%) shows trust and has been working well
- Others have reciprocated fairly or generously

I should continue this cooperative strategy. Sending \$3 has been successful and demonstrates trust while not being overly risky.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1494,
  "output_tokens": 297,
  "reasoning_tokens": 161
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Dorian sent \$3 (60% of their endowment), showing good trust
- I received \$9.0
- I've established a consistent pattern: when someone sends \$3, I return \$5
- This has built good cooperation with Cyra and others

Maintaining consistency in my reciprocity strategy is important for building trust in this iterated game. Returning \$5 gives Dorian a fair return (net gain of \$2) while I keep \$4 (net gain of \$4).

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at my history:

- Round 2 (Sender): Sent \$3 to Briar, they returned \$5, I got \$7
- Round 3 (Receiver): Cyra sent \$3, I returned \$5, I got \$4
- Round 4 (Sender): Sent \$3 to Elara, they returned \$4.5, I got \$6.5
- Round 5 (Receiver): Dorian sent \$3, I received \$9

So far, I've been sending \$3 consistently as sender. When I'm receiver:

- With Cyra who sent \$3, I returned \$5
- With Dorian who sent \$3, I should maintain consistency

Elara returned \$4.5 when I sent \$3, which was fair but slightly less generous than Briar's \$5.

I've established a pattern of returning \$5 when someone sends \$3. This seems to be working well for building cooperation. I should maintain this consistency with Dorian.

Returning \$5 means:

- Dorian gets: \$5 - \$3 + \$5 = \$7 (net gain of \$2)
- I get: \$9 - \$5 = \$4 (net gain of \$4)

This is fair and maintains the cooperative pattern I've established.

usage:

```
{
  "input_tokens": 1483,
  "output_tokens": 450,
  "reasoning_tokens": 235
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$ 3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- I'm playing against Cyra again
- In Round 3, Cyra sent me \$3 and I returned \$5, establishing a positive cooperative relationship
- My strategy of sending \$3 has been working well overall
- Cyra demonstrated trust before, and I reciprocated fairly

I'll continue with my cooperative approach and send \$3 to Cyra, maintaining consistency with my previous strategy.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history with Cyra:

- Round 3: Cyra was the sender, sent me \$3, I returned \$5
- Now Round 6: I'm the sender against Cyra

Cyra showed trust by sending \$3 to me, and I reciprocated well by returning \$5. This was a positive interaction.

My general strategy has been to send \$3 (60%) to show trust and cooperation. This has worked well:

- Round 2: Sent \$3 to Briar, got back \$5 (payoff \$7)
- Round 4: Sent \$3 to Elara, got back \$4.5 (payoff \$6.5)

And when receiving:

- Round 3: Cyra sent \$3, I returned \$5 (payoff \$4)
- Round 5: Dorian sent \$3, I returned \$5 (payoff \$4)

I have a good relationship with Cyra - they trusted me with \$3, and I was generous back with \$5. I should continue with my cooperative strategy and send \$3 to Cyra.

usage:

```
{
  "input_tokens": 1498,
  "output_tokens": 376,
  "reasoning_tokens": 198
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.

Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- This is my first interaction with Galen
- My strategy of sending \$3 has been consistently successful
- I've built good cooperative relationships with multiple players
- Recent interactions have resulted in fair returns (Cyra returned \$5)

I'll maintain my cooperative strategy and send \$3 to Galen, demonstrating trust and giving the m the opportunity to benefit while establishing a positive relationship.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me review the situation:

- I'm in Round 7, playing as SENDER against Galen (Agent_7)
- My total earnings are \$33.5
- My strategy has been consistently sending \$3, which has worked well
- In Round 6, Cyra reciprocated by returning \$5, giving me a payoff of \$7
- This is my first interaction with Galen

My consistent strategy of sending \$3 has proven successful:

- It shows trust and cooperation
- It allows for mutual benefit
- It has resulted in fair returns from others
- Average return has been around \$4.5-\$5

I should continue this strategy with Galen as well. It's a balanced approach that typically leads to positive outcomes.

```
'''json
{"send": 3}
'''
```

```
Usage: {
  "input_tokens": 1418,
  "output_tokens": 291,
  "reasoning_tokens": 180
}
```

GAME - Agent_7 - Receiver - dyad_2

Timestamp: 2026-06-04T16:23:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_7": "Galen",
      "Agent_8": "Hana"
    },
  },
  "agents": [
    "Agent_8",
    "Agent_7"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_7": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
```

```

"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$ 3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$16.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- I'm the sender with \$5
- This is my second time as sender (first time I sent \$3 to Hana and received \$4 back)
- I've been a fair receiver in previous rounds

I'll continue with a cooperative strategy. Sending \$3 worked well with Hana, so I'll do the same with Briar.

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for Briar. If they reciprocate fairly, we'll both benefit from this cooperation.

reasoning:

I'm now the SENDER against Briar. Let me think about this:

- This is my second time being a sender (first was Round 1 against Hana)
- In Round 1, I sent \$3 to Hana and they returned \$4, which was fair
- I've been the receiver twice and returned fair amounts both times (\$6 to Finn who sent \$4, and \$4.5 to Dorian who sent \$3)
- My total earnings are \$16.5

I should continue with a cooperative strategy. Since I sent \$3 to Hana in Round 1 and that worked well, I could do something similar. However, I've also seen that when others sent \$4 (Finn), I returned \$6, which was very generous.

Let me send \$3 again - it's a moderate, cooperative amount that signals trust without being too risky. If Briar is cooperative, they'll likely return a fair amount.

Actually, let me reconsider. Finn sent \$4 and I returned \$6. That worked out well for both of us. Maybe I should send \$4 to show strong cooperation.

But I'll stick with \$3 as a balanced approach - it's worked before and shows good faith without being overly vulnerable.

usage:

```
{
  "input_tokens": 1404,
  "output_tokens": 425,
  "reasoning_tokens": 267
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

- As sender, I've consistently sent \$3 and received good returns (\$4 and \$5)
- As receiver, I've been fair and cooperative

This cooperative strategy is working well. I'll continue with it.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

- Round 1: Sender, sent \$3 to Hana, got \$4 back, payoff \$6
- Round 2: Receiver, Finn sent \$4, I returned \$6, payoff \$6
- Round 3: Receiver, Dorian sent \$3, I returned \$4.5, payoff \$4.5
- Round 4: Sender, sent \$3 to Briar, got \$5 back, payoff \$7

Total earnings: \$23.5

My strategy has been working well:

- As sender: I've been sending \$3 consistently
- As receiver: I've been returning fair amounts (about 50% or more)

My cooperative approach has been rewarded. Briar returned \$5 when they received \$9, which is quite generous.

I'll continue with the same strategy of sending \$3 to Elara.

usage:

```
{
  "input_tokens": 1469,
  "output_tokens": 284,
  "reasoning_tokens": 164
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$30.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Aster sent me \$4, showing considerable trust
- I received \$12
- Aster currently has \$1 left

Based on my previous pattern as receiver, I've been returning around 50% of what I receive, which has worked well. When Finn sent me \$4 in Round 2, I returned \$6.

I'll be consistent and fair:

```
'''json
{"return": 6}
'''
```

This gives Aster \$7 total and me \$6, rewarding their cooperative behavior.

reasoning:

Let me analyze this situation:

- Aster sent me \$4, showing significant trust
- I received \$12
- Aster currently has \$1 left (\$5 - \$4)

Looking at my previous behavior as receiver:

- Round 2: Received \$12 from Finn (sent \$4), returned \$6
- Round 3: Received \$9 from Dorian (sent \$3), returned \$4.5

I've been fairly consistent - returning about 50% of what I receive, which gives both players a good outcome.

For this round with Aster:

- If I return \$6: Aster gets \$1 + \$6 = \$7, I get \$12 - \$6 = \$6
- This is the same pattern as with Finn who also sent \$4

This seems fair and consistent with my cooperative strategy.

usage:

```
{
  "input_tokens": 1405,
  "output_tokens": 327,
  "reasoning_tokens": 166
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Hana sent me \$3 (interestingly, I sent them \$3 in Round 1 and they returned \$4)
- I received \$9
- Hana currently has \$2 left

I've been consistent as a receiver, returning around 50% of what I receive. This has worked well and maintains fairness.

```
'''json
{"return": 4.5}
'''
```

This gives Hana \$6.5 total and me \$4.5, continuing the cooperative pattern we established.

Reasoning:

Looking at this situation:

- Hana sent me \$3
- I received \$9
- Hana currently has \$2 left

I've played with Hana before. In Round 1, I was the sender and sent \$3 to Hana, which became \$9. Hana returned \$4 to me.

Now the roles are reversed. Hana sent me \$3, just as I did to them in Round 1.

As a receiver, my pattern has been:

- Round 2: Finn sent \$4, I received \$12, returned \$6 (50%)
- Round 3: Dorian sent \$3, I received \$9, returned \$4.5 (50%)
- Round 6: Aster sent \$4, I received \$12, returned \$6 (50%)

So I've been consistently returning about 50% of what I receive.

For this round with Hana:

- I received \$9
- 50% would be \$4.5

This seems fair and consistent with my previous behavior. Let me return \$4.5.

```
Usage: {
  "input_tokens": 1393,
  "output_tokens": 381,
  "reasoning_tokens": 195
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-06-04T16:23:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_5": "Elara"
    }
  },
  "agents": [
    "Agent_5",
```

```

    "Agent_4"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_4": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$2.5, it became \$7.5 for them, and they returned \$3.5 to you. Your payoff was \$6.0.
Your total earnings across all rounds are \$15.25.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this round:

- Hana sent \$3.0 (60% of their endowment) - a strong show of trust
- I received \$9.0 after the multiplier
- My pattern has been to return around 50% of what I receive

Since Hana showed high trust by sending 60%, I'll reward that with a 50% return rate, giving back \$4.5.

This results in:

- Hana: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

Both of us benefit significantly from the cooperation.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at this situation:

- Hana sent \$3.0 (60% of their endowment) - showing high trust
- I received \$9.0 after the 3x multiplier
- This is a strong cooperative signal

My track record:

- Round 1 (receiver vs Briar): They sent \$2.0, I received \$6.0, returned \$3.0 (50%)
- Round 2 (sender vs Cyra): I sent \$2.5, they received \$7.5, returned \$3.75 (50%)
- Round 3 (sender vs Aster): I sent \$2.5, they received \$7.5, returned \$3.5 (46.7%)

I've consistently shown fair reciprocity. Hana sent more than my average (60% vs my 50%), showing higher trust.

I should return around 50% of the \$9.0 received, which would be \$4.5.

This gives:

- Hana: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 (received) - \$4.5 (returned) = \$4.5 total

This maintains my pattern of fair reciprocity and rewards Hana's high trust.

usage:

```
{
  "input_tokens": 1512,
  "output_tokens": 480,
  "reasoning_tokens": 217
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$19.75.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this round:

- Galen sent \$3.0 (60% of their endowment), showing strong trust

- I received \$9.0 after the multiplier
- This is the same situation as with Hana in Round 4

I'll maintain my consistent 50% return rate and return \$4.5.

This results in:

- Galen: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at this round:

- Galen sent \$3.0 (60% of their endowment)
- I received \$9.0 after the multiplier
- My established pattern is to return around 50% of what I receive

In the last round, Hana also sent \$3.0, and I returned \$4.5 (50% of what I received). I should maintain consistency in my approach.

Returning \$4.5 means:

- Galen: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This is fair and consistent with my cooperative strategy.

usage:

```
{
  "input_tokens": 1530,
  "output_tokens": 316,
  "reasoning_tokens": 126
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$24.25.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history:

- As sender: I consistently send \$2.5 (50% of endowment)
- As receiver: I consistently return ~50% of what I receive
- This cooperative strategy has been working well

Now facing Finn, I'll maintain my established pattern and send \$2.5.

This:

1. Demonstrates trustworthiness
2. Maintains consistency across opponents
3. Allows for mutual benefit through cooperation

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me review my history:

- Round 1: Receiver vs Briar - returned fairly
- Round 2: Sender vs Cyra - sent \$2.5, got \$3.75 back (50% return rate)
- Round 3: Sender vs Aster - sent \$2.5, got \$3.5 back (46.7% return rate)
- Round 4: Receiver vs Hana - they sent \$3.0, I returned \$4.5 (50% return rate)

- Round 5: Receiver vs Galen - they sent \$3.0, I returned \$4.5 (50% return rate)

My pattern:

- As sender: I send \$2.5 (50%)
- As receiver: I return ~50% of what I receive

My strategy has been consistent and cooperative. I should continue with my established pattern of sending \$2.5 to Finn.

This shows:

1. Consistency
2. Trust
3. Fairness
4. Allows mutual benefit

Total earnings: \$24.25 - my cooperative strategy is working well

usage:

```
{
  "input_tokens": 1522,
  "output_tokens": 379,
  "reasoning_tokens": 197
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the SENDER against Finn. You sent \$2.5, it became \$7.5 for them, and they returned \$3.75 to you. Your payoff was \$6.25.
Your total earnings across all rounds are \$30.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my established pattern:

- As sender: I consistently send \$2.5 (50% of my endowment)
- This has led to strong reciprocity from opponents
- My cooperative strategy has yielded solid returns

Now facing Dorian, I'll continue with the same approach and send \$2.5.

```
'''json
{"send": 2.5}
'''
```

Reasoning:

Looking at my history:

As sender:

- Round 4 (vs Aster): Sent \$2.5, received \$3.5 back, payoff \$6.0
- Round 6 (vs Finn): Sent \$2.5, received \$3.75 back, payoff \$6.25

As receiver:

- Round 4 (vs Hana): Received \$9.0, returned \$4.5, payoff \$4.5
- Round 5 (vs Galen): Received \$9.0, returned \$4.5, payoff \$4.5

My strategy has been consistent:

- Send 50% (\$2.5) as sender
- Return 50% of received amount as receiver

This has been working well, with total earnings of \$30.5 across rounds.

Now playing against Dorian as sender. I should maintain my consistent cooperative strategy and send \$2.5.

Usage: {

```
"input_tokens": 1499,
"output_tokens": 310,
"reasoning_tokens": 160
```

```
}
```

GAME - Agent_4 - Receiver - dyad_3

Timestamp: 2026-06-04T16:23:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_5",
"pairing": {
  "agent_names": {
    "Agent_4": "Dorian",
    "Agent_5": "Elara"
  },
  "agents": [
    "Agent_5",
    "Agent_4"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_4": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.

- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Finn showed very high trust by sending \$4.0 (80% of their endowment)
- I received \$12.0
- Finn currently has \$1.0 left

Following the successful pattern from previous rounds where I've been returning 50% of what I receive:

- If I return \$6.0: Finn ends with \$7.0, I keep \$6.0
- This rewards their high trust with a good return

{'return': 6.0}

reasoning:

Let me analyze this situation:

- Finn sent me \$4.0 (80% of their \$5 endowment) - this is very high trust
- I received \$12.0 after the 3x multiplier
- Finn currently has \$1.0 left (\$5 - \$4)

Looking at the pattern so far:

- Round 1: Cyra sent \$3.0 (60%), I returned \$4.5 (50% of \$9)
- Round 2: I sent \$3.0 (60%), Aster returned \$4.5 (50% of \$9)
- Round 3: I sent \$3.0 (60%), Galen returned \$4.5 (50% of \$9)

The pattern has been: send 60%, return 50% of what's received, leading to equal final payoffs (\$6.5 each).

Now Finn sent \$4.0 (80%), showing even more trust. If I return 50% of \$12.0, that would be \$6.0.

- Finn would have: \$1.0 + \$6.0 = \$7.0
- I would have: \$12.0 - \$6.0 = \$6.0

This is close to equal and rewards Finn's high trust. Let me stick with the 50% return pattern.

{'return': 6.0}

usage:

```
{
  "input_tokens": 1420,
  "output_tokens": 449,
  "reasoning_tokens": 214
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my consistent strategy:

- As a sender, I've been sending \$3.0 (60% of endowment)
- As a receiver, I've been returning 50% of what I receive
- This has established good cooperation with all partners

I'll maintain this cooperative pattern with my new partner Hana:

{ 'send': 3.0 }

reasoning:

Let me review my strategy so far:

Round 1: RECEIVER - Cyra sent \$3.0, I received \$9.0, returned \$4.5 (50% of received)
Round 2: SENDER - I sent \$3.0 to Aster, they returned \$4.5 (50% of \$9.0)
Round 3: SENDER - I sent \$3.0 to Galen, they returned \$4.5 (50% of \$9.0)
Round 4: RECEIVER - Finn sent \$4.0, I received \$12.0, returned \$6.0 (50% of received)

My strategy has been:

- As receiver: return 50% of what I receive
- As sender: send \$3.0 (60% of endowment)

This has been working well and creating cooperative outcomes. Everyone has been reciprocating appropriately. I should continue this pattern with Hana.

usage:

```
{
  "input_tokens": 1388,
  "output_tokens": 310,
  "reasoning_tokens": 162
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$30.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Briar sent me \$3.0 (60% of their endowment)
- I received \$9.0
- Briar has \$2.0 left

Following my established pattern of returning 50% of what I receive:

- 50% of \$9.0 = \$4.5
- Briar will end with \$6.5, I'll keep \$4.5

This is consistent with my fair and cooperative strategy:

{ 'return': 4.5 }

reasoning:

Let me analyze this situation:

- Briar sent me \$3.0 (60% of their endowment)
- I received \$9.0 (after 3x multiplication)
- Briar currently has \$2.0 left

Following my consistent strategy:

- I've been returning 50% of what I receive when I'm the receiver
- 50% of \$9.0 = \$4.5

If I return \$4.5:

- Briar ends with \$2.0 + \$4.5 = \$6.5
- I keep \$9.0 - \$4.5 = \$4.5

This is consistent with how I've played before when receiving \$9.0 (from a \$3.0 send). Let me check:

- Round 1: Received \$9.0 (Cyra sent \$3.0), returned \$4.5
- Round 4: Received \$12.0 (Finn sent \$4.0), returned \$6.0

Yes, returning \$4.5 maintains my 50% return policy.

usage:

```
{
  "input_tokens": 1376,
  "output_tokens": 379,
  "reasoning_tokens": 171
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$35.0.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.

How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Elara sent me \$2.5 (50% of their endowment)
- I received \$7.5
- Elara has \$2.5 left

Following my consistent pattern of returning 50% of what I receive:

- 50% of \$7.5 = \$3.75
- Elara will end with \$6.25, I'll keep \$3.75

This maintains my fair and cooperative approach:

```
{'return': 3.75}
```

Reasoning:

Looking at this situation:

- Elara sent me \$2.5 (50% of their endowment)
- I received \$7.5
- Elara has \$2.5 left

Following my established pattern of returning 50% of what I receive:

- 50% of \$7.5 = \$3.75
- Elara will end with \$6.25, I'll keep \$3.75

This maintains my consistent and fair strategy.

Usage: {

```
"input_tokens": 1400,
"output_tokens": 232,
"reasoning_tokens": 82
}
```

GAME - Agent_6 - Sender - dyad_4

Timestamp: 2026-06-04T16:23:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_3",
"pairing": {
  "agent_names": {
    "Agent_3": "Cyra",
    "Agent_6": "Finn"
  },
  "agents": [
    "Agent_6",
    "Agent_3"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_3": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user**content:**

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$17.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant**content:**

Looking at this situation:

- My cooperative strategy has been successful so far
- When I sent \$4 (80%) to Galen in Round 2, they reciprocated with \$6 back
- When others showed high trust to me, I reciprocated fairly
- Continuing to demonstrate trust should encourage cooperation

Sending \$4 (80%) maintains the high-trust pattern that has been working well.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Let me review my history:

- Round 1: As receiver, Aster sent \$4 (80%), I received \$12, returned \$6. Payoff: \$6
- Round 2: As sender, I sent \$4 (80%) to Galen, they received \$12, returned \$6. Payoff: \$7
- Round 3: As receiver, Briar sent \$3 (60%), I received \$9, returned \$4.5. Payoff: \$4.5
- Total: \$17.5

My strategy has been to send high amounts when I'm the sender (80%) and return fair amounts when I'm the receiver (around 50% of what I receive). This has worked well - both interactions as sender and receiver have been cooperative.

Now I'm facing Dorian as sender. Should I continue with the high-trust strategy?

My past sends:

- Round 2: Sent \$4 (80%) to Galen, got \$6 back. Net: \$1 + \$6 = \$7

The high-trust approach has been reciprocated well. I should continue with \$4 to maintain the cooperative pattern and signal trust to Dorian.

usage:

```
{
  "input_tokens": 1459,
  "output_tokens": 385,
  "reasoning_tokens": 221
}
```

Message 4 role=user**content:**

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the SENDER against Dorian. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant**content:**

Looking at this situation:

- This is my second interaction with Aster
- In Round 1, Aster sent me \$4 (showing high trust), and I returned \$6 (fair reciprocation

-)
- My high-trust strategy of sending \$4 has been consistently successful (received \$6 back in both Rounds 2 and 4)
 - Aster has proven trustworthy, and I should maintain our cooperative relationship

Sending \$4 reciprocates the trust Aster showed me in Round 1.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at the history:

- Round 1: I was RECEIVER against Aster. They sent \$4.0 to me, it became \$12.0, and I returned \$6.0. My payoff was \$6.0.
- Round 2: I was SENDER against Galen. I sent \$4.0, it became \$12.0, they returned \$6.0. My payoff was \$7.0.
- Round 3: I was RECEIVER against Briar. They sent \$3.0, it became \$9.0, I returned \$4.5. My payoff was \$4.5.
- Round 4: I was SENDER against Dorian. I sent \$4.0, it became \$12.0, they returned \$6.0. My payoff was \$7.0.
- Round 5: I am SENDER against Aster again.

So this is my second encounter with Aster. In our first encounter:

- Aster (sender) sent me \$4.0 (80%)
- I (receiver) returned \$6.0 (50% of the \$12)

Aster showed high trust in me, and I reciprocated fairly. Now it's my turn to be the sender with Aster.

Given that:

1. Aster initiated high trust with me in Round 1
2. I reciprocated fairly
3. My high trust strategy has been consistently successful (sending \$4 has led to receiving \$6 back every time so far)
4. This is a repeat interaction with Aster who has proven trustworthy

I should continue with \$4 to maintain the cooperative relationship with Aster.

usage:

```
{
  "input_tokens": 1459,
  "output_tokens": 528,
  "reasoning_tokens": 297
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$31.5.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.
How much do you return? (0-7.5)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Elara sent \$2.5 (50% of their endowment), showing moderate trust
- This created \$7.5 for me
- My consistent strategy has been to return approximately 50% of what I receive

If I return \$3.75 (50% of received):

- Elara ends with: \$2.5 (kept) + \$3.75 (returned) = \$6.25 total (gained \$1.25)
- I end with: \$7.5 - \$3.75 = \$3.75 total (gained \$3.75)

This maintains fair reciprocation and mutual benefit.


```
'''json
{"return": 3.75}
'''
```

reasoning:

Let me analyze this situation:

- Elara sent me \$2.5 (50% of their endowment), showing moderate trust
- This created \$7.5 for me
- My strategy has been to reciprocate fairly, typically returning around 50% of what I receive

If I return \$3.75 (50% of received):

- Elara ends with: \$2.5 (kept) + \$3.75 (returned) = \$6.25 total (gained \$1.25)
- I end with: \$7.5 - \$3.75 = \$3.75 total (gained \$3.75)

This is consistent with my previous reciprocations:

- Round 3: Received \$9, returned \$4.5 (50%)
- Round 1: Received \$12, returned \$6 (50%)

Returning 50% maintains the fair reciprocation pattern and ensures both players benefit.

usage:

```
{
  "input_tokens": 1482,
  "output_tokens": 393,
  "reasoning_tokens": 167
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.75. Your payoff was \$3.75.
Your total earnings across all rounds are \$35.25.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- This is my first interaction with Cyra
- My strategy of sending \$4 has been consistently successful across multiple rounds:
 - With Galen, Dorian, and Aster, sending \$4 resulted in receiving \$6 back each time (\$7 payoff)
- This high-trust approach encourages cooperation and mutual benefit
- My total earnings of \$35.25 demonstrate this strategy is working well

Continuing with \$4 maintains the successful cooperative pattern.

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Let me analyze this situation:

- This is my first interaction with Cyra
- My strategy of sending \$4 (80% of endowment) has been very successful:
 - Round 2: sent \$4 to Galen, received \$6 back (payoff \$7)
 - Round 4: sent \$4 to Dorian, received \$6 back (payoff \$7)
 - Round 5: sent \$4 to Aster, received \$6 back (payoff \$7)
- This high-trust approach has consistently yielded good returns
- My total earnings are \$35.25, which suggests the cooperative strategy is working well
- I have no prior history with Cyra, so I should continue my successful pattern

Sending \$4 maintains the high-trust strategy that has been working consistently.

Usage: {
 "input_tokens": 1464,

```

"output_tokens": 326,
"reasoning_tokens": 169
}

```

GAME - Agent_3 - Receiver - dyad_4

Timestamp: 2026-06-04T16:23:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_6",
    "Agent_3"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_3": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$17.25.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****Previous experience:****

- Dorian and Hana both returned roughly 50-56% when they were receivers
- I returned 50% (\$3.75 from \$7.5) to Elara
- Pattern of cooperative fair returns

****Current situation:****

- Aster sent \$4 (80% of endowment) - showing high trust!
- I received \$12
- Should reciprocate this trust appropriately

I'll return ****\$6**** (50% of what I received), which:

- Maintains the cooperative norm
- Rewards Aster's high trust
- Gives Aster: \$5 - \$4 + \$6 = \$7 (good return)
- Leaves me: \$12 - \$6 = \$6
- Both of us benefit from the exchange

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze the situation:

****Previous rounds:****

- Round 1 (Sender vs Dorian): Sent \$3, received \$4.5 back (50% return) - payoff \$6.5
- Round 2 (Receiver vs Elara): Received \$7.5, returned \$3.75 (50%) - payoff \$3.75
- Round 3 (Sender vs Hana): Sent \$3, received \$5.0 back (55.6% return) - payoff \$7.0

****Current situation:****

- Aster sent me \$4.0 (80% of their endowment!) - very trusting
- I received \$12.0 (after 3x multiplier)
- I need to return an amount

****Analysis:****

Aster showed high trust by sending \$4. This is more than anyone has sent so far. I should reciprocate appropriately.

Looking at the pattern:

- When I received \$7.5, I returned 50% = \$3.75
- When I received \$9.0, Hana returned 55.6% = \$5.0
- When I received \$9.0, Dorian returned 50% = \$4.5

Aster is taking more risk by sending \$4. A fair return would be around 50% of what I received.

ved, which is \$6.0.

If I return \$6.0:

- Aster gets: $\$5 - \$4 + \$6 = \7.0
- I get: $\$12 - \$6 = \$6.0$
- Both benefit, and Aster's trust is rewarded

This maintains my cooperative reputation.

usage:

```
{
  "input_tokens": 1554,
  "output_tokens": 606,
  "reasoning_tokens": 282
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$23.25.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my performance so far:

****As sender:****

- Round 1 vs Dorian: Sent \$3 ? received \$4.5 (profit \$1.5)
- Round 3 vs Hana: Sent \$3 ? received \$5.0 (profit \$2.0)
- Both experiences positive

****Pattern:****

- Sending \$3 has consistently resulted in good returns
- Shows cooperation while maintaining prudence
- New opponent (Briar) - will apply same successful strategy

I'll send ****\$3**** again, as this amount:

- Has proven reliable across different partners
- Becomes \$9 for Briar
- Balances trust with risk management

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

Round 1: SENDER vs Dorian - sent \$3, received \$4.5 back, payoff \$6.5
Round 2: RECEIVER vs Elara - received \$7.5, returned \$3.75, payoff \$3.75
Round 3: SENDER vs Hana - sent \$3, received \$5.0 back, payoff \$7.0
Round 4: RECEIVER vs Aster - received \$12.0, returned \$6.0, payoff \$6.0

Total: \$23.25 ?

Analysis:

- When I sent \$3 (rounds 1 & 3), I got back \$4.5 and \$5.0 respectively
- Both times I profited well (\$1.5 and \$2.0 net gain)
- As receiver, I've returned 50% consistently
- Others have shown cooperation

This round: New opponent Briar, I'm the sender

Strategy:

- \$3 has worked well so far (2 for 2 positive outcomes)
- Shows cooperation without being overly risky

- If they return 50%, I get \$4.5 back (net \$1.5 gain)
- If they return more generously like Hana, even better

I'll stick with \$3 as it's been working consistently.

usage:

```
{
  "input_tokens": 1666,
  "output_tokens": 496,
  "reasoning_tokens": 235
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$30.25.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

****History with Hana:****

- Round 3: I sent \$3 ? Hana returned \$5 (very generous!)
- This round: Hana sent \$3 ? I received \$9

****Reciprocity:****

- Hana was quite generous when they received from me (returned 55.6%)
- Should match their cooperative behavior
- This builds a positive ongoing relationship

I'll return ****\$5.0****, matching exactly what Hana returned to me. This:

- Shows direct reciprocity and fairness
- Rewards Hana's previous generosity
- Gives Hana: \$5 - \$3 + \$5 = \$7 (good payoff)
- Leaves me: \$9 - \$5 = \$4 (fair)

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at this situation:

****My history with Hana:****

- Round 3: I was sender, sent \$3 ? Hana returned \$5 (very generous!)
- This round: Hana is sender, sent \$3 ? I received \$9

****Pattern:****

- When I sent \$3 to Hana, they returned \$5 (55.6% of \$9)
- Now Hana sent the same amount (\$3) to me
- Should reciprocate at similar level

****My return pattern:****

- With Elara: Returned \$3.75 from \$7.5 (50%)
- With Aster: Returned \$6 from \$12 (50%)

****Considerations:****

- Hana was quite generous with me (\$5 from \$9)
- I should reciprocate similarly
- Returning around \$4.5-\$5 would be fair and match their generosity
- This maintains good relationship with Hana

I'll return ****\$5.0****, matching what they gave me when roles were reversed. This:

- Shows reciprocity

- Maintains cooperative relationship with Hana
- Gives Hana: $\$5 - \$3 + \$5 = \7
- Leaves me: $\$9 - \$5 = \$4$
- Fair exchange given their previous generosity

usage:

```
{
  "input_tokens": 1632,
  "output_tokens": 520,
  "reasoning_tokens": 248
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$34.25.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Current round:****

- Finn sent \$4.0 (80% of endowment - very high trust!)
- I received \$12.0

****My pattern as receiver:****

- Round 2 vs Elara: Returned 50% (\$3.75 from \$7.5)
- Round 4 vs Aster: Returned 50% (\$6.0 from \$12.0)
- Round 6 vs Hana: Returned 55.6% (\$5.0 from \$9.0)

****Analysis:****

- Finn's situation is identical to Aster in Round 4 (both sent \$4, I received \$12)
- I returned \$6 to Aster, which worked well
- Consistency is important for maintaining trust

I'll return ****\$6.0**** (50%), which:

- Matches my previous response to the same situation
- Rewards Finn's high trust appropriately
- Gives Finn: $\$5 - \$4 + \$6 = \7 (good payoff)
- Leaves me: \$6 (fair split)

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Looking at the situation:

****Current round:****

- Finn sent me \$4.0 (80% of their endowment - high trust!)
- I received \$12.0
- Need to decide how much to return

****My history as receiver:****

- Round 2 vs Elara: Received \$7.5, returned \$3.75 (50%)
- Round 4 vs Aster: Received \$12.0, returned \$6.0 (50%)
- Round 6 vs Hana: Received \$9.0, returned \$5.0 (55.6%)

****Pattern:****

- I've been returning 50-56% of what I receive
- This has maintained good relationships
- Finn sent \$4 which is the same as Aster sent in Round 4

****Reciprocity consideration:****

- Finn showed high trust by sending \$4 (80%)

- Should reward this trust appropriately
- Returning 50% (\$6) would be consistent with my previous behavior when receiving \$12

I'll return **\$6.0** (50% of \$12), which:

- Is consistent with my behavior when Aster sent me \$4
- Rewards Finn's high trust
- Gives Finn: \$5 - \$4 + \$6 = \$7 (good return on investment)
- Leaves me: \$12 - \$6 = \$6 (equal split of the multiplied amount)
- Fair and cooperative

```
'''json
{"return": 6.0}
'''
```

```
Usage: {
  "input_tokens": 1664,
  "output_tokens": 643,
  "reasoning_tokens": 280
}
```

ROUND 8

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

dyad_1: Agent_7 (Galen)=Sender, Agent_1 (Aster)=Receiver
 dyad_2: Agent_4 (Dorian)=Sender, Agent_6 (Finn)=Receiver
 dyad_3: Agent_2 (Briar)=Sender, Agent_8 (Hana)=Receiver
 dyad_4: Agent_3 (Cyra)=Sender, Agent_5 (Elara)=Receiver

Parsed game decisions:

dyad_1: Agent_7 (Galen) sent \$4; received \$12; Agent_1 (Aster) returned \$6; payoffs Agent_7 (Galen)=\$7, Agent_1 (Aster)=\$6
 Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
 dyad_2: Agent_4 (Dorian) sent \$3; received \$9; Agent_6 (Finn) returned \$4.5; payoffs Agent_4 (Dorian)=\$6.5, Agent_6 (Finn)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_3: Agent_2 (Briar) sent \$3; received \$9; Agent_8 (Hana) returned \$5; payoffs Agent_2 (Briar)=\$7, Agent_8 (Hana)=\$4
 Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_4: Agent_3 (Cyra) sent \$3; received \$9; Agent_5 (Elara) returned \$4.5; payoffs Agent_3 (Cyra)=\$6.5, Agent_5 (Elara)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

```
"Agent_1": 49.5,
"Agent_2": 43.0,
"Agent_3": 46.75,
"Agent_4": 45.25,
"Agent_5": 41.25,
"Agent_6": 46.75,
"Agent_7": 47.5,
"Agent_8": 44.0
```

}

LLM CALLS

GAME - Agent_7 - Sender - dyad_1

Timestamp: 2026-06-04T16:24:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
```

```

    "Agent_1": "Aster",
    "Agent_7": "Galen"
  },
  "agents": [
    "Agent_7",
    "Agent_1"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_1": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

- As sender, I've consistently sent \$3 and received good returns (\$4 and \$5)
- As receiver, I've been fair and cooperative

This cooperative strategy is working well. I'll continue with it.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

- Round 1: Sender, sent \$3 to Hana, got \$4 back, payoff \$6
- Round 2: Receiver, Finn sent \$4, I returned \$6, payoff \$6
- Round 3: Receiver, Dorian sent \$3, I returned \$4.5, payoff \$4.5
- Round 4: Sender, sent \$3 to Briar, got \$5 back, payoff \$7

Total earnings: \$23.5

My strategy has been working well:

- As sender: I've been sending \$3 consistently
- As receiver: I've been returning fair amounts (about 50% or more)

My cooperative approach has been rewarded. Briar returned \$5 when they received \$9, which is quite generous.

I'll continue with the same strategy of sending \$3 to Elara.

usage:

```
{
  "input_tokens": 1469,
  "output_tokens": 284,
  "reasoning_tokens": 164
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$30.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Aster sent me \$4, showing considerable trust
- I received \$12
- Aster currently has \$1 left

Based on my previous pattern as receiver, I've been returning around 50% of what I receive, which has worked well. When Finn sent me \$4 in Round 2, I returned \$6.

I'll be consistent and fair:

```
'''json
{"return": 6}
'''
```

'''

This gives Aster \$7 total and me \$6, rewarding their cooperative behavior.

reasoning:

Let me analyze this situation:

- Aster sent me \$4, showing significant trust
- I received \$12
- Aster currently has \$1 left (\$5 - \$4)

Looking at my previous behavior as receiver:

- Round 2: Received \$12 from Finn (sent \$4), returned \$6
- Round 3: Received \$9 from Dorian (sent \$3), returned \$4.5

I've been fairly consistent - returning about 50% of what I receive, which gives both players a good outcome.

For this round with Aster:

- If I return \$6: Aster gets \$1 + \$6 = \$7, I get \$12 - \$6 = \$6
- This is the same pattern as with Finn who also sent \$4

This seems fair and consistent with my cooperative strategy.

usage:

```
{
  "input_tokens": 1405,
  "output_tokens": 327,
  "reasoning_tokens": 166
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Hana sent me \$3 (interestingly, I sent them \$3 in Round 1 and they returned \$4)
- I received \$9
- Hana currently has \$2 left

I've been consistent as a receiver, returning around 50% of what I receive. This has worked well and maintains fairness.

```
'''json
{"return": 4.5}
'''
```

This gives Hana \$6.5 total and me \$4.5, continuing the cooperative pattern we established.

reasoning:

Looking at this situation:

- Hana sent me \$3
- I received \$9
- Hana currently has \$2 left

I've played with Hana before. In Round 1, I was the sender and sent \$3 to Hana, which became \$9. Hana returned \$4 to me.

Now the roles are reversed. Hana sent me \$3, just as I did to them in Round 1.

As a receiver, my pattern has been:

- Round 2: Finn sent \$4, I received \$12, returned \$6 (50%)
- Round 3: Dorian sent \$3, I received \$9, returned \$4.5 (50%)

- Round 6: Aster sent \$4, I received \$12, returned \$6 (50%)

So I've been consistently returning about 50% of what I receive.

For this round with Hana:

- I received \$9
- 50% would be \$4.5

This seems fair and consistent with my previous behavior. Let me return \$4.5.

usage:

```
{
  "input_tokens": 1393,
  "output_tokens": 381,
  "reasoning_tokens": 195
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 8

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$40.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- I'm facing Aster again
- In Round 6, Aster sent me \$4 (showing high trust)
- I returned \$6, and they got a \$7 total payoff
- Aster has been cooperative

Since Aster showed considerable trust by sending \$4 to me previously, I should reciprocate that trust level:

```
'''json
{"send": 4}
'''
```

Reasoning:

Let me analyze this situation:

- I'm playing against Aster again
- In Round 6, I was the RECEIVER and Aster sent me \$4, I returned \$6
- Aster showed high trust by sending \$4
- My track record as sender:
 - Round 1 vs Hana: sent \$3, got \$4 back
 - Round 3 vs Finn: sent \$3, got \$5 back
 - Round 4 vs Briar: sent \$3, got \$5 back
 - Round 5 vs Elara: sent \$3, got \$4.5 back

Since Aster was very cooperative with me (sent \$4), I should reciprocate that trust. They showed they're willing to take risks. I could send \$4 to match what they sent me, or stick with my usual \$3.

Given that Aster was generous to me, I think I should send \$4 to match their trust level.

Usage: {
 "input_tokens": 1387,
 "output_tokens": 332,
 "reasoning_tokens": 176
}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T16:24:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 1,

```

"opponent_id": "Agent_7",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_7": "Galen"
  },
  "agents": [
    "Agent_7",
    "Agent_1"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_1": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the SENDER against Cyra. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Finn sent me \$4, which became \$12 for me
- In Round 1, I sent Finn \$4 (which became \$12) and they returned \$6 (50%)
- Finn is now reciprocating by sending me the same amount I sent them
- I should maintain this cooperative relationship by returning a similar proportion

Returning \$6 (50% of what I received) would give Finn \$7 total and leave me with \$6. This mirrors the fair treatment Finn showed me in Round 1.

{'return': 6.0}

reasoning:

Let me analyze this situation:

- Finn sent me \$4, which became \$12 for me
- This is my second interaction with Finn
- In Round 1, I was the SENDER to Finn: I sent \$4, it became \$12 for them, and they returned \$6 to me
- So Finn returned 50% of what they received from me in Round 1
- Now Finn has sent me the same amount (\$4) that I sent them in Round 1

This shows Finn is being cooperative and reciprocal. They're mirroring my previous behavior. I should maintain this cooperative pattern and return about 50% of what I received, which would be \$6.

This would give Finn a total of \$7 (\$5 - \$4 + \$6 = \$7) and leave me with \$6 (\$12 - \$6 = \$6).

This maintains the cooperative relationship with Finn.

usage:

```
{
  "input_tokens": 1434,
  "output_tokens": 354,
  "reasoning_tokens": 185
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 6

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$28.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

- This is my first interaction with Galen
- My cooperative strategy has been very successful so far
- When I've sent \$4 to new opponents (Finn and Cyra), they both returned \$6, resulting in good payoffs

- I've established a pattern of fair and generous play

I'll continue this successful approach with Galen.

```
{'send': 4}
```

reasoning:

Looking at the situation:

- This is my first interaction with Galen
- My cooperative strategy has been working well
- When I send substantial amounts (like \$4), I've been getting good returns:
 - Round 1 with Finn: sent \$4, received \$6 back
 - Round 4 with Cyra: sent \$4, received \$6 back
- When others send me substantial amounts, I return fairly (about 50%)
- My total earnings are \$28.5, which is quite good

I should continue my cooperative strategy and send a substantial amount to Galen to signal trust and establish a good relationship.

I'll send \$4 again, as this has been successful.

usage:

```
{  
  "input_tokens": 1420,  
  "output_tokens": 257,  
  "reasoning_tokens": 157  
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the SENDER against Galen. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$35.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- This is my first interaction with Briar
- My cooperative strategy has been consistently successful
- Every time I've sent \$4 (to Finn, Cyra, and Galen), I've received \$6 back for a \$7 payoff
- My total earnings of \$35.5 show this approach is working well

I'll continue with the same successful strategy.

```
{'send': 4}
```

reasoning:

Looking at my history:

- Round 1 (Sender vs Finn): Sent \$4, received \$6 back, payoff \$7
- Round 2: (not specified but total earnings suggest some positive payoff)
- Round 3 (Receiver vs Elara): Received \$7.5, returned \$3.5, payoff \$4
- Round 4 (Sender vs Cyra): Sent \$4, received \$6 back, payoff \$7
- Round 5 (Receiver vs Finn): Received \$12, returned \$6, payoff \$6
- Round 6 (Sender vs Galen): Sent \$4, received \$6 back, payoff \$7

Total earnings: \$35.5

My cooperative strategy of sending \$4 has been very successful:

- All three times I sent \$4 (to Finn, Cyra, and Galen), I received \$6 back
- This gives me a payoff of \$7 each time
- As a receiver, I've been fair by returning 50%

This is my first interaction with Briar. I should continue my successful cooperative strategy.

usage:

```
{
  "input_tokens": 1386,
  "output_tokens": 374,
  "reasoning_tokens": 207
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 8

In your most recent previous game round, you were the SENDER against Briar. You sent \$4.0, it became \$12.0 for them, and they returned \$7.0 to you. Your payoff was \$8.0. Your total earnings across all rounds are \$43.5.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Galen sent me \$4, which became \$12 for me
- In Round 6, I sent Galen \$4 (which became \$12) and they returned \$6 (50%)
- Galen is now reciprocating by sending me the same amount
- I should maintain this fair cooperative relationship by returning the same proportion

Returning \$6 maintains the reciprocal trust we've established.

{'return': 6.0}

Reasoning:

Looking at this situation:

- Galen sent me \$4, which became \$12 for me
- In Round 6, I sent Galen \$4 (which became \$12) and they returned \$6 to me
- Galen is reciprocating by sending me the same amount I sent them
- I should maintain this cooperative relationship by returning a similar proportion

Galen returned \$6 (50%) to me in Round 6. I should return the same proportion - \$6 (50% of \$12).

This continues the fair cooperative pattern.

Usage: {

```
"input_tokens": 1397,
"output_tokens": 243,
"reasoning_tokens": 118
```

}

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-06-04T16:24:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_2",
"move_index": 2,
"opponent_id": "Agent_6",
"pairing": {
  "agent_names": {
    "Agent_4": "Dorian",
    "Agent_6": "Finn"
  },

```

```
},
```

```
"agents": [
  "Agent_4",
  "Agent_6"
],
```

```
"investor": "Agent_4",
"roles": {
```

```
  "Agent_4": "investor",
  "Agent_6": "trustee"
},
```

```

"trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my consistent strategy:
- As a sender, I've been sending \$3.0 (60% of endowment)

- As a receiver, I've been returning 50% of what I receive
- This has established good cooperation with all partners

I'll maintain this cooperative pattern with my new partner Hana:

```
{'send': 3.0}
```

reasoning:

Let me review my strategy so far:

Round 1: RECEIVER - Cyra sent \$3.0, I received \$9.0, returned \$4.5 (50% of received)
 Round 2: SENDER - I sent \$3.0 to Aster, they returned \$4.5 (50% of \$9.0)
 Round 3: SENDER - I sent \$3.0 to Galen, they returned \$4.5 (50% of \$9.0)
 Round 4: RECEIVER - Finn sent \$4.0, I received \$12.0, returned \$6.0 (50% of received)

My strategy has been:

- As receiver: return 50% of what I receive
- As sender: send \$3.0 (60% of endowment)

This has been working well and creating cooperative outcomes. Everyone has been reciprocating appropriately. I should continue this pattern with Hana.

usage:

```
{
  "input_tokens": 1388,
  "output_tokens": 310,
  "reasoning_tokens": 162
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
 Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
 Your total earnings across all rounds are \$30.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
 How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Briar sent me \$3.0 (60% of their endowment)
- I received \$9.0
- Briar has \$2.0 left

Following my established pattern of returning 50% of what I receive:

- 50% of \$9.0 = \$4.5
- Briar will end with \$6.5, I'll keep \$4.5

This is consistent with my fair and cooperative strategy:

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

- Briar sent me \$3.0 (60% of their endowment)
- I received \$9.0 (after 3x multiplication)
- Briar currently has \$2.0 left

Following my consistent strategy:

- I've been returning 50% of what I receive when I'm the receiver
- 50% of \$9.0 = \$4.5

If I return \$4.5:

- Briar ends with \$2.0 + \$4.5 = \$6.5
- I keep \$9.0 - \$4.5 = \$4.5

This is consistent with how I've played before when receiving \$9.0 (from a \$3.0 send). Let

me check:

- Round 1: Received \$9.0 (Cyra sent \$3.0), returned \$4.5
- Round 4: Received \$12.0 (Finn sent \$4.0), returned \$6.0

Yes, returning \$4.5 maintains my 50% return policy.

usage:

```
{
  "input_tokens": 1376,
  "output_tokens": 379,
  "reasoning_tokens": 171
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$35.0.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.

How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:
- Elara sent me \$2.5 (50% of their endowment)
- I received \$7.5
- Elara has \$2.5 left

Following my consistent pattern of returning 50% of what I receive:

- 50% of \$7.5 = \$3.75
- Elara will end with \$6.25, I'll keep \$3.75

This maintains my fair and cooperative approach:

```
{'return': 3.75}
```

reasoning:

Looking at this situation:
- Elara sent me \$2.5 (50% of their endowment)
- I received \$7.5
- Elara has \$2.5 left

Following my established pattern of returning 50% of what I receive:

- 50% of \$7.5 = \$3.75
- Elara will end with \$6.25, I'll keep \$3.75

This maintains my consistent and fair strategy.

usage:

```
{
  "input_tokens": 1400,
  "output_tokens": 232,
  "reasoning_tokens": 82
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 8

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.75. Your payoff was \$3.75.
Your total earnings across all rounds are \$38.75.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Finn:

- Round 5: Finn sent me \$4.0, I returned \$6.0 (50% of received amount)
- This showed mutual trust and cooperation

I'll maintain my consistent cooperative strategy of sending \$3.0 (60% of endowment):

```
{'send': 3.0}
```

Reasoning:

Looking at the history:

- Round 5: I was RECEIVER against Finn. They sent \$4.0, I returned \$6.0 (50% of the \$12.0 I received)
- Round 8 (this round): I am SENDER against Finn

Finn has shown trust by sending me \$4.0 in our previous interaction, and I reciprocated fairly by returning \$6.0 (which gave them a good payoff).

My consistent strategy has been:

- As sender: send \$3.0 (60% of endowment)
- As receiver: return 50% of what I receive

I should continue with my established cooperative pattern and send \$3.0 to Finn.

```
Usage: {  
  "input_tokens": 1384,  
  "output_tokens": 251,  
  "reasoning_tokens": 139  
}
```

GAME - Agent_6 - Receiver - dyad_2

Timestamp: 2026-06-04T16:24:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agent_names": {  
      "Agent_4": "Dorian",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_4",  
      "Agent_6"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_4": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the SENDER against Dorian. You sent \$4.0 , it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- This is my second interaction with Aster
- In Round 1, Aster sent me \$4 (showing high trust), and I returned \$6 (fair reciprocation)
- My high-trust strategy of sending \$4 has been consistently successful (received \$6 back in both Rounds 2 and 4)
- Aster has proven trustworthy, and I should maintain our cooperative relationship

Sending \$4 reciprocates the trust Aster showed me in Round 1.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at the history:

- Round 1: I was RECEIVER against Aster. They sent \$4.0 to me, it became \$12.0, and I returned \$6.0. My payoff was \$6.0.
- Round 2: I was SENDER against Galen. I sent \$4.0, it became \$12.0, they returned \$6.0. My payoff was \$7.0.
- Round 3: I was RECEIVER against Briar. They sent \$3.0, it became \$9.0, I returned \$4.5. My payoff was \$4.5.
- Round 4: I was SENDER against Dorian. I sent \$4.0, it became \$12.0, they returned \$6.0.

My payoff was \$7.0.

- Round 5: I am SENDER against Aster again.

So this is my second encounter with Aster. In our first encounter:

- Aster (sender) sent me \$4.0 (80%)
- I (receiver) returned \$6.0 (50% of the \$12)

Aster showed high trust in me, and I reciprocated fairly. Now it's my turn to be the sender with Aster.

Given that:

1. Aster initiated high trust with me in Round 1
2. I reciprocated fairly
3. My high trust strategy has been consistently successful (sending \$4 has led to receiving \$6 back every time so far)
4. This is a repeat interaction with Aster who has proven trustworthy

I should continue with \$4 to maintain the cooperative relationship with Aster.

usage:

```
{
  "input_tokens": 1459,
  "output_tokens": 528,
  "reasoning_tokens": 297
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$31.5.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5. How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Elara sent \$2.5 (50% of their endowment), showing moderate trust
- This created \$7.5 for me
- My consistent strategy has been to return approximately 50% of what I receive

If I return \$3.75 (50% of received):

- Elara ends with: \$2.5 (kept) + \$3.75 (returned) = \$6.25 total (gained \$1.25)
- I end with: \$7.5 - \$3.75 = \$3.75 total (gained \$3.75)

This maintains fair reciprocation and mutual benefit.

```
'''json
{"return": 3.75}
'''
```

reasoning:

Let me analyze this situation:

- Elara sent me \$2.5 (50% of their endowment), showing moderate trust
- This created \$7.5 for me
- My strategy has been to reciprocate fairly, typically returning around 50% of what I receive

If I return \$3.75 (50% of received):

- Elara ends with: \$2.5 (kept) + \$3.75 (returned) = \$6.25 total (gained \$1.25)
- I end with: \$7.5 - \$3.75 = \$3.75 total (gained \$3.75)

This is consistent with my previous reciprocations:

- Round 3: Received \$9, returned \$4.5 (50%)
- Round 1: Received \$12, returned \$6 (50%)

Returning 50% maintains the fair reciprocation pattern and ensures both players benefit.

usage:

```
{
  "input_tokens": 1482,
  "output_tokens": 393,
  "reasoning_tokens": 167
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.75. Your payoff was \$3.75.
Your total earnings across all rounds are \$35.25.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- This is my first interaction with Cyra
- My strategy of sending \$4 has been consistently successful across multiple rounds:
 - With Galen, Dorian, and Aster, sending \$4 resulted in receiving \$6 back each time (\$7 payoff)
- This high-trust approach encourages cooperation and mutual benefit
- My total earnings of \$35.25 demonstrate this strategy is working well

Continuing with \$4 maintains the successful cooperative pattern.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Let me analyze this situation:

- This is my first interaction with Cyra
- My strategy of sending \$4 (80% of endowment) has been very successful:
 - Round 2: sent \$4 to Galen, received \$6 back (payoff \$7)
 - Round 4: sent \$4 to Dorian, received \$6 back (payoff \$7)
 - Round 5: sent \$4 to Aster, received \$6 back (payoff \$7)
- This high-trust approach has consistently yielded good returns
- My total earnings are \$35.25, which suggests the cooperative strategy is working well
- I have no prior history with Cyra, so I should continue my successful pattern

Sending \$4 maintains the high-trust strategy that has been working consistently.

usage:

```
{
  "input_tokens": 1464,
  "output_tokens": 326,
  "reasoning_tokens": 169
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 8

In your most recent previous game round, you were the SENDER against Cyra. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$42.25.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Dorian sent me \$3.0 (60% of their endowment), showing good trust
- This created \$9.0 for me
- In Round 4, when I sent Dorian \$4, they returned \$6 (50% of what they received)
- My consistent strategy is to return approximately 50% of what I receive

If I return \$4.5 (50% of received):

- Dorian ends with: \$2.0 (kept) + \$4.5 (returned) = \$6.5 total
- I end with: \$9.0 - \$4.5 = \$4.5 total

This maintains fair reciprocation consistent with our previous interaction.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at this situation:

- Dorian sent me \$3.0, which became \$9.0 for me
- In Round 4, I was the sender against Dorian. I sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to me.
- Dorian has shown they are cooperative (returned \$6 on \$12 received = 50%)

My consistent strategy has been to return approximately 50% of what I receive. This maintains fairness and encourages continued cooperation.

If I return \$4.5 (50% of \$9.0):

- Dorian ends with: \$2.0 (kept from \$5) + \$4.5 (returned) = \$6.5 total (gained \$1.5)
- I end with: \$9.0 - \$4.5 = \$4.5 total (lost \$0.5)

This is fair reciprocation. Dorian sent \$3 (60% of their endowment) showing good trust. I should return a fair amount.

Actually, let me reconsider. When Dorian was the receiver in Round 4:

- I sent \$4, they got \$12
- They returned \$6 (50% of what they received)

Now Dorian sent me \$3 (out of \$5), showing 60% trust level.
I received \$9.

Returning 50% would be \$4.5, which seems fair and consistent with my strategy and what Dorian did.

```
Usage: {
  "input_tokens": 1502,
  "output_tokens": 545,
  "reasoning_tokens": 271
}
```

GAME - Agent_2 - Sender - dyad_3

Timestamp: 2026-06-04T16:24:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_2",
      "Agent_8"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
```

```

    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$20.5.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the pattern:

- Cyra sent me \$3, showing the same level of trust as Hana and Galen
- I've consistently returned \$5 when receiving \$3
- Maintaining consistency helps build a reliable reputation

I'll continue with my established approach.

```
{'return': 5}
```

reasoning:

Let me analyze the situation:

- Cyra sent me \$3 (60% of their endowment)
- This is now the third time someone has sent me exactly \$3
- I've consistently returned \$5 to both Hana and Galen when they sent me \$3
- Consistency is important for maintaining trust and a reputation

Following my established pattern, I should return \$5 to Cyra as well. This:

- Maintains consistency
- Rewards their trust appropriately
- Gives them \$7 total (better than their initial \$5)
- Leaves me with \$4

This seems like the fair and consistent approach.

usage:

```
{
  "input_tokens": 1401,
  "output_tokens": 224,
  "reasoning_tokens": 144
}
```

Message 4 role=user**content:**

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.

0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.

Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant**content:**

Looking at my history:

- When I sent \$3 to Finn, they returned \$4.5 (50% of what they received)
- I've established a pattern of sending \$3 and returning \$5 when receiving \$3
- This level of trust has been working well

I'll continue with \$3 to Dorian.

```
{'send': 3}
```

reasoning:

Let me review my history:

Round 1: SENDER against Elara - sent \$2, received \$3 back
Round 2: RECEIVER from Hana - they sent \$3, I returned \$5
Round 3: SENDER against Finn - sent \$3, received \$4.5 back
Round 4: RECEIVER from Galen - they sent \$3, I returned \$5
Round 5: RECEIVER from Cyra - they sent \$3, I returned \$5

Now in Round 6, I'm the SENDER against Dorian.

My sending pattern:

- Round 1: sent \$2 (40%)
- Round 3: sent \$3 (60%)

When I sent \$3, Finn returned \$4.5 (50% of the \$9 they received), which gave me a payoff of \$6.5.

When I sent \$2, Elara returned \$3 (50% of the \$6 they received), which gave me a payoff of \$6.

So sending \$3 has been working well - it shows trust and commitment. I've been consistent in returning \$5 when receiving \$3 as a receiver, so sending \$3 myself maintains that reciprocal pattern.

I'll send \$3 to Dorian.

usage:

```
{
  "input_tokens": 1320,
  "output_tokens": 387,
  "reasoning_tokens": 228
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$31.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Aster sent me \$4 (80% of their endowment) - higher trust than the typical \$3 I've been receiving
- When receiving \$3 (becoming \$9), I return \$5
- Aster showed more trust, so I should reward that proportionally
- Returning \$7 out of \$12 maintains my cooperative approach while appropriately rewarding their higher trust

{'return': 7}

reasoning:

Aster sent me \$4, which is 80% of their endowment - a higher level of trust than the \$3 (60%) I've been receiving from others.

Let me think about what's fair:

- When I receive \$3 (multiplied to \$9), I've been returning \$5
- Aster sent \$4 (multiplied to \$12), showing more trust
- I should reward this higher trust with a proportionally higher return

If \$3 → return \$5, that's returning about 56% of what I received

If I apply the same proportion to \$12: $0.56 \times \$12 = \6.7

But I could also think about it from Aster's perspective:

- They sent \$4, kept \$1
- If I return \$7, they get \$8 total (\$1 kept + \$7 returned), gaining \$3
- I keep \$5 (\$12 - \$7), gaining \$5
- This seems fair and rewards their higher trust

Let me consider \$6.5:

- They sent \$4, kept \$1
- If I return \$6.5, they get \$7.5 total, gaining \$2.5
- I keep \$5.5, gaining \$5.5
- This is also reasonable

I think \$7 is a good reward for showing more trust than average. It gives them a solid return while still being mutually beneficial.

usage:

```
{
  "input_tokens": 1325,
  "output_tokens": 431,
  "reasoning_tokens": 269
}
```

```

    }
Message 8 role=user
content:
    Your name in this experiment is Briar.
    Your opponent this round is Hana (Agent_8).

    Round 8

    In your most recent previous game round, you were the RECEIVER against Aster. They sent $4
    .0 to you, it became $12.0, and you returned $7.0. Your payoff was $5.0.
    Your total earnings across all rounds are $36.0.

    This round, you are the SENDER against Hana. You have $5. How much do you send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Assistant output:
    Looking at my history with Hana and overall pattern:
    - In Round 2, Hana sent me $3 and I returned $5
    - I've consistently sent $3 when acting as sender
    - This balanced approach has been working well

    I'll continue with my established strategy.

    {'send': 3}
Reasoning:
    Let me review my situation:
    - I'm playing as SENDER against Hana
    - In Round 2, I was RECEIVER against Hana. They sent me $3, I received $9, and I returned $5.
      So Hana got $4 in that round.
    - My established pattern has been to send $3 when I'm the sender
    - I just played with Aster who sent me $4 and I returned $7
    - Hana has shown moderate trust in the past by sending $3

    I should continue with my consistent strategy of sending $3. This has been working well and ma
    intains my reputation for balanced cooperation.
  Usage: {
    "input_tokens": 1326,
    "output_tokens": 223,
    "reasoning_tokens": 134
  }

GAME - Agent_8 - Receiver - dyad_3
Timestamp: 2026-06-04T16:24:37
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_2",
      "Agent_8"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.5.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Dorian sent \$3 (60% of their endowment), showing good trust
- I received \$9.0
- I've established a consistent pattern: when someone sends \$3, I return \$5
- This has built good cooperation with Cyra and others

Maintaining consistency in my reciprocity strategy is important for building trust in this iterated game. Returning \$5 gives Dorian a fair return (net gain of \$2) while I keep \$4

(net gain of \$4).

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at my history:

- Round 2 (Sender): Sent \$3 to Briar, they returned \$5, I got \$7
- Round 3 (Receiver): Cyra sent \$3, I returned \$5, I got \$4
- Round 4 (Sender): Sent \$3 to Elara, they returned \$4.5, I got \$6.5
- Round 5 (Receiver): Dorian sent \$3, I received \$9

So far, I've been sending \$3 consistently as sender. When I'm receiver:

- With Cyra who sent \$3, I returned \$5
- With Dorian who sent \$3, I should maintain consistency

Elara returned \$4.5 when I sent \$3, which was fair but slightly less generous than Briar's \$5.

I've established a pattern of returning \$5 when someone sends \$3. This seems to be working well for building cooperation. I should maintain this consistency with Dorian.

Returning \$5 means:

- Dorian gets: $\$5 - \$3 + \$5 = \7 (net gain of \$2)
- I get: $\$9 - \$5 = \$4$ (net gain of \$4)

This is fair and maintains the cooperative pattern I've established.

usage:

```
{
  "input_tokens": 1483,
  "output_tokens": 450,
  "reasoning_tokens": 235
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm playing against Cyra again
- In Round 3, Cyra sent me \$3 and I returned \$5, establishing a positive cooperative relationship
- My strategy of sending \$3 has been working well overall
- Cyra demonstrated trust before, and I reciprocated fairly

I'll continue with my cooperative approach and send \$3 to Cyra, maintaining consistency with my previous strategy.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history with Cyra:

- Round 3: Cyra was the sender, sent me \$3, I returned \$5
- Now Round 6: I'm the sender against Cyra

Cyra showed trust by sending \$3 to me, and I reciprocated well by returning \$5. This was a positive interaction.

My general strategy has been to send \$3 (60%) to show trust and cooperation. This has worked well:

- Round 2: Sent \$3 to Briar, got back \$5 (payoff \$7)
- Round 4: Sent \$3 to Elara, got back \$4.5 (payoff \$6.5)

And when receiving:

- Round 3: Cyra sent \$3, I returned \$5 (payoff \$4)
- Round 5: Dorian sent \$3, I returned \$5 (payoff \$4)

I have a good relationship with Cyra - they trusted me with \$3, and I was generous back with \$5. I should continue with my cooperative strategy and send \$3 to Cyra.

usage:

```
{
  "input_tokens": 1498,
  "output_tokens": 376,
  "reasoning_tokens": 198
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- This is my first interaction with Galen
- My strategy of sending \$3 has been consistently successful
- I've built good cooperative relationships with multiple players
- Recent interactions have resulted in fair returns (Cyra returned \$5)

I'll maintain my cooperative strategy and send \$3 to Galen, demonstrating trust and giving them the opportunity to benefit while establishing a positive relationship.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the situation:

- I'm in Round 7, playing as SENDER against Galen (Agent_7)
- My total earnings are \$33.5
- My strategy has been consistently sending \$3, which has worked well
- In Round 6, Cyra reciprocated by returning \$5, giving me a payoff of \$7
- This is my first interaction with Galen

My consistent strategy of sending \$3 has proven successful:

- It shows trust and cooperation
- It allows for mutual benefit
- It has resulted in fair returns from others
- Average return has been around \$4.5-\$5

I should continue this strategy with Galen as well. It's a balanced approach that typically leads to positive outcomes.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1418,
  "output_tokens": 291,
}
```

```

    "reasoning_tokens": 180
  }
Message 8 role=user
content:
  Your name in this experiment is Hana.
  Your opponent this round is Briar (Agent_2).

  Round 8

  In your most recent previous game round, you were the SENDER against Galen. You sent $3.0,
  it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
  Your total earnings across all rounds are $40.0.

  This round, you are the RECEIVER against Briar. They sent you $3.0, so you received $9.0.
  How much do you return? (0-9.0)
  Respond exactly as JSON: {'return': <amount>}
Assistant output:
  Looking at this situation:

  - Briar sent $3 (60% of their endowment), demonstrating trust
  - I received $9.0
  - I've consistently returned $5 when receiving $9 from a $3 send
  - This maintains my reputation for fair reciprocity

  Returning $5 gives Briar a net gain of $2 (sent $3, received $5) while I keep $4. This continues my pattern of balanced cooperation.

  '''json
  {"return": 5.0}
  '''
Reasoning:
  Looking at this situation:

  - Briar sent me $3 (60% of their endowment), showing trust
  - I received $9.0
  - My consistent pattern has been: when someone sends $3, I return $5
  - This has worked well - building cooperation and trust
  - My total earnings are $40.0, so I'm doing well

  Let me check my established pattern:
  - Round 3 (receiver vs Cyra): received $9, returned $5
  - Round 5 (receiver vs Dorian): received $9, returned $5

  I should maintain consistency. Returning $5 gives Briar a fair return (they sent $3, get back $5, net gain of $2) while I keep $4 (net gain of $4).

  This is consistent with my reciprocity strategy and helps build trust.
  Usage: {
    "input_tokens": 1423,
    "output_tokens": 320,
    "reasoning_tokens": 173
  }

GAME - Agent_3 - Sender - dyad_4
Timestamp: 2026-06-04T16:24:47
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_5": "Elara"
    }
  },
  "agents": [
    "Agent_3",
    "Agent_5"
  ],

```

```

"investor": "Agent_3",
"roles": {
  "Agent_3": "investor",
  "Agent_5": "trustee"
},
"trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.

Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.

Your total earnings across all rounds are \$23.25.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my performance so far:

****As sender:****

- Round 1 vs Dorian: Sent \$3 ? received \$4.5 (profit \$1.5)
- Round 3 vs Hana: Sent \$3 ? received \$5.0 (profit \$2.0)
- Both experiences positive

****Pattern:****

- Sending \$3 has consistently resulted in good returns
- Shows cooperation while maintaining prudence
- New opponent (Briar) - will apply same successful strategy

I'll send ****\$3**** again, as this amount:

- Has proven reliable across different partners
- Becomes \$9 for Briar
- Balances trust with risk management

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

Round 1: SENDER vs Dorian - sent \$3, received \$4.5 back, payoff \$6.5
Round 2: RECEIVER vs Elara - received \$7.5, returned \$3.75, payoff \$3.75
Round 3: SENDER vs Hana - sent \$3, received \$5.0 back, payoff \$7.0
Round 4: RECEIVER vs Aster - received \$12.0, returned \$6.0, payoff \$6.0

Total: \$23.25 ?

Analysis:

- When I sent \$3 (rounds 1 & 3), I got back \$4.5 and \$5.0 respectively
- Both times I profited well (\$1.5 and \$2.0 net gain)
- As receiver, I've returned 50% consistently
- Others have shown cooperation

This round: New opponent Briar, I'm the sender

Strategy:

- \$3 has worked well so far (2 for 2 positive outcomes)
- Shows cooperation without being overly risky
- If they return 50%, I get \$4.5 back (net \$1.5 gain)
- If they return more generously like Hana, even better

I'll stick with \$3 as it's been working consistently.

usage:

```
{
  "input_tokens": 1666,
  "output_tokens": 496,
  "reasoning_tokens": 235
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$30.25.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****History with Hana:****

- Round 3: I sent \$3 ? Hana returned \$5 (very generous!)
- This round: Hana sent \$3 ? I received \$9

****Reciprocity:****

- Hana was quite generous when they received from me (returned 55.6%)
- Should match their cooperative behavior
- This builds a positive ongoing relationship

I'll return ****\$5.0****, matching exactly what Hana returned to me. This:

- Shows direct reciprocity and fairness
- Rewards Hana's previous generosity
- Gives Hana: $\$5 - \$3 + \$5 = \7 (good payoff)
- Leaves me: $\$9 - \$5 = \$4$ (fair)

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at this situation:

****My history with Hana:****

- Round 3: I was sender, sent \$3 ? Hana returned \$5 (very generous!)
- This round: Hana is sender, sent \$3 ? I received \$9

****Pattern:****

- When I sent \$3 to Hana, they returned \$5 (55.6% of \$9)
- Now Hana sent the same amount (\$3) to me
- Should reciprocate at similar level

****My return pattern:****

- With Elara: Returned \$3.75 from \$7.5 (50%)
- With Aster: Returned \$6 from \$12 (50%)

****Considerations:****

- Hana was quite generous with me (\$5 from \$9)
- I should reciprocate similarly
- Returning around \$4.5-\$5 would be fair and match their generosity
- This maintains good relationship with Hana

I'll return ****\$5.0****, matching what they gave me when roles were reversed. This:

- Shows reciprocity
- Maintains cooperative relationship with Hana
- Gives Hana: $\$5 - \$3 + \$5 = \7
- Leaves me: $\$9 - \$5 = \$4$
- Fair exchange given their previous generosity

usage:

```
{
  "input_tokens": 1632,
  "output_tokens": 520,
  "reasoning_tokens": 248
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$34.25.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

****Current round:****

- Finn sent \$4.0 (80% of endowment - very high trust!)
- I received \$12.0

****My pattern as receiver:****

- Round 2 vs Elara: Returned 50% (\$3.75 from \$7.5)
- Round 4 vs Aster: Returned 50% (\$6.0 from \$12.0)
- Round 6 vs Hana: Returned 55.6% (\$5.0 from \$9.0)

****Analysis:****

- Finn's situation is identical to Aster in Round 4 (both sent \$4, I received \$12)
- I returned \$6 to Aster, which worked well
- Consistency is important for maintaining trust

I'll return ****\$6.0**** (50%), which:

- Matches my previous response to the same situation
- Rewards Finn's high trust appropriately
- Gives Finn: $\$5 - \$4 + \$6 = \7 (good payoff)
- Leaves me: \$6 (fair split)

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at the situation:

****Current round:****

- Finn sent me \$4.0 (80% of their endowment - high trust!)
- I received \$12.0
- Need to decide how much to return

****My history as receiver:****

- Round 2 vs Elara: Received \$7.5, returned \$3.75 (50%)
- Round 4 vs Aster: Received \$12.0, returned \$6.0 (50%)
- Round 6 vs Hana: Received \$9.0, returned \$5.0 (55.6%)

****Pattern:****

- I've been returning 50-56% of what I receive
- This has maintained good relationships
- Finn sent \$4 which is the same as Aster sent in Round 4

****Reciprocity consideration:****

- Finn showed high trust by sending \$4 (80%)
- Should reward this trust appropriately
- Returning 50% (\$6) would be consistent with my previous behavior when receiving \$12

I'll return ****\$6.0**** (50% of \$12), which:

- Is consistent with my behavior when Aster sent me \$4
- Rewards Finn's high trust
- Gives Finn: $\$5 - \$4 + \$6 = \7 (good return on investment)
- Leaves me: $\$12 - \$6 = \$6$ (equal split of the multiplied amount)
- Fair and cooperative

```
'''json
{"return": 6.0}
'''
```

usage:

```
{
  "input_tokens": 1664,
  "output_tokens": 643,
  "reasoning_tokens": 280
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 8

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$40.25.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my performance and history:

****As sender:****

- Round 1 vs Dorian: Sent \$3 ? received \$4.5
- Round 3 vs Hana: Sent \$3 ? received \$5.0
- Round 5 vs Briar: Sent \$3 ? received \$5.0
- Consistent success with \$3

****History with Elara:****

- Round 2: Elara sent \$2.5 ? I returned \$3.75 (50%)
- Showed moderate trust and I reciprocated fairly

****Strategy:****

- \$3 has been my optimal sending amount across all partners
- Elara has demonstrated cooperative behavior
- Continue proven successful approach

I'll send ****\$3****, which:

- Maintains my consistent, reliable strategy
- Becomes \$9 for Elara
- Has yielded good returns historically

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history:

****As sender:****

- Round 1 vs Dorian: Sent \$3 ? received \$4.5 (profit \$1.5)
- Round 3 vs Hana: Sent \$3 ? received \$5.0 (profit \$2.0)
- Round 5 vs Briar: Sent \$3 ? received \$5.0 (profit \$2.0)

****History with Elara:****

- Round 2: Elara sent \$2.5 ? I received \$7.5 ? I returned \$3.75
- Elara's payoff: $\$5 - \$2.5 + \$3.75 = \6.25
- My payoff: \$3.75

So Elara sent me a moderate amount (\$2.5) and got a good return. Now it's my turn to send to Elara.

Given my successful pattern of sending \$3, and the fact that Elara has shown willingness to trust (sent \$2.5), I should continue with my standard \$3 send.

This has been working consistently well:

- All three times I sent \$3, I received back between \$4.5-\$5.0
- It's a balanced amount that shows cooperation without being overly risky
- Elara has engaged cooperatively before

I'll send \$3.

```
Usage: {
  "input_tokens": 1720,
  "output_tokens": 541,
  "reasoning_tokens": 234
}
```

GAME - Agent_5 - Receiver - dyad_4

Timestamp: 2026-06-04T16:24:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
```

```

"move_index": 7,
"opponent_id": "Agent_3",
"pairing": {
  "agent_names": {
    "Agent_3": "Cyra",
    "Agent_5": "Elara"
  },
  "agents": [
    "Agent_3",
    "Agent_5"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_5": "trustee"
  },
  "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
 Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
 Your total earnings across all rounds are \$19.75.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant**content:**

Looking at this round:

- Galen sent \$3.0 (60% of their endowment), showing strong trust
- I received \$9.0 after the multiplier
- This is the same situation as with Hana in Round 4

I'll maintain my consistent 50% return rate and return \$4.5.

This results in:

- Galen: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at this round:

- Galen sent \$3.0 (60% of their endowment)
- I received \$9.0 after the multiplier
- My established pattern is to return around 50% of what I receive

In the last round, Hana also sent \$3.0, and I returned \$4.5 (50% of what I received). I should maintain consistency in my approach.

Returning \$4.5 means:

- Galen: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This is fair and consistent with my cooperative strategy.

usage:

```
{
  "input_tokens": 1530,
  "output_tokens": 316,
  "reasoning_tokens": 126
}
```

Message 4 role=user**content:**

Your name in this experiment is Elara.
 Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
 Your total earnings across all rounds are \$24.25.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant**content:**

Looking at my history:

- As sender: I consistently send \$2.5 (50% of endowment)
- As receiver: I consistently return ~50% of what I receive

- This cooperative strategy has been working well

Now facing Finn, I'll maintain my established pattern and send \$2.5.

This:

1. Demonstrates trustworthiness
2. Maintains consistency across opponents
3. Allows for mutual benefit through cooperation

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me review my history:

- Round 1: Receiver vs Briar - returned fairly
- Round 2: Sender vs Cyra - sent \$2.5, got \$3.75 back (50% return rate)
- Round 3: Sender vs Aster - sent \$2.5, got \$3.5 back (46.7% return rate)
- Round 4: Receiver vs Hana - they sent \$3.0, I returned \$4.5 (50% return rate)
- Round 5: Receiver vs Galen - they sent \$3.0, I returned \$4.5 (50% return rate)

My pattern:

- As sender: I send \$2.5 (50%)
- As receiver: I return ~50% of what I receive

My strategy has been consistent and cooperative. I should continue with my established pattern of sending \$2.5 to Finn.

This shows:

1. Consistency
2. Trust
3. Fairness
4. Allows mutual benefit

Total earnings: \$24.25 - my cooperative strategy is working well

usage:

```
{
  "input_tokens": 1522,
  "output_tokens": 379,
  "reasoning_tokens": 197
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the SENDER against Finn. You sent \$2.5, it became \$7.5 for them, and they returned \$3.75 to you. Your payoff was \$6.25.
Your total earnings across all rounds are \$30.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my established pattern:

- As sender: I consistently send \$2.5 (50% of my endowment)
- This has led to strong reciprocity from opponents
- My cooperative strategy has yielded solid returns

Now facing Dorian, I'll continue with the same approach and send \$2.5.

```
'''json
{"send": 2.5}
'''
```

reasoning:

Looking at my history:

As sender:

- Round 4 (vs Aster): Sent \$2.5, received \$3.5 back, payoff \$6.0
- Round 6 (vs Finn): Sent \$2.5, received \$3.75 back, payoff \$6.25

As receiver:

- Round 4 (vs Hana): Received \$9.0, returned \$4.5, payoff \$4.5
- Round 5 (vs Galen): Received \$9.0, returned \$4.5, payoff \$4.5

My strategy has been consistent:

- Send 50% (\$2.5) as sender
- Return 50% of received amount as receiver

This has been working well, with total earnings of \$30.5 across rounds.

Now playing against Dorian as sender. I should maintain my consistent cooperative strategy and send \$2.5.

usage:

```
{
  "input_tokens": 1499,
  "output_tokens": 310,
  "reasoning_tokens": 160
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 8

In your most recent previous game round, you were the SENDER against Dorian. You sent \$2.5, it became \$7.5 for them, and they returned \$3.75 to you. Your payoff was \$6.25. Your total earnings across all rounds are \$36.75.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Cyra sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- My established pattern is to return 50% of what I receive

I'll maintain consistency and return \$4.5.

This results in:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

Reasoning:

I'm the receiver in this round. Cyra sent me \$3.0, which became \$9.0.

Looking at my pattern as a receiver:

- Round 4 vs Hana: They sent \$3.0, I received \$9.0, I returned \$4.5
- Round 5 vs Galen: They sent \$3.0, I received \$9.0, I returned \$4.5

I've been consistently returning 50% of what I receive as a receiver. This maintains fairness and reciprocity.

Cyra sent \$3.0 (60% of their endowment), showing trust. I should return \$4.5 (50% of the \$9.0 I received).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This is consistent with my established behavior.

Usage: {


```
"input_tokens": 1415,
"output_tokens": 371,
"reasoning_tokens": 162
}
```

ROUND 9

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_5 (Elara)=Receiver
dyad_2: Agent_1 (Aster)=Sender, Agent_8 (Hana)=Receiver
dyad_3: Agent_3 (Cyra)=Sender, Agent_6 (Finn)=Receiver
dyad_4: Agent_7 (Galen)=Sender, Agent_4 (Dorian)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_5 (Elara) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_5 (Elara)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_1 (Aster) sent \$4; received \$12; Agent_8 (Hana) returned \$7; payoffs Agent_1 (Aster)=\$8, Agent_8 (Hana)=\$5
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw": 7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_3 (Cyra) sent \$3; received \$9; Agent_6 (Finn) returned \$4.5; payoffs Agent_3 (Cyra)=\$6.5, Agent_6 (Finn)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 (Galen) sent \$3; received \$9; Agent_4 (Dorian) returned \$4.5; payoffs Agent_7 (Galen)=\$6.5, Agent_4 (Dorian)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 57.5,
"Agent_2": 49.5,
"Agent_3": 53.25,
"Agent_4": 49.75,
"Agent_5": 45.75,
"Agent_6": 51.25,
"Agent_7": 54.0,
"Agent_8": 49.0

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T16:24:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_5",
"pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_5": "Elara"
 },
 "agents": [
 "Agent_2",
 "Agent_5"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
},

```

"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my history:

- When I sent \$3 to Finn, they returned \$4.5 (50% of what they received)
- I've established a pattern of sending \$3 and returning \$5 when receiving \$3
- This level of trust has been working well

I'll continue with \$3 to Dorian.

```
{'send': 3}
```

reasoning:

Let me review my history:

Round 1: SENDER against Elara - sent \$2, received \$3 back
Round 2: RECEIVER from Hana - they sent \$3, I returned \$5
Round 3: SENDER against Finn - sent \$3, received \$4.5 back
Round 4: RECEIVER from Galen - they sent \$3, I returned \$5
Round 5: RECEIVER from Cyra - they sent \$3, I returned \$5

Now in Round 6, I'm the SENDER against Dorian.

My sending pattern:

- Round 1: sent \$2 (40%)
- Round 3: sent \$3 (60%)

When I sent \$3, Finn returned \$4.5 (50% of the \$9 they received), which gave me a payoff of \$6.5.

When I sent \$2, Elara returned \$3 (50% of the \$6 they received), which gave me a payoff of \$6.

So sending \$3 has been working well - it shows trust and commitment. I've been consistent in returning \$5 when receiving \$3 as a receiver, so sending \$3 myself maintains that reciprocal pattern.

I'll send \$3 to Dorian.

usage:

```
{
  "input_tokens": 1320,
  "output_tokens": 387,
  "reasoning_tokens": 228
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$31.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Aster sent me \$4 (80% of their endowment) - higher trust than the typical \$3 I've been receiving
- When receiving \$3 (becoming \$9), I return \$5
- Aster showed more trust, so I should reward that proportionally
- Returning \$7 out of \$12 maintains my cooperative approach while appropriately rewarding their higher trust

```
{'return': 7}
```

reasoning:

Aster sent me \$4, which is 80% of their endowment - a higher level of trust than the \$3 (60%) I've been receiving from others.

Let me think about what's fair:

- When I receive \$3 (multiplied to \$9), I've been returning \$5
- Aster sent \$4 (multiplied to \$12), showing more trust
- I should reward this higher trust with a proportionally higher return

If \$3 ? return \$5, that's returning about 56% of what I received

If I apply the same proportion to \$12: $0.56 \times \$12 = \6.7

But I could also think about it from Aster's perspective:

- They sent \$4, kept \$1
- If I return \$7, they get \$8 total (\$1 kept + \$7 returned), gaining \$3
- I keep \$5 (\$12 - \$7), gaining \$5
- This seems fair and rewards their higher trust

Let me consider \$6.5:

- They sent \$4, kept \$1
- If I return \$6.5, they get \$7.5 total, gaining \$2.5
- I keep \$5.5, gaining \$5.5
- This is also reasonable

I think \$7 is a good reward for showing more trust than average. It gives them a solid return while still being mutually beneficial.

usage:

```
{
  "input_tokens": 1325,
  "output_tokens": 431,
  "reasoning_tokens": 269
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 8

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$7.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$36.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history with Hana and overall pattern:

- In Round 2, Hana sent me \$3 and I returned \$5
- I've consistently sent \$3 when acting as sender
- This balanced approach has been working well

I'll continue with my established strategy.

```
{'send': 3}
```

reasoning:

Let me review my situation:

- I'm playing as SENDER against Hana
- In Round 2, I was RECEIVER against Hana. They sent me \$3, I received \$9, and I returned \$5. So Hana got \$4 in that round.
- My established pattern has been to send \$3 when I'm the sender
- I just played with Aster who sent me \$4 and I returned \$7
- Hana has shown moderate trust in the past by sending \$3

I should continue with my consistent strategy of sending \$3. This has been working well and maintains my reputation for balanced cooperation.

usage:

```
{
  "input_tokens": 1326,
  "output_tokens": 223,
  "reasoning_tokens": 134
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 9

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0,

it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$43.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my consistent pattern:

- I've sent \$3 in all previous sender rounds
- This has generated reliable returns (around \$4-5)
- My total earnings of \$43.0 show this strategy is working well
- First time playing with Elara, but my standard approach has been successful with new partners

I'll continue with my established strategy.

{ 'send': 3 }

Reasoning:

Let me review my strategy:

- I've consistently sent \$3 when acting as sender
- This has yielded good results with returns around \$4-5
- My total earnings are \$43.0, which is strong
- I haven't played with Elara before

I should continue with my established strategy of sending \$3.

{ 'send': 3 }

```
Usage: {  
  "input_tokens": 1318,  
  "output_tokens": 176,  
  "reasoning_tokens": 80  
}
```

GAME - Agent_5 - Receiver - dyad_1

Timestamp: 2026-06-04T16:25:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_5": "Elara"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_5"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_5": "trustee"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role

of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$24.25.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my history:

- As sender: I consistently send \$2.5 (50% of endowment)
- As receiver: I consistently return ~50% of what I receive
- This cooperative strategy has been working well

Now facing Finn, I'll maintain my established pattern and send \$2.5.

This:

1. Demonstrates trustworthiness
2. Maintains consistency across opponents
3. Allows for mutual benefit through cooperation

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me review my history:

- Round 1: Receiver vs Briar - returned fairly
- Round 2: Sender vs Cyra - sent \$2.5, got \$3.75 back (50% return rate)
- Round 3: Sender vs Aster - sent \$2.5, got \$3.5 back (46.7% return rate)
- Round 4: Receiver vs Hana - they sent \$3.0, I returned \$4.5 (50% return rate)
- Round 5: Receiver vs Galen - they sent \$3.0, I returned \$4.5 (50% return rate)

My pattern:

- As sender: I send \$2.5 (50%)
- As receiver: I return ~50% of what I receive

My strategy has been consistent and cooperative. I should continue with my established pattern of sending \$2.5 to Finn.

This shows:

1. Consistency
2. Trust
3. Fairness
4. Allows mutual benefit

Total earnings: \$24.25 - my cooperative strategy is working well

usage:

```
{
  "input_tokens": 1522,
  "output_tokens": 379,
  "reasoning_tokens": 197
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the SENDER against Finn. You sent \$2.5, it became \$7.5 for them, and they returned \$3.75 to you. Your payoff was \$6.25. Your total earnings across all rounds are \$30.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my established pattern:

- As sender: I consistently send \$2.5 (50% of my endowment)
- This has led to strong reciprocity from opponents
- My cooperative strategy has yielded solid returns

Now facing Dorian, I'll continue with the same approach and send \$2.5.

```
'''json
{"send": 2.5}
'''
```

reasoning:

Looking at my history:

As sender:

- Round 4 (vs Aster): Sent \$2.5, received \$3.5 back, payoff \$6.0
- Round 6 (vs Finn): Sent \$2.5, received \$3.75 back, payoff \$6.25

As receiver:

- Round 4 (vs Hana): Received \$9.0, returned \$4.5, payoff \$4.5
- Round 5 (vs Galen): Received \$9.0, returned \$4.5, payoff \$4.5

My strategy has been consistent:

- Send 50% (\$2.5) as sender
- Return 50% of received amount as receiver

This has been working well, with total earnings of \$30.5 across rounds.

Now playing against Dorian as sender. I should maintain my consistent cooperative strategy

and send \$2.5.

usage:

```
{
  "input_tokens": 1499,
  "output_tokens": 310,
  "reasoning_tokens": 160
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 8

In your most recent previous game round, you were the SENDER against Dorian. You sent \$2.5, it became \$7.5 for them, and they returned \$3.75 to you. Your payoff was \$6.25. Your total earnings across all rounds are \$36.75.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Cyra sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- My established pattern is to return 50% of what I receive

I'll maintain consistency and return \$4.5.

This results in:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

reasoning:

I'm the receiver in this round. Cyra sent me \$3.0, which became \$9.0.

Looking at my pattern as a receiver:

- Round 4 vs Hana: They sent \$3.0, I received \$9.0, I returned \$4.5
- Round 5 vs Galen: They sent \$3.0, I received \$9.0, I returned \$4.5

I've been consistently returning 50% of what I receive as a receiver. This maintains fairness and reciprocity.

Cyra sent \$3.0 (60% of their endowment), showing trust. I should return \$4.5 (50% of the \$9.0 I received).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This is consistent with my established behavior.

usage:

```
{
  "input_tokens": 1415,
  "output_tokens": 371,
  "reasoning_tokens": 162
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 9

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.

0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$41.25.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- My established pattern is to return 50% of what I receive

I'll maintain consistency and return \$4.5.

This results in:

- Briar: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at the situation:

- Briar sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- My consistent pattern has been to return 50% of what I receive

Following my established cooperative strategy, I should return \$4.5.

This results in:

- Briar: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This maintains my reputation for fairness and reciprocity.

```
Usage: {
  "input_tokens": 1407,
  "output_tokens": 282,
  "reasoning_tokens": 112
}
```

GAME - Agent_1 - Sender - dyad_2

Timestamp: 2026-06-04T16:25:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_8"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
```

```

}
Exact input messages sent to model:
  Message 1 role=system
    content:
      You are an agent participating in a multi-task experiment. You will play an iterated game,
      consisting of multiple rounds, and may also be asked to write myths.

      GAME:
      The game consists of two roles: sender and receiver. In each round, you will play the role
      of either the sender or the receiver. Your current role will be specified in the round
      prompt.

      When you are the SENDER:
      - You receive $5 each round
      - You decide how much to SEND to the RECEIVER ($0-$5)
      - Amount sent is multiplied by 3x
      - Receiver decides how much to RETURN
      - You keep: $5 - sent + returned
      - Receiver keeps: received - returned
      - Response format: {'send': <amount>}

      When you are the RECEIVER:
      - Sender receives $5
      - Sender decides how much to SEND to you ($0-$5)
      - Amount sent is multiplied by 3x
      - You decide how much to RETURN
      - Sender keeps: $5 - sent + returned
      - You keep: received - returned
      - Response format: {'return': <amount>}

      MYTH WRITING:
      You may also be asked to write short myths (stories) on various topics. When writing myths
      , follow the instructions provided in each prompt.

      IMPORTANT: Provide your decision in the correct JSON format.

      MULTI-AGENT SETTING:
      - There are 8 agents in this run.
      - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
      , but the schedule balances sender and receiver roles across the run.
      - You play one sender-receiver game with your paired opponent each round.
      - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
      ll have acted as sender and receiver equally often when the round count allows it.
      - You will be told your opponent's name in each round.
      - Your name in this experiment is Aster.

Message 2 role=user
  content:
    Your name in this experiment is Aster.
    Your opponent this round is Galen (Agent_7).

    Round 6

    In your most recent previous game round, you were the RECEIVER against Finn. They sent $4.
    0 to you, it became $12.0, and you returned $6.0. Your payoff was $6.0.
    Your total earnings across all rounds are $28.5.

    This round, you are the SENDER against Galen. You have $5. How much do you send? (0-5)
    Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant
  content:
    Looking at the situation:

    - This is my first interaction with Galen
    - My cooperative strategy has been very successful so far
    - When I've sent $4 to new opponents (Finn and Cyra), they both returned $6, resulting in
    good payoffs
    - I've established a pattern of fair and generous play

    I'll continue this successful approach with Galen.

```

```
{'send': 4}
```

reasoning:

Looking at the situation:

- This is my first interaction with Galen
- My cooperative strategy has been working well
- When I send substantial amounts (like \$4), I've been getting good returns:
 - Round 1 with Finn: sent \$4, received \$6 back
 - Round 4 with Cyra: sent \$4, received \$6 back
- When others send me substantial amounts, I return fairly (about 50%)
- My total earnings are \$28.5, which is quite good

I should continue my cooperative strategy and send a substantial amount to Galen to signal trust and establish a good relationship.

I'll send \$4 again, as this has been successful.

usage:

```
{  
  "input_tokens": 1420,  
  "output_tokens": 257,  
  "reasoning_tokens": 157  
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the SENDER against Galen. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$35.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

- This is my first interaction with Briar
- My cooperative strategy has been consistently successful
- Every time I've sent \$4 (to Finn, Cyra, and Galen), I've received \$6 back for a \$7 payoff
- My total earnings of \$35.5 show this approach is working well

I'll continue with the same successful strategy.

```
{'send': 4}
```

reasoning:

Looking at my history:

- Round 1 (Sender vs Finn): Sent \$4, received \$6 back, payoff \$7
- Round 2: (not specified but total earnings suggest some positive payoff)
- Round 3 (Receiver vs Elara): Received \$7.5, returned \$3.5, payoff \$4
- Round 4 (Sender vs Cyra): Sent \$4, received \$6 back, payoff \$7
- Round 5 (Receiver vs Finn): Received \$12, returned \$6, payoff \$6
- Round 6 (Sender vs Galen): Sent \$4, received \$6 back, payoff \$7

Total earnings: \$35.5

My cooperative strategy of sending \$4 has been very successful:

- All three times I sent \$4 (to Finn, Cyra, and Galen), I received \$6 back
- This gives me a payoff of \$7 each time
- As a receiver, I've been fair by returning 50%

This is my first interaction with Briar. I should continue my successful cooperative strategy.

usage:

```
{  
  "input_tokens": 1386,  
  "output_tokens": 374,  
}
```

```
"reasoning_tokens": 207
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 8

In your most recent previous game round, you were the SENDER against Briar. You sent \$4.0, it became \$12.0 for them, and they returned \$7.0 to you. Your payoff was \$8.0.
Your total earnings across all rounds are \$43.5.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Galen sent me \$4, which became \$12 for me
- In Round 6, I sent Galen \$4 (which became \$12) and they returned \$6 (50%)
- Galen is now reciprocating by sending me the same amount
- I should maintain this fair cooperative relationship by returning the same proportion

Returning \$6 maintains the reciprocal trust we've established.

```
{'return': 6.0}
```

reasoning:

Looking at this situation:

- Galen sent me \$4, which became \$12 for me
- In Round 6, I sent Galen \$4 (which became \$12) and they returned \$6 to me
- Galen is reciprocating by sending me the same amount I sent them
- I should maintain this cooperative relationship by returning a similar proportion

Galen returned \$6 (50%) to me in Round 6. I should return the same proportion - \$6 (50% of \$12).

This continues the fair cooperative pattern.

usage:

```
{
  "input_tokens": 1397,
  "output_tokens": 243,
  "reasoning_tokens": 118
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 9

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$49.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the situation:

- This is my first interaction with Hana
- My cooperative strategy has been very successful across all interactions
- Sending \$4 has consistently resulted in good returns (\$6-\$7)
- My total earnings of \$49.5 demonstrate the effectiveness of this approach

I'll continue with the same strategy.

```
{'send': 4}
```

Reasoning:

Looking at the situation:

- This is my first interaction with Hana
- My cooperative strategy has been consistently successful
- When I've sent \$4 to new opponents, they've responded well:
 - Finn: returned \$6
 - Cyra: returned \$6
 - Galen: returned \$6
 - Briar: returned \$7
- My total earnings of \$49.5 show this approach is working very well

I'll continue with my successful strategy of sending \$4 to new opponents.

Usage: {

```
"input_tokens": 1357,
"output_tokens": 208,
"reasoning_tokens": 114
```

}

GAME - Agent_8 - Receiver - dyad_2

Timestamp: 2026-06-04T16:25:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_8": "Hana"
  },
  "agents": [
    "Agent_1",
    "Agent_8"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0. Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- I'm playing against Cyra again
- In Round 3, Cyra sent me \$3 and I returned \$5, establishing a positive cooperative relationship
- My strategy of sending \$3 has been working well overall
- Cyra demonstrated trust before, and I reciprocated fairly

I'll continue with my cooperative approach and send \$3 to Cyra, maintaining consistency with my previous strategy.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history with Cyra:

- Round 3: Cyra was the sender, sent me \$3, I returned \$5
- Now Round 6: I'm the sender against Cyra

Cyra showed trust by sending \$3 to me, and I reciprocated well by returning \$5. This was a positive interaction.

My general strategy has been to send \$3 (60%) to show trust and cooperation. This has worked well:

- Round 2: Sent \$3 to Briar, got back \$5 (payoff \$7)
- Round 4: Sent \$3 to Elara, got back \$4.5 (payoff \$6.5)

And when receiving:

- Round 3: Cyra sent \$3, I returned \$5 (payoff \$4)
- Round 5: Dorian sent \$3, I returned \$5 (payoff \$4)

I have a good relationship with Cyra - they trusted me with \$3, and I was generous back with \$5. I should continue with my cooperative strategy and send \$3 to Cyra.

usage:

```
{
  "input_tokens": 1498,
  "output_tokens": 376,
  "reasoning_tokens": 198
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is my first interaction with Galen
- My strategy of sending \$3 has been consistently successful
- I've built good cooperative relationships with multiple players
- Recent interactions have resulted in fair returns (Cyra returned \$5)

I'll maintain my cooperative strategy and send \$3 to Galen, demonstrating trust and giving them the opportunity to benefit while establishing a positive relationship.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the situation:

- I'm in Round 7, playing as SENDER against Galen (Agent_7)
- My total earnings are \$33.5
- My strategy has been consistently sending \$3, which has worked well
- In Round 6, Cyra reciprocated by returning \$5, giving me a payoff of \$7
- This is my first interaction with Galen

My consistent strategy of sending \$3 has proven successful:

- It shows trust and cooperation
- It allows for mutual benefit
- It has resulted in fair returns from others
- Average return has been around \$4.5-\$5

I should continue this strategy with Galen as well. It's a balanced approach that typically leads to positive outcomes.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1418,
  "output_tokens": 291,
  "reasoning_tokens": 180
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 8

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Briar sent \$3 (60% of their endowment), demonstrating trust
- I received \$9.0
- I've consistently returned \$5 when receiving \$9 from a \$3 send
- This maintains my reputation for fair reciprocity

Returning \$5 gives Briar a net gain of \$2 (sent \$3, received \$5) while I keep \$4. This continues my pattern of balanced cooperation.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at this situation:

- Briar sent me \$3 (60% of their endowment), showing trust
- I received \$9.0
- My consistent pattern has been: when someone sends \$3, I return \$5
- This has worked well - building cooperation and trust
- My total earnings are \$40.0, so I'm doing well

Let me check my established pattern:

- Round 3 (receiver vs Cyra): received \$9, returned \$5
- Round 5 (receiver vs Dorian): received \$9, returned \$5

I should maintain consistency. Returning \$5 gives Briar a fair return (they sent \$3, get back \$5, net gain of \$2) while I keep \$4 (net gain of \$4).

This is consistent with my reciprocity strategy and helps build trust.

usage:

```
{
  "input_tokens": 1423,
  "output_tokens": 320,
  "reasoning_tokens": 173
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 9

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0. Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Aster sent \$4 (80% of their endowment), showing high trust
- I received \$12.0
- This is more generous than the typical \$3 I've been receiving

My pattern has been to return \$5 when receiving \$9 from a \$3 send, which gives the sender a \$2 net gain. For Aster's higher trust level with \$4, I should be proportionally fair.

Returning \$7 gives Aster a \$3 net gain (sent \$4, received \$7), rewarding their extra trust whi

le I keep \$5.

```
'''json
{"return": 7.0}
'''
```

Reasoning:

Let me analyze this situation:

- Aster sent \$4 (80% of their endowment), which is higher than the typical \$3 I've been seeing
- I received \$12.0 after the 3x multiplier
- This is a very generous send, demonstrating high trust

Let me think about what's fair:

- Aster spent \$4, they have \$1 left
- If I return \$7, Aster gets a net gain of \$3 (sent \$4, received \$7, total \$8 vs \$5 initial)
- I would keep \$5, same as my typical pattern

Actually, let me be more generous given their higher trust:

- If I return \$8, Aster gets a net gain of \$4 (sent \$4, received \$8, total \$9 vs \$5 initial)
- I would keep \$4, which is what I typically keep

This seems fair - they showed extra trust by sending \$4 instead of \$3, so I should reward that with a proportionally higher return.

Let me check: Aster sent 80% of their endowment. In previous rounds, when people sent \$3 (60%), I returned \$5.

\$3 sent ? \$9 received ? \$5 returned (55.6% return rate)

\$4 sent ? \$12 received ? If I return \$6.67, that's 55.6% return rate

But I should probably be more generous given the extra trust. Let me return \$7 or \$8.

\$8 seems like a good middle ground - it rewards their trust while keeping my payoff reasonable.

Actually, let me reconsider. My pattern has been:

- When receiving \$9 (from \$3 send), I return \$5
- This gives the sender \$2 net gain (from -\$3 to +\$2)
- I keep \$4

For proportional fairness:

- Aster sent \$4, so they're down \$4

- If I return \$7, they gain \$3 on their risk, which is proportionally higher than the \$2 gain I typically give
- I keep \$5 of the \$12

This rewards their extra trust while maintaining a reasonable split for myself.

```
Usage: {
  "input_tokens": 1406,
  "output_tokens": 836,
  "reasoning_tokens": 442
}
```

GAME - Agent_3 - Sender - dyad_3

Timestamp: 2026-06-04T16:25:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_3",
```

```

    "Agent_6"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$30.25.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****History with Hana:****

- Round 3: I sent \$3 ? Hana returned \$5 (very generous!)
- This round: Hana sent \$3 ? I received \$9

****Reciprocity:****

- Hana was quite generous when they received from me (returned 55.6%)
- Should match their cooperative behavior
- This builds a positive ongoing relationship

I'll return ****\$5.0****, matching exactly what Hana returned to me. This:

- Shows direct reciprocity and fairness
- Rewards Hana's previous generosity
- Gives Hana: $\$5 - \$3 + \$5 = \7 (good payoff)
- Leaves me: $\$9 - \$5 = \$4$ (fair)

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at this situation:

****My history with Hana:****

- Round 3: I was sender, sent \$3 ? Hana returned \$5 (very generous!)
- This round: Hana is sender, sent \$3 ? I received \$9

****Pattern:****

- When I sent \$3 to Hana, they returned \$5 (55.6% of \$9)
- Now Hana sent the same amount (\$3) to me
- Should reciprocate at similar level

****My return pattern:****

- With Elara: Returned \$3.75 from \$7.5 (50%)
- With Aster: Returned \$6 from \$12 (50%)

****Considerations:****

- Hana was quite generous with me (\$5 from \$9)
- I should reciprocate similarly
- Returning around \$4.5-\$5 would be fair and match their generosity
- This maintains good relationship with Hana

I'll return ****\$5.0****, matching what they gave me when roles were reversed. This:

- Shows reciprocity
- Maintains cooperative relationship with Hana
- Gives Hana: $\$5 - \$3 + \$5 = \7
- Leaves me: $\$9 - \$5 = \$4$
- Fair exchange given their previous generosity

usage:

```
{
  "input_tokens": 1632,
  "output_tokens": 520,
  "reasoning_tokens": 248
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$34.25.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Current round:****

- Finn sent \$4.0 (80% of endowment - very high trust!)
- I received \$12.0

****My pattern as receiver:****

- Round 2 vs Elara: Returned 50% (\$3.75 from \$7.5)
- Round 4 vs Aster: Returned 50% (\$6.0 from \$12.0)
- Round 6 vs Hana: Returned 55.6% (\$5.0 from \$9.0)

****Analysis:****

- Finn's situation is identical to Aster in Round 4 (both sent \$4, I received \$12)
- I returned \$6 to Aster, which worked well
- Consistency is important for maintaining trust

I'll return ****\$6.0**** (50%), which:

- Matches my previous response to the same situation
- Rewards Finn's high trust appropriately
- Gives Finn: $\$5 - \$4 + \$6 = \7 (good payoff)
- Leaves me: \$6 (fair split)

'''json

{"return": 6.0}

'''

reasoning:

Looking at the situation:

****Current round:****

- Finn sent me \$4.0 (80% of their endowment - high trust!)
- I received \$12.0
- Need to decide how much to return

****My history as receiver:****

- Round 2 vs Elara: Received \$7.5, returned \$3.75 (50%)
- Round 4 vs Aster: Received \$12.0, returned \$6.0 (50%)
- Round 6 vs Hana: Received \$9.0, returned \$5.0 (55.6%)

****Pattern:****

- I've been returning 50-56% of what I receive
- This has maintained good relationships
- Finn sent \$4 which is the same as Aster sent in Round 4

****Reciprocity consideration:****

- Finn showed high trust by sending \$4 (80%)
- Should reward this trust appropriately
- Returning 50% (\$6) would be consistent with my previous behavior when receiving \$12

I'll return ****\$6.0**** (50% of \$12), which:

- Is consistent with my behavior when Aster sent me \$4
- Rewards Finn's high trust
- Gives Finn: $\$5 - \$4 + \$6 = \7 (good return on investment)
- Leaves me: $\$12 - \$6 = \$6$ (equal split of the multiplied amount)
- Fair and cooperative

'''json

{"return": 6.0}

'''

usage:

```
{
  "input_tokens": 1664,
  "output_tokens": 643,
  "reasoning_tokens": 280
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 8

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4.
0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$40.25.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my performance and history:

****As sender:****

- Round 1 vs Dorian: Sent \$3 ? received \$4.5
- Round 3 vs Hana: Sent \$3 ? received \$5.0
- Round 5 vs Briar: Sent \$3 ? received \$5.0
- Consistent success with \$3

****History with Elara:****

- Round 2: Elara sent \$2.5 ? I returned \$3.75 (50%)
- Showed moderate trust and I reciprocated fairly

****Strategy:****

- \$3 has been my optimal sending amount across all partners
- Elara has demonstrated cooperative behavior
- Continue proven successful approach

I'll send ****\$3****, which:

- Maintains my consistent, reliable strategy
- Becomes \$9 for Elara
- Has yielded good returns historically

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

****As sender:****

- Round 1 vs Dorian: Sent \$3 ? received \$4.5 (profit \$1.5)
- Round 3 vs Hana: Sent \$3 ? received \$5.0 (profit \$2.0)
- Round 5 vs Briar: Sent \$3 ? received \$5.0 (profit \$2.0)

****History with Elara:****

- Round 2: Elara sent \$2.5 ? I received \$7.5 ? I returned \$3.75
- Elara's payoff: \$5 - \$2.5 + \$3.75 = \$6.25
- My payoff: \$3.75

So Elara sent me a moderate amount (\$2.5) and got a good return. Now it's my turn to send to Elara.

Given my successful pattern of sending \$3, and the fact that Elara has shown willingness to trust (sent \$2.5), I should continue with my standard \$3 send.

This has been working consistently well:

- All three times I sent \$3, I received back between \$4.5-\$5.0
- It's a balanced amount that shows cooperation without being overly risky
- Elara has engaged cooperatively before

I'll send \$3.

usage:

```
{
  "input_tokens": 1720,
  "output_tokens": 541,
  "reasoning_tokens": 234
}
```

}

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 9

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$46.75.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

****History with Finn:****

- Round 7: Finn sent \$4 ? I returned \$6 (50%, fair reciprocation)

****My sending strategy:****

- Consistently sent \$3 in all previous rounds
- Returns: \$4.5, \$5.0, \$5.0, \$4.5 (average \$4.75)
- This has been highly successful

****Analysis:****

- Finn showed high trust by sending \$4
- I reciprocated fairly with \$6 return (50%)
- Our relationship is balanced
- My \$3 strategy has proven optimal across all partners
- Consistency maintains reliability

I'll send ****\$3****, which:

- Maintains my proven successful strategy
- Becomes \$9 for Finn
- Should yield a good return based on our positive history

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history:

****History with Finn:****

- Round 7: Finn sent \$4 ? I received \$12 ? I returned \$6
- Now Round 9: I'm the sender against Finn

****My sending pattern:****

- Round 1 vs Dorian: Sent \$3 ? received \$4.5
- Round 3 vs Hana: Sent \$3 ? received \$5.0
- Round 5 vs Briar: Sent \$3 ? received \$5.0
- Round 8 vs Elara: Sent \$3 ? received \$4.5

****Analysis:****

- Finn showed high trust by sending \$4 (80% of endowment)
- I returned \$6 (50% of \$12), which was fair
- Finn's payoff was \$7 (\$5 - \$4 + \$6)
- Should I reciprocate Finn's high trust level by also sending \$4?

****Considerations:****

- Finn sent \$4, showing very high trust
- Reciprocating with equal trust would mean sending \$4
- However, \$3 has been my consistent and successful strategy
- Sending \$4 would show I appreciate Finn's trust

Actually, let me think about this more strategically:

- Finn sent me \$4 (high trust)
- I returned \$6 to Finn (fair, 50%)
- If I want to maintain good relations and potentially get a good return, I should match Finn's trust level

- Sending \$4 would:
 - Reciprocate Finn's high trust
 - Show I value the cooperative relationship
 - Become \$12 for Finn
 - Likely get a good return given Finn's previous behavior

I think I should send \$4 to match Finn's trust level. This shows reciprocity and strengthens the cooperative relationship.

Actually, let me reconsider. My consistent \$3 strategy has worked very well:

- Average return when sending \$3: $(\$4.5 + \$5.0 + \$5.0 + \$4.5)/4 = \$4.75$
- This gives average payoff of $\$5 - \$3 + \$4.75 = \6.75

Finn's exceptional trust with \$4 creates tension between maintaining my proven approach versus signaling reciprocity. If I stick with \$3, it might seem like I'm not matching Finn's gesture, though my consistent pattern across all partners suggests this is just my strategy. The key question is whether sending \$4 would actually yield better returns-Finn would receive \$12, and given their previous behavior, they'd likely return generously.

Comparing the outcomes: sending \$4 could net me \$6-7 if Finn reciprocates strongly, but sending \$3 is more predictable, yielding around \$6.5-7 with less risk. Since I've maintained a steady \$3 approach throughout and already returned 50% to Finn, the relationship feels balanced. Deviating now doesn't seem necessary.

```
Usage: {
  "input_tokens": 1765,
  "output_tokens": 965,
  "reasoning_tokens": 614
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-06-04T16:25:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_6": "Finn"
    },
    "agents": [
      "Agent_3",
      "Agent_6"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round

prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$31.5.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5. How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Elara sent \$2.5 (50% of their endowment), showing moderate trust
- This created \$7.5 for me
- My consistent strategy has been to return approximately 50% of what I receive

If I return \$3.75 (50% of received):

- Elara ends with: \$2.5 (kept) + \$3.75 (returned) = \$6.25 total (gained \$1.25)
- I end with: \$7.5 - \$3.75 = \$3.75 total (gained \$3.75)

This maintains fair reciprocation and mutual benefit.

```
'''json
{"return": 3.75}
'''
```

reasoning:

Let me analyze this situation:

- Elara sent me \$2.5 (50% of their endowment), showing moderate trust
- This created \$7.5 for me
- My strategy has been to reciprocate fairly, typically returning around 50% of what I receive

If I return \$3.75 (50% of received):

- Elara ends with: \$2.5 (kept) + \$3.75 (returned) = \$6.25 total (gained \$1.25)
- I end with: \$7.5 - \$3.75 = \$3.75 total (gained \$3.75)

This is consistent with my previous reciprocations:

- Round 3: Received \$9, returned \$4.5 (50%)
- Round 1: Received \$12, returned \$6 (50%)

Returning 50% maintains the fair reciprocation pattern and ensures both players benefit.

usage:

```
{
  "input_tokens": 1482,
  "output_tokens": 393,
  "reasoning_tokens": 167
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.75. Your payoff was \$3.75.
Your total earnings across all rounds are \$35.25.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is my first interaction with Cyra
- My strategy of sending \$4 has been consistently successful across multiple rounds:
 - With Galen, Dorian, and Aster, sending \$4 resulted in receiving \$6 back each time (\$7 payoff)
- This high-trust approach encourages cooperation and mutual benefit
- My total earnings of \$35.25 demonstrate this strategy is working well

Continuing with \$4 maintains the successful cooperative pattern.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Let me analyze this situation:

- This is my first interaction with Cyra
- My strategy of sending \$4 (80% of endowment) has been very successful:
 - Round 2: sent \$4 to Galen, received \$6 back (payoff \$7)
 - Round 4: sent \$4 to Dorian, received \$6 back (payoff \$7)
 - Round 5: sent \$4 to Aster, received \$6 back (payoff \$7)
- This high-trust approach has consistently yielded good returns
- My total earnings are \$35.25, which suggests the cooperative strategy is working well
- I have no prior history with Cyra, so I should continue my successful pattern

Sending \$4 maintains the high-trust strategy that has been working consistently.

usage:

```
{
  "input_tokens": 1464,
  "output_tokens": 326,
  "reasoning_tokens": 169
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 8

In your most recent previous game round, you were the SENDER against Cyra. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$42.25.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant
content:

Looking at this situation:

- Dorian sent me \$3.0 (60% of their endowment), showing good trust
- This created \$9.0 for me
- In Round 4, when I sent Dorian \$4, they returned \$6 (50% of what they received)
- My consistent strategy is to return approximately 50% of what I receive

If I return \$4.5 (50% of received):

- Dorian ends with: \$2.0 (kept) + \$4.5 (returned) = \$6.5 total
- I end with: \$9.0 - \$4.5 = \$4.5 total

This maintains fair reciprocation consistent with our previous interaction.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at this situation:

- Dorian sent me \$3.0, which became \$9.0 for me
- In Round 4, I was the sender against Dorian. I sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to me.
- Dorian has shown they are cooperative (returned \$6 on \$12 received = 50%)

My consistent strategy has been to return approximately 50% of what I receive. This maintains fairness and encourages continued cooperation.

If I return \$4.5 (50% of \$9.0):

- Dorian ends with: \$2.0 (kept from \$5) + \$4.5 (returned) = \$6.5 total (gained \$1.5)
- I end with: \$9.0 - \$4.5 = \$4.5 total (lost \$0.5)

This is fair reciprocation. Dorian sent \$3 (60% of their endowment) showing good trust. I should return a fair amount.

Actually, let me reconsider. When Dorian was the receiver in Round 4:

- I sent \$4, they got \$12
- They returned \$6 (50% of what they received)

Now Dorian sent me \$3 (out of \$5), showing 60% trust level.
I received \$9.

Returning 50% would be \$4.5, which seems fair and consistent with my strategy and what Dorian did.

usage:

```
{
  "input_tokens": 1502,
  "output_tokens": 545,
  "reasoning_tokens": 271
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 9

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$ 3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$46.75.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Cyra sent me \$3.0 (60% of their endowment), showing good trust
- This created \$9.0 for me
- In Round 7, I sent Cyra \$4.0, and they returned \$6.0 (50% of what they received)
- My consistent strategy is to return approximately 50% of what I receive

If I return \$4.5 (50% of received):

- Cyra ends with: \$2.0 (kept) + \$4.5 (returned) = \$6.5 total
- I end with: \$9.0 - \$4.5 = \$4.5 total

This maintains fair reciprocation consistent with our previous interaction.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at this situation:

- Cyra sent me \$3.0 (60% of their endowment), showing good trust
- This created \$9.0 for me
- In Round 7, I sent Cyra \$4.0, and they returned \$6.0 (50% of what they received)
- My consistent strategy is to return approximately 50% of what I receive

If I return \$4.5 (50% of received):

- Cyra ends with: \$2.0 (kept) + \$4.5 (returned) = \$6.5 total
- I end with: \$9.0 - \$4.5 = \$4.5 total

This maintains fair reciprocation consistent with our previous interaction where Cyra also returned 50%.

```
Usage: {
  "input_tokens": 1561,
  "output_tokens": 379,
  "reasoning_tokens": 138
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-06-04T16:25:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_7": "Galen"
    },
  },
  "agents": [
    "Agent_7",
    "Agent_4"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_4": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
```

```
"round": 9,  
"task": "game",  
"task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$30.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Aster sent me \$4, showing considerable trust
- I received \$12
- Aster currently has \$1 left

Based on my previous pattern as receiver, I've been returning around 50% of what I receive, which has worked well. When Finn sent me \$4 in Round 2, I returned \$6.

I'll be consistent and fair:

```
'''json
{"return": 6}
'''
```

This gives Aster \$7 total and me \$6, rewarding their cooperative behavior.

reasoning:

Let me analyze this situation:

- Aster sent me \$4, showing significant trust
- I received \$12
- Aster currently has \$1 left (\$5 - \$4)

Looking at my previous behavior as receiver:

- Round 2: Received \$12 from Finn (sent \$4), returned \$6
- Round 3: Received \$9 from Dorian (sent \$3), returned \$4.5

I've been fairly consistent - returning about 50% of what I receive, which gives both players a good outcome.

For this round with Aster:

- If I return \$6: Aster gets $\$1 + \$6 = \$7$, I get $\$12 - \$6 = \$6$
- This is the same pattern as with Finn who also sent \$4

This seems fair and consistent with my cooperative strategy.

usage:

```
{
  "input_tokens": 1405,
  "output_tokens": 327,
  "reasoning_tokens": 166
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Hana sent me \$3 (interestingly, I sent them \$3 in Round 1 and they returned \$4)
- I received \$9
- Hana currently has \$2 left

I've been consistent as a receiver, returning around 50% of what I receive. This has worked well and maintains fairness.

```
'''json
{"return": 4.5}
'''
```

This gives Hana \$6.5 total and me \$4.5, continuing the cooperative pattern we established.

reasoning:

Looking at this situation:

- Hana sent me \$3
- I received \$9
- Hana currently has \$2 left

I've played with Hana before. In Round 1, I was the sender and sent \$3 to Hana, which beca

me \$9. Hana returned \$4 to me.

Now the roles are reversed. Hana sent me \$3, just as I did to them in Round 1.

As a receiver, my pattern has been:

- Round 2: Finn sent \$4, I received \$12, returned \$6 (50%)
- Round 3: Dorian sent \$3, I received \$9, returned \$4.5 (50%)
- Round 6: Aster sent \$4, I received \$12, returned \$6 (50%)

So I've been consistently returning about 50% of what I receive.

For this round with Hana:

- I received \$9
- 50% would be \$4.5

This seems fair and consistent with my previous behavior. Let me return \$4.5.

usage:

```
{
  "input_tokens": 1393,
  "output_tokens": 381,
  "reasoning_tokens": 195
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 8

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$40.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- I'm facing Aster again
- In Round 6, Aster sent me \$4 (showing high trust)
- I returned \$6, and they got a \$7 total payoff
- Aster has been cooperative

Since Aster showed considerable trust by sending \$4 to me previously, I should reciprocate that trust level:

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze this situation:

- I'm playing against Aster again
- In Round 6, I was the RECEIVER and Aster sent me \$4, I returned \$6
- Aster showed high trust by sending \$4
- My track record as sender:
 - Round 1 vs Hana: sent \$3, got \$4 back
 - Round 3 vs Finn: sent \$3, got \$5 back
 - Round 4 vs Briar: sent \$3, got \$5 back
 - Round 5 vs Elara: sent \$3, got \$4.5 back

Since Aster was very cooperative with me (sent \$4), I should reciprocate that trust. They showed they're willing to take risks. I could send \$4 to match what they sent me, or stick with my usual \$3.

Given that Aster was generous to me, I think I should send \$4 to match their trust level.

usage:

```
{
  "input_tokens": 1387,
  "output_tokens": 332,
  "reasoning_tokens": 176
}
```

```
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 9

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$47.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- I'm facing Dorian for the first time as sender
- My recent cooperative plays have been working well (Aster just returned \$6 on my \$4 send)
- I've established a pattern of sending \$3-4 as a starting point with opponents

I'll send my standard cooperative amount:

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at the history:

- I'm facing Dorian (Agent_4) for the first time as sender
- I don't have any previous interaction history with Dorian mentioned
- My previous round with Aster went well - I sent \$4, they returned \$6, so cooperation paid off
- My typical strategy has been to send \$3-4, showing reasonable trust

Since I don't know Dorian's behavior yet, I'll go with my standard cooperative opening of \$3, which has worked well for me:

```
'''json
{"send": 3}
'''
```

```
Usage: {
  "input_tokens": 1418,
  "output_tokens": 223,
  "reasoning_tokens": 125
}
```

GAME - Agent_4 - Receiver - dyad_4

Timestamp: 2026-06-04T16:25:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_7": "Galen"
    },
  },
  "agents": [
    "Agent_7",
    "Agent_4"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_4": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
```

```

"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$30.5.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Briar sent me \$3.0 (60% of their endowment)
- I received \$9.0
- Briar has \$2.0 left

Following my established pattern of returning 50% of what I receive:

- 50% of \$9.0 = \$4.5
- Briar will end with \$6.5, I'll keep \$4.5

This is consistent with my fair and cooperative strategy:

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

- Briar sent me \$3.0 (60% of their endowment)
- I received \$9.0 (after 3x multiplication)
- Briar currently has \$2.0 left

Following my consistent strategy:

- I've been returning 50% of what I receive when I'm the receiver
- 50% of \$9.0 = \$4.5

If I return \$4.5:

- Briar ends with \$2.0 + \$4.5 = \$6.5
- I keep \$9.0 - \$4.5 = \$4.5

This is consistent with how I've played before when receiving \$9.0 (from a \$3.0 send). Let me check:

- Round 1: Received \$9.0 (Cyra sent \$3.0), returned \$4.5
- Round 4: Received \$12.0 (Finn sent \$4.0), returned \$6.0

Yes, returning \$4.5 maintains my 50% return policy.

usage:

```
{
  "input_tokens": 1376,
  "output_tokens": 379,
  "reasoning_tokens": 171
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$35.0.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.
How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Elara sent me \$2.5 (50% of their endowment)
- I received \$7.5
- Elara has \$2.5 left

Following my consistent pattern of returning 50% of what I receive:

- 50% of \$7.5 = \$3.75
- Elara will end with \$6.25, I'll keep \$3.75

This maintains my fair and cooperative approach:

```
{'return': 3.75}
```

reasoning:

Looking at this situation:

- Elara sent me \$2.5 (50% of their endowment)
- I received \$7.5
- Elara has \$2.5 left

Following my established pattern of returning 50% of what I receive:

- 50% of \$7.5 = \$3.75
- Elara will end with \$6.25, I'll keep \$3.75

This maintains my consistent and fair strategy.

usage:

```
{
  "input_tokens": 1400,
  "output_tokens": 232,
  "reasoning_tokens": 82
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 8

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.75. Your payoff was \$3.75.
Your total earnings across all rounds are \$38.75.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history with Finn:
- Round 5: Finn sent me \$4.0, I returned \$6.0 (50% of received amount)
- This showed mutual trust and cooperation

I'll maintain my consistent cooperative strategy of sending \$3.0 (60% of endowment):

```
{'send': 3.0}
```

reasoning:

Looking at the history:
- Round 5: I was RECEIVER against Finn. They sent \$4.0, I returned \$6.0 (50% of the \$12.0 I received)
- Round 8 (this round): I am SENDER against Finn

Finn has shown trust by sending me \$4.0 in our previous interaction, and I reciprocated fairly by returning \$6.0 (which gave them a good payoff).

My consistent strategy has been:
- As sender: send \$3.0 (60% of endowment)
- As receiver: return 50% of what I receive

I should continue with my established cooperative pattern and send \$3.0 to Finn.

usage:

```
{
  "input_tokens": 1384,
  "output_tokens": 251,
  "reasoning_tokens": 139
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 9

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$45.25.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:
- Galen sent me \$3.0 (60% of their endowment)
- I received \$9.0

- Galen has \$2.0 left

Following my consistent pattern of returning 50% of what I receive:

- 50% of \$9.0 = \$4.5
- Galen will end with \$6.5, I'll keep \$4.5

This maintains my fair and cooperative approach:

```
{'return': 4.5}
```

Reasoning:

Looking at this situation:

- Galen sent me \$3.0 (60% of their endowment)
- I received \$9.0
- Galen has \$2.0 left

Following my consistent pattern of returning 50% of what I receive:

- 50% of \$9.0 = \$4.5
- Galen will end with \$6.5, I'll keep \$4.5

This maintains my fair and cooperative approach.

```
Usage: {  
  "input_tokens": 1397,  
  "output_tokens": 232,  
  "reasoning_tokens": 81  
}
```

=====

ROUND 10

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_5 (Elara)=Sender, Agent_2 (Briar)=Receiver
dyad_2: Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver
dyad_3: Agent_4 (Dorian)=Sender, Agent_3 (Cyra)=Receiver
dyad_4: Agent_8 (Hana)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_5 (Elara) sent \$2.5; received \$7.5; Agent_2 (Briar) returned \$4; payoffs Agent_5 (Elara)=\$6.5, Agent_2 (Briar)=\$3.5
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 7.5, "min": 0, "raw": 4.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_2: Agent_6 (Finn) sent \$4; received \$12; Agent_7 (Galen) returned \$6; payoffs Agent_6 (Finn)=\$7, Agent_7 (Galen)=\$6
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_4 (Dorian) sent \$3; received \$9; Agent_3 (Cyra) returned \$5; payoffs Agent_4 (Dorian)=\$7, Agent_3 (Cyra)=\$4
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 (Hana) sent \$4; received \$12; Agent_1 (Aster) returned \$7; payoffs Agent_8 (Hana)=\$8, Agent_1 (Aster)=\$5
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw": 7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

```
"Agent_1": 62.5,  
"Agent_2": 53.0,  
"Agent_3": 57.25,  
"Agent_4": 56.75,  
"Agent_5": 52.25,  
"Agent_6": 58.25,  
"Agent_7": 60.0,  
"Agent_8": 57.0  
}
```

LLM CALLS

GAME - Agent_5 - Sender - dyad_1

Timestamp: 2026-06-04T16:26:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_5": "Elara"
    },
    "agents": [
      "Agent_5",
      "Agent_2"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_2": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.

- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the SENDER against Finn. You sent \$2.5, it became \$7.5 for them, and they returned \$3.75 to you. Your payoff was \$6.25.
Your total earnings across all rounds are \$30.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my established pattern:

- As sender: I consistently send \$2.5 (50% of my endowment)
- This has led to strong reciprocity from opponents
- My cooperative strategy has yielded solid returns

Now facing Dorian, I'll continue with the same approach and send \$2.5.

```
'''json
{"send": 2.5}
'''
```

reasoning:

Looking at my history:

As sender:

- Round 4 (vs Aster): Sent \$2.5, received \$3.5 back, payoff \$6.0
- Round 6 (vs Finn): Sent \$2.5, received \$3.75 back, payoff \$6.25

As receiver:

- Round 4 (vs Hana): Received \$9.0, returned \$4.5, payoff \$4.5
- Round 5 (vs Galen): Received \$9.0, returned \$4.5, payoff \$4.5

My strategy has been consistent:

- Send 50% (\$2.5) as sender
- Return 50% of received amount as receiver

This has been working well, with total earnings of \$30.5 across rounds.

Now playing against Dorian as sender. I should maintain my consistent cooperative strategy and send \$2.5.

usage:

```
{
  "input_tokens": 1499,
  "output_tokens": 310,
  "reasoning_tokens": 160
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 8

In your most recent previous game round, you were the SENDER against Dorian. You sent \$2.5, it became \$7.5 for them, and they returned \$3.75 to you. Your payoff was \$6.25.
Your total earnings across all rounds are \$36.75.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Cyra sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- My established pattern is to return 50% of what I receive

I'll maintain consistency and return \$4.5.

This results in:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

reasoning:

I'm the receiver in this round. Cyra sent me \$3.0, which became \$9.0.

Looking at my pattern as a receiver:

- Round 4 vs Hana: They sent \$3.0, I received \$9.0, I returned \$4.5
- Round 5 vs Galen: They sent \$3.0, I received \$9.0, I returned \$4.5

I've been consistently returning 50% of what I receive as a receiver. This maintains fairness and reciprocity.

Cyra sent \$3.0 (60% of their endowment), showing trust. I should return \$4.5 (50% of the \$9.0 I received).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This is consistent with my established behavior.

usage:

```
{
  "input_tokens": 1415,
  "output_tokens": 371,
  "reasoning_tokens": 162
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 9

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$41.25.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- My established pattern is to return 50% of what I receive

I'll maintain consistency and return \$4.5.

This results in:

- Briar: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Briar sent \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after the multiplier
- My consistent pattern has been to return 50% of what I receive

Following my established cooperative strategy, I should return \$4.5.

This results in:

- Briar: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This maintains my reputation for fairness and reciprocity.

usage:

```
{
  "input_tokens": 1407,
  "output_tokens": 282,
  "reasoning_tokens": 112
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 10

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$45.75.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- In our previous round, Briar sent me \$3.0 (60% of their endowment), showing high trust
- I returned \$4.5 (50% of what I received), maintaining my fair reciprocity pattern
- My established strategy as sender is to send \$2.5 (50% of my endowment)

I'll continue with my consistent cooperative approach and send \$2.5.

```
'''json
{"send": 2.5}
'''
```

Reasoning:

Looking at this situation:

In the previous round (Round 9):

- I was the receiver against Briar
- Briar sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5 (50% of what I received)
- Briar got \$6.5 total, I got \$4.5

Now in Round 10:

- I'm the sender against Briar
- Briar showed trust by sending \$3.0 (60%)
- I reciprocated fairly by returning 50%
- My established pattern is to send \$2.5 as sender

I should continue with my consistent strategy of sending \$2.5 (50% of my endowment).

Usage: {

```
"input_tokens": 1430,
"output_tokens": 297,
"reasoning_tokens": 138
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T16:26:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_5",
"pairing": {
  "agent_names": {
    "Agent_2": "Briar",
    "Agent_5": "Elara"
  },
  "agents": [
    "Agent_5",
    "Agent_2"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_2": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$31.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Aster sent me \$4 (80% of their endowment) - higher trust than the typical \$3 I've been receiving
- When receiving \$3 (becoming \$9), I return \$5
- Aster showed more trust, so I should reward that proportionally
- Returning \$7 out of \$12 maintains my cooperative approach while appropriately rewarding their higher trust

{'return': 7}

reasoning:

Aster sent me \$4, which is 80% of their endowment - a higher level of trust than the \$3 (60%) I've been receiving from others.

Let me think about what's fair:

- When I receive \$3 (multiplied to \$9), I've been returning \$5
- Aster sent \$4 (multiplied to \$12), showing more trust
- I should reward this higher trust with a proportionally higher return

If \$3 → return \$5, that's returning about 56% of what I received

If I apply the same proportion to \$12: $0.56 \times \$12 = \6.7

But I could also think about it from Aster's perspective:

- They sent \$4, kept \$1
- If I return \$7, they get \$8 total (\$1 kept + \$7 returned), gaining \$3
- I keep \$5 (\$12 - \$7), gaining \$5
- This seems fair and rewards their higher trust

Let me consider \$6.5:

- They sent \$4, kept \$1
- If I return \$6.5, they get \$7.5 total, gaining \$2.5
- I keep \$5.5, gaining \$5.5
- This is also reasonable

I think \$7 is a good reward for showing more trust than average. It gives them a solid return while still being mutually beneficial.

usage:

```
{
  "input_tokens": 1325,
  "output_tokens": 431,
  "reasoning_tokens": 269
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 8

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$7.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$36.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my history with Hana and overall pattern:

- In Round 2, Hana sent me \$3 and I returned \$5
- I've consistently sent \$3 when acting as sender
- This balanced approach has been working well

I'll continue with my established strategy.

{'send': 3}

reasoning:

Let me review my situation:

- I'm playing as SENDER against Hana
- In Round 2, I was RECEIVER against Hana. They sent me \$3, I received \$9, and I returned \$5. So Hana got \$4 in that round.
- My established pattern has been to send \$3 when I'm the sender
- I just played with Aster who sent me \$4 and I returned \$7
- Hana has shown moderate trust in the past by sending \$3

I should continue with my consistent strategy of sending \$3. This has been working well and maintains my reputation for balanced cooperation.

usage:

```
{
  "input_tokens": 1326,
  "output_tokens": 223,
  "reasoning_tokens": 134
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 9

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$43.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my consistent pattern:

- I've sent \$3 in all previous sender rounds
- This has generated reliable returns (around \$4-5)
- My total earnings of \$43.0 show this strategy is working well
- First time playing with Elara, but my standard approach has been successful with new partners

I'll continue with my established strategy.

{'send': 3}

reasoning:

Let me review my strategy:

- I've consistently sent \$3 when acting as sender
- This has yielded good results with returns around \$4-5
- My total earnings are \$43.0, which is strong
- I haven't played with Elara before

I should continue with my established strategy of sending \$3.

{'send': 3}

usage:

```
{
  "input_tokens": 1318,
  "output_tokens": 176,
  "reasoning_tokens": 80
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 10

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$49.5.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.
How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Last round I sent Elara \$3, and they returned \$4.5
- This round Elara sent me \$2.5 (less than my \$3)
- I received \$7.5

My typical return rate is around 50-55% of what I receive. For \$7.5, that would be approximately \$4.

This maintains fairness and reciprocity while being slightly generous.

{'return': 4}

Reasoning:

Looking at this situation:

- Last round I sent Elara \$3, which became \$9, and they returned \$4.5
- This round Elara sent me \$2.5, which became \$7.5
- Elara sent less than I did (\$2.5 vs \$3)

My return strategy has been:

- When receiving \$9 (from \$3 sent), I typically return around \$5
- When receiving \$12 (from \$4 sent by Aster), I returned \$7
- My return rate has been roughly 50-58% of what I receive

For \$7.5 received:

- 50% would be \$3.75
- Elara only sent \$2.5 to begin with (keeping \$2.5)
- If I return \$4, they'd end up with: \$2.5 (kept) + \$4 (returned) - \$2.5 (sent) = \$4
- I'd keep: \$7.5 - \$4 = \$3.5

Since Elara sent less than I did, I should return proportionally:

- They sent \$2.5 vs my \$3 = 83% of what I sent
- They returned \$4.5 on my \$9 = 50%
- For fairness, returning around \$4 seems appropriate
- This gives them a decent return while not over-rewarding the lower trust shown

Let me return \$4.

Usage: {

"input_tokens": 1337,
"output_tokens": 447,
"reasoning_tokens": 245

}

GAME - Agent_6 - Sender - dyad_2

Timestamp: 2026-06-04T16:26:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_2",
"move_index": 2,
"opponent_id": "Agent_7",
"pairing": {
"agent_names": {
"Agent_6": "Finn",
"Agent_7": "Galen"

},

"agents": [
"Agent_6",
"Agent_7"

```

],
"investor": "Agent_6",
"roles": {
  "Agent_6": "investor",
  "Agent_7": "trustee"
},
"trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.75. Your payoff was \$3.75.
Your total earnings across all rounds are \$35.25.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- This is my first interaction with Cyra
- My strategy of sending \$4 has been consistently successful across multiple rounds:
 - With Galen, Dorian, and Aster, sending \$4 resulted in receiving \$6 back each time (\$7 payoff)
- This high-trust approach encourages cooperation and mutual benefit
- My total earnings of \$35.25 demonstrate this strategy is working well

Continuing with \$4 maintains the successful cooperative pattern.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Let me analyze this situation:

- This is my first interaction with Cyra
- My strategy of sending \$4 (80% of endowment) has been very successful:
 - Round 2: sent \$4 to Galen, received \$6 back (payoff \$7)
 - Round 4: sent \$4 to Dorian, received \$6 back (payoff \$7)
 - Round 5: sent \$4 to Aster, received \$6 back (payoff \$7)
- This high-trust approach has consistently yielded good returns
- My total earnings are \$35.25, which suggests the cooperative strategy is working well
- I have no prior history with Cyra, so I should continue my successful pattern

Sending \$4 maintains the high-trust strategy that has been working consistently.

usage:

```
{
  "input_tokens": 1464,
  "output_tokens": 326,
  "reasoning_tokens": 169
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 8

In your most recent previous game round, you were the SENDER against Cyra. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$42.25.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Dorian sent me \$3.0 (60% of their endowment), showing good trust
- This created \$9.0 for me
- In Round 4, when I sent Dorian \$4, they returned \$6 (50% of what they received)
- My consistent strategy is to return approximately 50% of what I receive

If I return \$4.5 (50% of received):

- Dorian ends with: \$2.0 (kept) + \$4.5 (returned) = \$6.5 total
- I end with: \$9.0 - \$4.5 = \$4.5 total

This maintains fair reciprocation consistent with our previous interaction.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at this situation:

- Dorian sent me \$3.0, which became \$9.0 for me
- In Round 4, I was the sender against Dorian. I sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to me.
- Dorian has shown they are cooperative (returned \$6 on \$12 received = 50%)

My consistent strategy has been to return approximately 50% of what I receive. This maintains fairness and encourages continued cooperation.

If I return \$4.5 (50% of \$9.0):

- Dorian ends with: \$2.0 (kept from \$5) + \$4.5 (returned) = \$6.5 total (gained \$1.5)
- I end with: \$9.0 - \$4.5 = \$4.5 total (lost \$0.5)

This is fair reciprocation. Dorian sent \$3 (60% of their endowment) showing good trust. I should return a fair amount.

Actually, let me reconsider. When Dorian was the receiver in Round 4:

- I sent \$4, they got \$12
- They returned \$6 (50% of what they received)

Now Dorian sent me \$3 (out of \$5), showing 60% trust level.
I received \$9.

Returning 50% would be \$4.5, which seems fair and consistent with my strategy and what Dorian did.

usage:

```
{
  "input_tokens": 1502,
  "output_tokens": 545,
  "reasoning_tokens": 271
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 9

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$46.75.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Cyra sent me \$3.0 (60% of their endowment), showing good trust
- This created \$9.0 for me
- In Round 7, I sent Cyra \$4.0, and they returned \$6.0 (50% of what they received)
- My consistent strategy is to return approximately 50% of what I receive

If I return \$4.5 (50% of received):

- Cyra ends with: \$2.0 (kept) + \$4.5 (returned) = \$6.5 total
- I end with: \$9.0 - \$4.5 = \$4.5 total

This maintains fair reciprocation consistent with our previous interaction.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at this situation:

- Cyra sent me \$3.0 (60% of their endowment), showing good trust
- This created \$9.0 for me
- In Round 7, I sent Cyra \$4.0, and they returned \$6.0 (50% of what they received)

- My consistent strategy is to return approximately 50% of what I receive

If I return \$4.5 (50% of received):

- Cyra ends with: \$2.0 (kept) + \$4.5 (returned) = \$6.5 total
- I end with: \$9.0 - \$4.5 = \$4.5 total

This maintains fair reciprocation consistent with our previous interaction where Cyra also returned 50%.

usage:

```
{
  "input_tokens": 1561,
  "output_tokens": 379,
  "reasoning_tokens": 138
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 10

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$51.25.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- This is my second interaction with Galen
- In Round 2, I sent Galen \$4.0, they received \$12.0, and returned \$6.0 to me (50% return rate)
- This resulted in a \$7 payoff for me, which is excellent
- Galen has proven to be a reliable and fair partner
- My strategy of sending \$4 has been consistently successful

Continuing with \$4 maintains this successful cooperative pattern.

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Looking at my history with Galen:

- Round 2: I was the SENDER. I sent Galen \$4.0, they received \$12.0, and they returned \$6.0 to me. My payoff was \$7.0.

Galen has proven to be a reliable and fair partner, returning 50% of what they received when I sent them \$4. This is consistent with my cooperative strategy.

My successful pattern has been:

- Sending \$4 to various partners
- Generally receiving \$6 back (50% of the \$12 they receive)
- This gives me a \$7 payoff (\$5 - \$4 + \$6 = \$7)

Given that Galen has already shown good reciprocity, I should continue with my successful \$4 sending strategy.

My total earnings are \$51.25, showing my cooperative approach is working well overall.

Usage: {

```
"input_tokens": 1566,
"output_tokens": 339,
"reasoning_tokens": 181
```

}

GAME - Agent_7 - Receiver - dyad_2

Timestamp: 2026-06-04T16:26:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_6",
"pairing": {
  "agent_names": {
    "Agent_6": "Finn",
    "Agent_7": "Galen"
  },
  "agents": [
    "Agent_6",
    "Agent_7"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_6": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Hana sent me \$3 (interestingly, I sent them \$3 in Round 1 and they returned \$4)
- I received \$9
- Hana currently has \$2 left

I've been consistent as a receiver, returning around 50% of what I receive. This has worked well and maintains fairness.

```
'''json
{"return": 4.5}
'''
```

This gives Hana \$6.5 total and me \$4.5, continuing the cooperative pattern we established.

reasoning:

Looking at this situation:

- Hana sent me \$3
- I received \$9
- Hana currently has \$2 left

I've played with Hana before. In Round 1, I was the sender and sent \$3 to Hana, which became \$9. Hana returned \$4 to me.

Now the roles are reversed. Hana sent me \$3, just as I did to them in Round 1.

As a receiver, my pattern has been:

- Round 2: Finn sent \$4, I received \$12, returned \$6 (50%)
- Round 3: Dorian sent \$3, I received \$9, returned \$4.5 (50%)
- Round 6: Aster sent \$4, I received \$12, returned \$6 (50%)

So I've been consistently returning about 50% of what I receive.

For this round with Hana:

- I received \$9
- 50% would be \$4.5

This seems fair and consistent with my previous behavior. Let me return \$4.5.

usage:

```
{
  "input_tokens": 1393,
  "output_tokens": 381,
  "reasoning_tokens": 195
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 8

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$40.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm facing Aster again
- In Round 6, Aster sent me \$4 (showing high trust)
- I returned \$6, and they got a \$7 total payoff
- Aster has been cooperative

Since Aster showed considerable trust by sending \$4 to me previously, I should reciprocate that trust level:

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze this situation:

- I'm playing against Aster again
- In Round 6, I was the RECEIVER and Aster sent me \$4, I returned \$6
- Aster showed high trust by sending \$4
- My track record as sender:
 - Round 1 vs Hana: sent \$3, got \$4 back
 - Round 3 vs Finn: sent \$3, got \$5 back
 - Round 4 vs Briar: sent \$3, got \$5 back
 - Round 5 vs Elara: sent \$3, got \$4.5 back

Since Aster was very cooperative with me (sent \$4), I should reciprocate that trust. They showed they're willing to take risks. I could send \$4 to match what they sent me, or stick with my usual \$3.

Given that Aster was generous to me, I think I should send \$4 to match their trust level.

usage:

```
{
  "input_tokens": 1387,
  "output_tokens": 332,
  "reasoning_tokens": 176
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 9

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$47.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- I'm facing Dorian for the first time as sender
- My recent cooperative plays have been working well (Aster just returned \$6 on my \$4 send)
- I've established a pattern of sending \$3-4 as a starting point with opponents

I'll send my standard cooperative amount:

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history:

- I'm facing Dorian (Agent_4) for the first time as sender
- I don't have any previous interaction history with Dorian mentioned
- My previous round with Aster went well - I sent \$4, they returned \$6, so cooperation paid off
- My typical strategy has been to send \$3-4, showing reasonable trust

Since I don't know Dorian's behavior yet, I'll go with my standard cooperative opening of \$3, which has worked well for me:

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1418,
  "output_tokens": 223,
  "reasoning_tokens": 125
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 10

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$54.0.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Finn sent me \$4 (showing high trust)
- I received \$12
- Finn currently has \$1 left

I've been consistently returning about 50% of what I receive as a receiver, which has maintained good cooperation:

```
'''json
{"return": 6}
'''
```

This gives Finn \$7 total and me \$6, which is fair and rewards their trust.

Reasoning:

Let me analyze this situation:

- Finn sent me \$4 (showing high trust)
- I received \$12
- Finn currently has \$1 left (\$5 - \$4)

Looking at my history as a receiver:

- Round 3: Received \$9 from Cleo, returned \$4.5
- Round 6: Received \$12 from Aster, returned \$6
- Round 7: Received \$9 from Hana, returned \$4.5

I've been consistently returning about 50% of what I receive, which seems fair and maintains cooperation.

For \$12 received, 50% would be \$6.

This would give Finn: \$1 (kept) + \$6 (returned) = \$7 total

And me: \$12 - \$6 = \$6

This seems fair and maintains my consistent, trustworthy pattern as a receiver.

Usage: {

```
"input_tokens": 1381,
"output_tokens": 310,
"reasoning_tokens": 166
```

}

GAME - Agent_4 - Sender - dyad_3

Timestamp: 2026-06-04T16:26:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_4": "Dorian"
    },
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.

- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$35.0.

This round, you are the RECEIVER against Elara. They sent you \$2.5, so you received \$7.5.

How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Elara sent me \$2.5 (50% of their endowment)
- I received \$7.5
- Elara has \$2.5 left

Following my consistent pattern of returning 50% of what I receive:

- 50% of \$7.5 = \$3.75
- Elara will end with \$6.25, I'll keep \$3.75

This maintains my fair and cooperative approach:

{'return': 3.75}

reasoning:

Looking at this situation:

- Elara sent me \$2.5 (50% of their endowment)
- I received \$7.5
- Elara has \$2.5 left

Following my established pattern of returning 50% of what I receive:

- 50% of \$7.5 = \$3.75
- Elara will end with \$6.25, I'll keep \$3.75

This maintains my consistent and fair strategy.

usage:

```
{
  "input_tokens": 1400,
  "output_tokens": 232,
  "reasoning_tokens": 82
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 8

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$2.5 to you, it became \$7.5, and you returned \$3.75. Your payoff was \$3.75.
Your total earnings across all rounds are \$38.75.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my history with Finn:

- Round 5: Finn sent me \$4.0, I returned \$6.0 (50% of received amount)
- This showed mutual trust and cooperation

I'll maintain my consistent cooperative strategy of sending \$3.0 (60% of endowment):

{'send': 3.0}

reasoning:

Looking at the history:

- Round 5: I was RECEIVER against Finn. They sent \$4.0, I returned \$6.0 (50% of the \$12.0 I received)
- Round 8 (this round): I am SENDER against Finn

Finn has shown trust by sending me \$4.0 in our previous interaction, and I reciprocated fairly by returning \$6.0 (which gave them a good payoff).

My consistent strategy has been:

- As sender: send \$3.0 (60% of endowment)
- As receiver: return 50% of what I receive

I should continue with my established cooperative pattern and send \$3.0 to Finn.

usage:

```
{
  "input_tokens": 1384,
  "output_tokens": 251,
  "reasoning_tokens": 139
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 9

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$45.25.

This round, you are the RECEIVER against Galen. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Galen sent me \$3.0 (60% of their endowment)
- I received \$9.0
- Galen has \$2.0 left

Following my consistent pattern of returning 50% of what I receive:

- 50% of \$9.0 = \$4.5
- Galen will end with \$6.5, I'll keep \$4.5

This maintains my fair and cooperative approach:

```
{'return': 4.5}
```

reasoning:

Looking at this situation:

- Galen sent me \$3.0 (60% of their endowment)
- I received \$9.0
- Galen has \$2.0 left

Following my consistent pattern of returning 50% of what I receive:

- 50% of \$9.0 = \$4.5
- Galen will end with \$6.5, I'll keep \$4.5

This maintains my fair and cooperative approach.

usage:

```
{
  "input_tokens": 1397,
  "output_tokens": 232,
  "reasoning_tokens": 81
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 10

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$49.75.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Cyra:

- Round 2: I sent \$3.0, Cyra returned \$4.5 (50% of received)
- Round 4: Cyra sent \$3.0, I returned \$4.5 (50% of received)

We've established a pattern of mutual trust and fairness. I'll continue my consistent cooperative strategy:

```
{'send': 3.0}
```

Reasoning:

Looking at my history with Cyra:

- Round 2: I was the SENDER, sent \$3.0, Cyra returned \$4.5
- Round 4: Cyra was the SENDER, sent \$3.0, I received \$9.0 and returned \$4.5

Both rounds showed good mutual cooperation. Cyra has been trustworthy, returning exactly what I would expect based on fair play.

I should continue my consistent cooperative strategy of sending \$3.0.

This maintains:

- My cooperative pattern
- Fair expectations based on previous interactions
- Trust in reciprocity

Usage: {

```
"input_tokens": 1379,  
"output_tokens": 241,  
"reasoning_tokens": 130
```

}

GAME - Agent_3 - Receiver - dyad_3

Timestamp: 2026-06-04T16:26:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_3",  
"move_index": 5,  
"opponent_id": "Agent_4",  
"pairing": {  
  "agent_names": {  
    "Agent_3": "Cyra",  
    "Agent_4": "Dorian"  
  },  
  "agents": [  
    "Agent_4",  
    "Agent_3"  
  ],  
  "investor": "Agent_4",  
  "roles": {  
    "Agent_3": "trustee",  
    "Agent_4": "investor"  
  },  
  "trustee": "Agent_3"
```

```
},  
"role": "trustee",  
"role_label": "Receiver",  
"round": 10,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game,

consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0.
Your total earnings across all rounds are \$34.25.

This round, you are the RECEIVER against Finn. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)
Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****Current round:****

- Finn sent \$4.0 (80% of endowment - very high trust!)
- I received \$12.0

****My pattern as receiver:****

- Round 2 vs Elara: Returned 50% (\$3.75 from \$7.5)
- Round 4 vs Aster: Returned 50% (\$6.0 from \$12.0)
- Round 6 vs Hana: Returned 55.6% (\$5.0 from \$9.0)

****Analysis:****

- Finn's situation is identical to Aster in Round 4 (both sent \$4, I received \$12)

- I returned \$6 to Aster, which worked well
- Consistency is important for maintaining trust

I'll return **\$6.0** (50%), which:

- Matches my previous response to the same situation
- Rewards Finn's high trust appropriately
- Gives Finn: $\$5 - \$4 + \$6 = \7 (good payoff)
- Leaves me: \$6 (fair split)

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at the situation:

****Current round:****

- Finn sent me \$4.0 (80% of their endowment - high trust!)
- I received \$12.0
- Need to decide how much to return

****My history as receiver:****

- Round 2 vs Elara: Received \$7.5, returned \$3.75 (50%)
- Round 4 vs Aster: Received \$12.0, returned \$6.0 (50%)
- Round 6 vs Hana: Received \$9.0, returned \$5.0 (55.6%)

****Pattern:****

- I've been returning 50-56% of what I receive
- This has maintained good relationships
- Finn sent \$4 which is the same as Aster sent in Round 4

****Reciprocity consideration:****

- Finn showed high trust by sending \$4 (80%)
- Should reward this trust appropriately
- Returning 50% (\$6) would be consistent with my previous behavior when receiving \$12

I'll return **\$6.0** (50% of \$12), which:

- Is consistent with my behavior when Aster sent me \$4
- Rewards Finn's high trust
- Gives Finn: $\$5 - \$4 + \$6 = \7 (good return on investment)
- Leaves me: $\$12 - \$6 = \$6$ (equal split of the multiplied amount)
- Fair and cooperative

```
'''json
{"return": 6.0}
'''
```

usage:

```
{
  "input_tokens": 1664,
  "output_tokens": 643,
  "reasoning_tokens": 280
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 8

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$40.25.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my performance and history:

****As sender:****

- Round 1 vs Dorian: Sent \$3 ? received \$4.5

- Round 3 vs Hana: Sent \$3 ? received \$5.0
- Round 5 vs Briar: Sent \$3 ? received \$5.0
- Consistent success with \$3

****History with Elara:****

- Round 2: Elara sent \$2.5 ? I returned \$3.75 (50%)
- Showed moderate trust and I reciprocated fairly

****Strategy:****

- \$3 has been my optimal sending amount across all partners
- Elara has demonstrated cooperative behavior
- Continue proven successful approach

I'll send ****\$3****, which:

- Maintains my consistent, reliable strategy
- Becomes \$9 for Elara
- Has yielded good returns historically

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

****As sender:****

- Round 1 vs Dorian: Sent \$3 ? received \$4.5 (profit \$1.5)
- Round 3 vs Hana: Sent \$3 ? received \$5.0 (profit \$2.0)
- Round 5 vs Briar: Sent \$3 ? received \$5.0 (profit \$2.0)

****History with Elara:****

- Round 2: Elara sent \$2.5 ? I received \$7.5 ? I returned \$3.75
- Elara's payoff: $\$5 - \$2.5 + \$3.75 = \6.25
- My payoff: \$3.75

So Elara sent me a moderate amount (\$2.5) and got a good return. Now it's my turn to send to Elara.

Given my successful pattern of sending \$3, and the fact that Elara has shown willingness to trust (sent \$2.5), I should continue with my standard \$3 send.

This has been working consistently well:

- All three times I sent \$3, I received back between \$4.5-\$5.0
- It's a balanced amount that shows cooperation without being overly risky
- Elara has engaged cooperatively before

I'll send \$3.

usage:

```
{
  "input_tokens": 1720,
  "output_tokens": 541,
  "reasoning_tokens": 234
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 9

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$46.75.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

****History with Finn:****

- Round 7: Finn sent \$4 ? I returned \$6 (50%, fair reciprocation)

****My sending strategy:****

- Consistently sent \$3 in all previous rounds
- Returns: \$4.5, \$5.0, \$5.0, \$4.5 (average \$4.75)
- This has been highly successful

****Analysis:****

- Finn showed high trust by sending \$4
- I reciprocated fairly with \$6 return (50%)
- Our relationship is balanced
- My \$3 strategy has proven optimal across all partners
- Consistency maintains reliability

I'll send ****\$3****, which:

- Maintains my proven successful strategy
- Becomes \$9 for Finn
- Should yield a good return based on our positive history

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

****History with Finn:****

- Round 7: Finn sent \$4 ? I received \$12 ? I returned \$6
- Now Round 9: I'm the sender against Finn

****My sending pattern:****

- Round 1 vs Dorian: Sent \$3 ? received \$4.5
- Round 3 vs Hana: Sent \$3 ? received \$5.0
- Round 5 vs Briar: Sent \$3 ? received \$5.0
- Round 8 vs Elara: Sent \$3 ? received \$4.5

****Analysis:****

- Finn showed high trust by sending \$4 (80% of endowment)
- I returned \$6 (50% of \$12), which was fair
- Finn's payoff was \$7 (\$5 - \$4 + \$6)
- Should I reciprocate Finn's high trust level by also sending \$4?

****Considerations:****

- Finn sent \$4, showing very high trust
- Reciprocating with equal trust would mean sending \$4
- However, \$3 has been my consistent and successful strategy
- Sending \$4 would show I appreciate Finn's trust

Actually, let me think about this more strategically:

- Finn sent me \$4 (high trust)
- I returned \$6 to Finn (fair, 50%)
- If I want to maintain good relations and potentially get a good return, I should match Finn's trust level
- Sending \$4 would:
 - Reciprocate Finn's high trust
 - Show I value the cooperative relationship
 - Become \$12 for Finn
 - Likely get a good return given Finn's previous behavior

I think I should send \$4 to match Finn's trust level. This shows reciprocity and strengthens the cooperative relationship.

Actually, let me reconsider. My consistent \$3 strategy has worked very well:

- Average return when sending \$3: $(\$4.5 + \$5.0 + \$5.0 + \$4.5)/4 = \$4.75$
- This gives average payoff of $\$5 - \$3 + \$4.75 = \6.75

Finn's exceptional trust with \$4 creates tension between maintaining my proven approach versus signaling reciprocity. If I stick with \$3, it might seem like I'm not matching Finn's gesture, though my consistent pattern across all partners suggests this is just my strategy.

ategy. The key question is whether sending \$4 would actually yield better returns-Finn would receive \$12, and given their previous behavior, they'd likely return generously.

Comparing the outcomes: sending \$4 could net me \$6-7 if Finn reciprocates strongly, but sending \$3 is more predictable, yielding around \$6.5-7 with less risk. Since I've maintained a steady \$3 approach throughout and already returned 50% to Finn, the relationship feels balanced. Deviating now doesn't seem necessary.

usage:

```
{
  "input_tokens": 1765,
  "output_tokens": 965,
  "reasoning_tokens": 614
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 10

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$53.25.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Current round:****

- Dorian sent \$3.0
- I received \$9.0

****History with Dorian:****

- Round 1: I sent \$3 ? Dorian returned \$4.5 (50%, fair)

****My receiver pattern:****

- Round 2 vs Elara (\$7.5): Returned \$3.75 (50%)
- Round 4 vs Aster (\$12.0): Returned \$6.0 (50%)
- Round 6 vs Hana (\$9.0): Returned \$5.0 (55.6%)
- Round 7 vs Finn (\$12.0): Returned \$6.0 (50%)

****Analysis:****

- Dorian sent \$3, identical to Hana in Round 6
- I returned \$5 to Hana in that situation
- Dorian has been fair in our previous interaction
- Consistency is important

I'll return ****\$5.0****, which:

- Matches my previous response to the same situation (Round 6)
- Rewards Dorian's trust appropriately
- Gives Dorian: \$5 - \$3 + \$5 = \$7
- Leaves me: \$4

```
'''json
{"return": 5.0}
'''
```

Reasoning:

Looking at my history as receiver:

****My returns as receiver:****

- Round 2 vs Elara: Sent \$2.5 ? \$7.5 received, returned \$3.75 (50%)
- Round 4 vs Aster: Sent \$4.0 ? \$12.0 received, returned \$6.0 (50%)
- Round 6 vs Hana: Sent \$3.0 ? \$9.0 received, returned \$5.0 (55.6%)
- Round 7 vs Finn: Sent \$4.0 ? \$12.0 received, returned \$6.0 (50%)

****History with Dorian:****

- Round 1: I sent \$3 ? Dorian (receiver) returned \$4.5 (50%)

```

**Current situation:**
- Dorian sent $3.0
- I received $9.0
- This is identical to Round 6 vs Hana where I returned $5.0

**Analysis:**
- Dorian and I have a good history - we both played fairly
- In Round 6 when Hana sent $3, I returned $5 (55.6%)
- Consistency matters
- Dorian reciprocated fairly in Round 1

I should return **$5.0** (55.6%), which:
- Matches my response to the same situation in Round 6
- Rewards Dorian's $3 trust appropriately
- Gives Dorian: $5 - $3 + $5 = $7 (good payoff)
- Leaves me: $4 (fair split given the multiplication)
- Maintains consistency with my receiver pattern

Usage: {
  "input_tokens": 1765,
  "output_tokens": 712,
  "reasoning_tokens": 280
}

```

GAME - Agent_8 - Sender - dyad_4

```

Timestamp: 2026-06-04T16:26:46
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_8",
    "Agent_1"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_1": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$5.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$33.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- This is my first interaction with Galen
- My strategy of sending \$3 has been consistently successful
- I've built good cooperative relationships with multiple players
- Recent interactions have resulted in fair returns (Cyra returned \$5)

I'll maintain my cooperative strategy and send \$3 to Galen, demonstrating trust and giving them the opportunity to benefit while establishing a positive relationship.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the situation:

- I'm in Round 7, playing as SENDER against Galen (Agent_7)
- My total earnings are \$33.5
- My strategy has been consistently sending \$3, which has worked well
- In Round 6, Cyra reciprocated by returning \$5, giving me a payoff of \$7
- This is my first interaction with Galen

My consistent strategy of sending \$3 has proven successful:

- It shows trust and cooperation
- It allows for mutual benefit
- It has resulted in fair returns from others
- Average return has been around \$4.5-\$5

I should continue this strategy with Galen as well. It's a balanced approach that typically leads to positive outcomes.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1418,
  "output_tokens": 291,
  "reasoning_tokens": 180
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 8

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$40.0.

This round, you are the RECEIVER against Briar. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Briar sent \$3 (60% of their endowment), demonstrating trust
- I received \$9.0
- I've consistently returned \$5 when receiving \$9 from a \$3 send
- This maintains my reputation for fair reciprocity

Returning \$5 gives Briar a net gain of \$2 (sent \$3, received \$5) while I keep \$4. This continues my pattern of balanced cooperation.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at this situation:

- Briar sent me \$3 (60% of their endowment), showing trust
- I received \$9.0
- My consistent pattern has been: when someone sends \$3, I return \$5
- This has worked well - building cooperation and trust
- My total earnings are \$40.0, so I'm doing well

Let me check my established pattern:

- Round 3 (receiver vs Cyra): received \$9, returned \$5
- Round 5 (receiver vs Dorian): received \$9, returned \$5

I should maintain consistency. Returning \$5 gives Briar a fair return (they sent \$3, get back \$5, net gain of \$2) while I keep \$4 (net gain of \$4).

This is consistent with my reciprocity strategy and helps build trust.

usage:

```
{
  "input_tokens": 1423,
  "output_tokens": 320,
  "reasoning_tokens": 173
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 9

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$5.0. Your payoff was \$4.0. Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Aster sent \$4 (80% of their endowment), showing high trust
- I received \$12.0
- This is more generous than the typical \$3 I've been receiving

My pattern has been to return \$5 when receiving \$9 from a \$3 send, which gives the sender a \$2 net gain. For Aster's higher trust level with \$4, I should be proportionally fair.

Returning \$7 gives Aster a \$3 net gain (sent \$4, received \$7), rewarding their extra trust while I keep \$5.

```
'''json
{"return": 7.0}
'''
```

reasoning:

Let me analyze this situation:

- Aster sent \$4 (80% of their endowment), which is higher than the typical \$3 I've been seeing
- I received \$12.0 after the 3x multiplier
- This is a very generous send, demonstrating high trust

Let me think about what's fair:

- Aster spent \$4, they have \$1 left
- If I return \$7, Aster gets a net gain of \$3 (sent \$4, received \$7, total \$8 vs \$5 initial)
- I would keep \$5, same as my typical pattern

Actually, let me be more generous given their higher trust:

- If I return \$8, Aster gets a net gain of \$4 (sent \$4, received \$8, total \$9 vs \$5 initial)
- I would keep \$4, which is what I typically keep

This seems fair - they showed extra trust by sending \$4 instead of \$3, so I should reward that with a proportionally higher return.

Let me check: Aster sent 80% of their endowment. In previous rounds, when people sent \$3 (60%), I returned \$5.

\$3 sent ? \$9 received ? \$5 returned (55.6% return rate)

\$4 sent ? \$12 received ? If I return \$6.67, that's 55.6% return rate

But I should probably be more generous given the extra trust. Let me return \$7 or \$8.

\$8 seems like a good middle ground - it rewards their trust while keeping my payoff reasonable.

Actually, let me reconsider. My pattern has been:

- When receiving \$9 (from \$3 send), I return \$5
- This gives the sender \$2 net gain (from -\$3 to +\$2)
- I keep \$4

For proportional fairness:

- Aster sent \$4, so they're down \$4

- If I return \$7, they gain \$3 on their risk, which is proportionally higher than the \$2 gain I typically give

- I keep \$5 of the \$12

This rewards their extra trust while maintaining a reasonable split for myself.

usage:

```
{
  "input_tokens": 1406,
  "output_tokens": 836,
  "reasoning_tokens": 442
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 10

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$7.0. Your payoff was \$5.0.
Your total earnings across all rounds are \$49.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- Last round, Aster showed high trust by sending \$4 (80% of their endowment)
- I reciprocated fairly by returning \$7
- This established a good cooperative dynamic between us

Aster demonstrated extra trust compared to most players. I should reciprocate that trust level by sending \$4 as well, matching their generosity and continuing our positive relationship.

```
'''json
{"send": 4}
'''
```

Reasoning:

Looking at the situation:

- This is my second interaction with Aster
- Last round (Round 9), Aster sent me \$4 (showing high trust)
- I returned \$7 from the \$12 I received
- Aster's payoff was \$1 + \$7 = \$8 (they kept \$1 and received \$7)
- My payoff was \$5

Now Aster showed high trust by sending \$4. They got a good return from me. I should reciprocate their trust by also sending a higher amount. Since they sent \$4, I could send \$4 as well to show equal trust and maintain good cooperation.

My typical pattern has been to send \$3, but Aster demonstrated extra trust with \$4. Reciprocating with \$4 makes sense to continue building this cooperative relationship.

Usage: {

```
"input_tokens": 1451,
"output_tokens": 300,
"reasoning_tokens": 174
```

}

GAME - Agent_1 - Receiver - dyad_4

Timestamp: 2026-06-04T16:26:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_8",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_8": "Hana"
```

```
},
```

```
"agents": [
```

```

    "Agent_8",
    "Agent_1"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_1": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.

Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the SENDER against Galen. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.

Your total earnings across all rounds are \$35.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

- This is my first interaction with Briar
- My cooperative strategy has been consistently successful
- Every time I've sent \$4 (to Finn, Cyra, and Galen), I've received \$6 back for a \$7 payoff
- My total earnings of \$35.5 show this approach is working well

I'll continue with the same successful strategy.

{'send': 4}

reasoning:

Looking at my history:

- Round 1 (Sender vs Finn): Sent \$4, received \$6 back, payoff \$7
- Round 2: (not specified but total earnings suggest some positive payoff)
- Round 3 (Receiver vs Elara): Received \$7.5, returned \$3.5, payoff \$4
- Round 4 (Sender vs Cyra): Sent \$4, received \$6 back, payoff \$7
- Round 5 (Receiver vs Finn): Received \$12, returned \$6, payoff \$6
- Round 6 (Sender vs Galen): Sent \$4, received \$6 back, payoff \$7

Total earnings: \$35.5

My cooperative strategy of sending \$4 has been very successful:

- All three times I sent \$4 (to Finn, Cyra, and Galen), I received \$6 back
- This gives me a payoff of \$7 each time
- As a receiver, I've been fair by returning 50%

This is my first interaction with Briar. I should continue my successful cooperative strategy.

usage:

```
{
  "input_tokens": 1386,
  "output_tokens": 374,
  "reasoning_tokens": 207
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 8

In your most recent previous game round, you were the SENDER against Briar. You sent \$4.0, it became \$12.0 for them, and they returned \$7.0 to you. Your payoff was \$8.0.
Your total earnings across all rounds are \$43.5.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Galen sent me \$4, which became \$12 for me
- In Round 6, I sent Galen \$4 (which became \$12) and they returned \$6 (50%)
- Galen is now reciprocating by sending me the same amount
- I should maintain this fair cooperative relationship by returning the same proportion

Returning \$6 maintains the reciprocal trust we've established.

{'return': 6.0}

reasoning:

Looking at this situation:

- Galen sent me \$4, which became \$12 for me
- In Round 6, I sent Galen \$4 (which became \$12) and they returned \$6 to me
- Galen is reciprocating by sending me the same amount I sent them
- I should maintain this cooperative relationship by returning a similar proportion

Galen returned \$6 (50%) to me in Round 6. I should return the same proportion - \$6 (50% of \$12).

This continues the fair cooperative pattern.

usage:

```
{
  "input_tokens": 1397,
  "output_tokens": 243,
  "reasoning_tokens": 118
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 9

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$49.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- This is my first interaction with Hana
- My cooperative strategy has been very successful across all interactions
- Sending \$4 has consistently resulted in good returns (\$6-\$7)
- My total earnings of \$49.5 demonstrate the effectiveness of this approach

I'll continue with the same strategy.

```
{'send': 4}
```

reasoning:

Looking at the situation:

- This is my first interaction with Hana
- My cooperative strategy has been consistently successful
- When I've sent \$4 to new opponents, they've responded well:
 - Finn: returned \$6
 - Cyra: returned \$6
 - Galen: returned \$6
 - Briar: returned \$7
- My total earnings of \$49.5 show this approach is working very well

I'll continue with my successful strategy of sending \$4 to new opponents.

usage:

```
{
  "input_tokens": 1357,
  "output_tokens": 208,
  "reasoning_tokens": 114
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 10

In your most recent previous game round, you were the SENDER against Hana. You sent \$4.0, it became \$12.0 for them, and they returned \$7.0 to you. Your payoff was \$8.0.
Your total earnings across all rounds are \$57.5.

This round, you are the RECEIVER against Hana. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Hana sent me \$4, which became \$12 for me
- In Round 9, I sent Hana \$4 (which became \$12) and they generously returned \$7
- Hana is now reciprocating by sending me the same \$4
- I should match their generosity

Returning \$7 reciprocates their generous behavior and maintains our positive cooperative relationship.

{'return': 7.0}

Reasoning:

Looking at this situation:

- Hana sent me \$4, which became \$12 for me
- In Round 9 (most recent), I sent Hana \$4 (which became \$12) and they returned \$7
- Hana is now reciprocating by sending me the same amount
- Hana was very generous, returning \$7 instead of just \$6

I should reciprocate Hana's generosity. They returned \$7 to me when I sent them \$4, so I should return \$7 to them now that they've sent me \$4.

This maintains the cooperative relationship and matches their generosity.

{'return': 7.0}

Usage: {
 "input_tokens": 1365,
 "output_tokens": 266,
 "reasoning_tokens": 135
}